# Experience Rating in Insurance Pricing

– Lecture Notes –

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# Preface and Terms of Use

**Lecture Notes.** These notes form the core of the lecture Experience Rating in Insurance Pricing at ETH Zurich. Prerequisite to these notes is the monograph Statistical Foundations of Actuarial Learning and its Applications [205].

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# Chapter 1

# Introduction

This lecture is part of the actuarial syllabus on Statistical Modeling in Non-Life Insurance. Subsequent to the introductory class on Non-Life Insurance: Mathematics & Statistics [200], we offer an advanced data science lecture Data Analytics for Non-Life Insurance Pricing [204]. This advanced lecture forms the core of the monograph Statistical Foundations of Actuarial Learning and its Applications [205]. This monograph considers generalized linear models and deep learning for insurance pricing. Such pricing approaches can be interpreted as static insurance pricing methods because they primarily rely on *prior rating information* which is typically available at the inception of an insurance contract. The next step is to integrate *past claims history* (if available) into the insurance pricing process, which can be seen as a dynamic view. In the past, this dynamic insurance pricing view was covered by the ETH Zurich lecture A Course in Credibility Theory and its Applications by Bühlmann and Gisler [41]. The goal of these notes is to enhance the static prior rating information pricing by a dynamic claims history approach, and our aim is to offer a lively and fresh view on credibility theory by exploiting credibility methods within a deep learning environment.

We consider discrete time $t \in \mathbb{N}$, and $Y_t$ denotes the insurance claim in period $t$. Typically, we think of calendar years, but more generally a period could be any other fixed time interval. The following distinction will be useful.

- **Homogeneous** or **null model** $\mathbb{E}[Y_{T+1}]$: In a null model we do not use any covariate information, and an insurance claim $Y_{T+1}$ in calendar year $T+1$ is predicted by its expected value $\mathbb{E}[Y_{T+1}]$. This is called the homogeneous or the null model because if we have multiple insurance contracts, we do not discriminate them but we assign the identical insurance premium to all of these contracts, under the assumption that the underlying insurance claims are independent and identically distributed (i.i.d.).

- **Fixed effects model** $\mathbb{E}[Y_{T+1}|\boldsymbol{x}_{T+1}]$: A fixed effects model considers covariate information $\boldsymbol{x}_{T+1}$, available at the beginning of calendar year $T+1$, to predict the insurance claim $Y_{T+1}$ by its conditional expectation $\mu(\boldsymbol{x}_{T+1}) := \mathbb{E}[Y_{T+1}|\boldsymbol{x}_{T+1}]$. We think of $\boldsymbol{x}_{T+1}$ being prior rating information (fixed effects) that is available at the inception of the insurance contract. The underlying insurance claims $Y_{T+1}$ are then

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assumed to be heterogeneous, with the covariates $\boldsymbol{x}_{T+1}$ allowing us to discriminate them. The main problem in this setup is to find the (unknown) regression function $\boldsymbol{x}_{T+1} \mapsto \mu(\boldsymbol{x}_{T+1})$ from past data that properly discriminates the contracts. This is done, e.g., by using generalized linear models or deep neural network regression models. This is the statistical problem that is going to be presented and solved in Chapter 2.

- **Random effects model** $\mathbb{E}[Y_{T+1}|Y_{1:T}]$: A random effects model aims at optimally predicting the insurance claim $Y_{T+1}$ in calendar year $T+1$ based on the past claims history $Y_{1:T} = (Y_1, \ldots, Y_T)$. This is the view presented in Bühlmann–Gisler [41] on credibility theory which studies the question of how much credibility should be assigned to the past claims history (linked by latent random effects) to predict the future insurance claim. We are going to present this credibility view in Sections 3.2 and 4.4.

- **Mixed effects model** $\mathbb{E}[Y_{T+1}|\boldsymbol{x}_{T+1}, Y_{1:T}]$: In a mixed effects model we combine the fixed effects and the random effects view by simultaneously considering the prior rating information $\boldsymbol{x}_{T+1}$ and the past claims history $Y_{1:T}$ to optimally predict the insurance claim $Y_{T+1}$ in calendar year $T+1$. We call this consideration of joint information *experience rating*. This mixed effects view is presented in Sections 3.3, 4.3 and 4.5.

- **Random and mixed effects models II**: There are different variants of the definition of random and mixed effects models which may make terminology confusing, but which are all valid definitions. We remark the following points:

  - In the classical literature in statistics, random effects are assumed to be latent (and unknown) variables, thus, their specific realisation is not available for modeling. In this terminology, the random effects account for unmeasured variability that affect certain groups in the data; see Fisher [79], Laird–Ware [128] and Robinson [175]. In this literature, these random effects often involve another regression structure, besides the regression structure of the (known) fixed effects. In our context, the (latent) random effects will not involve any additional regression structure. We will call these unknown background variables 'latent risk factors' and denote them by $\Theta$. In this sense, our terminology is consistent with classical statistics, only we assume a simpler structure for the random effects.

  - Unobserved heterogeneity can be treated in two different ways in statistical modeling, either we integrate it as a random variable $\Theta$ or it can be integrated as an unobserved fixed parameter $\theta$; see Boucher–Denuit [28]. The random variable case $\Theta$ leads to a Bayesian setup by assuming that $\Theta$ follows a prior distribution, and its estimation involves the classical toolbox from Bayesian statistics. In the fixed parameter case, $\theta$ is treated as an unknown (fixed) parameter that is estimated from past data. In the fixed and random effects terminology of Boucher–Denuit [28], the term fixed or random qualifies the

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specific statistical method that is used to deal with the very same phenomenon of unobserved heterogeneity in the data.

In these notes, we are going to use the first terminology (and modeling) of fixed and random effects. Namely, the fixed effects correspond to known covariates $\boldsymbol{x}_{T+1}$ that enter a regression structure, and the random effects (latent risk factors) $\Theta$ are used to connect the time-series of observations $(Y_1, \ldots, Y_T; Y_{T+1})$. If both features are present in the statistical model we call it a mixed effects model.

There is another distinction that we need to make. In order for random and mixed effects models to be effective and useful, we naturally need dependence within the responses $(Y_1, \ldots, Y_T; Y_{T+1})$, otherwise past claims history will not have any predictive power for future claims. It is precisely the random effects (latent risk factor) structure that provides us with this dependence along the time axis. The most popular approach is known as the Kalman filter [118] and the crucial mathematical technique behind this modeling approach is Bayes' rule [18]. We make the following distinction.

- **Static random/mixed effects model**: The static case assumes that there is a common latent risk factor $\Theta$, and conditionally, given this latent risk factor $\Theta$, the sequence of claims $(Y_1, \ldots, Y_T; Y_{T+1})$ decouples into (conditionally) independent components. This is the situation of the Bühlmann–Straub model [42], and we study this static case in Chapter 3.

- **Dynamic random/mixed effects model:** The dynamic case assumes that there is a latent risk factor process $(\Theta_t)_{t\geq 1}$ that drives the claims. This may, e.g., be an auto-regressive process of order one, known as AR(1) process. Conditionally given this latent risk factor process $(\Theta_t)_{t\geq 1}$, one then assumes that the claims $(Y_t)_{t\geq 1}$ are independent, and that the conditional distribution of the response $Y_t|_{(\Theta_t)_{t\geq 1}} \overset{(d)}{=} Y_t|_{\Theta_t}$ only depends on the $t$-th component of the latent risk factor process, for all $t \geq 1$. Such a dynamic random effects assumption is of Kalman filter [118] type. We are also going to present a slightly different dynamic approach, namely, in Chapter 4 we present models that have a feedback loop from the observations $(Y_t)_{t\geq 1}$ to the latent risk factor process $(\Theta_t)_{t\geq 1}$. Intuitively, this is a reasonable assumption in an insurance context, and practically, it leads to surprisingly tractable dynamic experience rating models.

- **Deep experience rating model**: The previous two items study full distributional models. In Chapter 5 we present non-parametric approaches that are not based on any distributional assumptions, but rather on deep learning algorithms that exploit flexible regression architectures to achieve maximal predictive power w.r.t. a given objective function. These deep experience rating models encompass all of the previously considered distributional experience rating models, but they are generally more flexible because they do not require analytical tractability. In fact, they are so flexible that they may integrate any information beyond classical fixed effects and past claims history, e.g., they allow one to consider telematics data or medical reports (using the corresponding time-series and text recognition features).

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Our models are supported by data and code. We use a synthetic data set that has been simulated from a statistical model that was fitted to real data. The synthetic data is described in Appendix A, and it can be downloaded from:

`https://github.com/wueth/Experience-Rating-in-Insurance-Pricing`

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# Chapter 2

# Prior Rating Information Regression Models

## 2.1 Literature overview

Fixed covariates regression models are the most commonly used models in statistical and predictive modeling. In actuarial pricing these fixed covariates regression models are also called *prior rating information regression models* because they only use information that is available at the inception of an insurance contract, such as age of policyholder, location and value of property insured, type and age of vehicle, etc. In particular, prior rating information regression models do not consider past claims experience (even if available). These regression models include generalized linear models (GLMs), generalized additive models (GAMs), neural network regression models and gradient boosting machines (GBMs). This chapter mainly follows our monograph Wüthrich–Merz [205], and we also use the notation of that monograph.

GLMs build the foundation of regression modeling. GLMs comprise the Gaussian linear regression model, the Poisson GLM, the gamma GLM, Tweedie's GLMs as well as logistic and multinomial classification models. GLMs have been introduced in the 1970s by Nelder–Wedderburn [151]; the classic reference on GLMs is the book of McCullagh–Nelder [146]. These days, GLMs are used in almost all fields of statistical modeling, both for inference and for predictive modeling, we refer to Shmueli [180] for an interesting discussion on explain vs. predict. Popular references in actuarial science on GLMs are, e.g., the two textbooks of Frees [84] and of Ohlsson–Johansson [154].

GLMs are based on the framework of the exponential dispersion family (EDF). The EDF is a very rich family of distributions. This family of distributions is particularly useful and interesting because its structural form is aligned with maximum likelihood estimation (MLE) of the model parameters. MLE is the crucial model fitting technique for GLMs. However, in actuarial science also the balance property and the property of unbiasedness are of crucial importance. These are not generally fulfilled in MLE (only asymptotic unbiasedness). We are going to discuss this in detail in the present chapter. The EDF has been introduced by Sir Fisher [80], and mathematical key references for the EDF are Jørgensen [113, 114, 115] and Barndorff-Nielsen [17], for the balance property we refer to Wüthrich [201] and Lindholm–Wüthrich [138].

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Neural network regression models can be seen as an extension of GLMs. Neural network regression models can cope with non-linear regression structures and interaction terms, as well as with time-series data and with unstructured data such as images and texts. Key references on the architecture and the use of neural networks in statistical modeling are Goodfellow et al. [95] and Hastie et al. [103]. Fitting of neural networks is typically based on the gradient descent algorithm using back-propagation; see Rumelhart et al. [176], which has been co-authored by the Noble Prize laureate Geoffrey Hinton. For applications and tutorials of neural networks in actuarial science we refer to Richman [167, 168], Noll et al. [153], Ferrario et al. [78] and Wüthrich–Merz [205].
Besides neural networks, GBMs belong to the most powerful machine learning techniques these days when dealing with tabular input data, we especially mention XGBoost of Chen–Guestrin [48] and LightGBM of Ke et al. [120]. GBMs have their roots in the work of Valiant [193], Kearns–Valiant [121, 122], Schapire [178], Freund [85] and Freund–Schapire [86], and they are based on the regression tree framework of Breiman et al. [37] For tutorials in actuarial science on GBMs we refer to Ferrario–Hämmerli [77], Lorentzen–Mayer [141] and Mayer et al. [145]. Moreover, we refer to our lecture notes Wüthrich–Buser [204] and to the textbook of Denuit et al. [63] for GBMs. In these notes, we are not going to study GBMs because they do not easily generalize to the use of time-series data or of unstructured data.

## 2.2 Exponential dispersion family

### 2.2.1 Definition and properties

We give a brief introduction to the EDF. This introduction is taken from our monograph Wüthrich–Merz [205, Chapter 2], and we also use the notation of that reference. The (single-parameter) EDF is given by random variables $Y$ having densities of the form

$$Y \;\sim\; f(y;\theta,v/\varphi)\,d\nu(y) = \exp\left\{\frac{y\theta-\kappa(\theta)}{\varphi/v} + a(y;v/\varphi)\right\}\,d\nu(y), \tag{2.1}$$

where $v>0$ is a given weight (volume, exposure), $\varphi>0$ is the dispersion parameter, $\theta\in\boldsymbol{\Theta}$ is the *canonical parameter*, $\kappa:\boldsymbol{\Theta}\to\mathbb{R}$ is the *cumulant function*, $\nu$ is a $\sigma$-finite measure on $\mathbb{R}$, and $a(\cdot;\cdot)$ is a normalizing function (not depending on $\theta$) so that this density (2.1) integrates to one w.r.t. $\nu$. The support of $Y$ is determined by the choice of the $\sigma$-finite measure $\nu$. This support is independent of the choice of the canonical parameter $\theta\in\boldsymbol{\Theta}$, but it may depend on the choice of $v/\varphi>0$. Thus, in fact, we have a $\sigma$-finite measure $\nu=\nu_{v/\varphi}$ on $\mathbb{R}$, but for notational convenience we drop the lower index.

We write $Y\sim\mathrm{EDF}(\theta;v/\varphi,\kappa,\nu)$ for a random variable $Y$ having EDF density (2.1).

The cumulant function $\kappa:\boldsymbol{\Theta}\to\mathbb{R}$ lives on the *effective domain* $\boldsymbol{\Theta}\subseteq\mathbb{R}$ which (by construction) is an interval; see Wüthrich–Merz [205, Section 2.2.1]. To exclude any trivial cases, we assume that this interval $\boldsymbol{\Theta}$ has a non-empty interior $\mathring{\boldsymbol{\Theta}}\neq\emptyset$. It then follows that the cumulant function $\kappa$ is *smooth* and *convex* on this interior $\mathring{\boldsymbol{\Theta}}$ of $\boldsymbol{\Theta}$. Moreover, if $\nu$ is not a single-point measure (which we will exclude throughout), we have *strict convexity* of $\kappa$ on $\mathring{\boldsymbol{\Theta}}$, that is, $\kappa''(\theta)>0$ for all $\theta\in\mathring{\boldsymbol{\Theta}}$. This implies that we can



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invert $\kappa'$ on $\mathring{\boldsymbol{\Theta}}$. This inverse $(\kappa')^{-1}$ is called *canonical link* of the chosen EDF, and we use the notation $h = (\kappa')^{-1}$ for this canonical link.
Choose canonical parameter $\theta \in \mathring{\boldsymbol{\Theta}}$ and $r \in \mathbb{R}$ such that $\theta + r\varphi/v \in \mathring{\boldsymbol{\Theta}}$. The random variable $Y \sim \mathrm{EDF}(\theta; v/\varphi, \kappa, \nu)$ has moment generating function

$$M_Y(r) = \mathbb{E}\left[\exp\{rY\}\right] = \exp\left\{\frac{v}{\varphi}\left(\kappa(\theta + r\varphi/v) - \kappa(\theta)\right)\right\}.$$

This implies for the first two moments of $Y$

$$\mu = \mu(\theta) = \mathbb{E}\left[Y\right] = \kappa'(\theta) \qquad \text{and} \qquad \mathrm{Var}\left(Y\right) = \frac{\varphi}{v}\kappa''(\theta) > 0. \tag{2.2}$$

The former is the motivation to consider the canonical link $h = (\kappa')^{-1}$. It satisfies

$$h(\mu) = h\left(\mathbb{E}\left[Y\right]\right) = h(\kappa'(\theta)) = \theta. \tag{2.3}$$

This gives us a one-to-one correspondence between the interior $\mathring{\boldsymbol{\Theta}}$ of the effective domain $\boldsymbol{\Theta}$ and the *mean parameter space* (also called *dual parameter space*) defined by

$$\mathcal{M} = \left\{\kappa'(\theta);\ \theta \in \mathring{\boldsymbol{\Theta}}\right\} = \kappa'\left(\mathring{\boldsymbol{\Theta}}\right). \tag{2.4}$$

**Remarks 2.1.**

- Often, it is more convenient to work with the mean parameter space $\mathcal{M}$ than with the interior of the effective domain $\mathring{\boldsymbol{\Theta}}$, because the former has an intuitive meaning. Due to the one-to-one correspondence, we can always switch between the two parametrizations.

- If we work with the random variable $Y \sim \mathrm{EDF}(\theta; v/\varphi, \kappa, \nu)$, we consider the so-called *reproductive form* of $Y$ which has mean and variance given by (2.2). We can also work with the *additive form* defined by $Z = vY/\varphi$. The additive form behaves as

$$\mathbb{E}\left[Z\right] = v\kappa'(\theta)/\varphi \qquad \text{and} \qquad \mathrm{Var}\left(Z\right) = v\kappa''(\theta)/\varphi > 0; \tag{2.5}$$

  we refer to Jørgensen [115, Section 3.1]. We give an example. Set $\varphi = 1$. We can have claim counts $Z$ (additive form) and claim frequencies $Y = Z/v$ (reproductive form), where a frequency is always understood relative to a suitable exposure $v > 0$. This is particularly useful in a Poisson claim counts model, where different insurance policies can have different time exposures $v > 0$, but the same expected frequency $\mu = \mu(\theta)$.

- By construction, the support of $Y \sim \mathrm{EDF}(\theta; v/\varphi, \kappa, \nu)$ is independent of the choice of the canonical parameter $\theta \in \boldsymbol{\Theta}$. Denote by $\overline{\mathrm{conv}}(Y)$ the convex closure of the support of the random variable $Y$. Barndorff-Nielsen [17, Theorem 9.2] proves that $\overline{\mathrm{conv}}(Y) = \overline{\mathcal{M}}$ if and only if the cumulant function $\kappa$ is so-called *steep* at the boundary of the effective domain $\boldsymbol{\Theta}$; for the precise definition of steepness we refer to Barndorff-Nielsen [17, formula (20) in Section 8.1]. All frequently used EDF examples fulfill this steepness property. Steepness is important in parameter estimation because it implies that the support of the observations $Y$ is aligned with the mean parameter space $\mathcal{M}$. That is, if $Y$ takes a value in the interior of $\overline{\mathrm{conv}}(Y)$, we can use $Y$ as a (proper) estimate $\widehat{\mu} = Y \in \mathcal{M}$ for $\mu$, with a possibly degenerate



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solution if $Y$ is in the boundary of $\overline{\text{conv}}(Y) = \overline{\mathcal{M}}$. For this reason, we throughout assume that all considered cumulant functions $\kappa$ are steep at the boundary of the effective domain $\boldsymbol{\Theta}$; Table 2.1, below, gives commonly used examples of steep cumulant functions.

- Within the EDF we have the *convolution formula* which plays a crucial role in analyzing *sufficient statistics*. The convolution formula states the following aggregation property; see Wüthrich–Merz [205, Corollary 2.15]. Assume that $Y_i \sim \text{EDF}(\theta; v_i/\varphi, \kappa, \nu_i)$ are independent, for $1 \leq i \leq n$, with common canonical parameter $\theta \in \mathring{\boldsymbol{\Theta}}$. Define the aggregated weight $v_+ = \sum_{i=1}^n v_i$. The aggregated response in its reproductive form satisfies

$$\bar{Y} = \frac{1}{v_+} \sum_{i=1}^n v_i Y_i \ \sim \ \text{EDF}(\theta; v_+/\varphi, \kappa, \nu_+). \tag{2.6}$$

  This is proved by calculating the corresponding moment generating function.

- There are different characterizations of the EDF. Assume a fixed $\sigma$-finite measure $\nu$ on $\mathbb{R}$, given mean parameter space $\mathcal{M}$ and a given variance function of an EDF

$$\mu \in \mathcal{M} \mapsto V(\mu) > 0. \tag{2.7}$$

  Then, the cumulant function $\kappa$ is fully determined by the variance function $V$ on $\mathcal{M}$, and we have $V(\mu) = \kappa''(h(\mu))$; see Jørgensen [115, Theorem 2.11] and Bar-Lev–Kokonendji [14, Sections 2 and 3]. For the variance function we also refer to (2.2) in combination with the canonical link (2.3), see also (2.15) below.

  Remark that the previous statement requires that $V$ is a (valid) variance function on $\mathcal{M}$ of an EDF, and in that case Bar-Lev–Kokonendji [14, formulas (9)-(10)] allow to reconstruct the cumulant function and the canonical parameter.

■

### 2.2.2 Maximum likelihood estimation

The functional form of the EDF is especially convenient for parameter estimation using MLE methods. Assume the responses $Y_i \sim \text{EDF}(\theta; v_i/\varphi, \kappa, \nu_i)$ are independent, for $1 \leq i \leq n$, with steep cumulant function $\kappa$. The latter implies that the convex closures $\overline{\text{conv}}(Y_i)$ of the supports of $Y_i$ coincide with the closure of the mean parameter space $\overline{\mathcal{M}}$, see Remarks 2.1. We consider the joint log-likelihood function of the given (independent) observations $\boldsymbol{Y} = (Y_1, \ldots, Y_n)^\top$

$$\theta \in \boldsymbol{\Theta} \ \mapsto \ \ell_{\boldsymbol{Y}}(\theta) = \sum_{i=1}^n \frac{Y_i\theta - \kappa(\theta)}{\varphi/v_i} + a(Y_i; v_i/\varphi). \tag{2.8}$$

The MLE $\widehat{\theta}^{\text{MLE}}$ of the canonical parameter $\theta$ is found by maximizing the log-likelihood function $\ell_{\boldsymbol{Y}}(\theta)$ in $\theta$, that is, subject to existence,

$$\widehat{\theta}^{\text{MLE}} \ \in \ \underset{\theta \in \boldsymbol{\Theta}}{\arg\max} \ \ell_{\boldsymbol{Y}}(\theta). \tag{2.9}$$



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This MLE does not necessarily exist. In fact, the effective domain $\boldsymbol{\Theta}$ can be an open set, and it may happen that the optimal parameter is at the boundary of this effective domain, hence, in this case it does not exist. This is the case, e.g., if all observations $Y_i$ are equal to zero in a Poisson EDF model. However, in the case the MLE exists, it is unique as we will just show. We solve (2.9). The *score equation* is given by

$$\frac{\partial}{\partial\theta}\ell_{\boldsymbol{Y}}(\theta) \;=\; \sum_{i=1}^n \frac{v_i}{\varphi}\left[Y_i-\kappa'(\theta)\right] \;=\; \sum_{i=1}^n \frac{v_i}{\varphi}\left[Y_i-\mu(\theta)\right] \;=\; 0. \tag{2.10}$$

we refer to (2.2) for the mean $\mu=\mu(\theta)=\kappa'(\theta)$. The second derivative is

$$\frac{\partial^2}{\partial\theta^2}\ell_{\boldsymbol{Y}}(\theta) \;=\; -\sum_{i=1}^n \frac{v_i}{\varphi}\kappa''(\theta)<0,$$

for all $\theta\in\mathring{\boldsymbol{\Theta}}$. Hence, we have a strictly concave maximization problem, which provides uniqueness of the MLE $\widehat{\theta}^{\mathrm{MLE}}$ of $\theta$ in case it exists. There remains to solve the score equation (2.10). This provides us with the following corollary.

**Corollary 2.2.** *The MLE $\widehat{\theta}^{\mathrm{MLE}}$ of $\theta$ under the log-likelihood function choice* (2.8) *is given by*

$$\widehat{\theta}^{\mathrm{MLE}}=h\left(\frac{\sum_{i=1}^n v_iY_i}{\sum_{i=1}^n v_i}\right),$$

*subject to existence, and with canonical link $h=(\kappa')^{-1}$.*

**Remark 2.3.** The proof of Corollary 2.2 is a straightforward consequence of the score equation (2.10). Under steepness of the cumulant function $\kappa$, we know that the convex closures $\overline{\mathrm{conv}}(Y_i)$ of the supports of $Y_i$ coincide with the closure of the mean parameter space $\overline{\mathcal{M}}$. If the weighted average $\bar{Y}=\sum_{i=1}^n v_iY_i/\sum_{i=1}^n v_i\in\mathcal{M}$, we have a proper MLE due to the one-to-one correspondence between $\mathcal{M}$ and $\mathring{\boldsymbol{\Theta}}$, see (2.4). If $\bar{Y}\in\overline{\mathcal{M}}\setminus\mathcal{M}$ we may only receive a degenerate solution (depending on the boundary of $\boldsymbol{\Theta}$). We are going to discuss this further in the mean parametrization. ■

In view of (2.2), we have the one-to-one correspondence between the canonical parameter and the mean parameter

$$\theta\in\mathring{\boldsymbol{\Theta}}\;\mapsto\;\mu(\theta)=\kappa'(\theta)\qquad\text{and}\qquad\mu\in\mathcal{M}\;\mapsto\;\theta(\mu)=h(\mu).$$

This allows us to also study the MLE $\widehat{\mu}^{\mathrm{MLE}}$ in the mean parametrization $\mu$, see (2.8)-(2.9),

$$\widehat{\mu}^{\mathrm{MLE}} \;=\; \underset{\mu\in\overline{\mathcal{M}}}{\arg\max}\;\sum_{i=1}^n \frac{Y_ih(\mu)-\kappa(h(\mu))}{\varphi/v_i}+a(Y_i;v_i/\varphi). \tag{2.11}$$

This provides us with score equation in the mean parametrization

$$\frac{\partial}{\partial\mu}\ell_{\boldsymbol{Y}}(\theta(\mu)) \;=\; \sum_{i=1}^n \frac{v_i}{\varphi}\left[Y_i-\mu\right]\frac{\partial h(\mu)}{\partial\mu} \;=\; \sum_{i=1}^n \frac{v_i}{\varphi}\left[Y_i-\mu\right]\frac{1}{\kappa''(h(\mu))} \;=\; 0, \tag{2.12}$$

recall $\kappa$ is strictly convex. This gives us the following corollary.



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**Corollary 2.4.** *Assume $\kappa$ is steep. The MLE $\widehat{\mu}^{\mathrm{MLE}}$ of $\mu$ for the log-likelihood function* (2.8) *with $\theta = \theta(\mu)$ is given by*

$$\widehat{\mu}^{\mathrm{MLE}} = \bar{Y} = \frac{\sum_{i=1}^n v_i Y_i}{\sum_{i=1}^n v_i} \ \in \ \overline{\mathcal{M}}. \tag{2.13}$$

From the convolution formula (2.6) we receive unbiasedness of the MLE $\widehat{\mu}^{\mathrm{MLE}}$ for $\mu \in \mathcal{M}$

$$\mathbb{E}\left[\widehat{\mu}^{\mathrm{MLE}}\right] = \kappa'(\theta) = \mu, \tag{2.14}$$

and we can quantify the estimation uncertainty by

$$\mathrm{Var}\left(\widehat{\mu}^{\mathrm{MLE}}\right) = \frac{\varphi}{\sum_{i=1}^n v_i}\kappa''(\theta) = \frac{\varphi}{\sum_{i=1}^n v_i}\kappa''(h(\mu)) \ > \ 0. \tag{2.15}$$

This estimation uncertainty goes to zero for $\sum_{i=1}^n v_i \to \infty$.

**Corollary 2.5** (balance property)**.** *The balance property holds*

$$\sum_{i=1}^n v_i \widehat{\mu}^{\mathrm{MLE}}/\varphi = \sum_{i=1}^n v_i Y_i/\varphi = \sum_{i=1}^n Z_i. \tag{2.16}$$

**Remark 2.6** (balance property)**.** This balance property (2.16) is very important in insurance pricing, and in financial applications more generally. It is a stronger (point-wise) property than the unbiasedness property (2.14). We briefly discuss this, and we will come back to this several times below. The notion of the *balance property* has been introduced and studied by Bühlmann–Gisler [41, Theorem 4.5] in the context of the homogeneous linear credibility estimator; we also refer to Wüthrich [201] and Lindholm–Wüthrich [138]. It states the fact that the totally charged premiums in the estimated model, left-hand side of (2.16), equals the totally observed claim amount, right-hand side of (2.16). This implies unbiasedness (2.14), but actually it says that the estimated premiums $(v_i\widehat{\mu}^{\mathrm{MLE}}/\varphi)_{i=1}^n$ are nothing else than a *reallocation* of the totally observed claims $(Z_i)_{i=1}^n$, keeping the total portfolio claim unchanged; here we use the additive form (observation scale). Under stationarity this furthermore implies that the premium level charged over the entire portfolio for the next period will be right on average. The balance property is a point-wise property because it holds for every realisation of the total claim amount in (2.16), and this implies unbiasedness which is an average property under the expectation operator $\mathbb{E}$ of $\mathbb{P}$.

MLE does not lead to unbiased models in general. This may question the MLE principle for insurance pricing as it does not generally give us any guarantees that the fitted model meets the correct premium level. MLE is asymptotically unbiased, but this may not be sufficient for insurance pricing on small or heterogeneous portfolios. From this viewpoint, the method of moments fitting procedure seems more suitable for insurance pricing as it enforces unbiasedness by construction. In other words, at this point, the objectives of statisticians and actuaries may differ, and one should always carefully analyze the bias property of fitted models in financial applications.

Another interesting point of the balance property is that it is model-free. Note that we estimate the premiums $\widehat{\mu}^{\mathrm{MLE}}$ with MLE within the EDF, and the balance property (2.16) gives us unbiasedness on the portfolio level. First, this unbiasedness is received

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*without* knowing the correct price level! Second, this unbiasedness is received even if the selected EDF is completely misspecified, i.e., if the claims $Z_i$ have been generated by a completely different model. This is precisely the strength of the balance property, i.e., we neither need to explicitly know the right price level nor the correct data generating mechanism to receive this unbiasedness. The key to this is that the estimated premiums are a reallocation of the totally observed claim amount. ■

**Remark 2.7** (Fisher's information)**.** *Fisher's information* w.r.t. $\theta \in \mathring{\boldsymbol{\Theta}}$ for responses $\boldsymbol{Y}$ is given by

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\ell_{\boldsymbol{Y}}(\theta)\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ell_{\boldsymbol{Y}}(\theta)\right] \;=\; \sum_{i=1}^n \frac{v_i}{\varphi}\kappa''(\theta) > 0.$$

We can also consider Fisher's information w.r.t. $\mu \in \mathcal{M}$. This requires a change of variable $\mu \mapsto \theta = \theta(\mu)$. The Jacobian of this variable transformation is given by

$$J(\mu) = \frac{\partial\theta(\mu)}{\partial\mu} = \frac{\partial h(\mu)}{\partial\mu} = \kappa''(h(\mu))^{-1}.$$

Fisher's information w.r.t. $\mu \in \mathcal{M}$ is then given by

$$I(\mu) = J(\mu)\, I(\theta(\mu))\, J(\mu) = \sum_{i=1}^n \frac{v_i}{\varphi}\kappa''(h(\mu))^{-1} > 0.$$

This allows us to express parameter estimation uncertainty in terms of the inverse of Fisher's information $I(\mu)$, see (2.15),

$$\mathrm{Var}\left(\widehat{\mu}^{\mathrm{MLE}}\right) = I(\mu)^{-1}.$$

In fact, this corresponds to the classical consistency and asymptotic normality results of MLE (as the number of observations and $\sum_{i=1}^n v_i$ go to infinity), with the asymptotic variance described by the inverse of Fisher's information $I(\mu)$ w.r.t. $\mu$. This is an infinite sample statement. On finite samples we have that the MLE (2.13) is uniformly minimum variance unbiased (UMVU) here, because it meets the Cramér–Rao information bound, see Lehmann [133, Section 2.6] and Wüthrich–Merz [205, Section 3.3]. However, we emphasize that this UMVU does not generally hold for MLE, and therefore also other methods may be suitable for model fitting in insurance pricing, see Remark 2.6. ■

MLE (2.11) considers a log-likelihood maximization problem. We can turn this into a loss minimization problem by considering the deviance loss function within the EDF. For given observation $Y_i$, we define the *unit deviance* in mean $\mu \in \mathcal{M}$ by

$$\begin{aligned} \mathfrak{d}\left(Y_i, \mu\right) &= 2\frac{\varphi}{v_i}\left(\ell_{Y_i}(Y_i) - \ell_{Y_i}(\mu)\right) \\ &= 2\left(Y_i h\left(Y_i\right) - \kappa\left(h\left(Y_i\right)\right) - Y_i h\left(\mu\right) + \kappa\left(h\left(\mu\right)\right)\right) \;\geq\; 0, \end{aligned} \tag{2.17}$$

where we interpret the log-likelihood $\mu \mapsto \ell_{Y_i}(\mu)$ of $Y_i$ as a function of the mean parameter $\mu$. The first term $\ell_{Y_i}(Y_i)$ on the right-hand side of (2.17) evaluates the log-likelihood in the MLE $\widehat{\mu}_i^{\mathrm{MLE}} = Y_i$ if we have only one observation $Y_i$ for model fitting, see also (2.13). Applying this to each response $Y_i$ individually gives us the so-called *saturated model*, where each observation has its own MLE parameter $\widehat{\mu}_i^{\mathrm{MLE}} = Y_i$; this justifies the lower bound zero in (2.17).



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**Remark 2.8.** To make the unit deviance (2.17) well-defined also in cases where $Y_i \in \overline{\mathcal{M}} \setminus \mathcal{M}$, and under the assumption that $\kappa$ is steep, we set at the boundary of the mean parameter space

$$\mathfrak{d}\left(Y_i, \mu\right) \;=\; 2\left(\sup_{\theta' \in \boldsymbol{\Theta}} \left(Y_i\theta' - \kappa\left(\theta'\right)\right) - Y_i h\left(\mu\right) + \kappa\left(h\left(\mu\right)\right)\right). \tag{2.18}$$

■

Using the unit deviances $\mathfrak{d}(Y_i, \mu)$ over all observations $1 \leq i \leq n$, we define the *deviance loss function* by

$$\begin{aligned} \mu \;\mapsto\; \mathfrak{D}(\boldsymbol{Y}, \mu) \;&=\; \frac{1}{n}\sum_{i=1}^{n} \frac{v_i}{\varphi}\, \mathfrak{d}\left(Y_i, \mu\right) \\ &=\; \frac{2}{n}\Big(\ell_{\boldsymbol{Y}}(\boldsymbol{Y}) - \ell_{\boldsymbol{Y}}(\mu)\Big) \\ &=\; \frac{2}{n}\sum_{i=1}^{n} \frac{v_i}{\varphi}\Big(Y_i h\left(Y_i\right) - \kappa\left(h\left(Y_i\right)\right) - Y_i h\left(\mu\right) + \kappa\left(h\left(\mu\right)\right)\Big) \;\geq\; 0, \end{aligned} \tag{2.19}$$

at the boundary we use (2.18). The lower bound zero is attained if and only if $\mu = Y_i$ for all $1 \leq i \leq n$. This allows us to formulate the MLE problem (2.11) equivalently as a loss minimization problem

$$\widehat{\mu}^{\mathrm{MLE}} \;=\; \arg\min_{\mu \in \overline{\mathcal{M}}}\; \mathfrak{D}(\boldsymbol{Y}, \mu). \tag{2.20}$$

An advantage of this loss minimization problem is that the loss function is naturally calibrated, meaning that it cannot be smaller than zero. In applications this may be an advantage as we may receive a ranking of models that is anchored in zero.

**Remark 2.9** (Bregman divergence)**.** The unit deviance (2.17) is a Bregman divergence (up to a multiplicative factor of 2). We set $\psi(\mu) = \mu h(\mu) - \kappa(h(\mu))$. This provides us with first and second derivatives in $\mu \in \mathcal{M}$

$$\psi'(\mu) = h(\mu) \qquad \text{and} \qquad \psi''(\mu) = h'(\mu) = 1/\kappa''(h(\mu)) > 0.$$

The Bregman divergence [36] for this convex function $\psi$ is given by

$$\begin{aligned} D_\psi(Y_i, \mu) \;&=\; \psi(Y_i) - \psi(\mu) - \psi'(\mu)\left(Y_i - \mu\right) \\ &=\; Y_i h(Y_i) - \kappa(h(Y_i)) + \kappa(h(\mu)) - h(\mu) Y_i \;=\; \frac{1}{2}\,\mathfrak{d}(Y_i, \mu) \;\geq\; 0. \end{aligned}$$

Thus, (2.20) can be interpreted as a Bregman divergence minimization problem, and this Bregman divergence, in case of the unit deviance, is equivalent to the Kullback–Leibler (KL) divergence of the corresponding EDF; we refer to Wüthrich–Merz [205, Section 2.3.2]. ■

### 2.2.3 Examples of the exponential dispersion family

Important examples of the EDF are the Bernoulli, the Poisson, the negative binomial, the Gaussian, the gamma, the inverse Gaussian and Tweedie's distributions $p > 1$. We



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| EDF distribution | $\kappa(\theta)$ | $\boldsymbol{\Theta}$ | $\mathfrak{d}(y,\mu)$ |
|---|---|---|---|
| Bernoulli | $\log(1+e^\theta)$ | $\mathbb{R}$ | $2\left(-y\log\mu-(1-y)\log(1-\mu)\right)$ |
| Poisson | $e^\theta$ | $\mathbb{R}$ | $2\left(\mu-y-y\log(\mu/y)\right)$ |
| negative binomial | $-\log(1-e^\theta)$ | $(-\infty,0)$ | $2\left(y\log\left(\frac{y}{\mu}\right)-(y+1)\log\left(\frac{y+1}{\mu+1}\right)\right)$ |
| Gaussian | $\theta^2/2$ | $\mathbb{R}$ | $(y-\mu)^2$ |
| gamma | $-\log(-\theta)$ | $(-\infty,0)$ | $2\left((y-\mu)/\mu+\log(\mu/y)\right)$ |
| inverse Gaussian | $-\sqrt{-2\theta}$ | $(-\infty,0]$ | $(y-\mu)^2/(\mu^2 y)$ |
| Tweedie's CP, $p\in(1,2)$ | $\frac{((1-p)\theta)^{\frac{2-p}{1-p}}}{2-p}$ | $(-\infty,0)$ | $2\left(y\frac{y^{1-p}-\mu^{1-p}}{1-p}-\frac{y^{2-p}-\mu^{2-p}}{2-p}\right)$ |
| Tweedie, $p>2$ | $\frac{((1-p)\theta)^{\frac{2-p}{1-p}}}{2-p}$ | $(-\infty,0]$ | $2\left(y\frac{y^{1-p}-\mu^{1-p}}{1-p}-\frac{y^{2-p}-\mu^{2-p}}{2-p}\right)$ |

Table 2.1: Selected EDF examples with steep cumulant functions $\kappa$.

briefly give their characteristics, this summary is taken from McCullagh–Nelder [146, Table 2.1], Jørgensen [115, Table 4.1] and Wüthrich–Merz [205, Tables 2.1 and 4.1].
Table 2.1 provides frequently used examples from the EDF. The first three examples are count random variables, the second three examples are absolutely continuous examples, Tweedie's CP is Tweedie's compound Poisson model which is absolutely continuous on $\mathbb{R}_+$ with a point mass in zero, and Tweedie's distributions with $p>2$ are absolutely continuous on $\mathbb{R}_+$. All these examples have a steep cumulant function $\kappa$, and, henceforth, the convex closure $\overline{\text{conv}}(Y)$ of the support of the responses $Y$ coincides with the closure of the mean parameter space $\overline{\mathcal{M}}$. Moreover, we highlight that the unit deviance of the Gaussian model is the square loss function, thus, if we perform model fitting under square losses we implicitly perform MLE within a Gaussian model.
We close this section with a short note on Tweedie's models. Tweedie's models have the property that the variance function introduced in (2.7) is a power of its mean, that is,

$$\text{Var}(Y)=\frac{\varphi}{v}\,\kappa''(h(\mu))=\frac{\varphi}{v}\,\mu^p.$$

Within the EDF all parameters $p\in\mathbb{R}\setminus(0,1)$ allow for an EDF distribution, see Jørgensen [114, Theorem 2]. Jørgensen [114] credits this EDF of power variance distributions to Tweedie [192], but it has been introduced simultaneously, independently and in a more complete manner by Bar-Lev–Enis [13], these authors also provide the correct results for $p<0$ (having non-steep cumulant functions $\kappa$).

## 2.3 Generalized linear model

### 2.3.1 Definition and estimation of generalized linear models

We lift the EDF models introduced in the previous section to GLMs. Assume that the responses $Y_i\sim\text{EDF}(\theta_i;v_i/\varphi,\kappa,\nu_i)$ are independent, $1\le i\le n$, with steep cumulant function $\kappa$ and with canonical parameters $\theta_i\in\boldsymbol{\Theta}$. We now allow for heterogeneity within the random vector $\boldsymbol{Y}=(Y_1,\dots,Y_n)^\top$ because the individual responses $Y_i$ may have different means

$$\mu_i=\mathbb{E}[Y_i]=\kappa'(\theta_i),$$



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for $1 \leq i \leq n$. Additionally, we assume that every response variable $Y_i$ is supported by *covariate*[1] information $\boldsymbol{x}_i \in \mathcal{X}$, for a given *covariate space* $\mathcal{X} \subseteq \{1\} \times \mathbb{R}^q$. The first component of $\boldsymbol{x}_i = (1, x_{i,1}, \ldots, x_{i,q})^\top$ is equal to 1, it will be used for taking care of the intercept[2] of the regression model. The remaining components of $\boldsymbol{x}_i$ describe the covariate information of instance $i$.[3] This covariate information is typically available at the inception of the insurance contract. Therefore, the resulting (regression) model is called a *prior (rating) information regression model*.
A GLM is received by choosing a smooth and strictly monotone link function $g$, and by assuming that all instances $1 \leq i \leq n$ have the regression structure

$$g(\mu_i) = g(\mu(\boldsymbol{x}_i)) = g(\mathbb{E}[Y_i | \boldsymbol{x}_i, v_i]) = \langle \boldsymbol{\beta}, \boldsymbol{x}_i \rangle = \beta_0 + \sum_{j=1}^{q} \beta_j x_{i,j}, \tag{2.21}$$

with GLM regression parameter $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_q)^\top \in \mathbb{R}^{q+1}$. In this conditional expectation notation we emphasize that the distribution of the response $Y_i$ depends on the covariates $\boldsymbol{x}_i \in \mathcal{X}$ and the exposure $v_i > 0$. It is convenient to define the *design matrix* that collects all covariate information (on the rows for the different instances $i$)

$$\mathfrak{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n]^\top = \begin{pmatrix} 1 & x_{1,1} & \cdots & x_{1,q} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n,1} & \cdots & x_{n,q} \end{pmatrix} \in \mathbb{R}^{n \times (q+1)}. \tag{2.22}$$

From (2.21), the *linear predictors* are given by

$$g(\mu_i) = g(\mu(\boldsymbol{x}_i)) = g(\mathbb{E}[Y_i | \boldsymbol{x}_i, v_i]) = \langle \boldsymbol{\beta}, \boldsymbol{x}_i \rangle = (\mathfrak{X}\boldsymbol{\beta})_i, \tag{2.23}$$

and we obtain conditional mean and variance, respectively,

$$\mu(\boldsymbol{x}_i) = \mathbb{E}[Y_i | \boldsymbol{x}_i, v_i] = g^{-1} \langle \boldsymbol{\beta}, \boldsymbol{x}_i \rangle, \tag{2.24}$$

$$\mathrm{Var}(Y_i | \boldsymbol{x}_i, v_i) = \frac{\varphi}{v_i} \kappa'' \left( h \left( g^{-1} \langle \boldsymbol{\beta}, \boldsymbol{x}_i \rangle \right) \right). \tag{2.25}$$

**Remarks 2.10.**

- In general, the covariates $\boldsymbol{x}_i$ of different instances $i$ will differ, because these covariates may comprise policyholder information like age of driver, vehicle age, vehicle type, value of building, etc. If we do not have any prior rating information ($q = 0$), we are back in the homogeneous case (2.8), with $\mu = \kappa'(\theta) = g^{-1}(\beta_0)$. Therefore, this case is called the *intercept model* or the *null model*, as it only includes the intercept parameter $\beta_0 \in \mathbb{R}$.

- The conditional distribution notation (2.24)-(2.25) is useful if we think of an insurance portfolio having different insurance policies $(\boldsymbol{x}_i, v_i)_{i=1}^n$. We typically assume

[1] Covariates are called features in the machine learning community or independent variables, predictor variables or explanatory variables in the statistics community.

[2] The intercept is called bias term in the machine learning community.

[3] Sometimes we call $i$ an instance and sometimes an insurance policy. The former is more neutral, as the presented theory is not specific to actuarial science.



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that these insurance policies are i.i.d. realizations from a population distribution $(\boldsymbol{X}, V) \sim \mathbb{P}$ reflecting the properties of the insurance portfolio. We then first simulate the insurance policy characteristics $(\boldsymbol{X}, V)$ from the population distribution $\mathbb{P}$, and based on this simulation we generate the claims

$$Y\,|_{(\boldsymbol{X},V)} \;\sim\; \mathrm{EDF}\left(h\left(g^{-1}\langle\boldsymbol{\beta},\boldsymbol{X}\rangle\right); V/\varphi, \kappa, \nu\right), \tag{2.26}$$

with canonical parameter $\theta = (h \circ g^{-1})\langle\boldsymbol{\beta},\boldsymbol{X}\rangle \in \boldsymbol{\Theta}$. This reflects how the synthetic data in Appendix A has been generated, see (A.1).

■

We generally assume that the design matrix $\mathfrak{X} \in \mathbb{R}^{n\times(q+1)}$ has *full rank* $q+1 \leq n$. This is a necessary condition to receive uniqueness of the MLE $\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}$ of the regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$. For categorical covariates this requires a full rank encoding, such as the dummy coding, which we generally apply for GLMs in these notes; we refer to Wüthrich–Merz [205, Section 5.2.2]. Having a single categorical covariate $x$ that takes $L$ levels $x \in A := \{a_1, \ldots, a_L\}$, dummy coding is given by the mapping

$$x \in A \;\mapsto\; \left(\mathbb{1}_{\{x=a_2\}}, \ldots, \mathbb{1}_{\{x=a_L\}}\right)^{\top} \in \mathbb{R}^{L-1}. \tag{2.27}$$

This reflects that we choose $a_1$ as the reference level, and the GLM describes differences to this reference level.

We now turn our attention to GLM fitting. In view of (2.19)-(2.20), for the MLE of $\boldsymbol{\beta}$ we can equivalently maximize the log-likelihood function $\boldsymbol{\beta} \mapsto \ell_{\boldsymbol{Y}}(\boldsymbol{\beta})$ or minimize the deviance loss function $\boldsymbol{\beta} \mapsto \mathfrak{D}(\boldsymbol{Y}, \boldsymbol{\beta})$ in $\boldsymbol{\beta}$. In a GLM, the latter is given by

$$\mathfrak{D}(\boldsymbol{Y}, \boldsymbol{\beta}) = \frac{2}{n}\sum_{i=1}^{n}\frac{v_i}{\varphi}\left[Y_i h\left(Y_i\right) - \kappa\left(h\left(Y_i\right)\right) - Y_i h\left(g^{-1}\langle\boldsymbol{\beta},\boldsymbol{x}_i\rangle\right) + \kappa\left(h\left(g^{-1}\langle\boldsymbol{\beta},\boldsymbol{x}_i\rangle\right)\right)\right]. \tag{2.28}$$

This requires to solve the score equations

$$\begin{aligned}\nabla_{\boldsymbol{\beta}}\,\mathfrak{D}(\boldsymbol{Y}, \boldsymbol{\beta}) \;&=\; -\frac{2}{n}\sum_{i=1}^{n}\frac{v_i}{\varphi}\,\frac{Y_i - \mu(\boldsymbol{x}_i)}{\kappa''\left(h\left(\mu(\boldsymbol{x}_i)\right)\right)}\,\frac{1}{g'(\mu(\boldsymbol{x}_i))}\,\boldsymbol{x}_i \\ &=\; -\frac{2}{n}\,\mathfrak{X}^{\top}\,W(\boldsymbol{\beta})\,\boldsymbol{R}(\boldsymbol{Y}, \boldsymbol{\beta}) \;=\; \boldsymbol{0},\end{aligned} \tag{2.29}$$

with diagonal working weight matrix

$$W(\boldsymbol{\beta}) = \mathrm{diag}\left(\left(\left.\frac{\partial g(\mu)}{\partial\mu}\right|_{\mu=\mu(\boldsymbol{x}_i)}\right)^{-2}\frac{v_i}{\varphi}\,\frac{1}{\kappa''\left(h\left(\mu(\boldsymbol{x}_i)\right)\right)}\right)_{1\leq i\leq n} \;\in\; \mathbb{R}^{n\times n},$$

and residual vector

$$\boldsymbol{R} = \boldsymbol{R}(\boldsymbol{Y}, \boldsymbol{\beta}) = \left(\left.\frac{\partial g(\mu)}{\partial\mu}\right|_{\mu=\mu(\boldsymbol{x}_i)}\left(Y_i - \mu(\boldsymbol{x}_i)\right)\right)^{\top}_{1\leq i\leq n} \;\in\; \mathbb{R}^{n}.$$

**Remarks 2.11.**

- A MLE $\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}$ of $\boldsymbol{\beta}$ fulfills the score equations (2.29). If we work with the canonical link $g = h$, the minimization problem is convex. This implies



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that a MLE is unique under the canonical link choice (and under a full rank design matrix $\mathfrak{X}$); see Proposition 2.12, below. If we work with link functions $g$ different from the canonical link $h$, convexity has to be checked case by case, e.g., an inverse Gaussian GLM with log-link does not provide a convex minimization problem, see Wüthrich–Merz [205, Example 5.6].

- The score equations are usually solved numerically with Fisher's scoring method or with the iterative re-weighted least squares (IRLS) algorithm; we refer to Nelder–Wedderburn [151]. These are quasi-Newton methods that involve the negative expected Hessian $\nabla^2_{\boldsymbol{\beta}}$ of the log-likelihood $\ell_{\boldsymbol{Y}}(\boldsymbol{\beta})$ which equals Fisher's information

$$I(\boldsymbol{\beta}) = \mathbb{E}_{\mathfrak{X},\boldsymbol{v}}\left[\nabla_{\boldsymbol{\beta}}\ell_{\boldsymbol{Y}}(\boldsymbol{\beta})\left(\nabla_{\boldsymbol{\beta}}\ell_{\boldsymbol{Y}}(\boldsymbol{\beta})\right)^\top\right] = -\mathbb{E}_{\mathfrak{X},\boldsymbol{v}}\left[\nabla^2_{\boldsymbol{\beta}}\ell_{\boldsymbol{Y}}(\boldsymbol{\beta})\right] = \mathfrak{X}^\top W(\boldsymbol{\beta})\mathfrak{X}, \quad (2.30)$$

where the expectations are conditional on the covariates $\mathfrak{X}$ and the weights $\boldsymbol{v} = (v_i)_{i=1}^n$; for the null model see Remark 2.7. Fisher's information $I(\boldsymbol{\beta})$ is a positive definite matrix under the full rank assumption on $\mathfrak{X}$. This provides us with the IRLS algorithm for computing the solution(s) to the score equations (2.29). Namely, iterate for $s \geq 0$

$$\widehat{\boldsymbol{\beta}}^{(s)} \;\mapsto\; \widehat{\boldsymbol{\beta}}^{(s+1)} = \widehat{\boldsymbol{\beta}}^{(s)} + I(\widehat{\boldsymbol{\beta}}^{(s)})^{-1}\mathfrak{X}^\top W(\widehat{\boldsymbol{\beta}}^{(s)})\boldsymbol{R}(\boldsymbol{Y},\widehat{\boldsymbol{\beta}}^{(s)}). \quad (2.31)$$

We refer to Wüthrich–Merz [205, Section 5.1.4].

■

### 2.3.2 Balance property

We have already emphasized the importance of the balance property in insurance pricing in Remark 2.6. The balance property means that the resulting individual premiums are obtained by a reallocation of the totally observed claims. The balance property caries over to the GLM case if we work with the canonical link $g = h$. We revisit Wüthrich–Merz [205, Corollary 5.7]; see also Wüthrich [201] and Lindholm–Wüthrich [138].

**Proposition 2.12** (balance property)**.** *Assume we have independent responses $Y_i \sim \mathrm{EDF}(\theta_i; v_i/\varphi, \kappa, \nu_i)$, $1 \leq i \leq n$, with steep cumulant function $\kappa$. Moreover, assume that each response $Y_i$ is equipped with a covariate $\boldsymbol{x}_i \in \mathcal{X} \subseteq \{1\} \times \mathbb{R}^q$ providing a full rank design matrix $\mathfrak{X}$. The linear predictors are assumed to satisfy the GLM equation* (2.23) *with canonical link $g = h = (\kappa')^{-1}$. Assume that the MLE $\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}$ of regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ exists, then it is unique, and we have the balance property on portfolio level*

$$\sum_{i=1}^n v_i\,\kappa'\langle\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}, \boldsymbol{x}_i\rangle/\varphi = \sum_{i=1}^n v_i Y_i/\varphi = \sum_{i=1}^n Z_i.$$

Under the canonical link choice $g = h$, the canonical parameter $\theta_i$ and the linear predictor $\langle\boldsymbol{\beta}, \boldsymbol{x}_i\rangle$ coincide

$$\theta_i = h\left(\mu_i\right) = h\left(\mu(\boldsymbol{x}_i)\right) = \langle\boldsymbol{\beta}, \boldsymbol{x}_i\rangle = (\mathfrak{X}\boldsymbol{\beta})_i.$$

**Proof.** We rewrite the score equations (2.29) under the canonical link choice

$$\nabla_{\boldsymbol{\beta}}\,\mathfrak{D}(\boldsymbol{Y},\boldsymbol{\beta}) = -\frac{2}{n}\sum_{i=1}^n \frac{v_i}{\varphi}\left(Y_i - \mu(\boldsymbol{x}_i)\right)\boldsymbol{x}_i = -\frac{2}{n}\,\mathfrak{X}^\top\,\mathrm{diag}(v_i/\varphi)_{1\leq i\leq n}\left(\boldsymbol{Y} - \kappa'(\mathfrak{X}\boldsymbol{\beta})\right) = \boldsymbol{0}, \quad (2.32)$$



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where $\kappa'(\mathfrak{X}\boldsymbol{\beta})$ is meant component-wise. This gives us Hessian

$$\nabla^2_{\boldsymbol{\beta}}\, \mathfrak{D}(\boldsymbol{Y}, \boldsymbol{\beta}) = \frac{2}{n}\, \mathfrak{X}^\top \operatorname{diag}\left(v_i \kappa''\langle \boldsymbol{\beta}, \boldsymbol{x}_i\rangle/\varphi\right)_{1\le i\le n} \mathfrak{X}.$$

The strict convexity of $\kappa$ and the full rank property of $\mathfrak{X}$ imply that this Hessian is positive definite. This provides the uniqueness of the MLE.
The first column of the design matrix $\mathfrak{X}$ is identically equal to 1 representing the intercept, see (2.22). Therefore, the score equations under the canonical link choice give us for this first column of $\mathfrak{X}$

$$(1,\dots,1)\ \operatorname{diag}(v_i/\varphi)_{1\le i\le n}\, \kappa'(\mathfrak{X}\widehat{\boldsymbol{\beta}}^{\text{MLE}}) = (1,\dots,1)\ \operatorname{diag}(v_i/\varphi)_{1\le i\le n}\, \boldsymbol{Y} = (1,\dots,1)\ \boldsymbol{Z}. \tag{2.33}$$

This proves the claim of the balance property. □

From Proposition 2.12 we conclude that a GLM with the canonical link choice provides a reallocation of the totally observed claims, see Remark 2.6, and this balance property provides unbiasedness in insurance pricing.

| EDF distribution | $\kappa(\theta)$ | $\boldsymbol{\Theta}$ | $h(\mu)$ | $\mathcal{M}$ |
|---|---|---|---|---|
| Bernoulli | $\log(1+e^\theta)$ | $\mathbb{R}$ | $\log(\mu/(1-\mu))$ | $(0,1)$ |
| Poisson | $e^\theta$ | $\mathbb{R}$ | $\log(\mu)$ | $(0,\infty)$ |
| negative binomial | $-\log(1-e^\theta)$ | $(-\infty,0)$ | $\log(\mu/(1+\mu))$ | $(0,\infty)$ |
| Gaussian | $\theta^2/2$ | $\mathbb{R}$ | $\mu$ | $\mathbb{R}$ |
| gamma | $-\log(-\theta)$ | $(-\infty,0)$ | $-1/\mu$ | $(0,\infty)$ |
| inverse Gaussian | $-\sqrt{-2\theta}$ | $(-\infty,0]$ | $-1/(2\mu^2)$ | $(0,\infty)$ |
| Tweedie's CP, $p\in(1,2)$ | $\frac{((1-p)\theta)^{\frac{2-p}{1-p}}}{2-p}$ | $(-\infty,0)$ | $\mu^{1-p}/(1-p)$ | $(0,\infty)$ |
| Tweedie, $p>2$ | $\frac{((1-p)\theta)^{\frac{2-p}{1-p}}}{2-p}$ | $(-\infty,0]$ | $\mu^{1-p}/(1-p)$ | $(0,\infty)$ |

Table 2.2: Canonical link of selected EDF examples.

Usually, in applications, only in the Gaussian, the Bernoulli and the Poisson GLMs the canonical link $h$ is used. The reason is that the linear predictor $\langle\boldsymbol{\beta},\boldsymbol{x}\rangle$ lives on the entire real line $\mathbb{R}$ for the covariates $\boldsymbol{x}\in\{1\}\times\mathbb{R}^q$, and under the canonical link choice this may lead to domain conflicts if the effective domain $\boldsymbol{\Theta}$ is not the entire real line, see third column of Table 2.2.

**Example 2.13** (balance property in GLMs)**.** We study a claim size example. We choose the motorcycle claims data of Ohlsson–Johansson [154]; this data is also used and described in Wüthrich–Merz [205, Section 13.2].[4]
For modeling the claim sizes, we restrict to the insurance policies of this motorcycle data that suffer claims. We select two covariate components for $\boldsymbol{x}$, the first one is an area code which we partition into 5 different categorical zones denoted by `Zone`, and the second one is the policyholder's age which we partition into 6 age classes denoted by `Age`. This results in 30 categorical risk cells $\{1,\dots,5\}\times\{1,\dots,6\}$. Table 2.3 illustrates this motorcycle claims data. The upper part of the table shows the observed aggregated claim amounts $Z_i$ in each risk cell and the lower part the numbers of claims $v_i\in\mathbb{N}$ in each risk cell $1\le i\le n$. We note that all risk cells suffer claims, thus, we have $n=30$ observations $(Z_i,v_i)_{i=1}^{30}$ in this example (with $v_i>0$).

[4]The full data is available from the R package `CASdatasets` [70].



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| | Age 1 | Age 2 | Age 3 | Age 4 | Age 5 | Age 6 |
|---|---|---|---|---|---|---|
| Zone 1 | 38,029 | 3,384,863 | 967,323 | 388,371 | 647,308 | 74,699 |
| Zone 2 | 15,416 | 1,640,018 | 590,738 | 1,868,273 | 484,749 | 160,225 |
| Zone 3 | 26,461 | 1,230,541 | 448,552 | 163,372 | 58,2095 | 7,459 |
| Zone 4 | 94,968 | 1,716,109 | 485,891 | 695,015 | 586,674 | 139,458 |
| Zone 5 | 68,000 | 247,540 | 43,164 | 13,715 | 19,415 | 1,600 |
| Zone 1 | 4 | 97 | 37 | 23 | 16 | 4 |
| Zone 2 | 2 | 61 | 22 | 42 | 25 | 11 |
| Zone 3 | 2 | 55 | 16 | 20 | 24 | 1 |
| Zone 4 | 13 | 69 | 18 | 45 | 36 | 12 |
| Zone 5 | 1 | 14 | 6 | 4 | 2 | 1 |

Table 2.3: Motorcycle example: the upper part shows the total claim amounts $Z_i$ and the lower part the observed numbers of claims $v_i$ for all risk cells $1 \leq i \leq n = 30$.

We transform the original claim amounts $Z_i$ to its reproductive form $Y_i = \varphi Z_i / v_i$, and since the dispersion parameter $\varphi > 0$ cancels in the score equations, see (2.29), for simplicity, we set it equal to 1. This does not influence MLE. Both categorical covariates are dummy coded which results in a regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ of dimension $q+1 = (5-1)+(6-1)+1 = 10$, see also (2.27).

We study three different examples within the EDF, namely, the Poisson example, the gamma example and the inverse Gaussian example. These three example have a power variance function $\mu \mapsto V(\mu) = \kappa''(h(\mu))$ given by

$$\kappa''(h(\mu)) = \mu, \qquad \kappa''(h(\mu)) = \mu^2 \qquad \text{and} \qquad \kappa''(h(\mu)) = \mu^3,$$

for the Poisson, the gamma and the inverse Gaussian case, respectively. Thus, the variance function $\mu \mapsto V(\mu) = \kappa''(h(\mu)) = \mu^p$ is a power function of its expected value, and the general class of distributions within the EDF that satisfies this power function property are Tweedie's distributions, which go back to Tweedie [192] and Bar-Lev–Enis [13], see Section 2.2.3. We fit the three EDF examples using a GLM regression function to the motorcycle data of Table 2.3. We use both the log-link and the canonical link for $g$ for the GLMs. In view of (2.29), we need to solve the score equations for the MLE of $\boldsymbol{\beta}$, that is,

$$\nabla_{\boldsymbol{\beta}} \mathfrak{D}(\boldsymbol{Y}, \boldsymbol{\beta}) = -\frac{2}{n} \sum_{i=1}^{n} \frac{v_i}{\varphi} \frac{Y_i - \mu(\boldsymbol{x}_i)}{\mu(\boldsymbol{x}_i)^p} \frac{1}{g'(\mu(\boldsymbol{x}_i))} \boldsymbol{x}_i = \boldsymbol{0}. \tag{2.34}$$

We observe that the power variance parameter $p$ of the chosen Tweedie's distribution influences how the residuals $Y_i - \mu(\boldsymbol{x}_i)$ are weighted in these score equations, with a linear weighting $p = 1$ in the Poisson case, a quadratic weighting $p = 2$ in the gamma case and cubic weighting $p = 3$ in the inverse Gaussian case.

The second column of Table 2.4 presents the estimated average claim amounts

$$\frac{1}{\sum_{i=1}^{n} v_i} \sum_{i=1}^{n} v_i \, g^{-1} \langle \widehat{\boldsymbol{\beta}}^{\text{MLE}}, \boldsymbol{x}_i \rangle,$$

in each of the considered models. This confirms that the balance property holds under



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| EDF distribution & GLM link | average estimate | balance property |
|---|---|---|
| empirical mean | 24,641 | yes |
| Poisson GLM – canonical link = log-link | 24,641 | yes |
| gamma GLM – log-link | 24,660 | no |
| gamma GLM – canonical link | 24,641 | yes |
| inverse Gaussian GLM – log-link | 25,018 | no |
| inverse Gaussian GLM – canonical link | 24,641 | yes |

Table 2.4: Balance property of the selected EDF examples with log-link $g = \log(\cdot)$ and with canonical link $h = (\kappa')^{-1}$.

the canonical link choice, and it fails to hold under the log-link choice in the gamma and the inverse Gaussian GLMs.

The balance property under the canonical link choice does not only hold on the portfolio level (global balance property), but if we work with dummy coded categorical levels, it holds on each categorical level (local balance property); the proof of this statement is identical to the one of Proposition 2.12, because the corresponding column of a dummy coded categorical variable in the design matrix $\mathfrak{X}$ exactly collects the parameters and claims of a fixed given level. More explicitly, the vector of 1's in (2.33) is replaced by the corresponding column of $\mathfrak{X}$ which has elements in $\{0,1\}^n$ under dummy coding, see (2.27). As a result, we find the method of the marginal total sums introduced by Bailey [12] and Jung [117] for a multiplicative tariff structure (log-link); we also refer to Denuit et al. [62] who call this the local balance property. Figure 2.1 (rhs) verifies this local balance property on all categorical levels, whereas Figure 2.1 (lhs) shows that this generally fails to hold under non-canonical link choices.

In Figure 2.1 (rhs) we verified that the local balance property is satisfied marginally for dummy coded categorical covariates (on each level), and Figure 2.2 shows that this is not the case within risk cells: Figure 2.2 shows the estimates (black/blue/red) within each risk cell, and it compares them to the observations (gray dots); the $x$-axis is ordered w.r.t. the sizes of the estimates in the Poisson GLM. Generally, we see more similarity under the canonical link choice (rhs), and higher fluctuations/differences under the log-link choice (lhs). This is explained by the impact of the weighting with the variance function in (2.34) which is not present in the canonical link case, see (2.32).

A quick ad-hoc solution to restore the global balance property is to adjust the intercept parameter estimate $\widehat{\beta}_0$ correspondingly. The adjusted model is no longer MLE, however, it satisfies the global balance property and, henceforth, it is unbiased, which is more important in insurance pricing and other financial applications. Though very simple, this quick ad-hoc solution is very effective, and Lindholm–Wüthrich [138] shows that it often outperforms more sophisticated balance correction methods. Lindholm–Wüthrich [138] also provide a tool for selecting a specific balance correction method by preferring the method that has the smallest asymptotic KL divergence. ■

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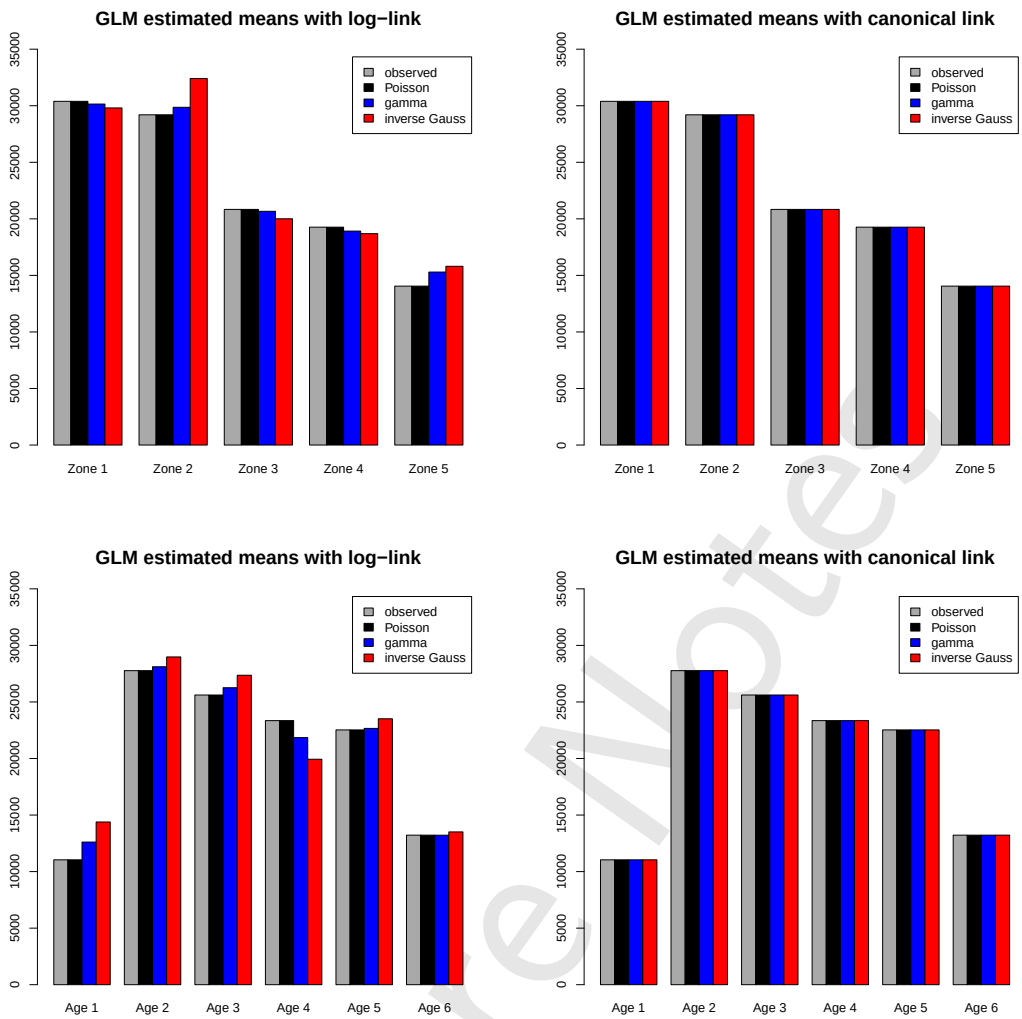


Figure 2.1: Marginal estimates using the different GLMs: local balance property on each level of the categorical covariates under the log-link choice (lhs) and the canonical link choice (rhs).

### 2.3.3 Model evaluation and model selection

#### Strictly consistency loss function for mean estimation

One of the most difficult tasks in statistical modeling is model selection. Generally, we try to find an estimated model that is as close as possible to the true (unknown) model. If we would know the true model, assume it is given by a density $f^*$ w.r.t. a $\sigma$-finite measure $\nu$ on $\mathbb{R}$, we could, e.g., calculate the KL divergence from the estimated model $\widehat{f}$ to the true model $f^*$. The KL divergence from $\widehat{f}$ to $f^*$ is given by

$$D_{\mathrm{KL}}\left(f^* \,\middle\|\, \widehat{f}\right) = \int_y f^*(y) \log\left(\frac{f^*(y)}{\widehat{f}(y)}\right) d\nu(y). \tag{2.35}$$

A well known result is that the KL divergence is zero if and only if $\widehat{f} = f^*$, $\nu$-a.e., see, e.g., Wüthrich–Merz [205, Lemma 2.21]. This makes the KL divergence a valid model



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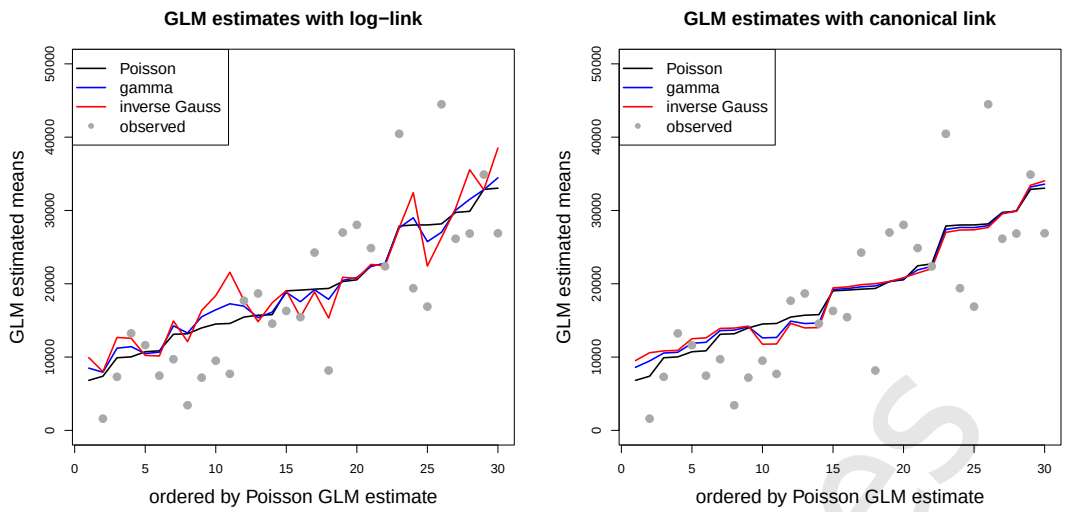


Figure 2.2: Predictions within risk cells: (lhs) log-link, and (rhs) canonical link.

selection criterion. Note, however, that the KL divergence is not a distance function because it neither satisfies the triangle inequality nor is it symmetric.

If $f^*$ and $\widehat{f}$ belong to the EDF with the same steep cumulant function $\kappa$, and if they only differ in their canonical parameters $\theta^* \in \boldsymbol{\Theta}$ and $\widehat{\theta} \in \boldsymbol{\Theta}$, respectively, the KL divergence has the following explicit form

$$\begin{aligned} D_{\mathrm{KL}}\left(f(\cdot;\theta^*,v/\varphi)\,\middle\|\,f(\cdot;\widehat{\theta},v/\varphi)\right) &= \frac{v}{\varphi}\left[\kappa'(\theta^*)\left(\theta^*-\widehat{\theta}\right)-\kappa\left(\theta^*\right)+\kappa\left(\widehat{\theta}\right)\right] \\ &= \frac{v}{\varphi}\left[\mu^*\left(h(\mu^*)-h(\widehat{\mu})\right)-\kappa\left(h(\mu^*)\right)+\kappa\left(h(\widehat{\mu})\right)\right] \\ &= \frac{1}{2}\frac{v}{\varphi}\,\mathfrak{d}\left(\mu^*,\widehat{\mu}\right) \;\geq\; 0, \end{aligned} \tag{2.36}$$

where we set $\mu^* = \kappa'(\theta^*)$ and $\widehat{\mu} = \kappa'(\widehat{\theta})$, and where $\mathfrak{d}$ is the unit deviance introduced in (2.17). Thus, the KL divergence from the estimated to the true EDF model can be expressed by the unit deviance $\mathfrak{d}$. This also links to Remark 2.9, saying that we have a Bregman divergence from the estimated to the true model.

MLE (2.28)-(2.29), minimizing the deviance loss function, now receives a different interpretation. Namely, in view of (2.36) and (2.19), we try to find the regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ such that we have minimal average KL divergence from the estimated to the true models

$$\begin{aligned} \overline{D}_{\mathrm{KL}}\left((\mu_i^*)_{i=1}^n \| \,\boldsymbol{\beta}\right) &:= \frac{1}{n}\sum_{i=1}^n D_{\mathrm{KL}}\left(f(\cdot;\theta_i^*,v_i/\varphi)\,\middle\|\,f\left(\cdot;h\left(g^{-1}\langle\boldsymbol{\beta},\boldsymbol{x}_i\rangle\right),v_i/\varphi\right)\right) \\ &= \frac{1}{n}\sum_{i=1}^n \frac{1}{2}\frac{v_i}{\varphi}\,\mathfrak{d}\left(\mu_i^*,g^{-1}\langle\boldsymbol{\beta},\boldsymbol{x}_i\rangle\right), \end{aligned} \tag{2.37}$$

where $\mu_i^* = \kappa'(\theta_i^*)$ are the true means of the observations $Y_i$ with covariates $\boldsymbol{x}_i$, $1 \leq i \leq n$. Since the true models and means $\mu_i^*$ are unknown, we simply replace them by their empirical counterparts $Y_i$ providing us with the deviance loss function

$$\frac{2}{n}\sum_{i=1}^n D_{\mathrm{KL}}\left(f(\cdot;h(Y_i),v_i/\varphi)\,\middle\|\,f\left(\cdot;h\left(g^{-1}\langle\boldsymbol{\beta},\boldsymbol{x}_i\rangle\right),v_i/\varphi\right)\right) = \mathfrak{D}(\boldsymbol{Y},\boldsymbol{\beta}), \tag{2.38}$$



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we refer to (2.28) and (2.19). In summary, the MLE $\widehat{\boldsymbol{\beta}}^{\text{MLE}}$ of $\boldsymbol{\beta}$ is the GLM parameter that minimizes the empirical average KL divergence from the estimated to the true models given by (2.38). We can even go one step further, namely, as described in (2.26), we can interpret the different insurance policies $(\boldsymbol{x}_i, v_i)_{i=1}^n$ as an i.i.d. sample of a population distribution $(\boldsymbol{X}, V) \sim \mathbb{P}$ describing the insurance portfolio. This motivates to replace the empirical version (2.38) by

$$2\mathbb{E}\left[D_{\text{KL}}\left(f(\cdot; h(Y), V/\varphi) \,\middle\|\, f\left(\cdot; h\left(g^{-1}\langle \boldsymbol{\beta}, \boldsymbol{X}\rangle\right), V/\varphi\right)\right)\right] = \mathbb{E}\left[\frac{V}{\varphi}\,\mathfrak{d}\left(Y, g^{-1}\langle \boldsymbol{\beta}, \boldsymbol{X}\rangle\right)\right]. \tag{2.39}$$

Mathematically speaking, under the suitable Cesàro sum generator property, the empirical version (2.38) converges to (2.39); see Gourieroux et al. [98].

Expression (2.39) now nicely connects to the theoretical foundation of estimation and forecast evaluation; we refer to Gneiting–Raftery [91] and Gneiting [90]. Namely, mean estimation in a given model for $Y$ should be done by minimizing the expected loss of a *strictly consistent loss function $L$ for mean estimation*, that is,

$$\widehat{\mu} \;=\; \arg\min_{\widetilde{\mu}} \mathbb{E}\left[L(Y, \widetilde{\mu})\right]. \tag{2.40}$$

Strict consistency of the loss function $L$ for mean estimation means that the *unique minimizer* of this optimization problem (2.40) is given by the *true mean* $\mu^*$ of $Y$. Savage [177] proved that the only strictly consistent loss functions for mean estimation (if and only if) are the Bregman divergences; see Gneiting [90, Theorem 7]. Thus, unit deviances $\mathfrak{d}$ are perfect candidates for mean estimation. In view of (2.39), we have

$$\boldsymbol{\beta}^* \;=\; \arg\min_{\widetilde{\boldsymbol{\beta}} \in \mathbb{R}^{q+1}} \mathbb{E}\left[\frac{V}{\varphi}\,\mathfrak{d}\left(Y, g^{-1}\langle \widetilde{\boldsymbol{\beta}}, \boldsymbol{X}\rangle\right)\right], \tag{2.41}$$

supposed that the true model is a GLM with mean structure (2.21) and true (and identifiable) GLM parameter $\boldsymbol{\beta}^* \in \mathbb{R}^{q+1}$. Thus, MLE considers an empirical version (2.38) of (2.41). Moreover, Gourieroux et al. [98] show that the specific choice of the strictly consistent loss function is asymptotically optimal (called best asymptotically normal) if it captures the right variance structure of the data.

These strict consistency results in mathematical statistics nicely summarize MLE (M-estimation and Z-estimation) and its properties. The caveat is that these are asymptotic results, and in insurance we often work on finite (small) samples. In fact, the entire MLE theory is centered around its asymptotic behavior (consistency and asymptotic normality) which may not be the most important requirements in insurance pricing. In many financial applications one should value higher a balance or unbiasedness property. This illustrates the conflicts of actuaries if they fully rely on classical statistical theory; we also refer to Lindholm–Wüthrich [138].

### Model selection

Since our examples consider simulated data, we know the true model $\mu^*$, we refer to (A.1) in the appendix. In such a situation, we can evaluate the KL divergence (2.36), and preference is given to the estimated model that has the smallest KL divergence to



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the true model $\mu^*$. Of course, this is a rather atypical situation because in most practical problems the true model is not known. In such more realistic cases one tries to mimic the KL divergence (2.36) by studying its empirical counterpart (2.38). However, in doing so, there is a critical point that needs to be considered. This is described next.

For given data $\boldsymbol{Y}$, the MLE $\widehat{\boldsymbol{\beta}}^{\text{MLE}}$ of $\boldsymbol{\beta}$ is found by solving

$$\widehat{\boldsymbol{\beta}}^{\text{MLE}} \in \underset{\widetilde{\boldsymbol{\beta}}\in\mathbb{R}^{q+1}}{\arg\min}\, \mathfrak{D}(\boldsymbol{Y}, \widetilde{\boldsymbol{\beta}}). \tag{2.42}$$

Deviance loss minimization (2.42) provides a biased version to the right-hand side of (2.39), we denote this by

$$\mathfrak{D}\left(\mathcal{L}, \widehat{\boldsymbol{\beta}}^{\text{MLE}}\right) := \mathfrak{D}\left(\boldsymbol{Y}, \widehat{\boldsymbol{\beta}}^{\text{MLE}}\right). \tag{2.43}$$

In fact, the expression in (2.43) is an *in-sample loss* because we use the same *learning sample*

$$\mathcal{L} := \{(Y_1, \boldsymbol{x}_1, v_1), \ldots, (Y_n, \boldsymbol{x}_n, v_n)\}, \tag{2.44}$$

for parameter estimation and for expected deviance loss approximation of (2.39). This in-sample bias has been subject to numerous research. A very popular piece of work is Akaike [8] who derives in Akaike's information criterion (AIC) an asymptotic correction term for this in-sample bias. Namely, the AIC value of a MLE estimated model is given by

$$\text{AIC} = -2\,\ell_{\boldsymbol{Y}}\left(\widehat{\boldsymbol{\beta}}^{\text{MLE}}\right) + 2\,\#\widehat{\boldsymbol{\beta}}^{\text{MLE}}, \tag{2.45}$$

that is, the log-likelihood in the MLE $\widehat{\boldsymbol{\beta}}^{\text{MLE}}$ needs to be corrected by the number of model parameters denoted by $\#\widehat{\boldsymbol{\beta}}^{\text{MLE}}$. This determines (under certain conditions) the asymptotic bias, if the sample size tends to infinity and if we use MLE for parameter estimation.

A second popular model selection method is to keep a hold-out sample, called *test sample*,

$$\mathcal{T} := \left\{(Y_1^\dagger, \boldsymbol{x}_1^\dagger, v_1^\dagger), \ldots, (Y_m^\dagger, \boldsymbol{x}_m^\dagger, v_m^\dagger)\right\}, \tag{2.46}$$

which is *independent* of the learning sample $\mathcal{L}$ (under $\mathbb{P}$), and whose components are independent and follow the same law as the ones of the learning sample $\mathcal{L}$. That is, $(Y_{i'}^\dagger, \boldsymbol{x}_{i'}^\dagger, v_{i'}^\dagger) \overset{(d)}{=} (Y_i, \boldsymbol{x}_i, v_i)$, for all $i$ and $i'$, and where the randomness includes the selection of the insurance policies $(\boldsymbol{X}, V) \sim \mathbb{P}$ from the population distribution. MLE is then solely performed on the learning sample $\mathcal{L}$, i.e., in-sample. This provides us with a valid out-of-sample approximation to (2.39)

$$\mathfrak{D}\left(\mathcal{T}, \widehat{\boldsymbol{\beta}}^{\text{MLE}}\right) := \mathfrak{D}\left(\boldsymbol{Y}^\dagger, \widehat{\boldsymbol{\beta}}^{\text{MLE}}\right). \tag{2.47}$$

The deviance loss (2.47) is called *out-of-sample loss* because it is calculated on the hold-out sample $\mathcal{T}$ that has not been seen during model fitting. This latter method is usually preferred in complex algorithmic models, because in these models the MLE cannot be found and, therefore, AIC is not a valid criterion. Moreover, in complex algorithmic models even the number of parameters can be unclear (because there may be redundancy); we refer to Abbas et al. [1] who discuss the effective dimension of neural networks.

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### 2.3.4 Lab: Motor claim frequencies

**In-sample and out-of-sample time-series data**

We study a car insurance claim frequency example. We use the synthetic data described in Appendix A. There is data available for calendar years $2011 \leq t \leq 2019$, and we have $266,975$ car drivers who are uniquely identified by a `PolicyID` running from $100,001$ to $366,975$. Since not all insurance contracts are active during the entire observation period (there are new inceptions and contract terminations), we denote by $\mathcal{I}_t$ all insurance policies that are active (have positive exposures) in calendar year $2011 \leq t \leq 2019$

$$\mathcal{I}_t = \Big\{\texttt{PolicyID}; \text{ the contract with } \texttt{PolicyID} \text{ is active in calendar year } t\Big\}. \tag{2.48}$$

Our general set-up of using the available data is as follows. We use the data of calendar years prior to $T$ to predict and price the contracts of year $T+1$. For a fixed $2014 \leq T \leq 2018$, this provides us with learning sample

$$\mathcal{L}_T = \Big\{\Big(Y_{i,t}, \boldsymbol{x}_{i,t}, v_{i,t}\Big);\ 2011 \leq t \leq T,\ i \in \mathcal{I}_t\Big\}, \tag{2.49}$$

and with test sample (to be forecast and for forecast evaluation)

$$\mathcal{T}_{T+1} = \Big\{\Big(Y_{i,T+1}, \boldsymbol{x}_{i,T+1}, v_{i,T+1}\Big);\ i \in \mathcal{I}_{T+1}\Big\},$$

where $Y_{i,t}$ is the claim of insurance policy $i$ in calendar year $t$, and $\boldsymbol{x}_{i,t} \in \mathcal{X}$ and $v_{i,t} \in (0,1]$ are the corresponding covariates and exposures. Figure 2.3 illustrates an example for calendar year $T = 2015$.

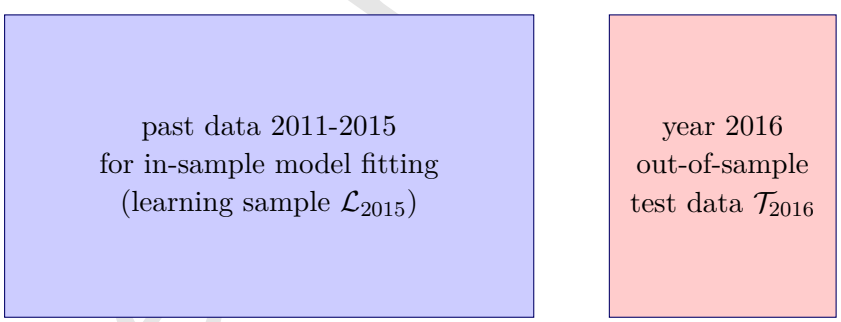


Figure 2.3: Example for calendar year $T = 2015$: past data $\mathcal{L}_{2015}$ of calendar years 2011-2015 for (in-sample) model fitting, and data $\mathcal{T}_{2016}$ of calendar year 2016 for out-of-sample forecasting.

In some examples, we do not use the entire past history $\mathcal{L}_T$ over all calendar years $2011 \leq t \leq T$, but only the last calendar year $T$ is considered. E.g., if we have non-stationarity in the data, it may not make sense to include the entire past history. For this, we define the smaller in-sample learning data

$$\mathcal{L}_T^{\circ} = \Big\{\Big(Y_{i,T}, \boldsymbol{x}_{i,T}, v_{i,T}\Big);\ i \in \mathcal{I}_T\Big\} \ \subset\ \mathcal{L}_T, \tag{2.50}$$



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giving us the identities

$$\mathcal{L}_T = \bigcup_{t=2011}^{T} \mathcal{L}_t^\circ \qquad \text{and} \qquad \mathcal{T}_{T+1} = \mathcal{L}_{T+1}^\circ. \tag{2.51}$$

Our first example in this section will only consider model fitting on the learning data $\mathcal{L}_T^\circ$ to forecast the test data $\mathcal{T}_{T+1}$, for calendar years $2014 \leq T \leq 2018$. Note that we have data starting from 2011, but we only start our predictions after calendar year 2014. The reason for this is that in later examples we need time-series data for experience rating. Starting in calendar year 2014 gives us four years 2011-2014 of historical data for the first prediction at the end of calendar year $T = 2014$. Thus, in later chapters we require some "burn-in" for time-series estimators to receive reasonable experience rating results.

### Generalized linear model for claim frequencies

Our first example considers a Poisson GLM to forecast the claims in $\mathcal{T}_{T+1}$, and we only use the last calendar year $\mathcal{L}_T^\circ$ to build this GLM. First, we note that we use a Poisson GLM to model the claim count responses, denoted by $N_{i,t}$. The Poisson assumption is not correct, because the data has been generated by a negative binomial model, see (A.1). However, at this stage, we typically do not have this knowledge, and a Poisson assumption is the first natural choice to model claim counts. We further analyze this in our examples below.

The Poisson distribution belongs to the EDF, and the count random variable $N_{i,t}$ is in its additive form,[5] see Remarks 2.1. We receive its reproductive form by setting $Y_{i,t} = N_{i,t}/v_{i,t}$; note that $\varphi = 1$ in the Poisson model. The Poisson EDF is then obtained by choosing the cumulant function $\kappa(\theta) = \exp\{\theta\}$ on the effective domain $\boldsymbol{\Theta} = \mathbb{R}$, see Table 2.1, dispersion parameter $\varphi = 1$, and $\nu$ being the counting measure on $\mathbb{N}_0/v_{i,t}$.

We have three continuous covariates `DrivAge`, `CatPrice`, `VehAge`. We have four binary covariates `Gender`, `Leasing`, `LowMileage` and `Deductible`, and we have one categorical covariate `Region` having 15 levels. We use dummy coding for `Region`, we implement `DrivAge` and `CatPrice` as continuous real-valued variables into the GLM regression function. Moreover, we build categorical classes for `VehAge` and integrate them by dummy coding. These choices are motivated by the figures in Appendix A. This provides us with covariates $\boldsymbol{x}_{i,t} \in \mathcal{X}$ of dimension $q + 1 = 28$; for a more detailed description of covariate engineering we refer to Wüthrich–Merz [205, Chapter 5].

We then set-up a Poisson GLM, that is, we assume that all responses $Y_{i,t}$ are independent and Poisson distributed. From the Poisson cumulant function $\kappa(\theta) = \exp\{\theta\}$, we obtain the log-link as the canonical link $h$, and for the GLM we choose this (canonical) log-link, i.e., we set $g(\mu) = h(\mu) = \log(\mu)$. Moreover, we choose the weights $v_{i,t} > 0$ equal to the time exposures given on line 6 of Listing A.1. This motivates us to consider the log-likelihood of the observations $\mathcal{L}_T^\circ$ at time $T$ as a function of the regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$

$$\boldsymbol{\beta} \;\mapsto\; \ell_{\mathcal{L}_T^\circ}(\boldsymbol{\beta}) \;\propto\; \sum_{i \in \mathcal{I}_T} v_{i,T} \left( Y_{i,T} \langle \boldsymbol{\beta}, \boldsymbol{x}_{i,T} \rangle - \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_{i,T} \rangle \right),$$

[5] Generally, we denote the additive form within the EDF by random variables $Z$, however, if we discuss claim counts, more commonly the notation $N$ is used.



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where we have dropped all terms that do not depend on $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ (this is indicated by the symbol $\propto$), and with GLM regression function (linear predictor)

$$\boldsymbol{x} \in \mathcal{X} \;\mapsto\; \log(\mu(\boldsymbol{x})) = \langle \boldsymbol{\beta}, \boldsymbol{x} \rangle. \tag{2.52}$$

This linear predictor is equal to the canonical parameter because we select the canonical link $g = h$ (which implies that the balance property will hold in this GLM).
As already mentioned, we are going to use the data $\mathcal{L}_T^\circ$ from calendar year $T$ to fit this GLM (in-sample), and we will forecast calendar year $T+1$ out-of-sample, i.e., we back-test on $\mathcal{T}_{T+1}$. The MLE is found by maximizing the log-likelihood or minimizing the Poisson deviance loss function

$$\widehat{\boldsymbol{\beta}}_T^{\text{MLE}} \;=\; \underset{\boldsymbol{\beta} \in \mathbb{R}^{q+1}}{\arg\max}\; \ell_{\mathcal{L}_T^\circ}(\boldsymbol{\beta}) \;=\; \underset{\boldsymbol{\beta} \in \mathbb{R}^{q+1}}{\arg\min}\; \mathfrak{D}(\mathcal{L}_T^\circ, \boldsymbol{\beta}), \tag{2.53}$$

where $\mathfrak{D}$ is the Poisson deviance loss obtained from the Poisson unit deviances given by, see Table 2.1,

$$\mathfrak{d}_{\text{Poi}}\left(Y_{i,T}, \mu(\boldsymbol{x}_{i,T})\right) = 2\left(\mu(\boldsymbol{x}_{i,T}) - Y_{i,T} - Y_{i,T}\log\left(\frac{\mu(\boldsymbol{x}_{i,T})}{Y_{i,T}}\right)\right). \tag{2.54}$$

At the boundary $Y_{i,T} = 0$ of the mean parameter space $\overline{\mathcal{M}}$ we use (2.18), giving us Poisson unit deviance $\mathfrak{d}_{\text{Poi}}(0, \mu(\boldsymbol{x}_{i,T})) = 2\mu(\boldsymbol{x}_{i,T})$. Recall that the solution to (2.53) is unique and it satisfies the balance property, see Proposition 2.12.

For model fitting we are going to use the Poisson unit deviances $\mathfrak{d}_{\text{Poi}}$, given in (2.54). The original data has been generated by a negative binomial distribution, see (A.1), and we briefly illustrate how we are going to use this negative binomial assumption for model validation. First, we need to bring the claim count observations $N_{i,t}$ into its reproductive form, see Remarks 2.1. In the negative binomial case this is received by setting

$$\widetilde{Y}_{i,t} = \varphi_{i,t} Y_{i,t} = \varphi_{i,t} N_{i,t}/v_{i,t} \qquad \text{and} \qquad \widetilde{\mu}(\boldsymbol{x}_{i,t}) = \varphi_{i,t}\mu(\boldsymbol{x}_{i,t}), \tag{2.55}$$

that is, compared to the Poisson case, where we work with $Y_{i,t}$ and $\mu(\boldsymbol{x}_{i,t})$, we need to additionally scale with the *unknown* dispersion parameter $\varphi_{i,t} > 0$ to receive the reproductive form of the negative binomial EDF.
The unit deviances in the negative binomial case are for the observations $\widetilde{Y}_{i,T}$ in calendar year $T$ given by, see Table 2.1,

$$\mathfrak{d}_{\text{NB}}\left(\widetilde{Y}_{i,T}, \widetilde{\mu}(\boldsymbol{x}_{i,T})\right) = 2\left(\widetilde{Y}_{i,T}\log\left(\frac{\widetilde{Y}_{i,T}}{\widetilde{\mu}(\boldsymbol{x}_{i,T})}\right) - \left(\widetilde{Y}_{i,T}+1\right)\log\left(\frac{\widetilde{Y}_{i,T}+1}{\widetilde{\mu}(\boldsymbol{x}_{i,T})+1}\right)\right), \tag{2.56}$$

we add tildes in (2.55) and (2.56) to highlight the reproductive form of the negative binomial EDF (in contrast to the Poisson EDF). These negative binomial unit deviances depend on the unknown dispersion parameters $\varphi_{i,T}$, see (2.55), which (unfortunately) does not simply cancel in (2.56). This causes some issues when working with the negative binomial EDF that cannot easily be resolved. Consider the mean and variance, respectively, of the negative binomial EDF in its reproductive form

$$\begin{aligned}
\mathbb{E}\left[\widetilde{Y}_{i,T}\middle|\, \boldsymbol{x}_{i,T}, v_{i,T}\right] &= \widetilde{\mu}(\boldsymbol{x}_{i,T}), \\
\text{Var}\left(\widetilde{Y}_{i,T}\middle|\, \boldsymbol{x}_{i,T}, v_{i,T}\right) &= \frac{\varphi_{i,T}}{v_{i,T}}\,\widetilde{\mu}(\boldsymbol{x}_{i,T})\left(1 + \widetilde{\mu}(\boldsymbol{x}_{i,T})\right),
\end{aligned}$$



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this is obtained from $\widetilde{\mu} = \kappa'(\theta)$ and $\kappa''(\theta) = \widetilde{\mu}(1+\widetilde{\mu})$ in the negative binomial model, see (2.2) and Table 2.1. This implies for the additive form, see (2.55),

$$\begin{aligned} \mathbb{E}\left[N_{i,T} \middle| \boldsymbol{x}_{i,T}, v_{i,T}\right] &= v_{i,T}\mu(\boldsymbol{x}_{i,T}), \\ \operatorname{Var}\left(N_{i,T} \middle| \boldsymbol{x}_{i,T}, v_{i,T}\right) &= v_{i,T}\mu(\boldsymbol{x}_{i,T})\left(1+\varphi_{i,T}\mu(\boldsymbol{x}_{i,T})\right). \end{aligned} \tag{2.57}$$

This now shows a next difficulty when working with the negative binomial model, namely, over-dispersion is often modeled by the variance divided by the mean being a constant factor bigger than 1. This has been used in the true model (A.2), namely, the variance scales with the constant factor $\varphi^* = 1.3 > 1$ with the true mean $\mu^*(\boldsymbol{x}_{i,T})$. This is quite different from the over-dispersion structure in the negative binomial EDF. Solving (2.57) for this constant over-dispersion (and the true mean) gives us EDF dispersion values

$$\varphi_{i,T} = \frac{\varphi^* - 1}{\mu^*(\boldsymbol{x}_{i,T})} \;>\; 0. \tag{2.58}$$

This is somewhat unpleasant as the unknown dispersion factor $\varphi_{i,T}$ depends on the unknown true mean $\mu^*(\boldsymbol{x}_{i,T})$, and we cannot easily disentangle this, even if $\varphi^*$ was known.

For these reasons, we prefer to work with the Poisson unit deviances $\mathfrak{d}_{\text{Poi}}$ for model fitting, though knowing that they do not reflect the true model. However, due to the strict consistency of the (Poisson) unit deviances, we still receive the true means, asymptotically, only the rate of convergence will not be optimal; we refer to the pseudo maximum likelihood method of Gourieroux et al. [98] who treat this problem rigorously.

Knowing the true model $\mu^*(\boldsymbol{x}_{i,t})$, we will still use the negative binomial unit deviances $\mathfrak{d}_{\text{NB}}$ to evaluate the KL divergence of the estimated model, we refer to (2.36). In its negative binomial reproductive form, see (2.55), an estimated model $\widetilde{\mu}(\boldsymbol{x}_{i,t})$ has weighted KL divergence to the true model $\widetilde{\mu}^*(\boldsymbol{x}_{i,t})$

$$\begin{aligned} &\frac{1}{2}\frac{v_{i,t}}{\varphi_{i,t}}\,\mathfrak{d}_{\text{NB}}\left(\widetilde{\mu}^*(\boldsymbol{x}_{i,t}), \widetilde{\mu}(\boldsymbol{x}_{i,t})\right) \\ &\quad= v_{i,t}\left(\mu^*(\boldsymbol{x}_{i,t})\log\left(\frac{\mu^*(\boldsymbol{x}_{i,t})}{\mu(\boldsymbol{x}_{i,t})}\right) - \left(\mu^*(\boldsymbol{x}_{i,t}) + \frac{1}{\varphi_{i,t}}\right)\log\left(\frac{\mu^*(\boldsymbol{x}_{i,t}) + \frac{1}{\varphi_{i,t}}}{\mu(\boldsymbol{x}_{i,t}) + \frac{1}{\varphi_{i,t}}}\right)\right) \\ &\quad= v_{i,t}\,\mu^*(\boldsymbol{x}_{i,t})\left(\log\left(\frac{\mu^*(\boldsymbol{x}_{i,t})}{\mu(\boldsymbol{x}_{i,t})}\right) - \frac{\varphi^*}{\varphi^*-1}\log\left(\frac{\varphi^*}{(\varphi^*-1)\frac{\mu(\boldsymbol{x}_{i,t})}{\mu^*(\boldsymbol{x}_{i,t})}+1}\right)\right), \end{aligned} \tag{2.59}$$

where in the last step we have inserted the true dispersion parameter choice (2.58). Mathematically, this is all fine, however, the true mean in the reproductive form of the negative binomial EDF looks a bit odd because we have a constant mean $\widetilde{\mu}^*(\boldsymbol{x}_{i,t}) = \varphi_{i,t}\mu^*(\boldsymbol{x}_{i,t}) = \varphi^* - 1$ to receive the constant over-dispersion factor $\varphi^*$ in (2.57) for the true model.

### Numerical results

As described in the previous section, we fit these Poisson GLMs for all calendar years $2014 \leq T \leq 2018$ separately, see (2.53). This produces five different predictive models, the corresponding R code [166] for calendar year $T = 2015$ is given in Listing 2.1.



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Listing 2.1: R code to fit the Poisson GLM to $\mathcal{L}_T^\circ$.

```
load(file=paste("./Data/DataSynthetic.rda", sep=""))
dat <- DataSynthetic
### pre-process VehAge
dat$VehAgeGLM   <- as.factor(cut(dat$VehAge, c(-1,0,1,2,3,4,9,12,21),
                        labels = c("0","1","2","3","4","5-9","10-12","13+"),
                        include.lowest = TRUE))
## set calendar year
T <- 2015
## fit GLM
d.glm <- glm(ClaimNb ~ DrivAge + Gender + Leasing + LowMileage +
                        Region + CatPrice + VehAgeGLM + Deductible,
                        data=dat[which(dat$Year==T),],
                        offset=log(Exposure), family=poisson())
## estimated means and forecasting
GLM.learn <- predict(d.glm, newdata=dat[which(dat$Year==T),], type="response")
GLM.test  <- predict(d.glm, newdata=dat[which(dat$Year==(T+1)),], type="response")
```

We use several different scores to analyze the fitting and predictive results. Some of these figures can only be computed because we know the true data generating model (ground truth). In the sequel, we highlight this fact by using green color for quantities that are based on the knowledge of the true model $\mu^*$.

(1) In-sample evaluation. We provide the following in-sample evaluations:

(a) in-sample Poisson deviance loss $\mathfrak{D}(\mathcal{L}_T^\circ, \widehat{\boldsymbol{\beta}}_T^{\text{MLE}})$ of the fitted model $\widehat{\boldsymbol{\beta}}_T^{\text{MLE}}$ against its learning data $\mathcal{L}_T^\circ$, using the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$, this quantity is prone to over-fitting for complex regression models;

(b) time $T$ Poisson deviance loss $\mathfrak{D}(\mathcal{L}_T^\circ, \mu_T^*)$ of the true model $\mu_T^* = \mu^*(\boldsymbol{x}_{i,T})$ against the learning data $\mathcal{L}_T^\circ$, using the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$, we highlight this quantity in green color as it can only be computed because we know the true models $\mu^*(\boldsymbol{x}_{i,T})$ of the policies $i \in \mathcal{I}_T$ at time $T$; this quantity is used to benchmark the in-sample loss received in item (a);

(c) time $T$ average KL divergence of the estimated model to the true model

$$\bar{D}_{\text{KL}}(\mu_T^* \| \widehat{\boldsymbol{\beta}}_T^{\text{MLE}}) := \frac{1}{2|\mathcal{I}_T|} \sum_{i \in \mathcal{I}_T} \frac{v_{i,T}}{\varphi_{i,T}} \, \mathfrak{d}_{\text{NB}} \left( \widetilde{\mu}^*(\boldsymbol{x}_{i,T}), \varphi_{i,T} \, g^{-1} \langle \widehat{\boldsymbol{\beta}}_T^{\text{MLE}}, \boldsymbol{x}_{i,T} \rangle \right),$$

we highlight this quantity in green color as it can only be computed because we know the true models $\mu^*(\boldsymbol{x}_{i,T})$ of policies $i \in \mathcal{I}_T$ at time $T$, and we use the negative binomial unit deviance $\mathfrak{d}_{\text{NB}}$ in its reproductive form, see (2.59); this quantity tells us how well we can discover the true model by MLE;

(2) Out-of-sample evaluation. We provide the following out-of-sample evaluations:

(a) out-of-sample Poisson deviance loss $\mathfrak{D}(\mathcal{T}_{T+1}, \widehat{\boldsymbol{\beta}}_T^{\text{MLE}})$ of the fitted model $\widehat{\boldsymbol{\beta}}_T^{\text{MLE}}$ on $\mathcal{L}_T^\circ$ against its test data $\mathcal{T}_{T+1}$, using the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$;

(b) time $T+1$ Poisson deviance loss $\mathfrak{D}(\mathcal{T}_{T+1}, \mu_{T+1}^*)$ of the true model $\mu_{T+1}^* = \mu^*(\boldsymbol{x}_{i,T+1})$ against the test data $\mathcal{T}_{T+1}$, using the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$; this quantity is used to benchmark the out-of-sample loss received in item (a);



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(c) time $T+1$ average KL divergence from the estimated model to the true model

$$\bar{D}_{\mathrm{KL}}(\mu^*_{T+1}\|\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}_T) := \frac{1}{2|\mathcal{I}_{T+1}|}\sum_{i\in\mathcal{I}_{T+1}}\frac{v_{i,T+1}}{\varphi_{i,T+1}}\,\mathfrak{d}_{\mathrm{NB}}\left(\widetilde{\mu}^*(\boldsymbol{x}_{i,T+1}), \varphi_{i,T+1}\, g^{-1}\langle\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}_T, \boldsymbol{x}_{i,T+1}\rangle\right),$$

we select the negative binomial unit deviance $\mathfrak{d}_{\mathrm{NB}}$ in its reproductive form.

The in-sample and out-of-sample results of the GLMs are presented in Table 2.5. In general, we observe that the Poisson deviance losses of the estimated GLMs are rather close to the ones of the true model, and the negative binomial KL divergence from the estimated GLMs to the true models live on a much smaller scale. This expresses that the main uncertainty driver is the irreducible risk coming from the randomness in the negative binomial distribution, and model uncertainty and systematic effects play a smaller role. Nevertheless, we would like to optimally capture these systematic effects to receive an appropriate price differentiation (risk-based pricing).

| | calendar years $T$ | | | | |
|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 |
| in-sample at time $T$: | | | | | |
| (1a) $\mathfrak{D}(\mathcal{L}^\circ_T, \widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}_T)$ | 45.36 | 43.87 | 43.67 | 43.55 | 43.94 |
| (1b) $\mathfrak{D}(\mathcal{L}^\circ_T, \mu^*_T)$ | 44.98 | 43.44 | 43.15 | 43.01 | 43.44 |
| (1c) $\bar{D}_{\mathrm{KL}}(\mu^*_T\|\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}_T)$ | 0.186 | 0.192 | 0.225 | 0.237 | 0.214 |
| out-of-sample for time $T+1$: | | | | | |
| (2a) $\mathfrak{D}(\mathcal{T}_{T+1}, \widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}_T)$ | 44.01 | 43.77 | 43.64 | 44.04 | 44.03 |
| (2b) $\mathfrak{D}(\mathcal{T}_{T+1}, \mu^*_{T+1})$ | 43.44 | 43.15 | 43.01 | 43.44 | 43.47 |
| (2c) $\bar{D}_{\mathrm{KL}}(\mu^*_{T+1}\|\widehat{\boldsymbol{\beta}}^{\mathrm{MLE}}_T)$ | 0.214 | 0.251 | 0.239 | 0.242 | 0.240 |

Table 2.5: GLMs fitted on the learning sample $\mathcal{L}^\circ_T$, and out-of-sample validation on the test sample $\mathcal{T}_{T+1}$ for calendar years $2014 \le T \le 2018$; units are $10^{-2}$.

Line (1a) gives the in-sample losses of the MLE estimated models on $\mathcal{L}^\circ_T$, and line (1b) gives the losses of the true models on the same data $\mathcal{L}^\circ_T$. The latter is the benchmark presenting the true model, and if the in-sample losses on line (1a) are smaller than line (1b), there is over-fitting. The main difference between blocks (1) and (2) is that the latter considers a one-period ahead forecast, adding the additional difficulty that the time-series from time period $T$ to $T+1$ may not be stationary. This is reflected by the fact that line (2c) generally is bigger than line (1c) because $\mu^*_T$ and $\mu^*_{T+1}$ slightly differ; this non-stationarity is verified by Figure A.1 (rhs) and we come back to this below.

Since the numbers in Table 2.5 are a bit difficult to read, we generally illustrate the results in graphs like Figure 2.4, and we complement these graphs by the results of the null model that does not consider any covariates. We observe from these plots that the GLM results are generally close to the true model, and it seems that there is not too much room for improvements. This means that covariate processing has been done in a sensible way. In the sequel we will generally focus on the KL divergences from the estimated to the true model, Figure 2.4 (bottom). The homogeneous model produces

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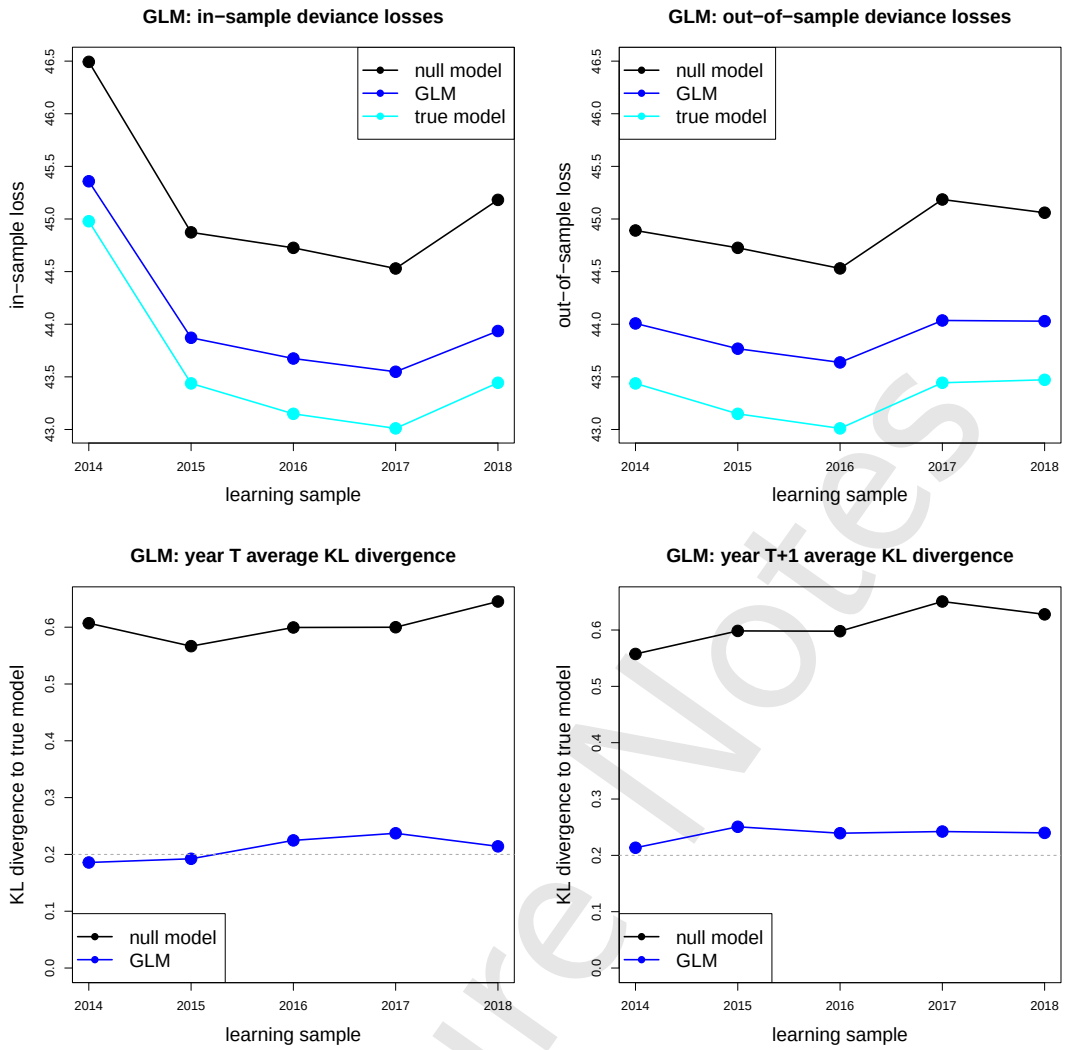


Figure 2.4: Fitted GLMs: (top-left) in-sample Poisson deviance losses, (top-right) out-of-sample Poisson deviance losses, this is the one-year ahead forecast, (bottom-left) year $T$ average KL divergences to the true models, (bottom-right) year $T+1$ average KL divergences to the true models, this is the one-year ahead forecast; units are $10^{-2}$.

an average KL divergence roughly of size $6 \cdot 10^{-3}$, and the GLM reduces this to roughly $2 \cdot 10^{-3}$. The main question is by how much it can further be reduced by using better predictive models, e.g., by integrating past claims history.

We could perform further model validation, variable selection, etc. We refrain from doing so and refer to Wüthrich–Merz [205, Section 5.3] for further statistical techniques.

### Considering all available past data

The models in the previous analysis of Table 2.5 satisfy the balance property because we used the canonical link of the Poisson EDF, see Proposition 2.12. Therefore, the estimated frequency over the entire portfolio equals the observed number of claims. An unfortunate fact is that this balance property is an in-sample property on the learning

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data $\mathcal{L}_T^\circ$ for the considered calendar year $2014 \leq T \leq 2018$. In our example we deal with the problem of a non-stationary time-series problem, this is indicated by the fact that the out-of-sample KL divergences $\bar{D}_{\mathrm{KL}}(\mu^*_{T+1} \| \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}})$ are consistently bigger than the corresponding in-sample figures $\bar{D}_{\mathrm{KL}}(\mu^*_T \| \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}})$, see Table 2.5 and the dotted horizontal lines in Figure 2.4 (bottom) help for orientation. In such a situation, the balance property may only play a subordinate role because non-stationarity is typically the dominant factor in model error, and the main problem is to get the trend right. This non-stationarity is also the reason for not considering all past years $\mathcal{L}_T$ for model fitting.

In the next analysis we use the entire learning data $\mathcal{L}_T$ for model fitting. For dealing with the non-stationarity, we add a categorical variable for the calendar years $t \leq T$, i.e., we extend the previous model by an additional categorical covariate. Then, we estimate the parameters of all covariates simultaneously over all past observations in $\mathcal{L}_T$. We give three remarks. First, having more past observations for model fitting hopefully increases model accuracy. This happens if the non-stationarity can be captured by a multiplicative (categorical) calendar year factor (since we work with the log-link). Second, using dummy coding (2.27) for the categorical calendar year variable implies that we have the (in-sample) balance property on each calendar year individually, though, the model is fitted jointly over all calendar years; the proof goes similarly to the one of Proposition 2.12, and it has been verified in an example, see Figure 2.1. Third, since calendar years are encoded categorically, we cannot easily extend the fitted model to forecast the next calendar year $T+1$, i.e., extrapolation of categorical variables needs to be done manually. To avoid this difficulty here, we only show in-sample results in this example.

Listing 2.2: R code to fit the Poisson GLM to $\mathcal{L}_T$.

```
### pre-process covariates
dat$VehAgeGLM   <- as.factor(cut(dat$VehAge, c(-1,0,1,2,3,4,9,12,21),
                       labels = c("0","1","2","3","4","5-9","10-12","13+"),
                       include.lowest = TRUE))
dat$YearGLM     <- as.factor(dat$Year)
## set calendar year
T <- 2015
## fit GLM
d.glm <- glm(ClaimNb ~ DrivAge + Gender + Leasing + LowMileage +
                       Region + CatPrice + VehAgeGLM + Deductible + YearGLM,
                       data=dat[which(dat$Year<=T),],
                       offset=log(Exposure), family=poisson())
## estimated means of last observed year
GLM.learn <- predict(d.glm, newdata=dat[which(dat$Year==T),], type="response")
```

Listing 2.2 gives the changes in the code (compared to Listing 2.1) when using all past observations $\mathcal{L}_T$ and adding a categorical variable for the calendar years. These estimated models provide the balance property on each observed calendar year.

Table 2.6 shows the resulting in-sample average KL divergences $\bar{D}_{\mathrm{KL}}(\mu^*_T \| \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}})$ of the models only fitted on the last observed year $\mathcal{L}_T^\circ$ on the first row, this is taken from Table 2.5, and the version fitted on all past years $\mathcal{L}_T$ on the second row. For calendar year $T = 2017$ we see a model improvement, for three years $T = 2015, 2016, 2018$ the model becomes worse, and for $T = 2014$ there is no essential change. These results do not support fitting this GLM an all past data simultaneously, i.e., it seems that the non-



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| | calendar years $T$ | | | | |
|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 |
| in-sample at time $T$: | | | | | |
| $\bar{D}_{\mathrm{KL}}(\mu_T^* \| \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}})$: learning sample $\mathcal{L}_T^\circ$ | 0.186 | 0.192 | 0.225 | 0.237 | 0.214 |
| $\bar{D}_{\mathrm{KL}}(\mu_T^* \| \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}})$: learning sample $\mathcal{L}_T$ | 0.186 | 0.194 | 0.232 | 0.231 | 0.226 |

Table 2.6: GLMs fitted on the learning samples $\mathcal{L}_T^\circ$ and $\mathcal{L}_T$, respectively; units are $10^{-2}$.

stationarity effect cannot solely be captured by a categorical calendar year variable that enters the regression function multiplicatively, but there also seem to be other interactions with the remaining covariates that change over time.

Listing 2.3: Estimated (categorical) regression parameters $\widehat{\beta}_t^{\mathrm{MLE}}$ in $\widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}}$ for the calendar years $2012 \leq t \leq 2018$, based on $\mathcal{L}_{T=2018}$; the reference level is calendar year $t = 2011$.

```
Coefficients:
                 Estimate Std. Error  z value Pr(>!z!)
YearGLM2012     3.180e-02  1.333e-02    2.386 0.017047 *
YearGLM2013    -3.400e-02  1.356e-02   -2.508 0.012158 *
YearGLM2014    -6.152e-03  1.348e-02   -0.457 0.648002
YearGLM2015    -5.192e-02  1.361e-02   -3.814 0.000137 ***
YearGLM2016    -5.862e-02  1.363e-02   -4.300 1.71e-05 ***
YearGLM2017    -7.224e-02  1.365e-02   -5.292 1.21e-07 ***
YearGLM2018    -5.390e-02  1.359e-02   -3.965 7.35e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimating all calendar years simultaneously brings the advantage that we can directly compare these calendar years within the same model through their regression parameter estimates in $\widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}}$. That is, through the categorical dummy coding of the calendar year variable, we receive the regression parameters $\widehat{\beta}_t^{\mathrm{MLE}}$ for the calendar years $2012 \leq t \leq 2018$. These describe the differences to the reference level being calendar year $t = 2011$ in our example. Listing 2.3 shows these regression parameters $(\widehat{\beta}_t^{\mathrm{MLE}})_{t=2012}^{2018}$. We observe significant differences in their values, i.e., it seems that this is an important covariate in joint regression estimation; this is further verified below.
Since we work with the log-link, we receive a multiplicative regression function structure having estimated means (based on the learning data $\mathcal{L}_T$ with $T = 2018$)

$$\begin{aligned}\widehat{\mu}(\boldsymbol{x}) &= \exp\left\langle \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}}, \boldsymbol{x} \right\rangle \\ &= \exp\left\{ \sum_{t=2012}^{2018} \widehat{\beta}_t^{\mathrm{MLE}} \mathbb{1}_{\{\boldsymbol{x} \text{ belongs to calendar year } t\}} \right\} \times \prod_{j \notin \{2012,\ldots,2018\}} \exp\left\{ \widehat{\beta}_j^{\mathrm{MLE}} x_j \right\},\end{aligned}$$

where on the second line not all notation has been properly defined, but its meaning should be intuitive and self-explanatory. The first term on the last line gives a multiplicative factor for each calendar year; these factors are illustrated in Figure 2.5 (lhs). They start at 1, being the reference level for calendar year 2011, and afterwards they fluctuate, describing the characteristics of the corresponding calendar years. It seems that these factors have a downward trend indicating that the average expected frequency



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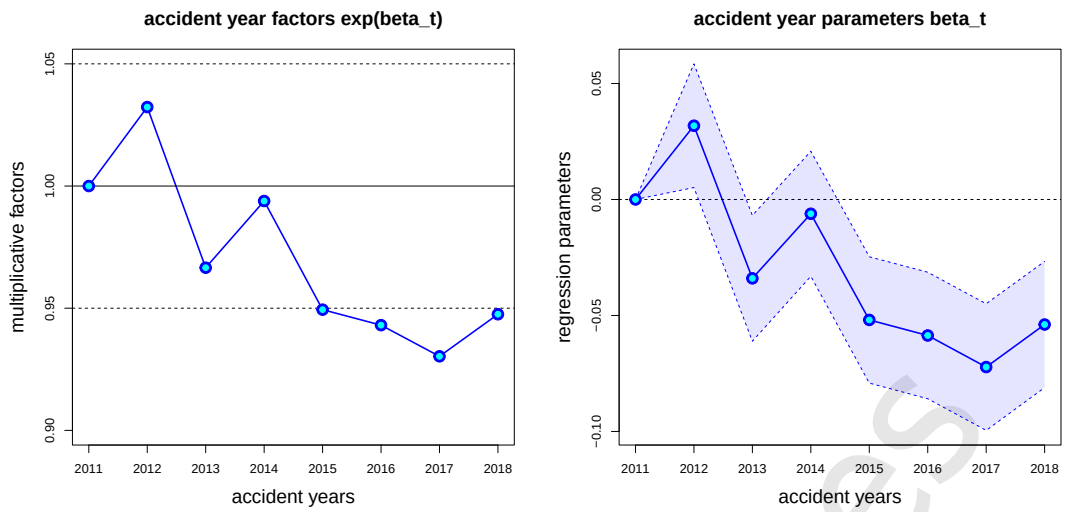


Figure 2.5: Calendar year differences (rhs) measured by the multiplicative GLM factors $\exp\{\widehat{\beta}_t^{\text{MLE}}\}$ and (lhs) estimated regression parameters $\widehat{\beta}_t^{\text{MLE}}$ with confidence bounds of two standard deviations for calendar years $2011 \leq t \leq 2018$ fitted on $\mathcal{L}_{2018}$.

decays over time. We give some remarks. First, for extrapolation we need to project these multiplicative factors. These factors are influenced, e.g., by external risk factors like whether conditions, inflation, etc., in future calendar years, and a time-series model capturing the underlying uncertainty seems most suitable for this extrapolation. Second, one is tempted to use a continuous encoding for the calendar year variable, however, in view of Figure 2.5 a time-series approach seems more suitable as we interpret the fluctuation as randomness and a continuous encoding approach is rather suitable for a deterministic trend modeling. That is, there seem to be latent (random) risk factors which rather require a random effects model than a smooth (deterministic) time trend model.

Listing 2.4: `drop1` analysis of the fitted Poisson GLM based on learning data $\mathcal{L}_{2018}$.

```
Single term deletions

            Df Deviance    AIC    LRT  Pr(>Chi)
<none>          522200 696411
DrivAge      1  522615 696824  415.1 < 2.2e-16 ***
Gender       1  522215 696424   15.2 9.707e-05 ***
Leasing      1  524740 698949 2540.3 < 2.2e-16 ***
LowMileage   1  522200 696409    0.2     0.682
Region      14  526486 700669 4285.7 < 2.2e-16 ***
CatPrice     1  524904 699113 2704.0 < 2.2e-16 ***
VehAgeGLM    7  522765 696962  564.5 < 2.2e-16 ***
Deductible   1  522696 696905  496.3 < 2.2e-16 ***
YearGLM      7  522311 696508  111.1 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Finally, we should study the question whether the calendar year differences in Listing 2.3 and Figure 2.5 are significant or whether they can be explained by (pure) noise (irre-



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ducible risk). For this we plot the estimated regression parameters $\widehat{\beta}_t^{\text{MLE}}$ with confidence bounds of two standard deviations (obtained from the inverse of Fisher's information matrix), see Figure 2.5 (rhs). This plot suggests that the estimated parameters are significantly different from zero, and they are also mutually significantly different. We verify this by applying a `drop1` analysis to the fitted model. A `drop1` analysis drops one variable at a time for its null hypotheses, and then performs a likelihood ratio test (LRT), additionally it calculates the AIC values, see (2.45). Listing 2.4 shows the results. These results only question the inclusion of the variable `LowMileage`, and all other variables should be kept in the model, also the categorical calendar year variable cannot be dropped. Inclusion of the calendar year variable reduces the Poisson deviance loss by 111.1 (with 7 regression parameters) which provides a $p$-value close to zero. This closes the example for the time being. ■

## 2.4 Fully-connected feed-forward neural network

### 2.4.1 Definition and model fitting

We introduce fully-connected feed-forward neural networks (FNNs) in this section. This section closely follows Wüthrich–Merz [205, Chapter 7], and it presents an extension of the GLMs considered in Section 2.3. The only difference to the previous section is that we replace the GLM regression function (2.21) by a FNN regression architecture

$$\boldsymbol{x}_i \;\mapsto\; g\left(\mu_i\right) = g\left(\mu(\boldsymbol{x}_i)\right) = g\left(\mathbb{E}[Y_i|\boldsymbol{x}_i, v_i]\right) = \left\langle \boldsymbol{\beta}, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_i)\right\rangle, \tag{2.60}$$

where $\boldsymbol{z}^{(d:1)} : \mathbb{R}^{q+1} \to \mathbb{R}^{q_d+1}$ is a FNN of depth $d \in \mathbb{N}$ that learns a new representation $\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_i) \in \mathbb{R}^{q_d+1}$ of the covariates $\boldsymbol{x}_i \in \mathbb{R}^{q+1}$, with readout parameter $\boldsymbol{\beta} \in \mathbb{R}^{q_d+1}$ and with smooth and strictly monotone link function $g$.
A FNN $\boldsymbol{z}^{(d:1)}$ of depth $d$ is a composition of $d$ FNN layers $\boldsymbol{z}^{(m)}$, $1 \le m \le d$,

$$\boldsymbol{x} \;\mapsto\; \boldsymbol{z}^{(d:1)}(\boldsymbol{x}) = \left(\boldsymbol{z}^{(d)} \circ \cdots \circ \boldsymbol{z}^{(1)}\right)(\boldsymbol{x}). \tag{2.61}$$

A FNN layer $\boldsymbol{z}^{(m)}$ with (non-linear) activation function $\phi : \mathbb{R} \to \mathbb{R}$ is a mapping

$$\begin{aligned} \boldsymbol{z}^{(m)} : \{1\} \times \mathbb{R}^{q_{m-1}} &\to \{1\} \times \mathbb{R}^{q_m} \\ \boldsymbol{z} &\mapsto \boldsymbol{z}^{(m)}(\boldsymbol{z}) = \left(1, z_1^{(m)}(\boldsymbol{z}), \ldots, z_{q_m}^{(m)}(\boldsymbol{z})\right)^\top, \end{aligned}$$

for layer dimensions $q_{m-1}, q_m \in \mathbb{N}$ and with neurons (ridge functions) for $1 \le j \le q_m$

$$\boldsymbol{z} \;\mapsto\; z_j^{(m)}(\boldsymbol{z}) = \phi\langle \boldsymbol{w}_j^{(m)}, \boldsymbol{z}\rangle,$$

having network weights $\boldsymbol{w}_j^{(m)} \in \mathbb{R}^{q_{m-1}+1}$. Thus, every neuron $\boldsymbol{z} \mapsto z_j^{(m)}(\boldsymbol{z})$ performs a GLM-operation using covariate $\boldsymbol{z}$ as input, inverse link $\phi$ and with regression parameter $\boldsymbol{w}_j^{(m)}$, compare to (2.23). The GLM-operation (that consists of the composition of an affine function $\boldsymbol{z} \mapsto \langle \boldsymbol{w}_j^{(m)}, \boldsymbol{z}\rangle$ with a univariate non-linear function $\phi$) is also called ridge function. This FNN has a network parameter $\boldsymbol{\vartheta} = (\boldsymbol{w}_1^{(1)}, \ldots, \boldsymbol{w}_{q_d}^{(d)}, \boldsymbol{\beta})^\top \in \mathbb{R}^r$ of dimension

$$r = \sum_{m=1}^{d} q_m(q_{m-1}+1) + (q_d+1).$$



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The main difficulty in FNN modeling is to fit this network architecture to the learning sample $\mathcal{L}_T$, similarly to the case of a GLM as shown in Figure 2.3. In principle, the MLE of $\boldsymbol{\vartheta}$ could be determined analogously to (2.53). However, due to the complexity of the problem (as the parameter dimension $r$ is usually rather large) and due to the non-convexity of this minimization problem (for non-linear activation functions $\phi$), typically, this MLE cannot be found explicitly. On the other hand, the MLE of $\boldsymbol{\vartheta}$ very likely (in-sample) over-fits to the learning data $\mathcal{L}_T$ because bigger networks provide very flexible regression models.[6] For this reason, one typically applies variants of the gradient descent algorithm and one exploits an early stopping strategy of the gradient descent algorithm. For this, the learning sample $\mathcal{L}_T$ is randomly partitioned into a training sample $\mathcal{U}_T$, that is used for network parameter $\boldsymbol{\vartheta}$ fitting, and a validation sample $\mathcal{V}_T$, which is used to exploit the early stopping rule, this is illustrated in Figure 2.6.

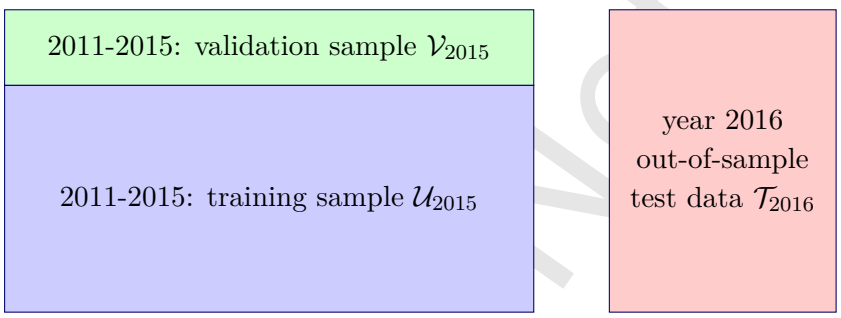


Figure 2.6: Example for year $T = 2015$: past data 2011-2015 partitioned into training sample $\mathcal{U}_{2015}$ and validation sample $\mathcal{V}_{2015}$ for model fitting, and calendar year 2016 for out-of-sample forecasting.

In applications, one typically uses variants of the stochastic gradient descent (SGD) algorithm, that partitions the training sample $\mathcal{U}_T$ at random into mini-batches, and for each gradient descent step only the data of a mini-batch is used. SGD is used for computational reasons (but not only), namely, having a large training sample $\mathcal{U}_T$ may lead to rather slow gradient descent computations because these involve large matrix multiplications which can be time consuming. For more on gradient descent fitting we refer to Goodfellow et al. [95, Sections 8.3 and 8.5], the back-propagation method of Rumelhart et al. [176], and to Wüthrich–Merz [205, Section 7.2.3]. In our applications below we use the `nadam` version of SGD, which is the Nesterov [152] accelerated version of the 'adaptive moment' estimation SGD introduced by Kingma–Ba [123]. The standard parametrization of `nadam` in the R package `keras` [51] often has a quite good performance without additional hyper-parameter fine-tuning.

[6] Over-fitting within large FNNs is attributed to the universality theorems that roughly say that compactly supported continuous functions can be approximated arbitrarily well (e.g., in supremum norm) by sufficiently large FNNs under a suitable activation function choice. In the case of $d = 1$, such results have been proved, e.g., by Cybenko [56] and Hornik et al. [106].



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### 2.4.2 Lab: Motor claim frequencies

We revisit the car insurance claim counts example of Section 2.3.4. We replace the GLM module by a FNN regression architecture of depth $d = 3$ with $(q_1, q_2, q_3) = (20, 15, 10)$ neurons in the three FNN layers and with hyperbolic tangent activation function for $\phi$. We use the covariate information as follows. We choose `DrivAge`, `CatPrice` and `VehAge` as continuous variables, and we normalize (standardize) them to being centered with unit variance, see Wüthrich–Merz [205, formula (7.30)]. The binary variables `Gender`, `Leasing`, `LowMileage` and `Deductible` are 0-1 (binary) encoded, and the categorical variable `Region` is implemented by an embedding layer with embedding dimension 2; embedding layers are treated in more detail in Section 2.6, below, and we also refer to Wüthrich–Merz [205, Section 7.4.1] and the references in Richman–Wüthrich [172] for embedding layers. This gives us a FNN architecture with a network parameter $\boldsymbol{\vartheta} \in \mathbb{R}^r$ of dimension $r = 716$; Listing 2.5 shows the implementation in the R package `keras` [51].

Listing 2.5: R code of a FNN architecture of depth $d = 3$ in `keras` [51].

```
library(keras)
library(tensorflow)
### function that defines a network of depth 3
network.embedding <- function(seed, q, lambda, b){
    k_clear_session()
    set.seed(seed)
    set_random_seed(seed)
    #
    Design   = layer_input(shape = c(7), dtype = 'float32')
    Cat      = layer_input(shape = c(1), dtype = 'int32')
    Exposure = layer_input(shape = c(1), dtype = 'float32')
    #
    CatEmb = Cat %>%
      layer_embedding(input_dim = 15, output_dim = b[1], input_length = 1) %>%
      layer_flatten()
    #
    Network = list(Design, CatEmb) %>% layer_concatenate() %>%
          layer_dense(units=q[1], activation='tanh') %>%
          layer_dense(units=q[2], activation='tanh') %>%
          layer_dense(units=q[3], activation='tanh') %>%
          layer_dense(units=1, activation='exponential',
             weights=list(array(0, dim=c(q[3],1)), array(log(lambda), dim=c(1))))
    #
    Response = list(Network, Exposure) %>% layer_multiply()
    #
    keras_model(inputs = c(Design, Cat, Exposure), outputs = c(Response))
    }
```

We fit this FNN regression model to the learning data $\mathcal{L}_T^\circ$, separately, for all calendar years $2014 \leq T \leq 2018$, and we forecast one-year ahead the test sample $\mathcal{T}_{T+1}$, this set-up is identical to the GLM procedure in Section 2.3.4. We use 20% of the learning sample $\mathcal{L}_T^\circ$ as validation sample $\mathcal{V}_T^\circ$ to exploit early stopping, and the remaining 80% as training sample $\mathcal{U}_T^\circ$. Using a callback we retrieve the parameter estimate $\widehat{\boldsymbol{\vartheta}}_T$ with the lowest validation loss; as loss function we throughout use the Poisson deviance loss based on the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$. This is illustrated in Listing 2.6.

The continuous and binary covariates of the learning sample $\mathcal{L}_T^\circ$ are contained in `X.learn`,



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Listing 2.6: R code for SGD fitting using `nadam` of a FNN with early stopping.

```
model <- network.embedding(100, c(20,15,10), lambda, 2)
path  <- paste("./Networks/FNN1_year",T,".h5", sep="")
CB    <- callback_model_checkpoint(path, monitor = "val_loss", verbose = 0,
                                   save_best_only = TRUE, save_weights_only = TRUE)
model %>% compile(loss = 'poisson', optimizer = 'nadam')
###
fit <- model %>% fit(list(X.learn, Re.learn, V.learn), Y.learn, validation_split=0.2,
                     batch_size=10^12, epochs=200, verbose=1, callbacks=CB)
###
load_model_weights_hdf5(model, path)
```

`Re.learn` contains the categorical `Region` information and `V.learn` the exposure information. `validation_split=0.2` partitions this learning sample at random into a validation sample $\mathcal{V}_T^\circ$ of 20% of the learning data and a training sample $\mathcal{U}_T^\circ$ of the remaining 80% of the learning data. The SGD algorithm is performed on mini-batches of size $2^{12} = 4,096$ over 200 epochs, and the parameter $\widehat{\boldsymbol{\vartheta}}_T$ with the lowest validation loss is retrieved by the callback `CB`. Because SGD fitting involves several elements of randomness, see Wüthrich–Merz [205, Remarks 7.9], we ensemble (average) the estimated FNN means over 10 different networks fittings; we refer to Richman–Wüthrich [169] and Wüthrich–Merz [205, Section 7.4.4]. We generally consider this ensembling in all network results presented in these notes.

A remark on choosing the training data $\mathcal{U}$ and the validation data $\mathcal{V}$ in a time-series context, i.e., not in the present modeling set-up: If one partitions time-series data, one has to ensure that an entire time-series is assigned to one of the two data sets, that is, one should not split the time-series and allocate one part to the training data and the other one to the validation data as this may induce unwanted correlations between the two data sets.

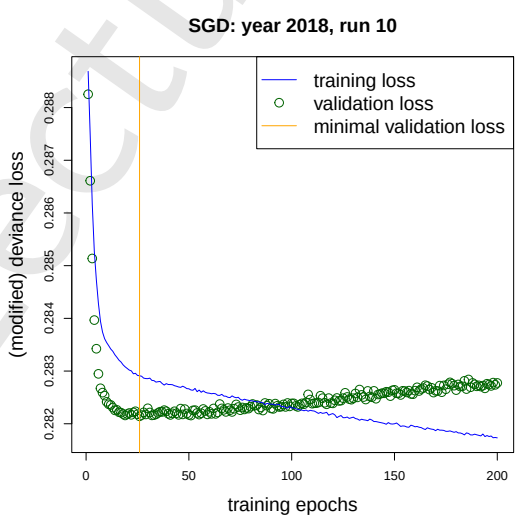


Figure 2.7: SGD algorithm showing training and validation losses on $\mathcal{U}_T^\circ$ and $\mathcal{V}_T^\circ$, respectively, for $T = 2018$ and a selected seed (starting value of the SGD algorithm, called run 10 in the title of the graph).

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Figure 2.7 shows one single SGD run, i.e., before ensembling over 10 fitted FNN fittings. The blue curve gives the training losses on $\mathcal{U}_T^\circ$ and the green dots show the validation losses on $\mathcal{V}_T^\circ$, respectively. We observe the lowest validation loss after 26 training epochs, which gives us the optimal network parameter $\widehat{\boldsymbol{\vartheta}}_T$ in this run (with this seed) of the SGD algorithm, and for prediction we average (ensemble) the results over 10 different seeds.

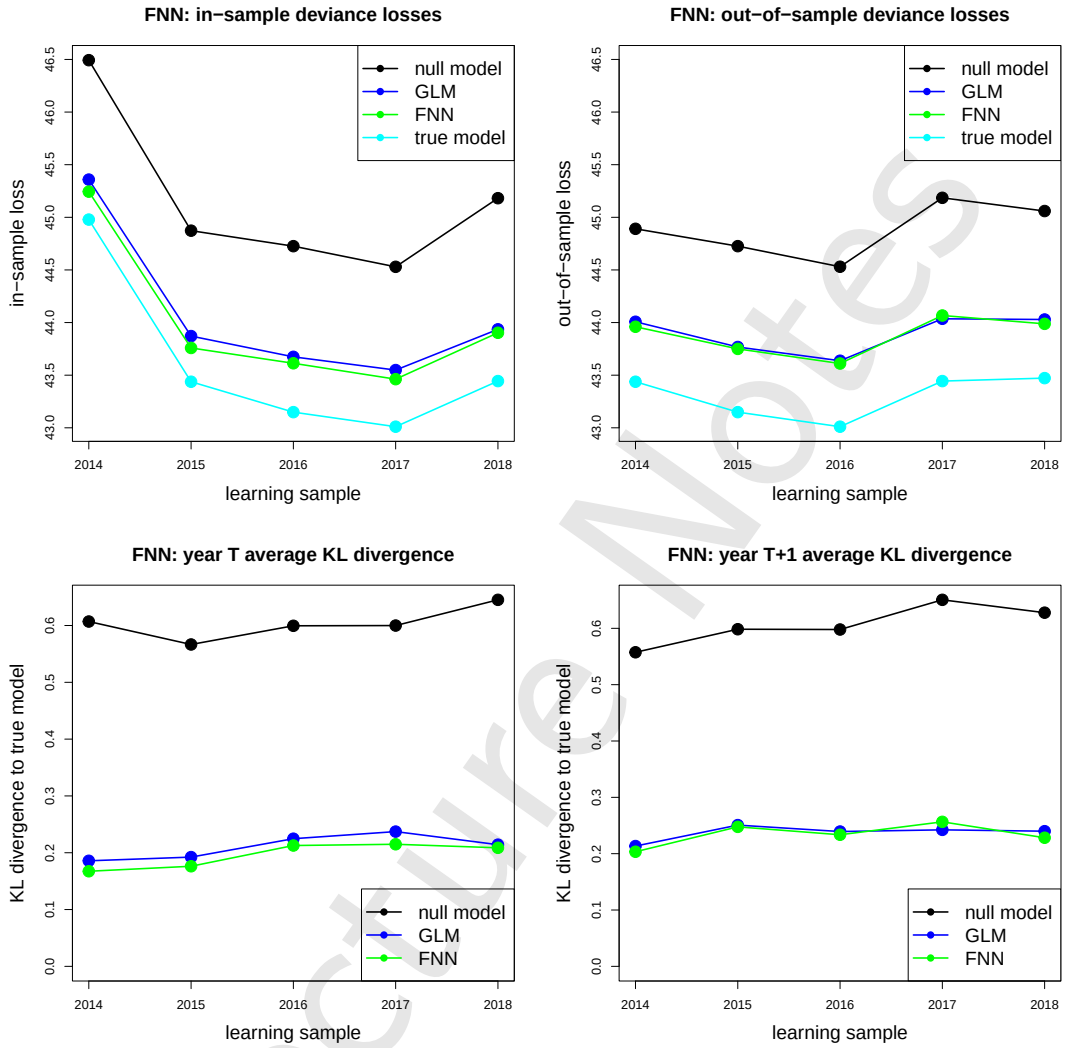


Figure 2.8: Fitted FNNs: (top-left) in-sample Poisson deviance losses, (top-right) out-of-sample Poisson deviance losses, (bottom-left) time $T$ average KL divergences to the true models, (bottom-right) time $T+1$ average KL divergences to the true models, this is the one-year ahead forecast; units are $10^{-2}$.

The results are presented in Figure 2.8. We observe an in-sample improvement of the FNN over the GLM regression models, especially, when considering the average KL divergence, see Figure 2.8 (bottom-left). This was expected, as typically FNNs allow for more modeling flexibility than GLMs, and they are only comparable if the covariates $(\boldsymbol{x}_{i,T})_{i\in\mathcal{I}_T}$ have optimally been feature-engineered for the GLMs. However, this more accurate modeling of the FNNs does not fully carry over to the one-year ahead forecasts, consider Figure 2.8 (rhs), where we compare the time $T$ fitted models to the true forecast

means $\mu^*(\boldsymbol{x}_{i,T+1})$ one-year ahead. The general difficulty here is again the non-stationarity of the time-series data that has already been discussed in Figure 2.5. As a result of this non-stationarity, the best model at time $T$ may not be the best model at time $T+1$ because the true means $\mu^*(\boldsymbol{x}_{i,T})$ may substantially change when going to the next calendar year $\mu^*(\boldsymbol{x}_{i,T+1})$, e.g., due to a changed global risk factors and trends. This is unpleasant, but it is a fact that we have to cope with. If we can anticipate structural changes, then we should try to bring them into the forecast model. However, typically, this is difficult because some global risk factors are just random like global weather conditions or changes in economic factors. This rather asks for a time-series approach, e.g., using random effects models, and to optimally anticipate (non-random) trends.

There are some points that we would like to raise. In most of the above examples we have fitted the models on calendar year $T$ data $\mathcal{L}_T^\circ$, only, and we have studied a one-year ahead forecast of calendar year $T+1$ data $\mathcal{T}_{T+1}$. In Listing 2.2 we have fitted a GLM on the entire past data $\mathcal{L}_T$, see (2.51). Since there are some differences between the calendar years, we have additionally integrated the calendar year information, `Year`, into the covariates $\boldsymbol{x}_{i,T}$, see lines 5 and 10 of Listing 2.2. This has been done by using a categorical dummy encoded variable. The difficulty with this approach is the extrapolation because the future calendar year's dummy level has not been trained. In FNNs, having more modeling flexibility, we can add the calendar year variable as a continuous input. This leads to an increase of the input dimension on line 9 in Listing 2.5 from 7 to 8. This increases the dimension of the network parameter $\boldsymbol{\vartheta}$ to $r = 736$, but all the remaining fitting procedure remains completely the same. Our goal is to see whether this increase of past information and continuous encoding of the calendar year variable leads to an enhanced forecast.

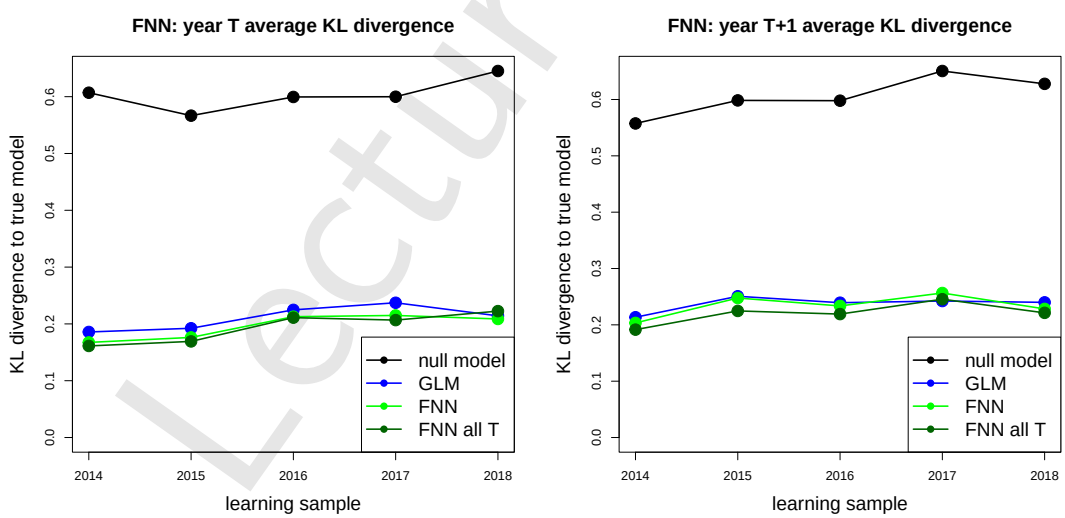


Figure 2.9: Fitted FNNs: (lhs) time $T$ average KL divergences to the true models of the FNNs fitted on all past data $\mathcal{L}_T$, (rhs) time $T+1$ average KL divergences to the true models, this is the one-year ahead forecast; units are $10^{-2}$.

The results are presented in Figure 2.9, where we consider the average KL divergences to the true models. The dark green lines show the results of the FNNs that have been



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fitted on the *entire past* information $\mathcal{L}_T$ using a continuous calendar year variable in the input. The left-hand side shows the in-sample average KL divergences $\bar{D}_{\mathrm{KL}}(\mu_T^* \| \widehat{\boldsymbol{\beta}}_T^{\mathrm{MLE}})$, note that for comparability reasons with the previous methods, we analyze this in-sample KL divergences *only* on the data $\mathcal{L}_T^\circ$, though the models have been fitted on $\mathcal{L}_T$. The right-hand side shows the one-year ahead average KL divergences. Indeed, we observe a general improvement of the predictive models when including all past observations into the modeling procedure and using a continuous calendar year variable, and this is also verified by Table 2.7 that shows the corresponding numbers.

| | calendar years $T$ | | | | |
|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 |
| (1c) in-sample $\bar{D}_{\mathrm{KL}}$: | | | | | |
| GLMs on $\mathcal{L}_T^\circ$ | 0.186 | 0.192 | 0.225 | 0.237 | 0.214 |
| GLMs on $\mathcal{L}_T$ | 0.186 | 0.194 | 0.232 | 0.231 | 0.226 |
| FNNs on $\mathcal{L}_T^\circ$ | 0.167 | 0.176 | 0.213 | 0.215 | 0.209 |
| FNNs on $\mathcal{L}_T$ | 0.161 | 0.169 | 0.211 | 0.207 | 0.222 |
| (2c) one-year ahead $\bar{D}_{\mathrm{KL}}$: | | | | | |
| GLMs on $\mathcal{L}_T^\circ$ | 0.214 | 0.251 | 0.239 | 0.242 | 0.240 |
| FNNs on $\mathcal{L}_T^\circ$ | 0.203 | 0.248 | 0.234 | 0.256 | 0.228 |
| FNNs on $\mathcal{L}_T$ | 0.192 | 0.225 | 0.219 | 0.246 | 0.221 |

Table 2.7: GLMs and FNNs fitted on the learning samples $\mathcal{L}_T^\circ$ and $\mathcal{L}_T$, respectively. The in-sample average KL divergences are evaluated on the data $\mathcal{L}_T^\circ$, and the one-year ahead forecasts correspond to the test samples $\mathcal{T}_{T+1}$ for $2014 \leq T \leq 2018$; units are $10^{-2}$.

We discuss the results of Table 2.7:

- *in-sample.* We see a general in-sample improvement in the FNNs by including all information $\mathcal{L}_T$ over only considering the latest information $\mathcal{L}_T^\circ$. This is different from the GLM results, and it tells us that there are non-multiplicative interactions with the calendar year variable that cannot be captured by the GLMs. Moreover, the FNNs outperform the GLMs which says that there is structure in the data not being captured by the (multiplicative) GLM regression function.

- *out-of-sample.* The preference of the FNN over the GLM carries over to the out-of-sample performance, though we have non-stationarity. Moreover, it is beneficial to train the FNN on the entire data $\mathcal{L}_T$, and since the calendar year variable is encoded continuously, it can be extrapolated. This is not the case in the GLM with the categorical calendar year variable, this is the reason for not showing the out-of-sample GLM results for $\mathcal{L}_T$.

We emphasize that we can extrapolate the calendar year variable in the fitted FNNs based all data $\mathcal{L}_T$ if we integrate this variable as a continuous input into the model. This seems very attractive, however, some care is needed. This extrapolation is only meaningful if the calendar year variable effect has the right slope and curvature at the observation boundary at time $T$. Often this is difficult to verify, and a similar procedure as for the GLM should/could be preferred, see Listing 2.2. For the FNN this would mean that we integrate the categorical calendar year variable as an entity embedded

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variable, similar to `Region`, see lines 13-15 of Listing 2.5. For extrapolation we then need to manually integrate/estimate an embedding value for any future calendar year. This would be similar to extrapolating the GLM of Listing 2.2. This will lead to more reliable and actuarially sound results, because the extrapolation values are manually verified, and these are not just (some) output of an algorithm.

An important point that has not been discussed so far is that the fitted FNNs typically do not have the (in-sample) balance property. This is a major deficiency of SGD fitting with early stopping, because the early stopped network calibration is not in a critical point of its deviance loss function, i.e., it does not have a vanishing gradient. If it would have a vanishing gradient under the canonical link choice, then a result similar to Proposition 2.12 would hold. This deficiency can be rectified by applying an additional GLM step with canonical link on the learned representations $(\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_i))_{i=1}^n$. This is the proposal of Wüthrich [201], and for more details we refer to that reference and to Wüthrich–Merz [205, Section 7.4.2].

The final remark is that all the above models have been fitted on prior rating information that is available to the insurer at the inception of the insurance contract (if we neglect potential changes of vehicles and similar effects). The core of these notes will be to extend this prior rating information view by past claims experience, using random effects models. Practically speaking, this seems quite reasonable as the prior rating information usually cannot fully capture the propensity to claims as, e.g., this prior rating information does not fully describe the risk profiles, driving styles, driving skills, etc. For this reason, we expect that past claims experience carries predictive information beyond these prior rating factors, and generally predictions should improve by integrating past claims experience. This is the hypothesis that we would like to verify in the sequel of this manuscript. ■

## 2.5 Selected further topics

### 2.5.1 Early contract termination

In practice, there are several reasons for not having a full calendar year exposure $v = 1$. One reason is that the inception of a new insurance contract can happen continuously over a calendar year. Our synthetic data considers this feature, as for new policies we have an arbitrary policy starting date. On real data there are further reasons for not having a full year of exposure, our data does not contain these other features. Nevertheless, we would briefly like to discuss these.

Exposures shorter than one year may also be caused by an (early) termination during the corresponding calendar year. There are different reasons for such early terminations.

(a) The contract is *regularly terminated* at the expiry date. This may happen, e.g., if the customer signs a contract on April 1, 2014, and this customer decides to not renew this contract on March 31, 2015. We call this a regular termination because the customer decides to stop the insurance coverage at expiry. Note that this is not the regular case, as can be seen from the red bars in Figure A.2 (lhs). In fact, in



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total, only between 10% and 15% of the policies are not renewed (automatically) on our data.

(b) The contract is *early terminated* because the customer has sold her/his vehicle. In this case the very same customer may sign a new contract with the same insurance company on a new car. In our case this is counted as a renewal with a new car, see yellow bars in Figure A.2 (lhs). Alternatively, this customer may decide to change the insurance company or to completely stop driving, then this customer belongs to the red bars in Figure A.2 (lhs).

(c) The customer has an accident and in this case both the insurance policyholder and the insurance company have the option to *early terminate* the contract; this is a common contract feature in many countries.

(d) A rather different reason of a non-full exposure is that the customer decides to suspend the insurance policy during the calendar year. This often happens with motorcycle insurance where bikers do not drive their motorcycles during winter season. We do not have this feature in our data. Finally, another difficulty that is not reflected in our synthetic data are multiple cars on the same insurance policy. This may happen if one registers with a multi-vehicle license plate that one can interchange between different cars, e.g., the customer drives a convertible car on sunny days and a sedan car on rainy days on the same license plate; in some countries this is legal feature exists. In this case, fixed covariates modeling becomes more complicated.

Coming back to item (1c). In case an early termination is associated with an accident, early termination carries some information about the response variable (claims). Therefore, such early terminations (which are random variables themselves) need a careful consideration and modeling, because they are correlated with claims. For a rigorous treatment of random exposures we refer to Lindholm et al. [137].

### 2.5.2 Auto-calibration

Besides the balance property discussed in Section 2.3.2, an important property in insurance pricing is the auto-calibration property. An early reference on auto-calibration is Schervish [179] who used this property in meteorology, references in the statistical and in the machine learning literature on auto-calibration include Zadrozny–Elkan [208], Menon et al. [148], Tsyplakov [191], Gneiting–Ranjan [92], Pohle [164] and Gneiting–Resin [93]. Recently, this property has also attracted some attention in the financial and actuarial literature, we mention Tasche [188], Krüger–Ziegel [125], Denuit et al. [62, 64], Fissler et al. [81], Lindholm et al. [137], Wüthrich [202, 203] and Wüthrich–Ziegel [206].

**Definition 2.14.** *A regression model $\mu : \mathcal{X} \to \mathbb{R}$ is auto-calibrated for $(Y, \boldsymbol{X})$ if, $\mathbb{P}$-a.s.,*

$$\mu(\boldsymbol{X}) = \mathbb{E}\left[Y \,|\, \mu(\boldsymbol{X})\right].$$

Intuitively speaking, this means that every (price) cohort $\mu(\boldsymbol{X})$ is on average self-financing, and we do not have any systematic cross-financing in our price system $\mu$.



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This is a quality criterion that price systems should generally fulfill, otherwise there is systematic cross-subsidization within the (insurance) portfolio. Basically, this is the very classical actuarial viewpoint of risk classification, namely, we build solidarity groups (in this case expressed by price cohorts) which should be able to cover their own (expected) claims.

First, typically, there are infinitely many auto-calibrated regression functions $\mu$ for the random tuple $(Y, \boldsymbol{X})$, and, in particular, the true regression function $\mu^*$ is an auto-calibrated example. This can be seen from the tower property, $\mathbb{P}$-a.s.,

$$\mathbb{E}\left[Y \middle| \mu^*(\boldsymbol{X})\right] = \mathbb{E}\left[\mathbb{E}\left[Y \middle| \boldsymbol{X}\right] \middle| \mu^*(\boldsymbol{X})\right] = \mathbb{E}\left[\mu^*(\boldsymbol{X}) \middle| \mu^*(\boldsymbol{X})\right] = \mu^*(\boldsymbol{X}),$$

since $\sigma(\mu^*(\boldsymbol{X})) \subset \sigma(\boldsymbol{X})$. Therefore, any regression function $\mu$ that does not satisfy auto-calibration for $(Y, \boldsymbol{X})$ cannot be the true one. On the other hand, any regression function

$$\mu_G(\boldsymbol{X}) := \mathbb{E}\left[Y \middle| G(\boldsymbol{X})\right], \tag{2.62}$$

is auto-calibrated for $(Y, \boldsymbol{X})$ for any measurable function $G$; this is again proved by applying the tower property. This statement justifies that we can have infinitely many auto-calibrated regression functions on regression problems and probability spaces that are not too simple.

Motivated by (2.62), a way to restore the auto-calibration property for an arbitrary regression model is to apply an isotonic regression step, this is called *isotonic recalibration*. Isotonic recalibration is non-parametric, it is rank-preserving, and it gives an in-sample (empirically) auto-calibrated regression function. This is the proposal presented in Wüthrich–Ziegel [206, Section 3.1]; for isotonic regression we refer to Ayer et al. [11], Barlow et al. [15] and Barlow–Brunk [16].

Isotonic recalibration is motivated by the following consideration. Assume we have a regression function $G : \mathcal{X} \to \mathbb{R}$ that gets the ranks right, i.e.,

$$G(\boldsymbol{x}) \leq G(\boldsymbol{x}') \iff \mathbb{E}\left[Y \middle| \boldsymbol{X} = \boldsymbol{x}\right] \leq \mathbb{E}\left[Y \middle| \boldsymbol{X} = \boldsymbol{x}'\right]. \tag{2.63}$$

Assume that the random variables $G(\boldsymbol{X})$ and $\mu^*(\boldsymbol{X}) = \mathbb{E}\left[Y \middle| \boldsymbol{X}\right]$ have continuous distribution functions. Then, property (2.63) gives us strict comonotonicity between $G(\boldsymbol{X})$ and the true regression mean $\mu^*(\boldsymbol{X})$. Consequently, there is a strictly increasing function $T$ with $\mu^*(\boldsymbol{X}) = T(G(\boldsymbol{X}))$, $\mathbb{P}$-a.s.[7] The isotonic recalibration step now precisely estimates this strictly increasing function $T$ and, thus, provides an estimate for the true regression function $\mu^*$ based on a correct ranking $G$. This is motivated by the following consideration

$$\mu_G(\boldsymbol{X}) = \mathbb{E}\left[Y \middle| G(\boldsymbol{X})\right] = \mathbb{E}\left[Y \middle| \mu^*(\boldsymbol{X})\right] = \mu^*(\boldsymbol{X}) = \mathbb{E}\left[Y \middle| \boldsymbol{X}\right],$$

$\mathbb{P}$-a.s., where we have used strict comonotonicity in the second step, and auto-calibration of $\mu^*$ in the third step. This gives us the following corollary.

**Corollary 2.15** (Wüthrich–Ziegel [206]). *Assume the regression function $G : \mathcal{X} \to \mathbb{R}$ gets the ranks right according to* (2.63)*, and that the considered random variables $G(\boldsymbol{X})$*

[7] The case of random variables $G(\boldsymbol{X})$ and $\mu^*(\boldsymbol{X})$ not having a continuous distribution function is more difficult because in that case comonotonicity and strict comonotonicity are not equivalent.



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*and $\mu^*(\boldsymbol{X})$ have continuous distribution functions. The true regression means are given by*

$$\mu^*(\boldsymbol{X}) = \mathbb{E}\left[Y \,|\, G(\boldsymbol{X})\right] \qquad \mathbb{P}\text{-}a.s.$$

Intuitively, this result reflects that $G(\boldsymbol{X})$ finds the right price cohorts (2.63), i.e., the same cohorts as the premium classes $\mu^*(\boldsymbol{X})$, but the level (calibration) of the cohorts $G(\boldsymbol{X})$ is not necessarily correct for the responses $Y$, yet. The isotonic recalibration step then precisely lifts these price cohorts to the right premium levels w.r.t. $Y$.

We close this subsection with the following remarks.

- Auto-calibration and isotonic regression are closely related to Murphy's score decomposition [150] and the $T$-reliability diagram discussed in Gneiting–Resin [93]; we also refer to Wüthrich–Buser [204, Section 2.5] and Delong–Wüthrich [61].

- Isotonic recalibration on data gives an empirical version of Corollary 2.15. This can be achieved by using the pool adjacent violators (PAV) algorithm of Ayer et al. [11], Miles [149] and Kruskal [126]. The PAV algorithm produces a partition of the portfolio so that adjacent blocks have isotonic empirical means, and on each such block the parameters are estimated by these empirical means. As a result, on each block we receive the balance property of Corollary 2.5, which is now termed *local balance property* because it holds locally (on each block). Thus, in this empirical construction local balance equals empirical auto-calibration, henceforth, we also fulfill the (global) balance property.

- The initial ranking function $G$ can be any regression function that provides the right ranks, e.g., if there are issues with large values in the observed responses $Y$, we may also fit a ranking function on logarithmized responses $\log(Y)$ as long as this provides the right ranking.

**Example 2.16.** We provide an example of an isotonic recalibration step that we apply to a previously fitted regression model. This isotonic recalibration step ensures (empirical) auto-calibration. As base model (that hopefully provides us with the correct ranking) we use a fitted FNN from Section 2.3.4. In particular, we select the FNN that has been trained on $\mathcal{L}^{\circ}_{2018}$ and which has been used to forecast $\mathcal{T}_{2019}$, this corresponds to the most right points in Figure 2.8. Denote this fitted FNN by $\widehat{\boldsymbol{z}}^{(d:1)}_{2018}$.
The isotonic recalibration step solves the following constraint minimization problem

$$\begin{aligned}\widehat{\boldsymbol{\mu}} \;&=\; \underset{\boldsymbol{\mu}\in\mathbb{R}^{|\mathcal{I}_T|}}{\arg\min} \sum_{i\in\mathcal{I}_T} v_{i,T}\left(Y_{i,T}-\mu_i\right)^2, \qquad (2.64)\\ &\qquad \text{subject to} \quad \mu_i \le \mu_k \iff \widehat{\boldsymbol{z}}^{(d:1)}_T(\boldsymbol{x}_{i,T}) \le \widehat{\boldsymbol{z}}^{(d:1)}_T(\boldsymbol{x}_{k,T}) \quad \text{for all } i,k\in\mathcal{I}_T,\end{aligned}$$

and where we set $T = 2018$. Note that (2.64) is a non-parametric weighted square loss minimization problem that preserves the ranks assigned by the FNN $\widehat{\boldsymbol{z}}^{(d:1)}_T$ to the instances $i \in \mathcal{I}_T$. It provides empirical (in-sample) auto-calibration and the balance property. Moreover, the square loss in (2.64) can be replaced by any other unit deviance $\mathfrak{d}$ (having the required support) with exactly the same optimal solution.
For the isotonic recalibration step (2.64) we use the R package `monotone` [43] which is an implementation of the PAV algorithm. This R package can deal with different exposures



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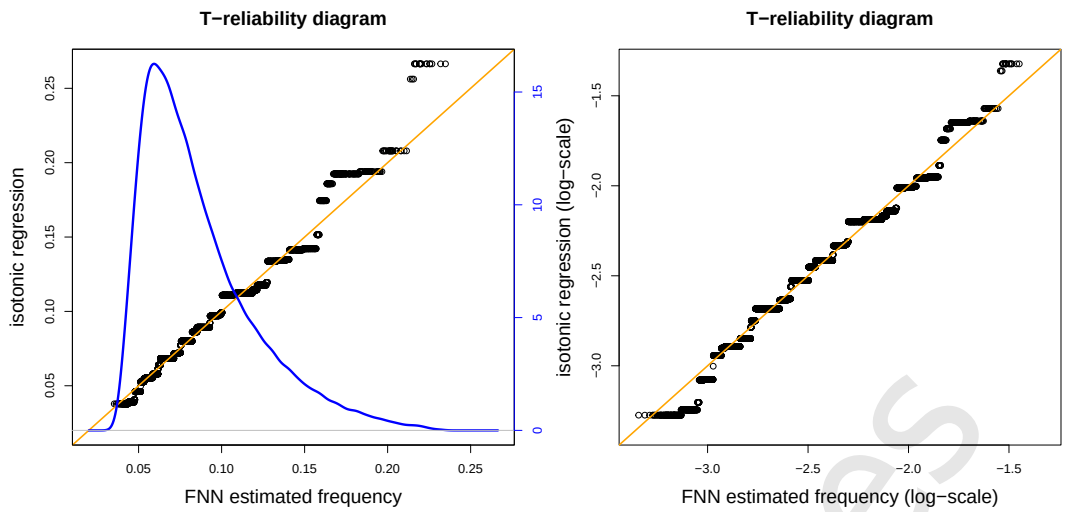


Figure 2.10: $T$-reliability diagram based on an isotonic recalibration step: (lhs) original scale, (rhs) log-scale.

$v > 0$ and it provides us with the isotonic recalibration estimate $\widehat{\boldsymbol{\mu}}$. There is one caveat though, isotonic regression tends to over-fit at the boundaries. For this reason, we merge the three smallest and the three largest isotonic regression classes to one group each, and we re-estimate the corresponding parameters by setting them equal to the observed frequencies in this new smallest and new largest frequency group, respectively. For the prediction we use the R function `stepfun` which connects the discrete values $\widehat{\boldsymbol{\mu}}$ to a step function, this is necessary to forecast on the test data $\mathcal{T}_{T+1}$.

Figure 2.10 gives the results, the left-hand side shows the figures on the original scale and the right-hand side on the log-scale. These plots are also called CORP estimates of the $T$-reliability diagram; see Gneiting–Resin [93]. CORP stands for 'consistent - optimally binned - reproducible - PAV based'; this has been introduced in Dimitriadis et al. [67]. A $T$-reliability diagram plots the isotonic regression estimates $\widehat{\mu}_i$ against its rankings $\widehat{\boldsymbol{z}}_T^{(d:1)}(\boldsymbol{x}_{i,T})$, for $i \in \mathcal{I}_T$. If the rankings are auto-calibrated, then all black dots should lie on the orange diagonal line in this $T$-reliability diagram. From Figure 2.10 (lhs) we observe some under-estimation in the upper tail and from Figure 2.10 (rhs) some over-estimation in the lower tail, telling us that the FNN cannot fully correctly capture the extremely good and bad insurance policies. The blue line in Figure 2.10 (lhs) gives the empirical density of the FNN predictions $\widehat{\boldsymbol{z}}_T^{(d:1)}(\boldsymbol{x}_{i,T})$, for $i \in \mathcal{I}_T$. This shows that in the main body of the FNN predictors we have auto-calibration, and the upper tail, suffering from under-estimation, belongs to only very few insurance policies; this also explains why model fitting is not very accurate in this part of the covariate space $\mathcal{X}$.

Finally, we remark that there is a certain sparsity of different values in the isotonic regression estimate $\widehat{\boldsymbol{\mu}}$. In fact, we have precisely 46 different values, i.e., all insurance policies are binned into 46 different price categories. This is a common phenomenon in isotonic regression under a low signal-to-noise ratio, and it is precisely the topic that has been investigated in Wüthrich–Ziegel [206]. ■

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### 2.5.3 Over-dispersion and zero-inflation

In our above examples, we have used the Poisson model and the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$ for model fitting. Following Gneiting [90] and Gneiting–Raftery [91], we should use a strictly consistent loss function for mean estimation, see (2.40), and due to Savage [177] we know that these strictly consistent loss functions are given by the Bregman divergences (if and only if). Since every unit deviance $\mathfrak{d}$ is a Bregman divergence, see Remark 2.9, indeed, the Poisson unit deviance $\mathfrak{d}_{\text{Poi}}$ is a valid choice for mean estimation. Since this strict consistency is an infinite sample (asymptotic) statement, we can still fine-tune the speed of convergence, and Gourieroux et al. [98, Theorem 4] tell us that we have optimal speed of convergence within the EDF if we also get the variance modeling right. This distinguishes more or less optimal unit deviance choices $\mathfrak{d}$ for finite sample estimation; we also refer to Lindholm–Wüthrich [138] who exploit this question in a GLM context.
Our data has been generated by a negative binomial model, see (A.1), with the first two moments given by

$$\mathbb{E}\left[Y \,|\, \boldsymbol{x}, v\right] = \mu^*(\boldsymbol{x}) \qquad \text{and} \qquad \operatorname{Var}\left(Y \,|\, \boldsymbol{x}, v\right) = \frac{\varphi^*}{v}\, \mu^*(\boldsymbol{x}).$$

If we get this variance right, we have optimal speed of convergence (best asymptotically normal). Unfortunately, this is non-trivial as it involves the unknown mean parameter $\mu^*(\boldsymbol{x})$. Also, the (over-)dispersion $\varphi^*$ is typically unknown and it may have a more complicated form involving covariates, too. In this case, an iterative estimation scheme can be explored similar to the one discussed in Delong–Wüthrich [61], and optimality (in a GLM context) is proved in Gourieroux et al. [98] (under certain assumptions). The iterative scheme of Delong–Wüthrich [61] involves an isotonic regression, described in the previous section, and it is based on the assumption that the variance behavior is monotone in the mean. This is often a reasonable assumption. Alternatively, one may also try different deviance loss functions. If we work within Tweedie's family, see Tweedie [192] and Bar-Lev–Enis [13], we have mean and variance

$$\mathbb{E}\left[Y \,|\, \boldsymbol{x}, v\right] = \mu(\boldsymbol{x}) \qquad \text{and} \qquad \operatorname{Var}\left(Y \,|\, \boldsymbol{x}, v\right) = \frac{\varphi}{v}\, \mu(\boldsymbol{x})^p,$$

thus, the variance is a power function of the mean with power variance parameter $p \in \{0\} \cup [1, \infty)$. This gives different weightings to the residuals $Y - \mu(\boldsymbol{x})$ when using different Tweedie's unit deviances, see (2.34). If we consider as variance functions polynomials of maximal cubic order

$$\operatorname{Var}\left(Y \,|\, \boldsymbol{x}, v\right) = \frac{1}{v}\, \left(a_0 + a_1 \mu(\boldsymbol{x}) + a_2 \mu(\boldsymbol{x})^2 + a_3 \mu(\boldsymbol{x})^3\right),$$

then there are exactly 12 different (at most cubic) polynomials that lead to distributions within the EDF: the absolutely continuous Gaussian, the integer valued Poisson, four quadratic examples (binomial, negative binomial, gamma and hyperbolic cosine), and six cubic examples (two of them are absolutely continuous and four of them are integer valued); see Letac–Mora [136, Table 2].
All these considerations are only based on mean estimation. If one wants to simulate data from such a model, or if one needs to evaluate solvency requirements, then, of



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course, higher moments and reasonable dispersion estimates are inevitable, and an isotonic regression approach may be a reasonable estimation framework. Alternatively, double regression models, such as double GLMs, may be useful; see Smyth [183], Smyth–Verbyla [185], Jørgensen–de Souza [116] or Smyth–Jørgensen [184]. Efficient estimation in a double GLM is often achieved by a saddlepoint approximation, see Wüthrich–Merz [205, Section 5.5]. All this is very much focused on MLE, and the global balance property my further constrain model estimation.

An other interesting problem to be considered is that typical claim counts data often has more zeros than a Poisson or a negative binomial model can accommodate. For instance, a conditional Poisson assumption with mean $v\mu > 0$ may look quite reasonable for $N|_{\{N>0\}}$, that is,

$$\mathbb{P}\left[N = k|\, N > 0\right] = \frac{1}{1-e^{-v\mu}}\, e^{-v\mu}\frac{(v\mu)^k}{k!} \qquad \text{for } k \geq 1, \tag{2.65}$$

but the data requires $\mathbb{P}[N = 0] \gg \exp\{-v\mu\}$. This phenomenon is called *zero-inflation*, because we have more zeros than expected under a certain (classical) model assumption. There are two different modeling approaches for dealing with this problem, either one considers a zero-inflated Poisson (ZIP) model defined by

$$\mathbb{P}\left[N = k\right] = \begin{cases} p_0 + (1-p_0)e^{-v\mu} & \text{for } k = 0, \\ (1-p_0)e^{-v\mu}\frac{(v\mu)^k}{k!} & \text{for } k \geq 1, \end{cases}$$

for $p_0 \in (0,1)$; we refer to Lambert [129], Boucher et al. [32], Calderín-Ojeda et al. [44] and Lee [131]. Alternatively, hurdle Poisson models are studied which are defined by

$$\mathbb{P}\left[N = k\right] = \begin{cases} p_0 & \text{for } k = 0, \\ \frac{1-p_0}{1-e^{-v\mu}}\, e^{-v\mu}\frac{(v\mu)^k}{k!} & \text{for } k \geq 1, \end{cases}$$

for $p_0 \in (0,1)$; we refer to Cragg [55] and Boucher et al. [30, 31]. A general difficulty with zero-inflated and hurdle Poisson models is the integration of the exposure $v > 0$, i.e., what is the right functional form to integrate an exposure into the zero-inflation probability $p_0 \in (0,1)$? Apart from that, dealing with these models seems rather straightforward. The model in (2.65) is called zero-truncated Poisson (ZTP) model, and if the exposure $v$ is equal to 1, one can easily show that the ZTP distribution belongs to the EDF with cumulant function $\kappa(\theta) = e^{\theta} + \log(1 - e^{-e^{\theta}})$ on the effective domain $\boldsymbol{\Theta} = \mathbb{R}$; see Wüthrich–Merz [205, Remark 6.20].

## 2.6 Regularization of high-cardinality categorical features

Above, we have been considering covariates as fixed (known) effects emphasizing that we treat prior rating information $\boldsymbol{x}$ as fixed covariates in our regression models. In the next chapters, we are going to add *random effects* to make claims time-series $(Y_t)_{t\leq T+1}$ dependent (for the same policyholder). This will allow us to use past claims experience $(Y_t)_{t\leq T}$ at time $T$ to forecast the next observation $Y_{T+1}$ of that policyholder. Integrating time-series information into the models leads to *panel data* (also called *longitudinal data*) considerations. Before discussing panel data, we discuss a static random effects model



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that is used for regularization in case of high-cardinality categorical covariates. This approach can be seen as a preparation for the next chapters as it uses similar techniques, but it solves a different problem. For references on high-cardinality categorical covariates see Campo–Antonio [45, 46], Delong–Kozak [60] Simchoni–Rosset [181], Avanzi et al. [10] and Richman–Wüthrich [172].

In this section, we discuss the proposal of Avanzi et al. [10] and Richman–Wüthrich [172]. If we have a (univariate) categorical covariate $x$ that takes many different levels $x \in A = \{a_1, \ldots, a_L\}$, we can result in very high-dimensional models. E.g., in the GLM framework of Section 2.3, we implement categorical covariates by dummy coding which results in $L-1$ regression parameters for a categorical covariate taking the $L$ different values in $A$, see (2.27). If we think of vehicle brands and vehicle models, this can easily result in hundreds of levels and regression parameters, respectively. Since some levels may only have scarce observations, e.g., there are some very rare vehicle models, we can have a big uncertainty in the corresponding parameter estimates: if we work with the canonical link in a GLM we have the local balance property on each level $a_j$, see Figure 2.1 (rhs), and if some levels only have a few observations with possible outliers, the corresponding regression parameters may be completely misspecified (not very credible). This is the issue studied in Campo–Antonio [45, 46], Avanzi et al. [10] and Richman–Wüthrich [172]. If we estimate an individual parameter for each level, this is called *no pooling*, see Antonio–Zhang [9], and if categorical covariates are completely disregarded we are in the case of *complete pooling*. Random effects modeling aims at finding a balance between these two extreme cases.

In a nutshell, we randomize the categorical regression parameters by using a prior distribution on the parameter space, and instead of estimating a parameter for each level, we compute their posterior distributions from the prior distribution and the observations using Bayes' rule [18]. This leads to a regularized estimation, and the number of parameters is determined by the complexity of the prior distribution choice. This requires to embed categorical levels into low-dimensional spaces, and then to choose a prior distribution on this low-dimensional space. This (entity) embedding is motivated by similar approaches in natural language processing (NLP), see Bengio et al. [19, 20, 21]. These ideas have been transferred to statistical learning, e.g., by Brébisson et al. [35], Guo–Berkhahn [99], Richman [167, 168], and we have already used it in the entity embedding approach in Section 2.4.2, but not in a Bayesian setting.

Choose a fixed embedding dimension $b \in \mathbb{N}$ and assign to each level $a_j \in A$ an embedding vector $\boldsymbol{u}_j \in \mathbb{R}^b$

$$a_j \in A \;\mapsto\; \boldsymbol{u}_j \in \mathbb{R}^b.$$

The goal is to learn these embedding vectors such that they are optimal for the prediction task to be solved. In particular, we may imagine that if we have two vehicle brands $a_j$ and $a_{j'}$ that have a similar claims pattern, there should be some proximity in their embedding vectors $\boldsymbol{u}_j$ and $\boldsymbol{u}_{j'}$ reflecting this similarity in their respective claims patterns.

We start from different insurance policies $1 \leq i \leq n$ having covariates $(\boldsymbol{x}_i, x_i)$, where $\boldsymbol{x}_i \in \{1\} \times \mathbb{R}^{q-1}$ are the continuous covariates of insurance policy $i$, and there is assumed to be one categorical covariate $x_i \in A$ that reflects the level of insurance policy $i$ in this categorical covariate; generalization to multiple categorical covariates is straightforward



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but notation gets more cumbersome. We adopt the following notation from Avanzi et al. [10]. We say that categorical covariate $x_i \in A$ of insurance policy $i$ takes level $a_{j[i]} \in A$, by defining the lower index as follows

$$i \;\mapsto\; j[i] \;=\; \{j' \in \{1,\ldots,L\} \text{ with } x_i = a_{j'}\} \;=\; \left(\mathbb{1}_{\{x_i=a_1\}},\ldots,\mathbb{1}_{\{x_i=a_L\}}\right)(1,\ldots,L)^\top.$$

Using the above embedding vectors $\boldsymbol{U} := (\boldsymbol{u}_j)_{j=1}^L$, we receive the encoded (embedded) covariate of insurance policy $i$

$$(\boldsymbol{x}_i, x_i) \;\mapsto\; (\boldsymbol{x}_i, \boldsymbol{u}_{j[i]}) \in \{1\} \times \mathbb{R}^{q-1+b}.$$

Assume we have independent (in $i$) EDF responses $Y_i \sim \mathrm{EDF}(\theta_i; v_i/\varphi, \kappa, \nu_i)$ with canonical parameters $\theta_i \in \mathring{\boldsymbol{\Theta}}$ being expressed by the regression function

$$(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]}) \;\mapsto\; \theta_i = h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right),$$

with canonical link $h = (\kappa')^{-1}$ and with regression function $\mu(\cdot) = \mu_{\boldsymbol{\vartheta}}(\cdot)$ depending on the model parameter $\boldsymbol{\vartheta}$. For given embedding vectors $\boldsymbol{U} = (\boldsymbol{u}_j)_{j=1}^L$, we receive the joint log-likelihood function of the independent EDF responses in $\boldsymbol{Y} = (Y_1,\ldots,Y_n)^\top$

$$\boldsymbol{\vartheta} \;\mapsto\; \ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}|\boldsymbol{U}) \;\propto\; \sum_{i=1}^n \frac{Y_i\, h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right) - \kappa\left(h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right)\right)}{\varphi/v_i},$$

the symbol $\propto$ expresses that we drop terms that do not depend on $\boldsymbol{\vartheta}$ and $\boldsymbol{U}$. If we jointly maximize this over $(\boldsymbol{\vartheta}, \boldsymbol{U})$ we have the no pooling case giving us the joint log-likelihood of $(\boldsymbol{\vartheta}, \boldsymbol{U})$. This no pooling case involves $Lb$ parameters from the categorical embeddings $\boldsymbol{U}$; in the terminology of Boucher–Denuit [28], this no pooling case is also called a fixed effects treatment of the latent variables $\boldsymbol{U}$. If we set $\boldsymbol{U} \equiv 0$, only maximizing over $\boldsymbol{\vartheta}$, we have the complete pooling case giving us the MLE of $\boldsymbol{\vartheta}$ neglecting the categorical covariate information $x_i$, and providing us with zero parameters from this categorical covariate. Random effects treatment of $\boldsymbol{U}$ chooses a compromise by assuming a prior density $\pi$ on $\boldsymbol{U}$ which gives us the joint log-density of $(\boldsymbol{Y}, \boldsymbol{U})$

$$\begin{aligned}\log f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{U}) \;&=\; \ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}|\boldsymbol{U}) + \log \pi(\boldsymbol{U}) \\ &\propto \left[\sum_{i=1}^n \frac{Y_i\, h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right) - \kappa\left(h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right)\right)}{\varphi/v_i}\right] + \log \pi(\boldsymbol{U}).\end{aligned} \tag{2.66}$$

In the literature, this is called the *complete log-likelihood* because it assumes that we have observed the responses $\boldsymbol{Y}$ *and* the random effects $\boldsymbol{U}$. However, $\boldsymbol{U}$ is latent (unobservable), therefore, we need to discuss how this estimation problem can be solved.
A closed form approach would integrate out the random effects, which results in considering the *marginal log-likelihood*

$$\ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}) = \log f_{\boldsymbol{\vartheta},\pi}(\boldsymbol{Y}) = \log\left(\int_{\boldsymbol{u}} f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{u})\, d\boldsymbol{u}\right) = \log\left(\int_{\boldsymbol{u}} \exp\left\{\ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}|\boldsymbol{u})\right\} \pi(\boldsymbol{u})\, d\boldsymbol{u}\right), \tag{2.67}$$

for simplicity we assume that $\pi$ is a density w.r.t. the Lebesgue measure on $\mathbb{R}^b$. This marginal log-likelihood (2.67) involves the parameters $\boldsymbol{\vartheta}$ and the parameters of the prior



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density $\pi$, i.e., compared to the no pooling case with $Lb$ parameters it (only) involves the parameters of the prior density $\pi$. Unfortunately, in most cases this marginalization cannot be solved explicitly and numerical methods or approximations need to be employed. It is important to realize that all observations in $\boldsymbol{Y}$ share the same random effect $\boldsymbol{U}$ and, thus, the integration in (2.67) makes the observations in $\boldsymbol{Y}$ dependent by this marginalization, i.e., we only have conditional independence in $\boldsymbol{Y}$, given $\boldsymbol{U}$.

A basic way to solve the above estimation problem is to maximize the joint log-density (2.66) in $(\boldsymbol{\vartheta}, \boldsymbol{U})$ which gives us the maximal a posteriori (MAP) estimator of $\boldsymbol{U}$. Choosing a Gaussian prior for $\pi(\boldsymbol{U})$ results in the classical ridge regularization of Tikhonov [190], we also refer to Hastie et al. [104]. Compared to the no pooling case, this still involves $Lb$ parameters, however, as we are just going to show, the choice of the prior density $\pi$ regularizes this parameter estimation procedure; this is also called *shrinkage* in the machine learning literature.

We make a specific prior choice. Assume that all components of $\boldsymbol{U} = (\boldsymbol{u}_j)_{j=1}^L$ are i.i.d. centered Gaussian with variance $\tau^2 > 0$. In view of (2.66), this gives us joint log-density for $(\boldsymbol{Y}, \boldsymbol{U})$

$$\begin{aligned}
&\log f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{U}) \\
&\quad \propto \sum_{j'=1}^{L} \left( \left[ \sum_{i=1}^{n} \mathbb{1}_{\{j[i]=j'\}} \frac{Y_i\, h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j'})\right) - \kappa\left(h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j'})\right)\right)}{\varphi / v_i} \right] - \frac{1}{2\tau^2} \left\| \boldsymbol{u}_{j'} \right\|^2 \right) \\
&\quad = \frac{1}{\varphi} \sum_{i=1}^{n} v_i \left[ Y_i\, h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right) - \kappa\left(h\left(\mu(\boldsymbol{x}_i, \boldsymbol{u}_{j[i]})\right)\right) - \frac{1}{w_{j[i]}} \frac{\varphi}{2\tau^2} \left\| \boldsymbol{u}_{j[i]} \right\|^2 \right], \qquad (2.68)
\end{aligned}$$

with *aggregated case weights* for the levels $1 \leq j' \leq L$

$$w_{j'} = \sum_{i=1}^{n} \mathbb{1}_{\{j[i]=j'\}}\, v_i. \qquad (2.69)$$

There are two important observations from (2.68). First, large values of $\boldsymbol{u}_j$ are penalized because they lead to large squared norms $\|\boldsymbol{u}_j\|^2$, henceforth, to small log-likelihoods. This is ridge ($L^2$-)regularization (shrinkage) towards zero. Second, regularization is inversely proportional to the aggregated case weights $w_j$, i.e., the more observations we have for a certain level $a_j$, the less the regularization of $\|\boldsymbol{u}_j\|^2/w_j$ impacts the log-likelihood (2.68). The MAP estimator is then received by solving

$$\left( \widehat{\boldsymbol{\vartheta}}^{\text{MAP}}, \widehat{\boldsymbol{U}}^{\text{MAP}} \right) \in \underset{\boldsymbol{\vartheta}, \boldsymbol{U}}{\arg\max}\ \log f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{U}). \qquad (2.70)$$

The MAP estimator (2.70) is not a full Bayesian estimator, as it does not use the full posterior distribution. Markov chain Monte Carlo (MCMC) methods may be useful to compute the posterior distribution, but the curse of dimensionality often prevents from performing such a full Bayesian analysis in regression modeling (already in comparably small GLMs). Variational Bayesian (VB) methods are often preferred. These try to approximate the posterior distribution. We briefly describe this.

The posterior density of $\boldsymbol{U}$, given observations $\boldsymbol{Y}$, is obtained from Bayes' rule [18] by

$$\pi_{\boldsymbol{\vartheta}}\left(\boldsymbol{U} | \boldsymbol{Y}\right) = \frac{f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{U})}{f_{\boldsymbol{\vartheta}, \pi}(\boldsymbol{Y})} = \frac{f_{\boldsymbol{\vartheta}}(\boldsymbol{Y} | \boldsymbol{U})\, \pi(\boldsymbol{U})}{\int_{\boldsymbol{u}} f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{u})\, d\boldsymbol{u}}.$$



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This posterior density contains the intractable integral (2.67) in the nominator. In VB methods we approximate this posterior density by a so-called *variational density*. Let $\mathcal{P} = \{p_\psi(\boldsymbol{U})\}$ be the family of candidate variational densities on the support of $\boldsymbol{U}$ and being parametrized by $\psi$. We try to find the variational density in $\mathcal{P}$ such that the true posterior $\pi_{\boldsymbol{\vartheta}}(\cdot|\boldsymbol{Y})$ has minimal KL divergence to this variational density. That is, we try to solve, see also (2.35),

$$\begin{aligned} \psi^* &\in \arg\min_{\psi} D_{\mathrm{KL}}\left(p_\psi \| \, \pi_{\boldsymbol{\vartheta}}(\cdot|\boldsymbol{Y})\right) \\ &= \arg\min_{\psi} \int_{\boldsymbol{u}} p_\psi(\boldsymbol{u}) \log\left(\frac{p_\psi(\boldsymbol{u})}{\pi_{\boldsymbol{\vartheta}}(\boldsymbol{u}|\boldsymbol{Y})}\right) d\boldsymbol{u}. \end{aligned} \tag{2.71}$$

Naturally, if the family $\mathcal{P}$ of variational densities contains the true posterior density $\pi_{\boldsymbol{\vartheta}}(\cdot|\boldsymbol{Y})$, the above minimization finds this true posterior, because the KL divergence is zero if and only if the two densities coincide, a.e., see Wüthrich–Merz [205, Lemma 2.21]. We rewrite (2.71) by using the evidence lower bound (ELBO) defined by

$$\mathcal{E}\left(\psi|\, \boldsymbol{Y}, \boldsymbol{\vartheta}\right) = \int_{\boldsymbol{u}} p_\psi(\boldsymbol{u}) \log\left(\frac{f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{u})}{p_\psi(\boldsymbol{u})}\right) d\boldsymbol{u}.$$

We have the following (useful) standard lemma in VB.

**Lemma 2.17.** *Using any variational density $p_\psi$ we can rewrite the marginal log-likelihood*

$$\ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}) = \mathcal{E}\left(\psi|\, \boldsymbol{Y}, \boldsymbol{\vartheta}\right) + D_{\mathrm{KL}}\left(p_\psi \| \, \pi_{\boldsymbol{\vartheta}}(\cdot|\boldsymbol{Y})\right).$$

**Proof of Lemma 2.17.** We have for the variational density $p_\psi$, using Bayes' rule on the second line which brings in $\boldsymbol{u}$ into the marginal log-likelihood,

$$\begin{aligned} \ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}) &= \ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}) \int_{\boldsymbol{u}} p_\psi(\boldsymbol{u})\, d\boldsymbol{u} \;=\; \int_{\boldsymbol{u}} \log f_{\boldsymbol{\vartheta},\pi}(\boldsymbol{Y})\, p_\psi(\boldsymbol{u})\, d\boldsymbol{u} \\ &= \int_{\boldsymbol{u}} \log\left(\frac{f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{u})}{\pi_{\boldsymbol{\vartheta}}(\boldsymbol{u}|\boldsymbol{Y})}\right) p_\psi(\boldsymbol{u})\, d\boldsymbol{u} \\ &= \int_{\boldsymbol{u}} \log\left(\frac{f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{u})/p_\psi(\boldsymbol{u})}{\pi_{\boldsymbol{\vartheta}}(\boldsymbol{u}|\boldsymbol{Y})/p_\psi(\boldsymbol{u})}\right) p_\psi(\boldsymbol{u})\, d\boldsymbol{u} \\ &= \mathcal{E}\left(\psi|\, \boldsymbol{Y}, \boldsymbol{\vartheta}\right) + D_{\mathrm{KL}}\left(p_\psi \| \, \pi_{\boldsymbol{\vartheta}}(\cdot|\boldsymbol{Y})\right). \end{aligned}$$

This proves the claim. □

Crucially, the left-hand side in the statement of Lemma 2.17 does not depend on $\psi$. Therefore, we can reformulate the optimization problem (2.71) as follows

$$\psi^* \in \arg\min_{\psi} D_{\mathrm{KL}}\left(p_\psi \| \, \pi_{\boldsymbol{\vartheta}}(\cdot|\boldsymbol{Y})\right) \;=\; \arg\max_{\psi} \mathcal{E}\left(\psi|\, \boldsymbol{Y}, \boldsymbol{\vartheta}\right). \tag{2.72}$$

The nice thing about the ELBO is that it does not involve the intractable posterior, but it only involves the joint density of $(\boldsymbol{Y}, \boldsymbol{U})$. Naturally, an optimal parameter $\psi^* = \psi^*(\boldsymbol{Y}, \boldsymbol{\vartheta})$ of (2.72) is a function of the observations $\boldsymbol{Y}$ and the network parameter $\boldsymbol{\vartheta}$. Using the non-negativity of the KL divergence, the ELBO satisfies

$$\ell_{\boldsymbol{Y}}(\boldsymbol{\vartheta}) \;\geq\; \max_{\psi} \mathcal{E}\left(\psi|\, \boldsymbol{Y}, \boldsymbol{\vartheta}\right). \tag{2.73}$$



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We have an equality if and only if the optimal KL divergence in (2.71) is zero, i.e., if and only if the family of variational densities $\mathcal{P}$ contains the true posterior. Moreover, this provides uniqueness because the KL divergence is zero if and only if the optimal variational density equals the true posterior density, a.e.
The ELBO (2.73) then motivates the VB estimator, subject to existence,

$$\left(\widehat{\boldsymbol{\vartheta}}^{\mathrm{VB}}, \widehat{\psi}^{\mathrm{VB}}\right) \in \underset{\boldsymbol{\vartheta},\psi}{\arg\max}\, \mathcal{E}\left(\psi \middle| \boldsymbol{Y}, \boldsymbol{\vartheta}\right). \tag{2.74}$$

Such a VB estimator can be received recursively by alternating estimation of $\psi$, given $\boldsymbol{\vartheta}$, and $\boldsymbol{\vartheta}$, given $\psi$, in both steps maximizing the right-hand side of (2.73). This provides us with a sequence of estimators $(\widehat{\psi}_s, \widehat{\boldsymbol{\vartheta}}_s)_{s\geq 1}$ giving us the following monotonicity

$$\ldots \leq \mathcal{E}\left(\widehat{\psi}_s \middle| \boldsymbol{Y}, \widehat{\boldsymbol{\vartheta}}_s\right) \leq \mathcal{E}\left(\widehat{\psi}_{s+1} \middle| \boldsymbol{Y}, \widehat{\boldsymbol{\vartheta}}_s\right) \leq \mathcal{E}\left(\widehat{\psi}_{s+1} \middle| \boldsymbol{Y}, \widehat{\boldsymbol{\vartheta}}_{s+1}\right) \leq \ldots \leq \ell_{\boldsymbol{Y}}(\widehat{\boldsymbol{\vartheta}}^{\mathrm{MLE}}).$$

If the upper bound is finite, this monotone sequence converges. But, this does not necessarily imply that the parameter values $(\widehat{\psi}_s, \widehat{\boldsymbol{\vartheta}}_s)_{s\geq 1}$ converge, this requires stronger assumptions, we also refer to Wu [199] for the equivalent problem in the Expectation-Maximization (EM) algorithm. However, in most cases we are satisfied by just choosing one parameter value $\widehat{\boldsymbol{\vartheta}}_{s^*}$ close to convergence, $s^* \gg 1$, and we use this parameter value $\widehat{\boldsymbol{\vartheta}}^{\mathrm{VB}} = \widehat{\boldsymbol{\vartheta}}_{s^*}$ as an approximation to the MLE $\widehat{\boldsymbol{\vartheta}}^{\mathrm{MLE}}$.

The remaining difficulty is to efficiently calculate the ELBO on the right-hand side of (2.73). Similar to Lemma 2.17, the ELBO satisfies

$$\begin{aligned}\mathcal{E}\left(\psi \middle| \boldsymbol{Y}, \boldsymbol{\vartheta}\right) &= \int_{\boldsymbol{u}} p_\psi(\boldsymbol{u}) \log\left(\frac{f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}, \boldsymbol{u})/\pi(\boldsymbol{u})}{p_\psi(\boldsymbol{u})/\pi(\boldsymbol{u})}\right) d\boldsymbol{u} \\ &= -D_{\mathrm{KL}}\left(p_\psi \,\|\pi\right) + \int_{\boldsymbol{u}} \log f_{\boldsymbol{\vartheta}}(\boldsymbol{Y}|\boldsymbol{u})\, p_\psi(\boldsymbol{u})\, d\boldsymbol{u}. \end{aligned} \tag{2.75}$$

First, if both the variational densities $p_\psi \in \mathcal{P}$ and the prior density $\pi$ are multivariate Gaussian with independent components, the negative KL divergence in (2.75) can be calculated in closed form, this is a well known result and it is called the *Gaussian mean field case*. In this Gaussian mean field case, the KL divergence is given by

$$D_{\mathrm{KL}}\left(p_\psi \,\|\pi\right) = \sum_{k=1}^{b}\sum_{j=1}^{L} \frac{1}{2}\left(\frac{m_{k,j}^2 + \sigma_{k,j}^2}{\tau^2} - \log\left(\frac{\sigma_{k,j}^2}{\tau^2}\right) - 1\right), \tag{2.76}$$

where $\tau^2 > 0$ is the variance of the independent centered Gaussian components of $\boldsymbol{U}$ under $\pi$, see (2.68), and $\psi = (m_{k,j}, \sigma_{k,j}^2)_{1\leq k\leq b, 1\leq j\leq L}$ is the variational parameter of the Gaussian mean field variational density $p_\psi \in \mathcal{P}$. This makes the first term on the right-hand side of (2.75) tractable. Second, for Gaussian mean field variational densities $p_\psi$, one can use that $\boldsymbol{U} \sim p_\psi$ is equal in distribution to

$$\boldsymbol{U} \overset{(\mathrm{d})}{=} \boldsymbol{a}(\psi) + A(\psi)\,\boldsymbol{\varepsilon} = \boldsymbol{m} + \mathrm{diag}(\sigma_{k,j}^2)_{k,j}\,\boldsymbol{\varepsilon},$$

for a multivariate standard Gaussian random vector $\boldsymbol{\varepsilon}$ with independent components, for mean vector $\boldsymbol{a}(\psi) = \boldsymbol{m} = (m_{k,j})_{k,j}$ and diagonal covariance matrix $A(\psi) = \mathrm{diag}(\sigma_{k,j}^2)_{k,j}$. This reformulation is called the reparametrization trick of Kingma–Welling [124]. It



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allows one to evaluate the second term in (2.75) empirically because simulation of the multivariate standard Gaussian random vector $\boldsymbol{\varepsilon}$ no longer involves (hyper-)parameters because $\psi$ only enters the deterministic functions $\psi \mapsto \boldsymbol{a}(\psi)$ and $\psi \mapsto A(\psi)$. Precisely this fact then allows one to solve the $\psi$-maximization (2.73) with gradient ascent from an empirical version of the integral in (2.75). Simulate i.i.d. standard Gaussian random variables $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbb{1})$, then we have an empirical version of the ELBO

$$\begin{aligned}\widehat{\mathcal{E}}\,(\psi|\,\boldsymbol{Y},\boldsymbol{\vartheta}) \;&=\; \frac{1}{\varphi}\sum_{i=1}^n v_i \Bigg\{ Y_i\, h\left(\mu(\boldsymbol{x}_i, \boldsymbol{m}_{j[i]} + \Sigma_{j[i]}^{1/2}\boldsymbol{\varepsilon}_i)\right) - \kappa\left(h\left(\mu(\boldsymbol{x}_i, \boldsymbol{m}_{j[i]} + \Sigma_{j[i]}^{1/2}\boldsymbol{\varepsilon}_i)\right)\right) \\ &\qquad - \frac{1}{w_{j[i]}}\sum_{k=1}^b \frac{\varphi}{2}\left(\frac{m_{k,j[i]}^2 + \sigma_{k,j[i]}^2}{\tau^2} - \log\left(\frac{\sigma_{k,j[i]}^2}{\tau^2}\right) - 1\right)\Bigg\}, \quad (2.77)\end{aligned}$$

with $\boldsymbol{m}_j = (m_{1,j}, \ldots, m_{b,j})^\top \in \mathbb{R}^b$ and $\Sigma_j^{1/2} = \operatorname{diag}(\sigma_{1,j}, \ldots, \sigma_{b,j}) \in \mathbb{R}^{b\times b}$ being the mean and covariance matrix of the embedding vector $\boldsymbol{u}_j$ under the Gaussian mean field variational density $p_\psi$; for a more detailed derivation we refer to Richman–Wüthrich [172, (2.17)-(2.18)]. We highlight the similarity of this last expression to the ridge regularization case (2.68) giving the MAP estimator (2.70). This empirical version of the ELBO (2.77) can now be maximized in $(\boldsymbol{\vartheta}, \psi)$ using a gradient ascent algorithm. This gives an approximative solution to (2.74).

Richman–Wüthrich [172] provide a discussion about hyper-parameter selection, and an extension to multiple categorical covariates which may even have a hierarchical structure. This hierarchical structure estimation problem can then be solved by using recurrent neural networks (RNNs) or Transformers. For more details, implementation and an explicit example in motor insurance pricing, we refer to this reference.

**Example 2.18.** We present an example that is based on a Poisson GLM for claim counts. We select the canonical log-link of the Poisson GLM and we assume that the regression function $\mu(\boldsymbol{x}, u)$ is given by the following GLM

$$(\boldsymbol{x}_i, u_{j[i]}) \;\mapsto\; \log\mu(\boldsymbol{x}_i, u_{j[i]}) = \langle\boldsymbol{\beta}, \boldsymbol{x}_i\rangle + u_{j[i]},$$

for regression parameter $\boldsymbol{\beta} \in \{0\} \times \mathbb{R}^{q-1}$ and for $b = 1$ dimensional embeddings $u_j \in \mathbb{R}$; this is why we do not use bold-face notation in the embedding parameters $u_1, \ldots, u_L$. To make the parameters identifiable, we set $\beta_0 = 0$; note that we do not use dummy coding for the categorical covariate here, but rather one-hot encoding. This GLM gives identical predictive results to the previously studied GLMs with one categorical covariate. The joint log-likelihood (2.68) of $(\boldsymbol{Y}, \boldsymbol{U})$ under a centered and independent Gaussian prior assumption for $\pi$ is given by

$$\log f_{\boldsymbol{\beta}}(\boldsymbol{Y}, \boldsymbol{U}) \;\propto\; \sum_{i=1}^n v_i \left\{ Y_i\left(\langle\boldsymbol{\beta}, \boldsymbol{x}_i\rangle + u_{j[i]}\right) - \exp\left(\langle\boldsymbol{\beta}, \boldsymbol{x}_i\rangle + u_{j[i]}\right) - \frac{1}{w_{j[i]}}\,\frac{u_{j[i]}^2}{2\tau^2}\right\}.$$

This formula expresses ridge regularization of the embedding parameters $u_1, \ldots, u_L \in \mathbb{R}$ with regularization parameter $1/(2\tau^2)$. Maximizing this quantity gives us the MAP estimator (2.70) of $(\boldsymbol{\beta}, \boldsymbol{U})$. For $\tau^2 = \infty$, we have a classical GLM with one-hot encoding and excluding the intercept; this is the no pooling case. For $\tau^2 = 0$, we completely drop



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the categorical covariate, complete pooling case, and this requires to add an intercept instead.
The Gaussian mean field VB estimation provides empirical ELBO (2.77) in this Poisson GLM case

$$\begin{aligned}\widehat{\mathcal{E}}\left(\psi|\,\boldsymbol{Y},\boldsymbol{\beta}\right) &= \sum_{i=1}^{n} v_i \Bigg\{ Y_i \left(\langle \boldsymbol{\beta}, \boldsymbol{x}_i\rangle + m_{j[i]} + \sigma_{j[i]}\varepsilon_i\right) - \exp\left(\langle \boldsymbol{\beta}, \boldsymbol{x}_i\rangle + m_{j[i]} + \sigma_{j[i]}\varepsilon_i\right) \\ &\qquad - \frac{1}{w_{j[i]}}\frac{1}{2}\left(\frac{m_{j[i]}^2}{\tau^2} + \frac{\sigma_{j[i]}^2}{\tau^2} - \log\left(\frac{\sigma_{j[i]}^2}{\tau^2}\right) - 1\right)\Bigg\},\end{aligned}$$

for i.i.d. standard Gaussian random variables $\varepsilon_i$ and Gaussian mean field variational parameters $m_j \in \mathbb{R}$ and $\sigma_j > 0$. Now the similarities between the MAP estimation (ridge regularization) and the Gaussian mean field VB estimation become apparent. The variational mean $m_{j[i]}$ plays the role of the embedding parameter $u_{j[i]}$ of instance $i$. In the Gaussian mean field case this embedding parameter is perturbed by Gaussian noise $\sigma_{j[i]}\varepsilon_i$ to exploit the neighborhood of mean parameter $m_{j[i]}$. At the same time, this perturbation is penalized by additional terms through the KL divergence (2.76) to ensure that the perturbation remains on a similar scale than the prior uncertainty $\tau$; note that we have second order Taylor expansion

$$\frac{\sigma_{j[i]}^2}{\tau^2} - \log\left(\frac{\sigma_{j[i]}^2}{\tau^2}\right) - 1 \;\approx\; \frac{1}{2}\left(\frac{\sigma_{j[i]}^2}{\tau^2} - 1\right)^2,$$

for the ratio being close to one. The VB approach has the advantage that it provides an approximation to the posterior distribution. This allows one to study parameter estimation uncertainty quantified by the estimated posterior standard deviations $\sigma_j$. This closes this example. ■

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# Chapter 3

# Static Mixed Effects Models

## 3.1 Literature overview

We start with a literature overview on *static mixed effects* models for *experience rating*. Generally speaking, the models presented in Chapter 2 have only been considering prior rating information which has been integrated as fixed effects into the regression functions, see, e.g., regression assumption (2.21). This prior rating information can be complemented by past claims experience that is connected by latent random factors. Essentially, these latent random factors make claims observations dependent. If we only consider latent random factors and past claims experience we have a *random effects model*. If we combine latent random factors, past claims experience and prior rating information, we obtain a *mixed effects model* using both fixed effects (through prior rating information) and latent random effects (exploiting past claims experience).

We start this chapter by considering *(homogeneous) static random effects models*, homogeneous means that we do not consider any fixed effects nor systematic insurance policy heterogeneity in the random effects, and *static* refers to a latent random factor model that does *not* have a temporal component. Thus, in a first step we assume only having response variables $Y_{1:T+1} = (Y_1, \ldots, Y_T, Y_{T+1})$, and not having any insurance policy specific covariates in terms of fixed effects. These response variables $Y_{1:T+1}$ share a common static latent (random) factor $\Theta$ which follows the same law for all insurance policies. The goal is to optimally predict $Y_{T+1}$, given the (past) observations $Y_{1:T} = (Y_1, \ldots, Y_T)$; for the moment we only consider one insurance policy, therefore we do not add a lower index $i$ for the insurance policy considered, see Section 2.3.4. If there is a common static latent (random) factor $\Theta$ that influences all responses $Y_{1:T+1} = (Y_1, \ldots, Y_T, Y_{T+1})$, they become dependent (associated), and we can learn from past observations $Y_{1:T}$ for predicting the future claim $Y_{T+1}$ through this dependence. *Experience rating* deals with the problem of optimally forecasting $Y_{T+1}$, based on past claims experience $Y_{1:T}$; therefore, it is also called *posterior ratemaking* (in contrast to prior information rating). Essentially, experience rating should result in a higher prediction for $Y_{T+1}$, if past claims history $Y_{1:T}$ is not positive and vice versa, if the claims are positively associated. Such problems can be solved using suitable stochastic models, and the main tool for predicting $Y_{T+1}$, based on past claims experience $Y_{1:T}$, is Bayes' rule [18]. Often the resulting formulas turn out to be rather complex. Therefore, there are simplified commercial schemes for insurance un-

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derwriting that can more easily be explained to stakeholders. Such simplified commercial schemes are called *bonus-malus systems* (BMS); see page xix of Denuit et al. [65].

Early work in the 1960s on experience rating includes the Poisson-gamma model; for this early work on experience rating in motor insurance pricing we refer to Bichsel [24, 25], Franckx [82], Martin [144], Philipson [157] and Thyrion [189]. These considerations also build an intuitive foundation of the linear credibility theory framework introduced by Bühlmann [39] and Bühlmann–Straub [42]; for an extended essay on linear credibility theory, called *Bühlmann credibility*, we refer to the monograph of Bühlmann–Gisler [41], for BMS we refer to Lemaire [134, 135].

There are many variations of the Poisson-gamma model, e.g., a Poisson-inverse Gaussian variant, see Dean et al. [59], Lemaire [135], Brouhns et al. [38], Boucher et al. [30], and Sundt [187]. Moreover, Walhin–Paris [198] and Boucher–Denuit [28] encompass more models of the mixed Poisson family type. Pinquet [158] considers mixed gamma and mixed log-normal distributions for claim size modeling, and Frangos–Vrontos [83] emphasize the importance of joint claim counts and claim size modeling for BMS considerations. Typically, if the response variables $Y_t$ belong to the EDF and if one chooses a *conjugate prior* for the static random effects, one receives closed from solutions for the posterior; this has been considered by Jewell [109, 110, 111], Diaconis–Ylvisaker [66] and Landsman–Makov [130], see also Section 3.2.2, below. Most other cases cannot be solved explicitly, and one needs to rely on approximations, e.g., the Bühlmann–Straub [42] linear credibility framework, or numerical schemes like MCMC simulations; see Gilks et al. [89].

Another interesting idea to model static random effects, in case of intractable models, is the so-called *empirical Bayesian approach*. The non-parametric empirical Bayes' formula of Robbins [174] came as a surprise to the statistical community because it can be used without specifying the law of the static random effects. Robbins' formula has been developed in communication with Maurice Kenneth Tweedie, therefore, it also is called Tweedie's formula, see Efron [72]. In a parametric empirical Bayesian approach, this is closely related to the James–Stein estimator [107] which was another surprising result to the statistical community and questioned the use of MLE, see Efron [74]. For an introduction and examples to empirical Bayesian methods, we also refer to Casella [47].

While most of the discussion so far has been focusing on pure static random effects models, there is also a huge variety of models considering *static mixed effects models*, i.e., including covariates (prior rating information) as fixed effects. That is, in addition to considering the claims history $Y_{1:T}$ for forecasting $Y_{T+1}$, static mixed effects models include prior rating information in terms of (fixed effects) covariates $\boldsymbol{x}_{T+1}$, see Chapter 2 for fixed effects examples. Posterior ratemaking in that case means that we predict $Y_{T+1}$, based on $(Y_{1:T}, \boldsymbol{x}_{T+1})$. The previously mentioned work of Pinquet [158] and Frangos–Vrontos [83] consider such static mixed effects models where the covariates enter a GLM term. Fundamental work in this field has been done by Dionne–Vanasse [68, 69], Pinquet [158, 159, 160], Brouhns et al. [38], Boucher–Denuit [28], Denuit et al. [65] and Lee et al. [132]. Furthermore, we mention Boucher et al. [29, 30, 31] who consider in this context the modeling of excess zeros either using zero-inflated Poisson models or hurdle models, see Section 2.5. Corradin et al. [52], Duval et al. [71] and Yanez et al. [207]

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extend experience rating beyond past claims history $Y_{1:T}$ by including additional past information that may come, e.g., from telematics car driving data. That is, we may, e.g., have a history of night-driving, of extreme braking events or of speeding events. Having a static latent factor model, we can not only receive an association between the claims $Y_{1:T+1}$, but we may receive dependence between any useful response variable, e.g., an extreme braking index measuring the frequency and severity of extreme braking events.

Besides the above mentioned literature on experience rating, there is a second stream of literature that designs BMS. As mentioned in Denuit et al. [65] and Boucher [27], BMS are simplified (commercial) schemes of experience rating models. Instead of calculating full posterior expected claims $\mathbb{E}[Y_{T+1}|Y_{1:T}, \boldsymbol{x}_{T+1}]$, BMS rather try to design (simple) transition rules (Markov chains) on a finite state space, with levels (states) referring to the quality of the drivers. E.g., an accident-free calendar year leads to a decrease of one level in a BMS scheme, and, in contrast, a calendar year with an accident leads to an increase of, say, three levels in that scheme. Such BMS schemes then summarize past claims history in one single summary statistics which is used as a score/covariate in a regression model. This allows the insurance company to weight past observations by seniority depending on how transition rules are defined. Early work on such BMS schemes has been done by Loimaranta [139] and De Pril [57] who studied the efficiency and the asymptotic behavior of BMS schemes. More recent work includes Denuit et al. [65], Boucher–Inoussa [33], Verschuren [195] and Ágoston–Gyetvai [2], the last reference considering mixed integer linear programming for finding optimal transition rules and premium scales. We are rather going to study the full experience rating problem using random and mixed effects, and we do not discuss any further its BMS discretized commercial schemes in the present notes.

## 3.2 Static random effects model

### 3.2.1 Bühlmann–Straub credibility framework

The starting point of experience rating is the Bühlmann [39] and the Bühlmann–Straub [42] linear credibility framework. Assume that the responses $Y_{1:T+1}$ depend on a (static) latent factor $\Theta$, and conditionally, given this latent factor $\Theta$, all components of $Y_{1:T+1}$ are independent. This latent factor $\Theta$ may reflect unknown (or unobservable) common risk characteristics of the considered insurance policy and, in particular, it will make the components of the claims vector $Y_{1:T+1}$ associated.

A static random effects model assumes that this random factor $\Theta$ has a prior distribution $\pi(\theta)$.[1] This prior distribution may either reflect uncertainty in the "right" choice of $\Theta$ for the considered insurance policy, but it may also be interpreted as the random selection of a policyholder from an insurance portfolio with risk characteristics being described by this prior distribution over the entire portfolio; this is similar to the population distribution $\mathbb{P}$ discussed in Remarks 2.10. We note that this is a homogeneous consideration in the sense that all policyholders of the insurance portfolio share the same prior distribution $\pi(\theta)$. This is different from the literature in statistics that usually imposes a regression

[1] In Bayesian theory one often uses prior distribution $\pi(\theta)$ interchangeably for distribution and density.



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structure on the unobserved factors (random effects); see, e.g., Laird–Ware [128] and Robinson [175]

**Definition 3.1** (static random effects EDF model)**.** *The tuple* $(Y_{1:T+1}, \Theta)$ *follows a (homogeneous) static random effects EDF model if, given* $\Theta \sim \pi(\theta)$*, the real-valued components of* $Y_{1:T+1}$ *are conditionally independent with conditional EDF distributions* $Y_t|_\Theta \sim \mathrm{EDF}(\Theta; v_t/\varphi, \kappa, \nu_t)$ *for* $1 \leq t \leq T+1$*.*

Remark that we use the notation $\Theta$ for the (homogeneous) static random effects. The support of this random variable is the effective domain $\boldsymbol{\Theta} \subseteq \mathbb{R}$ of the chosen EDF, and it takes values (canonical parameters) $\theta \in \boldsymbol{\Theta}$; the effective domain $\boldsymbol{\Theta} \subseteq \mathbb{R}$ is an interval and we assume that this interval has a non-empty interior $\mathring{\boldsymbol{\Theta}} \neq \emptyset$; see Section 2.2.1. Realizations $\Theta = \theta$ should not be confused with $\boldsymbol{\vartheta}$, the latter is typically used to denote the parameters of the selected (regression) model, e.g., the FNN parameter, see Section 2.4.1.

The general goal in this section is to forecast $Y_{T+1}$, given $Y_{1:T}$, under the assumption of having a (homogeneous) static random effects EDF model. That is, we aim at computing the *predictive mean* (*pure risk premium*, *one-period ahead prediction*)

$$\begin{aligned}
\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} &:= \mathbb{E}\left[Y_{T+1}|\, Y_{1:T}\right] \;=\; \mathbb{E}\left[\mathbb{E}\left[Y_{T+1}|\,\Theta\right]|\, Y_{1:T}\right] \\
&= \int_\theta \int_y y\, f(y;\theta, v_{T+1}/\varphi)\, d\nu_{T+1}(y)\, d\pi(\theta|Y_{1:T}) \qquad (3.1)\\
&= \int_\theta \kappa'(\theta)\, d\pi(\theta|Y_{1:T}) \;=\; \mathbb{E}\left[\kappa'(\Theta)\big|\, Y_{1:T}\right],
\end{aligned}$$

where $\pi(\theta|Y_{1:T})$ is the posterior distribution of the latent factor $\Theta$, given $Y_{1:T}$, and we use conditional independence between $Y_{1:T}$ and $Y_{T+1}$, given $\Theta$, and the EDF mean property (2.2). Prediction uncertainty in terms of the (conditional) mean squared error (MSE) can also be calculated as follows

$$\begin{aligned}
\mathbb{E}\left[\left(Y_{T+1} - \mu^{\text{post}}_{Y_{T+1}|Y_{1:T}}\right)^2 \Big| Y_{1:T}\right] &= \mathrm{Var}\left(Y_{T+1}|\, Y_{1:T}\right) \qquad (3.2)\\
&= \mathbb{E}\left[\mathrm{Var}\left(Y_{T+1}|\,\Theta\right)|\, Y_{1:T}\right] + \mathrm{Var}\left(\mathbb{E}\left[Y_{T+1}|\,\Theta\right]|\, Y_{1:T}\right)\\
&= \frac{\varphi}{v_{T+1}} \mathbb{E}\left[\kappa''(\Theta)\big|\, Y_{1:T}\right] + \mathrm{Var}\left(\kappa'(\Theta)\big|\, Y_{1:T}\right).
\end{aligned}$$

The first term is a Bayesian (average) prediction uncertainty (irreducible risk) measured by the variance, and the second term is a Bayesian (average) parameter uncertainty.
The predictive mean (3.1) can be calculated in selected cases, but, generally, it has to be computed numerically, e.g., using MCMC methods, or it can be approximated either by the linear credibility estimator of Bühlmann–Straub [42] or by VB methods, see Section 2.6. Under Definition 3.1, we have conditional moments, see also (3.2),

$$\mathbb{E}\left[Y_t|\,\Theta\right] = \kappa'(\Theta) \qquad \text{and} \qquad \mathrm{Var}\left(Y_t|\,\Theta\right) = \frac{\varphi}{v_t}\,\kappa''(\Theta).$$

These are the Bühlmann–Straub [42] assumptions of a single insurance policy, and extending this to multiple independent policies gives us an example of the classical Bühlmann–Straub [42] model.



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**Model 3.2** (Bühlmann–Straub model [42])**.** *Assume we have $n \in \mathbb{N}$ insurance policies (or risk classes) and $T+1$ responses per insurance policy. Assume fixed volumes $v_{i,t} > 0$, $1 \leq i \leq n$ and $1 \leq t \leq T+1$, are given.*

- *Conditionally, given $\Theta_i$, the components of $Y_{i,1:T+1}$ are independent with the first two conditional moments given by*

$$\begin{aligned} \mathbb{E}\left[Y_{i,t} \middle| \Theta_i\right] &= \mu(\Theta_i), \\ \mathrm{Var}\left(Y_{i,t} \middle| \Theta_i\right) &= \frac{\sigma^2(\Theta_i)}{v_{i,t}}, \end{aligned}$$

  *for measurable functions $\mu : \boldsymbol{\Theta} \to \mathbb{R}$ and $\sigma^2 : \boldsymbol{\Theta} \to \mathbb{R}_+$.*

- *The pairs $(\Theta_1, Y_{1,1:T+1}), \ldots, (\Theta_n, Y_{n,1:T+1})$ are independent and $\Theta_1, \ldots, \Theta_n$ are i.i.d.*

*Moreover, all considered random variables are assumed to have finite second moments.*

The Bühlmann–Straub Model 3.2 is essentially based on first and second moment assumptions. This is sufficient for considering the method of moments estimators or MSE minimizations. For MLE we need full distributional assumptions, similar to Definition 3.1. From an actuarial viewpoint the latter is not necessarily preferred as MLE does not automatically imply the balance property, which is important in financial applications.

To get a better understanding of (3.1)-(3.2) we solve the Poisson-gamma example of Bichsel [25] which has the advantage of a closed form solution, though one should be aware of the fact that intractability and computational issues are very prohibitive in this field of Bayesian inference.

**Example 3.3** (Poisson-gamma model)**.** Assume the responses $N_t$, $1 \leq t \leq T+1$, are conditionally independent and Poisson distributed with conditional means $v_t\Lambda$, for given exposures $v_t > 0$ and given static latent factor $\Lambda \sim \Gamma(\alpha, \alpha/\mu_0)$ with prior parameters $\mu_0 > 0$ and $\alpha > 0$. This gamma distributed random variable has prior mean and variance

$$\mathbb{E}\left[\Lambda\right] = \mu_0 \qquad \text{and} \qquad \mathrm{Var}\left(\Lambda\right) = \mu_0^2/\alpha.$$

In this example it is more convenient to directly work with the means $\Lambda v_t$ of the Poisson random variables $N_t$, and not the canonical parameter of the underlying EDF. The cumulant function of the Poisson model is given by $\kappa(\theta) = \exp\{\theta\}$ for canonical parameter $\theta \in \boldsymbol{\Theta} = \mathbb{R}$. For the static random effects $\Theta$ this gives us mean in the reproductive version

$$\kappa'(\Theta) = \exp\{\Theta\} = \Lambda \ \sim \ \Gamma(\alpha, \alpha/\mu_0), \tag{3.3}$$

i.e., the canonical parametrization is obtain by the log-link $\Theta = \log(\Lambda)$. The observations $N_t$ are in the additive form, and the reproductive form is in the Poisson model given by $Y_t = N_t/v_t$. We have the following first two moments

$$\mathbb{E}\left[N_t\right] = \mathbb{E}\left[\mathbb{E}\left[N_t \middle| \Lambda\right]\right] = v_t\mathbb{E}\left[\Lambda\right] = v_t\mu_0,$$

and

$$\mathrm{Var}\left(N_t\right) = \mathbb{E}\left[\mathrm{Var}\left(N_t \middle| \Lambda\right)\right] + \mathrm{Var}\left(\mathbb{E}\left[N_t \middle| \Lambda\right]\right) = v_t\mu_0\left(1 + \frac{v_t\mu_0}{\alpha}\right).$$



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In fact, this is the variance of the negative binomial model, see (2.57), and the unconditional claim counts $N_t$ are negative binomially distributed with a dependence along the time axis $t \geq 1$ through the common static latent factor $\Lambda \sim \Gamma(\alpha, \alpha/\mu_0)$. Moreover, this model satisfies the Bühlmann–Straub Model 3.2 (for a single insurance policy we drop index $i$) with conditional mean and variance

$$\begin{aligned}
\mathbb{E}\left[Y_t|\,\Theta\right] &= \mathbb{E}\left[Y_t|\,\Lambda\right] = \Lambda = \exp\{\Theta\} = \kappa'(\Theta), \\
\mathrm{Var}\left(Y_t|\,\Theta\right) &= \mathrm{Var}\left(Y_t|\,\Lambda\right) = \frac{1}{v_t}\,\Lambda = \frac{1}{v_t}\exp\{\Theta\} = \frac{1}{v_t}\,\kappa''(\Theta).
\end{aligned}$$

We calculate the posterior distribution of $\Lambda$, given the observations $N_{1:T}$. Using Bayes' rule [18], we have posterior density

$$\begin{aligned}
\pi(\Lambda|N_{1:T}) &\propto f(N_{1:T}|\Lambda, v_{1:T})\,\pi(\Lambda) \\
&= \left[\prod_{t=1}^{T} e^{-\Lambda v_t}\frac{(\Lambda v_t)^{N_t}}{N_t!}\right]\frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)}\Lambda^{\alpha-1}e^{-\alpha\Lambda/\mu_0} \\
&\propto \Lambda^{\alpha+\sum_{t=1}^T N_t - 1}\,e^{-\left(\alpha/\mu_0+\sum_{t=1}^T v_t\right)\Lambda}.
\end{aligned}$$

This gives us posterior distribution

$$\Lambda|_{N_{1:T}} \sim \Gamma\left(\alpha + \sum_{t=1}^T N_t, \alpha/\mu_0 + \sum_{t=1}^T v_t\right).$$

This provides for the predictive mean (3.1) the following Bühlmann credibility formula

$$\begin{aligned}
\mu^{\text{post}}_{N_{T+1}|N_{1:T}} &= \mathbb{E}\left[N_{T+1}|\,N_{1:T}\right] = v_{T+1}\,\mathbb{E}\left[\Lambda|\,N_{1:T}\right] \\
&= v_{T+1}\left[\frac{\sum_{t=1}^T v_t}{\alpha/\mu_0+\sum_{t=1}^T v_t}\,\frac{\sum_{t=1}^T N_t}{\sum_{t=1}^T v_t} + \frac{\alpha}{\alpha/\mu_0+\sum_{t=1}^T v_t}\right] \\
&= v_{T+1}\left[\omega_T\,\bar{Y}_T + (1-\omega_T)\,\mu_0\right] \qquad (3.4) \\
&= v_{T+1}\mu_0\left[\omega_T\,\bar{\Lambda}_T + (1-\omega_T)\right],
\end{aligned}$$

with credibility weight

$$\omega_T = \frac{\sum_{t=1}^T v_t}{\sum_{t=1}^T v_t + \alpha/\mu_0} \in (0,1), \qquad (3.5)$$

and with empirical means

$$\bar{Y}_T = \frac{1}{\sum_{t=1}^T v_t}\sum_{t=1}^T N_t = \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s}\,Y_t \qquad \text{and} \qquad \bar{\Lambda}_T = \frac{\bar{Y}_T}{\mu_0} = \frac{\sum_{t=1}^T v_t Y_t}{\sum_{t=1}^T v_t\mu_0}.$$

Formula (3.4) is formula (1) in Bichsel [25] adjusted for different time exposures $v_t > 0$. ■

**Remarks 3.4.**

- We observe that the claims history $N_{1:T}$ directly influences the pure risk premium $\mu^{\text{post}}_{N_{T+1}|N_{1:T}}$ via the empirical mean $\bar{Y}_T$. This implies that we use the past claims experience $N_{1:T}$ for insurance rating, and higher past empirical means lead to higher premiums. This is a consequence of the (positive) association through the common static risk factor $\Lambda$.



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- For increasing past experience the influence of the prior mean $\mathbb{E}[\Lambda] = \mu_0$ vanishes because the credibility weight $\omega_T$ converges to 1 for $\sum_{t=1}^T v_t \to \infty$.
- The static random effects $\Theta$ of the Poisson-gamma model is determined by (3.3), i.e., $\Theta = \log(\Lambda)$ with $\Lambda \sim \Gamma(\alpha, \alpha/\mu_0)$. This random variable corresponds to the second component of the vector-valued version of the exponential family (EF) gamma model, see Wüthrich–Merz [205, Section 2.1.3] with the first moment given by $\mathbb{E}[\Theta] = \Gamma'(\alpha)/\Gamma(\alpha) - \log(\alpha/\mu_0)$. The moment generating function is for $r > -\alpha$ given by
$$M_\Theta(r) = \mathbb{E}\left[\exp\{r\Theta\}\right] = \mathbb{E}\left[\Lambda^r\right] = \frac{\Gamma(\alpha+r)}{\Gamma(\alpha)}\,(\alpha/\mu_0)^{-r}. \tag{3.6}$$

■

Coming back to (3.2), we can also conclude that the Bayesian premium $\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}}$ has minimal (conditional) MSE prediction uncertainty, because
$$\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} \;=\; \underset{\widehat{\mu}\in\mathbb{R}}{\arg\min}\; \mathbb{E}\left[(Y_{T+1}-\widehat{\mu})^2 \middle| Y_{1:T}\right], \qquad \mathbb{P}\text{-a.s.}$$
We can equivalently solve
$$\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} \;=\; \underset{\widehat{\mu}(\cdot)}{\arg\min}\; \mathbb{E}\left[(Y_{T+1}-\widehat{\mu}(Y_{1:T}))^2\right], \tag{3.7}$$
where the minimization runs over all measurable functions $\widehat{\mu}: \mathbb{R}^T \to \mathbb{R}$ such that $\widehat{\mu}(Y_{1:T})$ is square-integrable. This minimization is also the foundation of the Bühlmann–Straub estimator discussed below, and basically it tells us that the posterior mean $\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}}$ is received by the orthogonal projection of $Y_{T+1}$ onto the subspace generated by $Y_{1:T}$ in the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$. We come back to this in Section 3.2.3, below.

**Example 3.5** (gamma-gamma model)**.** As a second example we consider a gamma model for insurance claim size modeling. Assume that the responses $Y_t$, $1 \le t \le T+1$, are conditionally independent and gamma distributed $Y_t|_\Lambda \sim \Gamma(v_t/\varphi, v_t\Lambda/\varphi)$, given $\Lambda > 0$, with shape parameter $1/\varphi > 0$ and exposure $v_t \in \mathbb{N}$. The conditional mean and variance satisfy for $1 \le t \le T+1$
$$\mathbb{E}\left[Y_t|\,\Lambda\right] = 1/\Lambda \qquad \text{and} \qquad \text{Var}\left(Y_t|\,\Lambda\right) = \frac{\varphi}{v_t}\,\Lambda^{-2},$$
this is again a Bühlmann–Straub Model 3.2. This is the reproductive form of the gamma distribution. We motivate this and we give an explicit meaning to the exposure $v_t \in \mathbb{N}$. Assume that we have $v_t$ individual claims $Z_{t,k}$, $1 \le k \le v_t$, in calendar year $t$. Given $\Lambda$, these claims are assumed to be conditionally i.i.d. gamma distributed with shape parameter $1/\varphi > 0$ and scale parameter $\Lambda > 0$. Since the gamma distribution is closed under aggregation of i.i.d. gamma random variables, we obtain total claim in this calendar year $t$
$$Z_t = \sum_{k=1}^{v_t} Z_{t,k}\Big|_\Lambda \;\sim\; \Gamma\left(v_t/\varphi, \Lambda\right).$$
Bringing this into the reproductive form requires setting
$$Y_t = \varphi Z_t/v_t|_\Lambda \;\sim\; \Gamma\left(v_t/\varphi, v_t\Lambda/\varphi\right),$$



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this is precisely our assumption from above. Essentially, we have only used the convolution formula within the EDF for this aggregation, see (2.6). This consideration tells us that we do not need to know the individual claims $Z_{t,k}$, but it is sufficient to know the total annual claim $Z_t$, because this aggregated claim amount builds a sufficient statistics for canonical parameter estimation within the EDF framework.
There remains the choice of the common static random effects. We choose a gamma density $\Lambda \sim \Gamma(\alpha, \mu_0(\alpha-1))$ with given parameters $\mu_0 > 0$ and $\alpha > 1$. This gamma distributed random variable has moments

$$\mathbb{E}\left[\Lambda^{-1}\right] = \mu_0 \qquad \text{and} \qquad \mathbb{E}\left[\Lambda^{-2}\right] = \frac{\alpha-1}{\alpha-2}\,\mu_0^2.$$

This follows similarly to (3.6), the existence of the second expected value requires $\alpha > 2$. The cumulant function of the gamma EDF is given by $\kappa(\theta) = -\log(-\theta)$ for canonical parameter $\theta \in \boldsymbol{\Theta} = (-\infty, 0)$. For the static random effects $\Theta$ we have

$$\kappa'(\Theta) = -1/\Theta = 1/\Lambda \qquad \text{with } \ \Lambda \ \sim \ \Gamma(\alpha, \mu_0(\alpha-1)). \tag{3.8}$$

The individual observations $Z_{t,k}$ have the following first two moments

$$\mathbb{E}\left[Z_{t,k}\right] = \mathbb{E}\left[\mathbb{E}\left[\,Z_{t,k}|\,\Lambda\right]\right] = \mathbb{E}\left[(\Lambda\varphi)^{-1}\right] = \mu_0/\varphi,$$

and

$$\begin{aligned}
\mathrm{Var}\left(Z_{t,k}\right) &= \mathbb{E}\left[\mathrm{Var}\left(\,Z_{t,k}|\,\Lambda\right)\right] + \mathrm{Var}\left(\mathbb{E}\left[\,Z_{t,k}|\,\Lambda\right]\right) \\
&= \mathbb{E}\left[(\Lambda^2\varphi)^{-1}\right] + \mathrm{Var}\left((\Lambda\varphi)^{-1}\right) \\
&= \mathbb{E}\left[\Lambda^{-2}\right]\left(1/\varphi + 1/\varphi^2\right) + \mathbb{E}\left[\Lambda^{-1}\right]^2/\varphi^2 \\
&= \left(\frac{\mu_0}{\varphi}\right)^2\left(\frac{\alpha-1}{\alpha-2}(\varphi+1)+1\right) \ = \ \left(\frac{\mu_0}{\varphi}\right)^2\frac{(\alpha-1)\varphi-1}{\alpha-2}.
\end{aligned}$$

We calculate the posterior distribution of $\Lambda$, given the observations $Y_{1:T}$. Using Bayes' rule [18], we have posterior density

$$\begin{aligned}
\pi(\Lambda|Y_{1:T}) &\propto f(Y_{1:T}|\Lambda, v_{1:T})\,\pi(\Lambda) \\
&= \left[\prod_{t=1}^{T}\frac{(v_t\Lambda/\varphi)^{v_t/\varphi}}{\Gamma(v_t/\varphi)}Y_t^{v_t/\varphi-1}\exp\left\{-Y_t v_t\Lambda/\varphi\right\}\right]\frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)}\Lambda^{\alpha-1}e^{-\mu_0(\alpha-1)\Lambda} \\
&\propto \Lambda^{\alpha+\sum_{t=1}^T v_t/\varphi-1}\,e^{-\left(\mu_0(\alpha-1)+\sum_{t=1}^T v_tY_t/\varphi\right)\Lambda}.
\end{aligned}$$

This implies that we have a posterior gamma distribution

$$\Lambda|_{Y_{1:T}} \sim \Gamma\left(\alpha + \sum_{t=1}^T v_t/\varphi,\ \mu_0(\alpha-1) + \sum_{t=1}^T v_tY_t/\varphi\right).$$

This provides for the posterior mean (3.1) the following Bühlmann credibility formula

$$\begin{aligned}
\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} &= \mathbb{E}\left[\,Y_{T+1}|\,Y_{1:T}\right] \ = \ \mathbb{E}\left[\Lambda^{-1}\middle|\,Y_{1:T}\right] \\
&= \frac{1}{\alpha+\sum_{t=1}^T v_t/\varphi-1}\left(\mu_0(\alpha-1)+\sum_{t=1}^T v_tY_t/\varphi\right) \qquad (3.9)\\
&= \omega_T\,\bar{Y}_T + (1-\omega_T)\,\mu_0,
\end{aligned}$$



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with credibility weight and weighted mean

$$\omega_T = \frac{\sum_{t=1}^T v_t}{\sum_{t=1}^T v_t + \varphi(\alpha-1)} \;\in\; (0,1) \qquad \text{and} \qquad \bar{Y}_T = \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s} Y_t. \tag{3.10}$$

We obtain a credibility result that has the same structure as in the Poisson-gamma case of Example 3.3, and essentially the same remarks apply. ■

### 3.2.2 Exponential dispersion family with conjugate priors

We extend the previous examples to the EDF with conjugate priors; this is based on Bühlmann–Gisler [41, Chapter 2], and it has been introduced by Jewell [109, 110, 111], Diaconis–Ylvisaker [66] and Landsman–Makov [130]. Assume the static random effects EDF model of Definition 3.1 holds with a steep cumulant function $\kappa$. The latter is equivalent to $\mathcal{M} = \kappa'(\mathring{\mathbf{\Theta}})$ being equal to the interior of the convex hull of the supports of the responses $Y_t$; see Barndorff-Nielsen [17, Theorem 9.2]. Because we exclude any trivial case, the effective domain has a non-empty interior and, thus, $\mathcal{M}$ is not the empty set. Choose the (Lebesgue) prior density $\pi$ on the interval $\mathbf{\Theta} \subseteq \mathbb{R}$ as

$$\pi(\theta) = \pi_{\mu_0,\tau_0}(\theta) = \exp\left\{\frac{\mu_0\theta - \kappa(\theta)}{\tau_0^2} + d(\mu_0,\tau_0)\right\}, \tag{3.11}$$

for $\mu_0 \in \mathcal{M}$ and $\tau_0 > 0$. Diaconis–Ylvisaker [66, Theorem 1] proved that (3.11) gives a proper density that can be normalized to one by a suitable constant $d(\mu_0,\tau_0)$, i.e., this requires to prove integrability on the effective domain $\mathbf{\Theta}$. We calculate the posterior mean of the static random effects $\Theta$, given observations $Y_{1:T}$. We have posterior density

$$\begin{aligned}
\pi\left(\theta \middle| Y_{1:T}\right) &\propto \left[\prod_{t=1}^T \exp\left\{\frac{Y_t\theta - \kappa(\theta)}{\varphi/v_t}\right\}\right] \exp\left\{\frac{\mu_0\theta - \kappa(\theta)}{\tau_0^2}\right\} \\
&= \exp\left\{\left[\sum_{t=1}^T \frac{Y_t v_t}{\varphi} + \frac{\mu_0}{\tau_0^2}\right]\theta - \left[\sum_{t=1}^T \frac{v_t}{\varphi} + \frac{1}{\tau_0^2}\right]\kappa(\theta)\right\} \\
&= \exp\left\{\frac{\left[\sum_{t=1}^T \frac{v_t}{\varphi} + \frac{1}{\tau_0^2}\right]^{-1}\left[\sum_{t=1}^T \frac{Y_t v_t}{\varphi} + \frac{\mu_0}{\tau_0^2}\right]\theta - \kappa(\theta)}{\left[\sum_{t=1}^T \frac{v_t}{\varphi} + \frac{1}{\tau_0^2}\right]^{-1}}\right\} \\
&\propto \pi_{\mu_T^{\text{post}},\tau_T}(\theta),
\end{aligned}$$

with posterior dispersion parameter

$$\tau_T = \left[\sum_{t=1}^T \frac{v_t}{\varphi} + \frac{1}{\tau_0^2}\right]^{-1/2} \;\in\; (0,\tau_0),$$

and with posterior mean parameter

$$\mu_T^{\text{post}} = \omega_T\,\bar{Y}_T + (1-\omega_T)\,\mu_0 \;\in\; \mathcal{M}, \qquad \mathbb{P}\text{-a.s.}, \tag{3.12}$$

this is again a Bühlmann credibility formula with credibility weights and weighted mean

$$\omega_T = \frac{\sum_{t=1}^T v_t}{\sum_{t=1}^T v_t + \varphi/\tau_0^2} \;\in\; (0,1) \qquad \text{and} \qquad \bar{Y}_T = \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s} Y_t.$$



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Moreover, this posterior $\pi_{\mu_T^{\text{post}},\tau_T}(\theta)$ is a proper (integrable) density on the effective domain $\mathbf{\Theta}$; see Diaconis–Ylvisaker [66, Theorem 1].

**Proposition 3.6.** *Assume we have a static random effects EDF model with a steep cumulant function $\kappa$ and with conjugate prior density $\pi_{\mu_0,\tau_0}$ given by* (3.11)*. Additionally, assume that the conjugate densities vanish at the boundary of the effective domain $\mathbf{\Theta}$. We have for all $1 \leq t \leq T+1$ prior and posterior means, respectively,*

$$\mathbb{E}[Y_t] = \mu_0 \qquad \textit{and} \qquad \mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} = \mathbb{E}\left[Y_{T+1}\middle| Y_{1:T}\right] = \mu_T^{\text{post}},$$

*where $\mu_T^{\text{post}}$ was defined in* (3.12)*.*

**Proof of Proposition 3.6.** We have using the tower property

$$\mathbb{E}[Y_t] = \mathbb{E}[\kappa'(\Theta)] \qquad \text{and} \qquad \mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} = \mathbb{E}\left[Y_{T+1}\middle| Y_{1:T}\right] = \mathbb{E}\left[\kappa'(\Theta)\middle| Y_{1:T}\right], \tag{3.13}$$

based on the proper densities $\pi_{\mu_0,\tau_0}$ and $\pi_{\mu_T^{\text{post}},\tau_T}$, respectively. Since these two densities have the same structural form, it suffices to prove one of the two claims. We show the second one

$$\begin{aligned}
\mathbb{E}\left[\kappa'(\Theta)\middle| Y_{1:T}\right] &= \int_{\mathbf{\Theta}} \kappa'(\theta)\, \pi_{\mu_T^{\text{post}},\tau_T}(\theta)\, d\theta \\
&= \int_{\mathbf{\Theta}} \kappa'(\theta) \exp\left\{\frac{\mu_T^{\text{post}}\theta - \kappa(\theta)}{\tau_T^2} + d(\mu_T^{\text{post}},\tau_T)\right\} d\theta \\
&= \mu_T^{\text{post}} - \tau_T^2 \int_{\mathbf{\Theta}} \frac{\mu_T^{\text{post}} - \kappa'(\theta)}{\tau_T^2} \exp\left\{\frac{\mu_T^{\text{post}}\theta - \kappa(\theta)}{\tau_T^2} + d(\mu_T^{\text{post}},\tau_T)\right\} d\theta \\
&= \mu_T^{\text{post}} - \tau_T^2 \exp\left\{\frac{\mu_T^{\text{post}}\theta - \kappa(\theta)}{\tau_T^2} + d(\mu_T^{\text{post}},\tau_T)\right\}\Bigg|_{\partial\mathbf{\Theta}} \;=\; \mu_T^{\text{post}},
\end{aligned}$$

by the assumption that the density $\pi_{\mu_T^{\text{post}},\tau_T}(\theta)$ vanishes at the boundary of the effective domain $\mathbf{\Theta}$. □

The proof of Proposition 3.6 is taken from Bühlmann–Gisler [41, Theorem 2.20]. It follows in a quite elegant way from the assumption that the density vanishes at the boundary of the effective domain $\mathbf{\Theta}$. Diaconis–Ylvisaker [66, Theorem 2] prove the same claim under the assumption that the effective domain $\mathbf{\Theta}$ is an open set. We conclude from Proposition 3.6 that the predictive mean $\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}}$ is a Bühlmann credibility formula in the EDF with conjugate priors.

**Example 3.7** (Poisson-gamma model)**.** We revisit Example 3.3. Formula (3.3) gives the distribution of the common static random effects in the Poisson-gamma model. We have for $\Lambda \sim \Gamma(\alpha, \alpha/\mu_0)$ and $\theta \in \mathbb{R}$

$$\begin{aligned}
\pi(\theta) &= \frac{d}{d\theta}\mathbb{P}\left[\Theta \leq \theta\right] \;=\; \frac{d}{d\theta}\mathbb{P}\left[\log(\Lambda) \leq \theta\right] \;=\; \frac{d}{d\theta}\int_0^{e^\theta} \frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\alpha/\mu_0} dx \\
&= \frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)} \exp\left\{\alpha\theta - e^\theta \alpha/\mu_0\right\} \qquad (3.14)\\
&= \exp\left\{\frac{\mu_0\theta - \kappa(\theta)}{\mu_0/\alpha} + \log\left(\frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)}\right)\right\},
\end{aligned}$$

with cumulant function $\kappa(\theta) = \exp\{\theta\}$ and parameters $\mu_0 > 0$ and $\tau_0^2 = \mu_0/\alpha > 0$. This is a proper density on the effective domain $\mathbf{\Theta} = \mathbb{R}$ which vanishes for $\theta \to \pm\infty$. Therefore, Proposition 3.6 applies and we obtain the results derived in Example 3.3. ■



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**Example 3.8** (gamma-gamma model)**.** We revisit Example 3.5. Formula (3.8) gives the distribution of the static random effects in the gamma-gamma model. We have for $\Lambda \sim \Gamma(\alpha, \mu_0(\alpha-1))$, $\alpha > 1$, $\mu_0 > 0$, and $\theta < 0$

$$\begin{aligned}
\pi(\theta) &= \frac{d}{d\theta}\mathbb{P}\left[\Theta \leq \theta\right] = \frac{d}{d\theta}\mathbb{P}\left[\Lambda \geq -\theta\right] = \frac{d}{d\theta}\int_{-\theta}^{\infty} \frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\mu_0(\alpha-1)} dx \\
&= \frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)} \exp\left\{(\alpha-1)\log(-\theta) + \theta\mu_0(\alpha-1)\right\} \\
&= \exp\left\{\frac{\mu_0\theta - \kappa(\theta)}{(\alpha-1)^{-1}} + \log\left(\frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)}\right)\right\},
\end{aligned}$$

with cumulant function $\kappa(\theta) = -\log(-\theta)$ and dispersion parameter $\tau_0^2 = (\alpha-1)^{-1}$. This is a proper density on the effective domain $\boldsymbol{\Theta} = (-\infty, 0)$ which vanishes for $\theta \uparrow 0$ and for $\theta \to -\infty$. Therefore, Proposition 3.6 applies and we obtain the results derived in Example 3.5. ■

### 3.2.3 Inhomogeneous Bühlmann–Straub formula

The EDF with conjugate priors gives an attractive and tractable family of models for which we can analytically perform experience rating and for which the resulting posterior mean satisfies the Bühlmann credibility formula (3.12). Generally, this is not the case and the posterior mean cannot explicitly be calculated. In that case, the Bühlmann–Straub framework can be used to compute a *linear approximation* to the (optimal) Bayesian posterior estimator. Under Model 3.2, we can work in the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$ of square-integrable random variables on the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The Bayesian posterior estimator is, under square-integrability, obtained by an orthogonal projection of $Y_{T+1}$ onto the subspace in $L^2(\Omega, \mathcal{F}, \mathbb{P})$ that is generated by all $\sigma(Y_{1:T})$-measurable and square-integrable random variables. In the sequel we denote this subspace by $\mathcal{G}(\boldsymbol{Y}) \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$, and the orthogonal projection is expressed by the MSE minimization (3.7).

Bühlmann–Straub [42] require that the resulting credibility formula is linear in the observations $\boldsymbol{Y} = (Y_{i,1:T})_{i=1}^n$. This motivates to define the following closed subspace of affine functions in $\boldsymbol{Y}$

$$\mathcal{L}(\boldsymbol{Y}, 1) = \left\{\widehat{\mu} = a_0 + \sum_{i=1}^n \sum_{t=1}^T a_{i,t} Y_{i,t};\ a_0 \in \mathbb{R}, a_{i,t} \in \mathbb{R} \text{ for all } i, t\right\} \ \subset\ \mathcal{G}(\boldsymbol{Y}),$$

where $\mathcal{G}(\boldsymbol{Y}) \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$ is the subspace of all $\sigma(\boldsymbol{Y})$-measurable and square-integrable random variables. Note that $\mathcal{G}(\boldsymbol{Y})$ and $\mathcal{L}(\boldsymbol{Y}, 1)$ are closed subspaces in $L^2(\Omega, \mathcal{F}, \mathbb{P})$.

The *inhomogeneous credibility estimator* of $\mu(\Theta_i)$, given $\boldsymbol{Y}$, is defined by

$$\widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y}) = \underset{\widehat{\mu} \in \mathcal{L}(\boldsymbol{Y},1)}{\arg\min}\ \mathbb{E}\left[(\widehat{\mu} - \mu(\Theta_i))^2\right] = \underset{\widehat{\mu} \in \mathcal{L}(\boldsymbol{Y},1)}{\arg\min}\ \mathbb{E}\left[(\widehat{\mu} - \mathbb{E}[Y_{i,T+1}|\Theta_i])^2\right].$$

We refer to the first line of (3.1) that motivates the projection of $\mathbb{E}[Y_{i,T+1}|\Theta_i]$ onto the subspace $\mathcal{L}(\boldsymbol{Y}, 1)$. That is, we determine the closest approximation in the subspace $\mathcal{L}(\boldsymbol{Y}, 1)$ to the true conditional mean $\mu(\Theta_i) \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Thus, we project to a smaller subspace $\mathcal{L}(\boldsymbol{Y}, 1) \subset \mathcal{G}(\boldsymbol{Y})$ admitting for non-optimality compromising for the intractability



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of the optimal solution in $\mathcal{G}(\boldsymbol{Y})$. The approximation error that we make by this non-optimal projection is obtained by the Pythagorean theorem, by first projecting $\mu(\Theta_i)$ to the bigger space $\mathcal{G}(\boldsymbol{Y})$ and then to the smaller space $\mathcal{L}(\boldsymbol{Y},1) \subset \mathcal{G}(\boldsymbol{Y})$, resulting in the approximation error (from the second step)

$$\mathbb{E}\left[\left(\widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y}) - \mathbb{E}[\mu(\Theta_i)|\boldsymbol{Y}]\right)^2\right] = \mathbb{E}\left[\left(\widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y}) - \mu^{\text{post}}_{Y_{i,T+1}|\boldsymbol{Y}}\right)^2\right].$$

**Theorem 3.9** (inhomogeneous Bühlmann–Straub formula)**.** *Assume the Bühlmann–Straub Model 3.2 is fulfilled. The inhomogeneous Bühlmann credibility estimator is given by*

$$\widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y}) = \omega_{i,T}\bar{Y}_{i,T} + (1-\omega_{i,T})\,\mu_0,$$

*with credibility weights $\omega_{i,T} \in [0,1]$ and observation based estimators $\bar{Y}_{i,T}$, respectively,*

$$\omega_{i,T} = \frac{\sum_{t=1}^T v_{i,t}}{\sum_{t=1}^T v_{i,t} + \sigma^2/\tau_0^2} \qquad \textit{and} \qquad \bar{Y}_{i,T} = \sum_{t=1}^T \frac{v_{i,t}}{\sum_{s=1}^T v_{i,s}}\, Y_{i,t},$$

*and with collective mean, heterogeneity between risks and volatility within risks*

$$\mu_0 = \mathbb{E}[\mu(\Theta_1)], \qquad \tau_0^2 = \mathrm{Var}(\mu(\Theta_1)) \qquad \textit{and} \qquad \sigma^2 = \mathbb{E}[\sigma^2(\Theta_1)].$$

**Proof of Theorem 3.9.** There are different proofs of this theorem. Bühlmann–Gisler [41, Theorem 4.2] prove this result using methods from orthogonal projections and the resulting normal equations. The brute force method just solves the quadratic minimization problem over $\boldsymbol{a} = (a_0, a_{1,1}, \ldots, a_{n,T})^\top \in \mathbb{R}^{nT+1}$ of the objective function

$$\boldsymbol{a} \;\mapsto\; \mathbb{E}\left[\left(a_0 + \sum_{l=1}^n \sum_{s=1}^T a_{l,s} Y_{l,s} - \mu(\Theta_i)\right)^2\right].$$

This is a bit tedious but solves the problem. On the other, the Bühlmann–Straub model assumptions are non-distributional and we can take any model that satisfies these moment and independence conditions to calculate the corresponding projections. An insightful example is a multi-variate Gaussian one because this can be fully calculated, see next example. □

**Example 3.10.** We study a Gaussian example that fulfills the Bühlmann–Straub Model 3.2. Assume that the components of the vector $\underline{\Theta} = (\Theta_1, \ldots, \Theta_n)^\top$ are i.i.d. Gaussian distributed with mean $\mu_0 \in \mathbb{R}$ and variance $\tau_0^2 > 0$.[2] Conditionally, given $\underline{\Theta}$, $Y_{i,t}$ and $Y_{j,s}$ are assumed to be independent for $(i,t) \neq (j,s)$ with conditional Gaussian distributions

$$Y_{i,t}|_{\underline{\Theta}} \;\overset{(d)}{=}\; Y_{i,t}|_{\Theta_i} \;\sim\; \mathcal{N}(\Theta_i, \sigma^2/v_{i,t}),$$

with variance parameter $\sigma^2 > 0$ and weights $v_{i,t} > 0$, for $1 \le i \le n$ and $1 \le t \le T+1$. This model fulfills the Bühlmann–Straub Model 3.2 with $\mu(\Theta_i) = \Theta_i$ and $\sigma^2(\Theta_i) \equiv \sigma^2$. Moreover, it belongs to the EDF with conjugate priors and cumulant function $\kappa(\theta) = \theta^2/2$ on the effective domain $\boldsymbol{\Theta} = \mathbb{R}$. This implies $\mu(\Theta_i) = \kappa'(\Theta_i) = \Theta_i$.

[2] We use notation $\underline{\Theta}$ to denote a vector with components $(\Theta_i)_{i=1}^n$, because the bold face notation $\boldsymbol{\Theta} \subseteq \mathbb{R}$ is already occupied by the effective domain of the chosen EDF.



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Consider the vector $\boldsymbol{Y} = (Y_{i,1:T})_{i=1}^n \in \mathbb{R}^{nT}$ of all observations at time $T$. Conditionally, given $\underline{\Theta}$, this vector $\boldsymbol{Y}$ has a multivariate Gaussian distribution

$$\boldsymbol{Y}|_{\underline{\Theta}} \;\sim\; \mathcal{N}\left(\mathfrak{Z}\underline{\Theta},\, \Sigma = \operatorname{diag}(\sigma^2/v_{i,t})_{i,t}\right),$$

with $\mathfrak{Z} \in \{0,1\}^{nT\times n}$ being a one-hot encoded design matrix indicating to which latent factor in $\underline{\Theta}$ the entries of $\boldsymbol{Y}$ belong to, and $\Sigma \in \mathbb{R}^{nT\times nT}$ is the diagonal covariance matrix.[3] The vector of latent factors has also a multivariate Gaussian distribution

$$\underline{\Theta} \;\sim\; \mathcal{N}\left(\boldsymbol{\mu}_0 = (\mu_0,\ldots,\mu_0)^\top,\, \mathfrak{T} = \operatorname{diag}(\tau_0^2,\ldots,\tau_0^2)\right),$$

with diagonal covariance matrix $\mathfrak{T} \in \mathbb{R}^{n\times n}$. This allows us to calculate the posterior distribution of the latent factors $\underline{\Theta}$, given $\boldsymbol{Y}$. This posterior density is proportional to

$$\begin{aligned}\pi(\underline{\Theta}|\boldsymbol{Y}) \;&\propto\; \exp\left\{-\frac{1}{2}\underline{\Theta}^\top\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}\underline{\Theta} + \underline{\Theta}^\top\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y}\right\}\exp\left\{-\frac{1}{2}\underline{\Theta}^\top\mathfrak{T}^{-1}\underline{\Theta} + \underline{\Theta}^\top\mathfrak{T}^{-1}\boldsymbol{\mu}_0\right\}\\ &=\; \exp\left\{-\frac{1}{2}\underline{\Theta}^\top\left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)\underline{\Theta} + \underline{\Theta}^\top\left(\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y}+\mathfrak{T}^{-1}\boldsymbol{\mu}_0\right)\right\}. \qquad (3.15)\end{aligned}$$

This is a multivariate Gaussian posterior distribution

$$\underline{\Theta}|_{\boldsymbol{Y}} \;\sim\; \mathcal{N}\left(\left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\left(\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y}+\mathfrak{T}^{-1}\boldsymbol{\mu}_0\right),\, \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\right).$$

This provides us with posterior mean vector

$$\begin{aligned}\mathbb{E}\left[(\mu(\Theta_1),\ldots,\mu(\Theta_n))^\top\middle|\boldsymbol{Y}\right] \;&=\; \mathbb{E}\left[\underline{\Theta}|\,\boldsymbol{Y}\right] \qquad (3.16)\\ &=\; \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\left(\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y}+\mathfrak{T}^{-1}\boldsymbol{\mu}_0\right).\end{aligned}$$

This posterior mean vector is equal to the orthogonal projections of the components of $(\mu(\Theta_1),\ldots,\mu(\Theta_n))^\top$ onto the subspace $\mathcal{G}(\boldsymbol{Y}) \subset L^2(\Omega,\mathcal{F},\mathbb{P})$. Because these posterior means are affine linear in the observations $\boldsymbol{Y}$, they are also equal to the orthogonal projections onto the linear subspace $\mathcal{L}(\boldsymbol{Y},1) \subset \mathcal{G}(\boldsymbol{Y})$. Thus, in this multivariate Gaussian conjugate prior case, the Bayesian estimator (posterior mean) is equal to the inhomogeneous Bühlmann credibility estimator

$$\mathbb{E}\left[\underline{\Theta}|\,\boldsymbol{Y}\right] = \mathbb{E}\left[(\mu(\Theta_1),\ldots,\mu(\Theta_n))^\top\middle|\boldsymbol{Y}\right] = \left(\widehat{\widehat{\mu(\Theta_1)}}(\boldsymbol{Y}),\ldots,\widehat{\widehat{\mu(\Theta_n)}}(\boldsymbol{Y})\right)^\top.$$

We analyze this posterior mean vector. First, we remark that across index $1 \le i \le n$, the random variables are independent. Therefore we also have posterior independence across this index $1 \le i \le n$. Consequently, the matrix multiplications in (3.16) only aggregate over the calendar years $1 \le t \le T$ for every $1 \le i \le n$. Thus, the term of the posterior mean corresponding to the $i$-th component is given by the Bühlmann credibility

$$\mathbb{E}\left[\mu(\Theta_i)|\,\boldsymbol{Y}\right] = \left(\sum_{t=1}^T \frac{v_{i,t}}{\sigma^2} + \frac{1}{\tau_0^2}\right)^{-1}\left(\sum_{t=1}^T \frac{v_{i,t}}{\sigma^2}\,Y_{i,t} + \frac{1}{\tau_0^2}\,\mu_0\right) = \omega_{i,T}\bar{Y}_{i,T} + (1-\omega_{i,T})\mu_0.$$

This proves the Bühlmann–Straub Theorem 3.9. ■

[3] In contrast to the design matrix $\mathfrak{X}$ introduced in the GLM Section 2.3, the design matrix $\mathfrak{Z}$ does not have full rank because we use one-hot encoding. In a Bayesian context this is fine because the (non-degenerate) prior provides identifiability.



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Remark that formula (3.16) holds for any positive definite matrices $\Sigma$ and $\mathfrak{T}$, and for any design matrix $\mathfrak{Z}$. That is, in fact, we have proved the following more general result.

**Corollary 3.11.** *Assume the $n$-dimensional vector $\underline{\Theta}$ is multivariate Gaussian distributed with mean $\boldsymbol{\mu}_0 \in \mathbb{R}^n$ and positive definite covariance matrix $\mathfrak{T} \in \mathbb{R}^{n\times n}$. Conditionally, given $\underline{\Theta}$, the $m$-dimensional vector $\boldsymbol{Y}$ is assumed to be multivariate Gaussian distributed with mean $\mathfrak{Z}\underline{\Theta}$ for design matrix $\mathfrak{Z} \in \mathbb{R}^{m\times n}$ and positive definite covariance matrix $\Sigma \in \mathbb{R}^{m\times m}$. We have multivariate Gaussian posterior distribution*

$$\underline{\Theta}|_{\boldsymbol{Y}} \;\sim\; \mathcal{N}\left(\left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\left(\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y}+\mathfrak{T}^{-1}\boldsymbol{\mu}_0\right),\left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\right).$$

*The posterior mean vector is given by the Bühlmann credibility*

$$\mathbb{E}\left[\underline{\Theta}|\,\boldsymbol{Y}\right] = A\boldsymbol{Y} + (\mathbb{1} - A\mathfrak{Z})\,\boldsymbol{\mu}_0,$$

*with $n\times n$-dimensional unit matrix $\mathbb{1}$ and with credibility matrix*

$$A = \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\mathfrak{Z}^\top\Sigma^{-1} \;\in\; \mathbb{R}^{n\times m}.$$

**Proof of Corollary 3.11.** The first part of the claim of the corollary follows from Example 3.10 and there only remains to prove the credibility formula. We have posterior mean

$$\mathbb{E}\left[\underline{\Theta}|\,\boldsymbol{Y}\right] = \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\left(\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y}+\mathfrak{T}^{-1}\boldsymbol{\mu}_0\right).$$

The following computation has first been considered by Jewell [112], see also De Vylder [58]. We set $B = (\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1})^{-1}\mathfrak{T}^{-1}$. We have, massaging matrix $B$ from the right to the left,

$$\begin{aligned}
\mathbb{1} - B &= \left(\mathfrak{T} - \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\right)\mathfrak{T}^{-1} \\
&= \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\left(\left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)\mathfrak{T} - \mathbb{1}\right)\mathfrak{T}^{-1} \\
&= \left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z}+\mathfrak{T}^{-1}\right)^{-1}\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z} \;=\; A\mathfrak{Z}.
\end{aligned}$$

This proves the claim. □

**Remark 3.12.** Corollary 3.11 allows us to compute any inhomogeneous Bühlmann credibility formula by disentangling the corresponding matrix multiplications, such as the Bühlmann–Straub formula [42], the hierarchical credibility formula, see Bühlmann–Gisler [41, Chapter 6] or Hachemeister's regression credibility formula [100], that is, it fully solves the general linear Bühlmann credibility case by choosing the corresponding covariance matrices $\mathfrak{T}$ and $\Sigma$, design matrix $\mathfrak{Z}$ and prior mean vector $\boldsymbol{\mu}_0$. ■

From Theorem 3.9 we see that the inhomogeneous Bühlmann credibility estimator is unbiased

$$\mathbb{E}\left[\widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y})\right] = \mu_0.$$

The MSE prediction uncertainty is given by, see Bühlmann–Gisler [41, Theorem 4.3],

$$\begin{aligned}
\mathbb{E}\left[\left(Y_{i,T+1} - \widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y})\right)^2\right] &= \mathbb{E}\left[\left(Y_{i,T+1} - \mu(\Theta_i)\right)^2\right] + \mathbb{E}\left[\left(\mu(\Theta_i) - \widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y})\right)^2\right] \\
&= \frac{\sigma^2}{v_{i,T+1}} + (1-\omega_{i,T})\,\tau_0^2.
\end{aligned}$$



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### 3.2.4 Homogeneous Bühlmann–Straub formula

The difficulty in the application of the inhomogeneous Bühlmann credibility formula is that it requires the knowledge of the collective/prior mean $\mu_0$. Often this is not readily available and needs to be elicited from the data, too; this can be interpreted as an *empirical Bayesian approach*, we come back to this in Section 3.2.5, below. Remark that in the inhomogeneous Bühlmann credibility formula, only the information of insurance policy $i$ is used in the credibility estimator of instance $i$, i.e., we have Bühlmann credibility

$$\widehat{\widehat{\mu(\Theta_i)}}(\boldsymbol{Y}) = \widehat{\widehat{\mu(\Theta_i)}}(Y_{i,1:T}) = \omega_{i,T}\bar{Y}_{i,T} + (1-\omega_{i,T})\,\mu_0.$$

To elicit the collective mean $\mu_0$ from data we need multiple insurance policies $1 \le i \le n$. Define the subspace

$$\mathcal{L}_e(\boldsymbol{Y}) = \left\{ \widehat{\mu} = \sum_{i=1}^n \sum_{t=1}^T a_{i,t} Y_{i,t};\ a_{i,t} \in \mathbb{R} \text{ for all } i,t \text{ and } \mathbb{E}[\widehat{\mu}] = \mathbb{E}[\mu(\Theta_1)] \right\} \ \subset\ \mathcal{L}(\boldsymbol{Y},1).$$

$\mathcal{L}_e(\boldsymbol{Y})$ is a closed affine subspace in $L^2(\Omega,\mathcal{F},\mathbb{P})$; see Bühlmann–Gisler [41, Appendix B]. We consider the hierarchy of orthogonal projections onto $\mathcal{G}(\boldsymbol{Y}) \supset \mathcal{L}(\boldsymbol{Y},1) \supset \mathcal{L}_e(\boldsymbol{Y})$, respectively, each providing a smaller subspace, and distances between resulting projections can be obtained by the Pythagorean theorem. Importantly, the smallest subspace $\mathcal{L}_e(\boldsymbol{Y})$ does not require any knowledge about the specific value of the collective mean $\mu_0$, it only states unbiasedness.

The homogeneous credibility estimator of $\mu(\Theta_i)$ based on $\boldsymbol{Y}$ is defined by

$$\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) = \underset{\widehat{\mu}\in\mathcal{L}_e(\boldsymbol{Y})}{\arg\min}\ \mathbb{E}\left[(\widehat{\mu}-\mu(\Theta_i))^2\right].$$

This homogeneous credibility estimator is not based on (the explicit knowledge of) $\mu_0$.

**Theorem 3.13** (homogeneous Bühlmann–Straub formula)**.** *Under the same assumptions and notation as in Theorem 3.9, we have the homogeneous Bühlmann credibility estimator*

$$\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) = \omega_{i,T}\bar{Y}_{i,T} + (1-\omega_{i,T})\,\widehat{\widehat{\mu}}_T^{\text{hom}},$$

*with collective mean estimate*

$$\widehat{\widehat{\mu}}_T^{\text{hom}} = \frac{1}{\sum_{i=1}^n \omega_{i,T}} \sum_{i=1}^n \omega_{i,T}\bar{Y}_{i,T}. \tag{3.17}$$

**Proof of Theorem 3.13.** There are again different proofs for this statement. Bühlmann–Gisler [41, Theorem 4.4] present a proof that is based on orthogonal projects and normal equations in Hilbert spaces. Essentially we need to project the inhomogeneous Bühlmann credibility estimators to the affine subspace $\mathcal{L}_e(\boldsymbol{Y})$. The brute force proof just considers a quadratic minimization problem with a linear side constraint. This can be solved with the method of Lagrange. □

From Theorem 3.13 we see again the unbiasedness

$$\mathbb{E}\left[\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y})\right] = \mu_0,$$



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with MSE prediction uncertainty given by, see Bühlmann–Gisler [41, Theorem 4.6],

$$\mathbb{E}\left[\left(Y_{i,T+1} - \widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y})\right)^2\right] = \frac{\sigma^2}{v_{i,T+1}} + (1-\omega_{i,T})\,\tau_0^2\left(1 + \frac{1-\omega_{i,T}}{\sum_{k=1}^n \omega_{k,T}}\right).$$

An important observation from Theorem 3.13 is that we do not choose the sample mean for the portfolio mean estimation $\mu_0$, but rather a credibility weighted average $\widehat{\widehat{\mu}}_T^{\text{hom}}$. This can be justified as follows. We calculate the variance of $\bar{Y}_{i,T}$. It is given by

$$\begin{aligned}
\text{Var}\left(\bar{Y}_{i,T}\right) &= \mathbb{E}\left[\text{Var}\left(\bar{Y}_{i,T}\middle|\,\Theta_i\right)\right] + \text{Var}\left(\mathbb{E}\left[\bar{Y}_{i,T}\middle|\,\Theta_i\right]\right) \\
&= \frac{1}{\sum_{t=1}^T v_{i,t}}\mathbb{E}\left[\sigma^2(\Theta_i)\right] + \text{Var}\left(\mu(\Theta_i)\right) \qquad (3.18)\\
&= \frac{\sigma^2}{\sum_{t=1}^T v_{i,t}} + \tau_0^2 \;=\; \frac{\tau_0^2}{\omega_{i,T}}.
\end{aligned}$$

It is a well known result that the linear unbiased estimator with minimal MSE is inversely proportional to the variances of $\bar{Y}_{i,T}$, see, e.g., Wüthrich [200, Lemma 2.26]. This gives the MSE-optimal linear unbiased estimator for the prior mean

$$\widehat{\mu}_0 = \frac{1}{\sum_{i=1}^n \omega_{i,T}/\tau_0^2}\sum_{i=1}^n \frac{\omega_{i,T}}{\tau_0^2}\bar{Y}_{i,T} = \widehat{\widehat{\mu}}_T^{\text{hom}}.$$

Another important observation is that the homogeneous Bühlmann credibility estimator fulfills the balance property; see Bühlmann–Gisler [41, Theorem 4.5].

**Corollary 3.14** (balance property)**.** *The homogeneous Bühlmann credibility satisfies the global balance property*

$$\sum_{i=1}^n\sum_{t=1}^T v_{i,t}\,\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) \;=\; \sum_{i=1}^n\sum_{t=1}^T v_{i,t}Y_{i,t} \;=\; \sum_{i=1}^n\sum_{t=1}^T v_{i,t}\bar{Y}_{i,T}.$$

**Remark 3.15.** This result is another strong argument for taking the credibility weighted average (3.17) for the prior mean estimation $\mu_0$ in this credibility approach. Of course, the same argument applies to the EDF with conjugate priors case, see (3.12), and the balance property applies to *any choice* of $\tau_0^2 > 0$, $\sigma^2 > 0$ and $\varphi > 0$, respectively. ■

**Proof of Corollary 3.14.** We calculate the difference between the left- and the right-hand side of the statement

$$\begin{aligned}
\sum_{i=1}^n\sum_{t=1}^T v_{i,t}\left(\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) - Y_{i,t}\right) &= \sum_{i=1}^n\sum_{t=1}^T v_{i,t}\left(\omega_{i,T}\,\bar{Y}_{i,T} + (1-\omega_{i,T})\,\widehat{\widehat{\mu}}_T^{\text{hom}} - \bar{Y}_{i,T}\right)\\
&= \sum_{i=1}^n\sum_{t=1}^T v_{i,t}\left(-(1-\omega_{i,T})\,\bar{Y}_{i,T} + (1-\omega_{i,T})\,\widehat{\widehat{\mu}}_T^{\text{hom}}\right).
\end{aligned}$$

Note that $(1-\omega_{i,T})\sum_{t=1}^T v_{i,t} = \omega_{i,T}\sigma^2/\tau_0^2$. This provides

$$\sum_{i=1}^n\sum_{t=1}^T v_{i,t}\left(\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) - Y_{i,t}\right) = \frac{\sigma^2}{\tau_0^2}\sum_{i=1}^n \omega_{i,T}\left(\widehat{\widehat{\mu}}_T^{\text{hom}} - \bar{Y}_{i,T}\right) = 0. \qquad (3.19)$$

This proves the claim. □



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In view of (3.19) we also conclude that the prior estimator $\widehat{\widehat{\mu}}_T^{\text{hom}}$ minimizes the weighted MSE of the weighted sample means $(\bar{Y}_{i,T})_{i=1}^n$ with weights $(\omega_{i,T})_{i=1}^n$. Of course, this again relates to (3.18).

We conclude this section on the homogeneous Bühlmann credibility by a different view and a new derivation. This requires to come back to the multivariate Gaussian setting of Corollary 3.11 and Example 3.10. This example can be interpreted as a linear regression model with one categorical covariate indicating to which insurance policy $i$ the observations $Y_{i,t}$ belong to. This is precisely what is encoded in the design matrix $\mathfrak{Z}$ of Example 3.10. In the next example we extend this set-up by a common latent factor $\Theta_0$ across all instances $1 \leq i \leq n$. To the best of our knowledge the following result using a non-informative prior is new and unpublished at the time of writing the first version of these notes. We will also see that it can be interpreted in the hierarchical credibility sense; see Bühlmann–Gisler [41, Chapter 6] for hierarchical credibility.

**Theorem 3.16** (homogeneous Bühlmann credibility, 2nd version)**.** *Choose a $(n+1)$-dimensional random vector $\underline{\Theta} = (\Theta_0, \Theta_1, \ldots, \Theta_n)^\top$ with multivariate Gaussian distribution*

$$\underline{\Theta} \;\sim\; \mathcal{N}\left(\boldsymbol{\mu}_0 = (\mu_0, 0, \ldots, 0)^\top, \mathfrak{T} = \operatorname{diag}(\tau_{00}^2, \tau_0^2, \ldots, \tau_0^2)\right),$$

*with $\mu_0 \in \mathbb{R}$ and $\tau_{00}^2, \tau_0^2 > 0$. Conditionally, given $\underline{\Theta}$, the observations $Y_{i,t}$ are independent Gaussian with*

$$Y_{i,t}|_{\underline{\Theta}} \;\sim\; \mathcal{N}\left(\Theta_0 + \Theta_i,\, \sigma^2/v_{i,t}\right),$$

*with $\sigma^2 > 0$ and $v_{i,t} > 0$ for all $1 \leq i \leq n$ and $1 \leq t \leq T+1$. Set $\boldsymbol{Y} = (Y_{i,1:T})_{i=1}^n$, then*

$$\lim_{\tau_{00}^2 \to \infty} \mu^{\text{post}}_{Y_{i,T+1}|\boldsymbol{Y}} = \lim_{\tau_{00}^2 \to \infty} \mathbb{E}\left[Y_{i,T+1}|\,\boldsymbol{Y}\right] = \omega_{i,T}\bar{Y}_{i,T} + (1-\omega_{i,T})\,\widehat{\widehat{\mu}}_T^{\text{hom}},$$

*where the latter is the homogeneous Bühlmann credibility estimator of Theorem 3.13.*

In fact, the posterior mean $\mu^{\text{post}}_{Y_{i,T+1}|\boldsymbol{Y}}$ in the above theorem can explicitly be computed, not only in the limit, and its formula is given in (3.24) below.

**Proof of Theorem 3.16.** Similar to GLMs, component $\Theta_0$ plays the role of the intercept that applies to all observations $Y_{i,t}$. We consider a one-hot encoded design matrix $\mathfrak{Z} \in \{0,1\}^{nT \times (n+1)}$, where the first column is identical equal to 1 and the remaining columns one-hot encode the insurance policies $i$ the individual observations $Y_{i,t}$ belong to. We set $\Sigma = \operatorname{diag}(\sigma^2/v_{i,t})$ and apply Corollary 3.11. This requires computation of the following positive definite matrix

$$\mathfrak{Z}^\top \Sigma^{-1} \mathfrak{Z} + \mathfrak{T}^{-1} = \begin{pmatrix} \frac{\sum_i \sum_t v_{i,t}}{\sigma^2} + \frac{1}{\tau_{00}^2} & \frac{\sum_t v_{1,t}}{\sigma^2} & \frac{\sum_t v_{2,t}}{\sigma^2} & \cdots & \frac{\sum_t v_{n,t}}{\sigma^2} \\ \frac{\sum_t v_{1,t}}{\sigma^2} & \frac{\sum_t v_{1,t}}{\sigma^2} + \frac{1}{\tau_0^2} & 0 & \cdots & 0 \\ \frac{\sum_t v_{2,t}}{\sigma^2} & 0 & \frac{\sum_t v_{2,t}}{\sigma^2} + \frac{1}{\tau_0^2} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\sum_t v_{n,t}}{\sigma^2} & 0 & 0 & \cdots & \frac{\sum_t v_{n,t}}{\sigma^2} + \frac{1}{\tau_0^2} \end{pmatrix}, \tag{3.20}$$

where the summations run over the time indices $1 \leq t \leq T$. Coming back to the posterior $\pi(\underline{\Theta}|\boldsymbol{Y})$ in (3.15), we have log-likelihood

$$\log \pi(\underline{\Theta}|\boldsymbol{Y}) \;\propto\; -\frac{1}{2}\underline{\Theta}^\top \left(\mathfrak{Z}^\top \Sigma^{-1} \mathfrak{Z} + \mathfrak{T}^{-1}\right) \underline{\Theta} + \underline{\Theta}^\top \left(\mathfrak{Z}^\top \Sigma^{-1} \boldsymbol{Y} + \mathfrak{T}^{-1} \boldsymbol{\mu}_0\right).$$



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This is a Gaussian posterior, and Gaussian (posterior) densities are symmetric around its means, with posterior means equal to the posterior a posteriori (MAP) estimators. The MAP estimator (and posterior mean) is received by solving the score equations

$$\nabla_{\underline{\Theta}} \log \pi(\underline{\Theta}|\boldsymbol{Y}) \;\propto\; -\left(\mathfrak{Z}^\top \Sigma^{-1} \mathfrak{Z} + \mathfrak{T}^{-1}\right) \underline{\Theta} + \mathfrak{Z}^\top \Sigma^{-1} \boldsymbol{Y} + \mathfrak{T}^{-1} \boldsymbol{\mu}_0 \;=\; \mathbf{0}.$$

In view of (3.20), the first component of the score equation thus requires us to solve

$$\left(\sum_{i=1}^n \sum_{t=1}^T v_{i,t} + \frac{\sigma^2}{\tau_{00}^2}\right) \widehat{\Theta}_0^{\text{MAP}} + \sum_{i=1}^n \sum_{t=1}^T v_{i,t} \widehat{\Theta}_i^{\text{MAP}} = \sum_{i=1}^n \sum_{t=1}^T v_{i,t} Y_{i,t} + \frac{\sigma^2}{\tau_{00}^2} \mu_0, \tag{3.21}$$

and for the remaining components $1 \leq i \leq n$ we need to solve

$$\sum_{t=1}^T v_{i,t} \widehat{\Theta}_0^{\text{MAP}} + \left(\sum_{t=1}^T v_{i,t} + \frac{\sigma^2}{\tau_0^2}\right) \widehat{\Theta}_i^{\text{MAP}} = \sum_{t=1}^T v_{i,t} Y_{i,t}. \tag{3.22}$$

This is reformulated using the definitions of the credibility weights $\omega_{i,T}$ and the weighted means $\bar{Y}_{i,T}$

$$\widehat{\Theta}_i^{\text{MAP}} = \frac{1}{\sum_{t=1}^T v_{i,t} + \sigma^2/\tau_0^2} \sum_{t=1}^T v_{i,t} Y_{i,t} - \frac{\sum_{t=1}^T v_{i,t}}{\sum_{t=1}^T v_{i,t} + \sigma^2/\tau_0^2} \widehat{\Theta}_0^{\text{MAP}} = \omega_{i,T} \bar{Y}_{i,T} - \omega_{i,T} \widehat{\Theta}_0^{\text{MAP}}, \tag{3.23}$$

for all $1 \leq i \leq n$. Because posterior means are equal to posterior MAP estimators in the Gaussian case, we thus have

$$\mathbb{E}\left[Y_{i,T+1} | \boldsymbol{Y}\right] = \mathbb{E}\left[\Theta_0 + \Theta_i | \boldsymbol{Y}\right] = \widehat{\Theta}_0^{\text{MAP}} + \widehat{\Theta}_i^{\text{MAP}} = \omega_{i,T} \bar{Y}_{i,T} + (1 - \omega_{i,T}) \widehat{\Theta}_0^{\text{MAP}}.$$

Finally, we need to compute $\widehat{\Theta}_0^{\text{MAP}}$ in terms of $(\bar{Y}_{i,T})_{i=1}^n$. If we aggregate (3.22) over $1 \leq i \leq n$ and insert (3.21) on the right-hand side, we receive

$$\sum_{i=1}^n \sum_{t=1}^T v_{i,t} \widehat{\Theta}_0^{\text{MAP}} + \left(\sum_{t=1}^T v_{i,t} + \frac{\sigma^2}{\tau_0^2}\right) \widehat{\Theta}_i^{\text{MAP}} = \left(\sum_{i=1}^n \sum_{t=1}^T v_{i,t} + \frac{\sigma^2}{\tau_{00}^2}\right) \widehat{\Theta}_0^{\text{MAP}} + \sum_{i=1}^n \sum_{t=1}^T v_{i,t} \widehat{\Theta}_i^{\text{MAP}} - \frac{\sigma^2}{\tau_{00}^2} \mu_0.$$

This provides identity

$$\sum_{i=1}^n \widehat{\Theta}_i^{\text{MAP}} = \frac{\tau_0^2}{\tau_{00}^2} \left(\widehat{\Theta}_0^{\text{MAP}} - \mu_0\right).$$

Inserting (3.23) into this identity gives us

$$\frac{\tau_0^2}{\tau_{00}^2} \left(\widehat{\Theta}_0^{\text{MAP}} - \mu_0\right) = \sum_{i=1}^n \omega_{i,T} \bar{Y}_{i,T} - \omega_{i,T} \widehat{\Theta}_0^{\text{MAP}} = \sum_{i=1}^n \omega_{i,T} \bar{Y}_{i,T} - \sum_{i=1}^n \omega_{i,T} \widehat{\Theta}_0^{\text{MAP}}.$$

We solve this for $\widehat{\Theta}_0^{\text{MAP}}$ and let $\tau_{00}^2 \to \infty$,

$$\lim_{\tau_{00}^2 \to \infty} \widehat{\Theta}_0^{\text{MAP}} = \lim_{\tau_{00}^2 \to \infty} \frac{1}{\sum_{i=1}^n \omega_{i,T} + \tau_0^2/\tau_{00}^2} \left(\sum_{i=1}^n \omega_{i,T} \bar{Y}_{i,T} + \frac{\tau_0^2}{\tau_{00}^2} \mu_0\right) = \widehat{\bar{\mu}}_T^{\text{hom}}.$$

This complete the proof of the claim. □

**Remarks 3.17.**

- Theorem 3.16 provides the homogeneous Bühlmann credibility formula under different assumptions than the projection argument onto the affine linear subspace $\mathcal{L}_e(\boldsymbol{Y})$. The crucial difference is that we assume $\mu_0$ being unknown and therefore we model it by a hierarchical (non-informative prior) Bayesian approach under the assumptions of Theorem 3.16. This non-informative prior on the collective mean acts similar as the intercept under the canonical link choice, which is the main argument for the balance property to hold here. Thus, this non-informative prior Bayesian component allows the model to self-calibrate to the



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global balance, and the resulting homogeneous Bühlmann credibility estimator is a reallocation of the totally observed claim.

We believe that this is a fairly interesting result, and it gives a different interpretation to the homogeneous Bühlmann credibility not relying on a projection argument onto the affine linear subspace $\mathcal{L}_e(\boldsymbol{Y})$.

- For finite $\tau_{00}^2 \in (0,\infty)$ we obtain the Bayesian posterior estimator

$$\mu^{\text{post}}_{Y_{i,T+1}|\boldsymbol{Y}} = \omega_{i,T}\bar{Y}_{i,T} + (1-\omega_{i,T})\left[\alpha_{i,T}\widehat{\widehat{\mu}}^{\text{hom}}_T + (1-\alpha_{i,T})\,\mu_0\right], \tag{3.24}$$

with credibility weight

$$\alpha_{i,T} = \frac{\sum_{i=1}^n \omega_{i,T}}{\sum_{i=1}^n \omega_{i,T} + \tau_0^2/\tau_{00}^2} \;\in\; (0,1),$$

this can be seen from the proof of Theorem 3.16. In the inhomogeneous Bühlmann credibility estimator we have absolute certainty about the prior mean $\mu_0$, and we let $\tau_{00}^2 \to 0$ and $\alpha_{i,T} \to 0$. For the homogeneous Bühlmann credibility estimator we do not have any information about $\mu_0$, thus, we let $\tau_{00}^2 \to \infty$ and $\alpha_{i,T} \to 1$. This indicates that partial knowledge about $\mu_0$ can be quantified through the variance parameter $\tau_{00}^2$.

- A side result of Theorem 3.16 is that the homogeneous Bühlmann credibility estimator, which looks a bit cumbersome in Theorem 3.13, can be calculated by simple matrix operations. We use Corollary 3.11 to obtain

$$\widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) \;=\; \boldsymbol{z}_i^\top\left(\mathfrak{Z}^\top\Sigma^{-1}\mathfrak{Z} + \mathfrak{T}^{-1}\right)^{-1}\mathfrak{Z}^\top\Sigma^{-1}\boldsymbol{Y},$$

where $\boldsymbol{z}_i^\top = (1,0,\ldots,0,1,0,\ldots,0) \in \{0,1\}^{n+1}$ with the second 1 in the $(i+1)$-st component encoding instance $i$, and where we set $\tau_{00}^2 = \infty$ in $\mathfrak{T}^{-1}$.

### 3.2.5 Empirical Bayesian estimation

The previous section has provided the homogeneous Bühlmann credibility formula which satisfies the balance property, see Corollary 3.14, and this for *any choice* of $\tau_0^2 > 0$ and $\sigma^2 > 0$, see Remark 3.15. In mathematical statistics one often considers a different form of empirical Bayesian estimation which is based on MLE. We discuss this in the present section. This will give us Tweedie's formula and Robbins' formula [174]; see Efron [72]. In large sample situations these empirical Bayesian estimations seem fine, however, on a small portfolio they may be problematic as the balance property typically fails to hold, this questions this approach in actuarial applications. This approach is also questioned in Bayesian statistics, it violates the Bayesian principle in its pure sense because the Bayesian methodology does not allow to elicit the prior from observations. Nevertheless, we discuss this empirical Bayesian approach in the present section because it is interesting.



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**Tweedie's formula**

Choose a response variable $Y \sim \text{EDF}(\theta; v/\varphi, \kappa, \nu)$ having EDF density

$$f(y;\theta,v/\varphi) = \exp\left\{\frac{y\theta-\kappa(\theta)}{\varphi/v} + a(y;v/\varphi)\right\},$$

w.r.t. the given $\sigma$-finite measure $\nu$ on $\mathbb{R}$ not depending on the canonical parameter $\theta \in \mathbf{\Theta} \subseteq \mathbb{R}$. We assume throughout $\mathring{\mathbf{\Theta}} \neq \emptyset$. Under the static random effects EDF model assumptions of Definition 3.1, we choose a prior density $\Theta \sim \pi(\theta)$ on $\mathbf{\Theta}$ for the (unknown) canonical parameter $\theta$ reflecting the prior knowledge. The marginal density of $Y$ is given by

$$f(y;v/\varphi) = \int_{\mathbf{\Theta}} f(y;\theta,b/\varphi)\pi(\theta)\,d\theta = \int_{\mathbf{\Theta}} \exp\left\{\frac{y\theta-\kappa(\theta)}{\varphi/v} + a(y;v/\varphi)\right\}\pi(\theta)\,d\theta,$$

where we assume that this integral exists for all $y$ in the convex hull of the support of $Y$. In the Poisson-gamma conjugate prior case this is the negative binomial model in case of a single year of observation $Y$, see (3.30)-(3.31), below. Using Bayes' rule [18], we have posterior density, we now write all terms of Bayes' formula,

$$\begin{aligned}
\pi(\theta|Y) &= \frac{f(Y;\theta,v/\varphi)\pi(\theta)}{f(Y;v/\varphi)} \\
&= \exp\left\{\frac{Y\theta-\kappa(\theta)}{\varphi/v} + a(Y;v/\varphi) + \log\pi(\theta) - \log f(Y;v/\varphi)\right\} \\
&= \exp\left\{\frac{\theta Y - \frac{\varphi}{v}\left(\log f(Y;v/\varphi) - a(Y;v/\varphi)\right)}{\varphi/v} - \frac{v}{\varphi}\kappa(\theta) + \log\pi(\theta)\right\}.
\end{aligned}$$

This looks pretty interesting and it gives us the following result.

**Theorem 3.18** (Tweedie's formula)**.** *Choose $Y \sim \text{EDF}(\theta; v/\varphi, \kappa, \nu)$ with effective domain $\mathbf{\Theta}$ having a non-empty interior. Choose a prior density $\pi$ on $\mathbf{\Theta}$ such that for all $y \in \text{conv}(Y)$ in the convex hull of the support of $Y$*

$$\mathfrak{L}(y) := \int_{\mathbf{\Theta}} \exp\left\{\frac{vy}{\varphi}\,\theta - \frac{v}{\varphi}\kappa(\theta) + \log\pi(\theta)\right\}\,d\theta < \infty.$$

*This implies that $\Theta|_Y \sim \text{EDF}(Y; v/\varphi, b, \widetilde{\nu})$ follows an EDF with cumulant function*

$$y \in \mathcal{Y} \;\mapsto\; b(y) = \log\mathfrak{L}(y) = \frac{\varphi}{v}\left(\log f(y;v/\varphi) - a(y;v/\varphi)\right), \tag{3.25}$$

*on an effective domain $\mathcal{Y} \supseteq \text{conv}(Y)$ and for Lebesgue measure $\widetilde{\nu}$ on $\mathbf{\Theta}$.*

The exciting observation of Theorem 3.18 is that we remain in the EDF for the posterior $\pi(\theta|Y)$ of the canonical parameter $\Theta$, given observation $Y$. Moreover, the effective domain $\mathcal{Y}$ is a possibly infinity interval in $\mathbb{R}$, and the cumulant function $b$ is smooth and convex on the interior $\mathring{\mathcal{Y}}$ of $\mathcal{Y}$. These latter properties follow because the cumulant function $b$ is the logarithm of the Laplace transform $\mathfrak{L}$; see Jørgensen [113, 114, 115] and Wüthrich–Merz [205, Section 2.1.1]. This also provides a smooth extension in case of discrete observations $Y$, e.g., in the Poisson model.

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**Corollary 3.19.** *Under the assumptions of Theorem 3.18 we have posterior moment generating function of $\Theta$, given $Y \in \mathring{\mathcal{Y}}$,*

$$M_{\Theta|Y}(r) = \mathbb{E}\left[\exp\{r\Theta\}|\, Y\right] = \exp\left\{\frac{v}{\varphi}\left[b(Y + r\varphi/v) - b(Y)\right]\right\},$$

*for $|r|$ sufficiently small such that $Y + r\varphi/v \in \mathring{\mathcal{Y}}$.*

Corollary 3.19 gives another remarkable result, namely, we can compute all posterior moments of $\Theta$, given $Y$, without explicitly specifying the prior density $\pi(\theta)$, but only the marginal of $Y$ is necessary, see (3.25), to obtain for all $k \in \mathbb{N}$

$$\begin{aligned}\mathbb{E}\left[\Theta^k \middle| Y\right] &= \frac{d^k}{dr^k} M_{\Theta|Y}(r)\bigg|_{r=0} = \frac{d^k}{dr^k}\exp\left\{\frac{v}{\varphi}\left[b(Y + r\varphi/v) - b(Y)\right]\right\}\bigg|_{r=0} \\ &= \frac{d^k}{dr^k}\frac{f(Y + r\varphi/v; v/\varphi)}{f(Y; v/\varphi)}\exp\left\{a(Y; v/\varphi) - a(Y + r\varphi/v; v/\varphi)\right\}\bigg|_{r=0}.\end{aligned}$$

For the first two moments we have

$$\mathbb{E}\left[\Theta|\, Y\right] = \frac{\varphi}{v}\left[\frac{f'(Y; v/\varphi)}{f(Y; v/\varphi)} - a'(Y; v/\varphi)\right],$$

and

$$\mathbb{E}\left[\Theta^2 \middle| Y\right] = \frac{\varphi^2}{v^2}\left[\frac{f''(Y; v/\varphi)}{f(Y; v/\varphi)} - 2\,\frac{f'(Y; v/\varphi)}{f(Y; v/\varphi)}\,a'(Y; v/\varphi) + a''(Y; v/\varphi)\right].$$

These formulas do not explicitly involve the prior $\pi(\theta)$, only implicitly through the marginal $f(y; v/\varphi)$. One can try to elicit this marginal directly without explicitly specifying the prior $\pi(\theta)$. This is emphasized in Efron [73, Section 5], namely, it does not require to deconvolute the random parts coming from priors and responses.

**Example 3.20** (Poisson-gamma model)**.** We revisit the Poisson-gamma Example 3.7 in its canonical form for the reproductive version of the response. For the Poisson EDF we set $\varphi = 1$, we have cumulant function $\kappa(\theta) = \exp\{\theta\}$, $\theta \in \boldsymbol{\Theta} = \mathbb{R}$, and we have normalizing function

$$y \geq 0 \;\mapsto\; a(y; v/\varphi) = \log\left(\frac{v^{vy}}{\Gamma(vy+1)}\right) = vy\log v - \log\Gamma(vy+1).$$

The gamma prior in the canonical parameter representation is given by, see (3.14),

$$\pi(\theta) = \exp\left\{\frac{\mu_0\theta - \kappa(\theta)}{\mu_0/\alpha} + \log\left(\frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)}\right)\right\}.$$

This gives the negative binomial marginal for $Y$

$$f(y; v/\varphi) = \frac{\Gamma(\alpha + vy)}{\Gamma(\alpha)\Gamma(vy+1)}\left(\frac{v\mu_0}{\alpha + v\mu_0}\right)^{vy}\left(\frac{\alpha}{\alpha + v\mu_0}\right)^{\alpha},$$

for the derivation see (3.30)-(3.31), below. Collecting all the terms we obtain cumulant function in the posterior Poisson-gamma case

$$b(y) = \frac{1}{v}\log\left(\frac{\Gamma(\alpha + vy)}{\Gamma(\alpha)}\left(\frac{\mu_0}{\alpha + v\mu_0}\right)^{vy}\left(\frac{\alpha}{\alpha + v\mu_0}\right)^{\alpha}\right). \tag{3.26}$$



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This gives us posterior moment generating function

$$M_{\Theta|Y}(r) = \mathbb{E}\left[\exp\{r\Theta\}\middle| Y\right] = \frac{\Gamma(\alpha + vY + r)}{\Gamma(\alpha + vY)} \left(\frac{\mu_0}{\alpha + v\mu_0}\right)^r,$$

and the first posterior moment

$$\mathbb{E}\left[\Theta\middle| Y\right] = \frac{\Gamma'(\alpha + vY)}{\Gamma(\alpha + vY)} - \log\left(\frac{\alpha + v\mu_0}{\mu_0}\right),$$

we also refer to the text before (3.6). This provides a Bayesian estimate for the canonical parameter. The issue with this is, in general, we are not interested into the posterior mean of $\Theta$ but rather into the posterior mean of $\Lambda = \kappa'(\Theta)$, given $Y$, see (3.13). In the Poisson case with exponential cumulant function this is a rather straightforward computation from the moment generating function

$$\begin{aligned}\mathbb{E}\left[\Lambda\middle| Y\right] &= \mathbb{E}\left[\kappa'(\Theta)\middle| Y\right] = M_{\Theta|Y}(1) = (\alpha + vY)\frac{\mu_0}{\alpha + v\mu_0} \\ &= \frac{v}{v + \alpha/\mu_0} Y + \left(1 - \frac{v}{v + \alpha/\mu_0}\right)\mu_0.\end{aligned}$$

This is the well-known Bühlmann credibility in the Poisson-gamma model, see (3.4). ■

**Parametric empirical Bayesian estimation**

Up to know, we have not discussed the choice of the prior $\pi(\theta)$. If its functional form is known then it requires determination of its parameters. This then allows one to calculate the marginal $f(y; v/\varphi)$ and the cumulant function $b$ given in (3.25), as well as any posterior moment of $\Theta$, given $Y$, or the Bühlmann credibility (3.12) in the conjugate prior case. Denote the parameters in $\pi$ by $\gamma$, i.e., assume $\pi(\theta) = \pi(\theta|\gamma)$. If $\gamma$ is unknown, things start to become difficult. Taking the viewpoint of a hierarchical Bayesian model, one could choose another prior on $\gamma$, maybe letting it be a weakly informative prior, this relates to Remarks 3.17. An empirical Bayesian viewpoint would try to directly estimate $\gamma$ from data. This approach is quite controversial within the Bayesian community as it mixes Bayesian with frequentist's elements. Nevertheless, it provides a feasible way of eliciting (somehow) the prior model to calculate experience rating premiums. For later purposes, we revisit the Poisson-gamma example.

**Example 3.20, revisited.** We restrict to one period and we drop the time index $t$. Having multiple independent observations $\boldsymbol{Y} = (Y_1, \ldots, Y_n)^\top$ following the same Poisson-gamma model with potentially different exposures $v_i$, we have the joint negative binomial log-likelihood function, see Example 3.20,

$$\ell_{\boldsymbol{Y}}(\mu_0, \alpha) = \sum_{i=1}^n \log\left[\frac{\Gamma(\alpha + v_i Y_i)}{\Gamma(\alpha)\Gamma(v_i Y_i + 1)} \left(\frac{v_i\mu_0}{\alpha + v_i\mu_0}\right)^{v_i Y_i} \left(\frac{\alpha}{\alpha + v_i\mu_0}\right)^{\alpha}\right].$$

In empirical Bayesian estimation, one maximizes this log-likelihood in the prior parameters $\alpha$ and $\mu_0$. For given shape parameter $\alpha > 0$, the score equation is given by

$$\frac{\partial}{\partial\mu_0}\ell_{\boldsymbol{Y}}(\mu_0, \alpha) = \sum_{i=1}^n \frac{v_i Y_i}{\mu_0} - v_i \frac{\alpha + v_i Y_i}{\alpha + v_i\mu_0} = 0.$$



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We solve this for $\mu_0$ to receive the MLE

$$\widehat{\mu}_0^{\text{MLE}} = \frac{\sum_{i=1}^n v_i Y_i}{\sum_{i=1}^n v_i}. \tag{3.27}$$

This MLE does not depend on $\alpha$ which allows us to consider the second score equation

$$\frac{\partial}{\partial \alpha} \ell_{\boldsymbol{Y}}(\widehat{\mu}_0^{\text{MLE}}, \alpha) = 0, \tag{3.28}$$

to estimate the shape parameter $\alpha$ with MLE in an empirical Bayesian fashion. The empirical Bayesian posterior estimator for the expected number of Poisson claims is then given by estimated Bühlmann credibility

$$\widehat{\mathbb{E}}\left[\kappa'(\Theta) \middle| Y\right] = \frac{v}{v + \widehat{\alpha}^{\text{MLE}}/\widehat{\mu}_0^{\text{MLE}}} Y + \left(1 - \frac{v}{v + \widehat{\alpha}^{\text{MLE}}/\widehat{\mu}_0^{\text{MLE}}}\right) \widehat{\mu}_0^{\text{MLE}}. \tag{3.29}$$

We emphasize that we put a hat $\widehat{\mathbb{E}}$ on top of the expectation operator $\mathbb{E}$ to indicate that this is an empirical Bayesian estimate. ■

This seems all fine, but in Remark 3.22, below, will indicate that there is an issue with the balance property of this empirical Bayesian approach which requires special attention in actuarial pricing.

### Robbins' formula

The previous example shows an empirical Bayesian posterior estimator in an example where the marginal density $f(y; v/\varphi)$ is analytically tractable. When $f(y; v/\varphi)$ is not analytically tractable and $\pi$ unknown, one can estimate this marginal density non-parametrically, e.g., using splines; see Efron [72]. Importantly, such a non-parametric empirical Bayesian approach does not require an explicit specification of the prior $\pi$. We give an example which is known as Robbins' formula [174].

**Example 3.21** (Robbins' formula [174])**.** We consider the Poisson example, but not in its canonical form. Assume, given $\Lambda$, that $Y$ is Poisson distributed with mean $\Lambda$, we set $v = 1$. The posterior mean of $\Lambda$, given $Y$, is

$$\mathbb{E}\left[\Lambda \middle| Y\right] = \frac{\int_0^\infty \lambda e^{-\lambda} \frac{\lambda^Y}{Y!} \pi(\lambda)\, d\lambda}{\int_0^\infty e^{-\lambda} \frac{\lambda^Y}{Y!} \pi(\lambda)\, d\lambda} = (Y+1) \frac{p_{Y+1}}{p_Y},$$

where we have defined the (marginal) probability weights for $k \in \mathbb{N}_0$ by

$$p_k = \mathbb{P}[Y = k] = \mathbb{E}\left[\mathbb{P}[Y = k|\Lambda]\right] = \int_0^\infty e^{-\lambda} \frac{\lambda^k}{k!} \pi(\lambda)\, d\lambda.$$

There remains the estimation of these marginal probability weights. Assuming i.i.d. copies $(Y_i)_{i=1}^n$ of $Y$, we empirically estimate

$$\widehat{p}_k = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{Y_i = k\}}.$$



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This then provides non-parametric empirical Bayesian estimate

$$\widehat{\mathbb{E}}\left[\Lambda \middle| Y=k\right] = (k+1)\,\frac{\widehat{p}_{k+1}}{\widehat{p}_k}.$$

Note that this does not require any specific form of the prior distribution $\pi$.

We give a numerical example to Robbins' formula and to the empirical Poisson-gamma Bayesian estimation; the data is taken from Efron–Hastie [75, Table 6.1]. For this data, we set $T = 1$ (one year of observations) and unit exposures $v_{i,1} = 1$ for all instances $1 \le i \le n = 9,461$. The data is given in Table 3.1.

| number of claims $k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| counts $n_k$ | 7,840 | 1,317 | 239 | 42 | 14 | 4 | 4 | 1 |
| total claims $kn_k$ | 0 | 1,317 | 478 | 126 | 56 | 20 | 44 | 7 |
| probabilities $\widehat{p}_k$ | 0.8287 | 0.1392 | 0.0253 | 0.0044 | 0.0015 | 0.0004 | 0.0004 | 0.0001 |
| Robbins $\widehat{\mathbb{E}}[\Lambda\|Y=k]$ | 0.1680 | 0.3629 | 0.5272 | 1.3333 | 1.4286 | 6.0000 | 1.7500 | – |
| $k\widehat{\mathbb{E}}[\Lambda\|Y=k]$ | 1,317 | 478 | 126 | 56 | 20 | 44 | 7 | – |
| empirical Bayes | 0.1642 | 0.3982 | 0.6323 | 0.8663 | 1.1004 | 1.3344 | 1.5685 | 1.8025 |

Table 3.1: Claim count example taken from Efron–Hastie [75, Table 6.1].

We start with Robbins' formula. Let $k \in \{0,\ldots,7\}$ be the number of observed claims and $n_k$ the number of policies that have $k$ claims

$$n_k = \sum_{i=1}^{n} \mathbb{1}_{\{Y_i=k\}} = n\widehat{p}_k.$$

Table 3.1 gives Robbins' formula $\widehat{\mathbb{E}}[\Lambda|Y=k]$ for $k \in \{0,\ldots,6\}$. Note that we cannot calculate a sensible price for $k=7$ because $\widehat{p}_8 = 0$, that is, we do not have a sensible estimate for this probability; we highlight this by using the symbol – in Table 3.1. This shows the issue of Robbins' formula, namely, that it is very sensitive in the estimates $\widehat{p}_k$, which are typically volatile (non-credible) for larger $k$'s, and for the largest class there is no sensible estimate. This missing robustness prevents the Robbins' formula from practical applications in insurance, and, typically, parametric models for the prior $\pi$ are preferred. Another interesting observation is the total premium charged. We have, we omit the claims class $Y=7$ in the following computation,

$$\sum_{k=0}^{7-1} n_k\,\widehat{\mathbb{E}}\left[\Lambda \middle| Y=k\right] = \sum_{k=0}^{7-1} n_k\,(k+1)\,\frac{\widehat{p}_{k+1}}{\widehat{p}_k} = \sum_{k=0}^{7-1} n_{k+1}\,(k+1) = \sum_{k=1}^{7} n_k\,k.$$

Thus, the global balance property holds if we do not charge any positive price to the insurance policies with $Y=7$ claims, thus, Robbins' formula is a reallocation of the claims, not assigning any claims to the insurance policies with the highest number of claims. Of course, we will always charge a positive price to these highest claims policies. Henceforth, the Robbins' method always has a positive bias if we just add a positive price for $Y=7$. The reallocation implied by the previous computation is nicely indicated by the second last line of Table 3.1, namely, one just shifts the observed claims one column to the left $n_k\widehat{\mathbb{E}}[\Lambda|Y=k] = n_{k+1}(k+1)$.

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Since Robbins' method is not robust, we turn our attention to the Poisson-gamma model using an empirical Bayesian estimation, see (3.29). This requires to estimate the mean parameter $\mu_0$ and the shape parameter $\alpha$ with MLE. The mean is rather straightforward in our example, see (3.27), and we obtain $\widehat{\mu}_0^{\text{MLE}} = 0.2144$. MLE of the shape parameter $\alpha$ is performed numerically using the function `optim` from the `R` package `stats` [166] to solve score equation (3.28). We receive $\widehat{\alpha}^{\text{MLE}} = 0.7015$. This gives us estimated credibility weights for $v_{i,1} = 1$, see (3.29),

$$\widehat{\omega}_{i,1} = \frac{v_{i,1}}{v_{i,1} + \widehat{\alpha}^{\text{MLE}}/\widehat{\mu}_0^{\text{MLE}}} = 0.2340.$$

Using these credibility weights, we can calculate the empirical Bayesian estimates. The results are shown on the last line of Table 3.1, and we observe that this parametric empirical Bayesian estimation is more robust than Robbins' formula. ■

We close this section on the empirical Bayesian estimation with the following (important) remark; the following result has been highlighted by Bühlmann [40].

**Remark 3.22** (failure of the balance property)**.** Typically, the balance property fails to hold under the empirical Bayesian estimation framework. Therefore, this method has to be used with great care for actuarial pricing, and a bias correction is necessary. We briefly highlight the failure of the balance property in the Poisson-gamma model related to formula (3.29). We only consider one calendar year and, therefore, drop the time index $t$. Assume we have $n \geq 2$ insurance policies with exposures $v_i > 0$. We set for the estimated credibility weights

$$\widehat{\omega}_i = \frac{v_i}{v_i + \widehat{\alpha}^{\text{MLE}}/\widehat{\mu}_0^{\text{MLE}}}.$$

The totally charged empirical Bayesian premium is equal to

$$\begin{aligned}
\sum_{i=1}^n v_i \widehat{\mathbb{E}}\left[\kappa'(\Theta_i) \middle| Y_i\right] &= \sum_{i=1}^n v_i \left(\widehat{\omega}_i Y_i + (1-\widehat{\omega}_i)\, \widehat{\mu}_0^{\text{MLE}}\right) \\
&= \sum_{i=1}^n v_i \left(\widehat{\omega}_i Y_i + (1-\widehat{\omega}_i)\, \widehat{\widehat{\mu}}_1^{\text{hom}}\right) + \sum_{i=1}^n v_i\, (1-\widehat{\omega}_i) \left(\widehat{\mu}_0^{\text{MLE}} - \widehat{\widehat{\mu}}_1^{\text{hom}}\right) \\
&= \sum_{i=1}^n v_i\, \widehat{\widehat{\mu(\Theta_i)}}^{\text{hom}}(\boldsymbol{Y}) + \sum_{i=1}^n v_i\, (1-\widehat{\omega}_i) \left(\widehat{\mu}_0^{\text{MLE}} - \widehat{\widehat{\mu}}_1^{\text{hom}}\right),
\end{aligned}$$

with collective mean estimate from the homogeneous Bühlmann credibility formula, see Theorem 3.13,

$$\widehat{\widehat{\mu}}_1^{\text{hom}} = \frac{1}{\sum_{i=1}^n \widehat{\omega}_i} \sum_{i=1}^n \widehat{\omega}_i Y_i.$$

Since the homogeneous Bühlmann credibility satisfies the balance property, see Corollary 3.14, the empirical Bayesian estimate fulfills the balance property if and only if $\widehat{\mu}_0^{\text{MLE}} = \widehat{\widehat{\mu}}_1^{\text{hom}}$. But this typically fails to hold for exposures $v_i > 0$ varying over the different insurance policies $1 \leq i \leq n$. ■



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### 3.2.6 Lab: Static random effects Poisson-gamma example

We fit the static random effects Poisson-gamma model of Example 3.3 to our claim counts data. This model has two parameters $\mu_0 > 0$ and $\alpha > 0$ that need to be determined. We fit this model in four different ways (a), (a0), (b) and (b0), this is discussed next.

*(a) Full MLE: empirical Bayesian approach.* We calculate the posterior mean resulting in the inhomogeneous Bühlmann credibility formula (3.4), and we replace the unknown parameters by estimates received as follows. For the parameters involved, we exploit an empirical Bayesian estimation using MLE at time $T$ and based on the entire available data $\mathcal{L}_T$, see (2.49). The log-likelihood function of a single insurance policy for given observations $(N_{1:T}, v_{1:T})$ reads as, we drop the insurance policy index $i$ for simplicity,

$$\begin{aligned}
\ell_{N_{1:T}}(\mu_0,\alpha) &= \log\left(\int_0^\infty \prod_{t=1}^T \left(e^{-v_t\lambda}\frac{(v_t\lambda)^{N_t}}{N_t!}\right)\frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\alpha\lambda/\mu_0}\,d\lambda\right)\\
&\propto \log\left(\int_0^\infty \frac{(\alpha/\mu_0)^\alpha}{\Gamma(\alpha)}\lambda^{\alpha+\sum_{t=1}^T N_t-1}e^{-\left(\alpha/\mu_0+\sum_{t=1}^T v_t\right)\lambda}\,d\lambda\right) \qquad (3.30)\\
&= \log\left(\frac{\Gamma\left(\alpha+\sum_{t=1}^T N_t\right)}{\Gamma(\alpha)}\frac{(\alpha/\mu_0)^\alpha}{(\alpha/\mu_0+\sum_{t=1}^T v_t)^{\alpha+\sum_{t=1}^T N_t}}\right).
\end{aligned}$$

Thus, we have a closed form solution for the log-likelihood function in $(\mu_0,\alpha)$, and maximization can be done by the Newton–Rapshon algorithm. We are going to use the function `optim` from the R package `stats` [166], see Listing 3.1, below.
We rewrite the previous log-likelihood function in a more compact form. Since this aggregation provides a sufficient statistics in the static random effects case, we receive the same MLE for $(\mu_0,\alpha)$ from the last line of (3.30) as from the negative binomial model

$$\ell_{N_{1:T}}(\mu_0,\alpha) \propto \log\left(\frac{\Gamma\left(\alpha+\sum_{t=1}^T N_t\right)}{\Gamma(\alpha)\left(\sum_{t=1}^T N_t\right)!}\left(\frac{\alpha}{\alpha+\sum_{t=1}^T v_t\mu_0}\right)^\alpha\left(\frac{\sum_{t=1}^T v_t\mu_0}{\alpha+\sum_{t=1}^T v_t\mu_0}\right)^{\sum_{t=1}^T N_t}\right). \qquad (3.31)$$

We implement this log-likelihood function and maximize it in $(\mu_0,\alpha)$ over all observations at time $T$ and using that different insurance policies $i$ are assumed to be independent, that is, we solve

$$\left(\widehat{\mu}_{0,1:T}^{\text{MLE}},\widehat{\alpha}_{1:T}^{\text{MLE}}\right) = \underset{(\mu_0,\alpha)}{\arg\max}\sum_i \ell_{N_{i,1:T}}(\mu_0,\alpha). \qquad (3.32)$$

We perform this for a fixed calendar year $2014 \le T \le 2018$, and we consider all insurance policies $i$ to obtain this MLE. Observe that we only require $\sum_{t=1}^T v_{i,t} > 0$, and some of the underlying calendar years $2011 \le t \le T$ may have a zero exposure $v_{i,t} = 0$. Using this estimated model, we then calculate the predictive means (3.4) for $N_{i,T+1}$, given observations $N_{i,1:T}$, and using this empirical Bayesian approach (3.32) for parameter estimation of $(\mu_0,\alpha)$.

*(a0) All accident years: homogeneous Bühlmann credibility approach.* As mentioned in Remark 3.22, typically, the empirical Bayesian fitted model fails to fulfill the (in-sample) balance property. On the other hand, the homogeneous Bühlmann credibility estimator fulfills the balance property for any choice of structural parameters $\sigma^2$ and $\tau_0^2$ or, in other

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words, for *any* credibility coefficient $\sigma^2/\tau_0^2$, see Corollary 3.14. In the Poisson-gamma case this credibility coefficient is given by $\alpha/\mu_0$, see (3.5). We use the MLEs from the previous solution, case (a), to estimate this credibility coefficient by $\widehat{\alpha}_{1:T}^{\text{MLE}}/\widehat{\mu}_{0,1:T}^{\text{MLE}}$. This provides us with estimated credibility weights for insurance policy $i$

$$\widehat{\omega}_{i,T} = \frac{\sum_{t=1}^T v_{i,t}}{\sum_{t=1}^T v_{i,t} + \widehat{\alpha}_{1:T}^{\text{MLE}}/\widehat{\mu}_{0,1:T}^{\text{MLE}}}.$$

Using these estimated credibility weights we then use the homogeneous Bühlmann credibility estimator which replaces in the empirical Bayesian estimator (3.29) the last MLE term $\widehat{\mu}_0^{\text{MLE}}$ by the estimate

$$\widehat{\widehat{\mu}}_T^{\text{hom}} = \frac{1}{\sum_{i=1}^n \widehat{\omega}_{i,T}} \sum_{i=1}^n \widehat{\omega}_{i,T} \bar{Y}_{i,T}.$$

The resulting fitted model satisfies the (in-sample) balance property on $\mathcal{L}_T$.

*(b) One-year ahead MLE approach: empirical Bayesian approach.* The previous model used in (3.30) may be too rigid, especially, if the data is non-stationary. In such situations it may be beneficial to only consider the MLE from the last responses $N_{i,T}$, $i \in \mathcal{I}_T$, conditionally given all previous observations $N_{i,1:T-1}$. This is more in the spirit of Section 2.3.4, where we only performed one-year ahead predictions based on information $\mathcal{L}_T^\circ$, see (2.53). In that case, the log-likelihood (3.30) then needs to be replaced by the conditional log-likelihood, we drop index $i$ for the moment,

$$\begin{aligned}
\ell_{N_T|N_{1:T-1}}(\mu_0,\alpha) &= \log\left(\int_0^\infty e^{-v_T\lambda}\frac{(v_T\lambda)^{N_T}}{N_T!}\frac{(\alpha/\mu_0+\sum_{t=1}^{T-1}v_t)^{\alpha+\sum_{t=1}^{T-1}N_t}}{\Gamma(\alpha+\sum_{t=1}^{T-1}N_t)}\right.\\
&\qquad\left.\times\,\lambda^{\alpha+\sum_{t=1}^{T-1}N_t-1}e^{-(\alpha/\mu_0+\sum_{t=1}^{T-1}v_t)\lambda}\,d\lambda\right) \qquad (3.33)\\
&= \log\left(\frac{\Gamma\left(\alpha+\sum_{t=1}^T N_t\right)}{\Gamma(\alpha+\sum_{t=1}^{T-1}N_t)N_T!}\left(\frac{\alpha+\sum_{t=1}^{T-1}v_t\mu_0}{\alpha+\sum_{t=1}^T v_t\mu_0}\right)^{\alpha+\sum_{t=1}^{T-1}N_t}\left(\frac{v_T\mu_0}{\alpha+\sum_{t=1}^T v_t\mu_0}\right)^{N_T}\right),
\end{aligned}$$

This is similar to (3.31). The basic idea applied is that the Bayesian estimation is recursive with $\Lambda|_{N_{1:T-1}}$ having a gamma distribution with the parameters given in (3.33). This gamma distribution is then used as the prior of the latent gamma risk factor for the next observation $N_T$. This allows to perform a recursive Bayesian inference step, or to integrate out the latent factor resulting in the negative binomial distribution for the last observation $N_T$, conditionally given $N_{1:T-1}$. Aggregating over all accident years $i \in \mathcal{I}_T$, we solve in this one-period ahead approach the MLE problem

$$\left(\widehat{\mu}_{0,T}^{\text{MLE}}, \widehat{\alpha}_T^{\text{MLE}}\right) = \underset{(\mu_0,\alpha)}{\arg\max} \sum_{i\in\mathcal{I}_T} \ell_{N_{i,T}|N_{i,1:T-1}}(\mu_0,\alpha). \qquad (3.34)$$

Note that we have changed the lower index in the MLEs to distinguish them from the full MLE approach (a), compare to (3.32).

*(b0) One-year ahead: homogeneous Bühlmann credibility approach.* As mentioned in Remark 3.22, also the one-year ahead MLE approach of item (b) does not satisfy the balance



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property, now w.r.t. $\mathcal{L}_T^\circ$. Proceeding as in (a0) we also study the homogeneous Bühlmann credibility approach in this one-year ahead situation, just replacing the estimates from (a) by the ones from (b). The resulting fitted model satisfies the balance property, this time on $\mathcal{L}_T^\circ$.

**Remarks 3.23.**

- Bühlmann–Gisler [41, Chapter 4] present a different estimation method not being based on MLE. Namely, the basic method of credibility estimation is to retain the moments (method of moments). Thus, one can also estimate the heterogeneity between risks $\tau_0^2$ and the volatility within risks $\sigma^2$ by unbiased estimators, and then use those to calculate the credibility weights $\omega_{i,T}$. Moreover, in the Poisson-gamma case we have $\sigma^2 = \mu_0$, which allows for an iterative estimation scheme; see Bühlmann–Gisler [41, Section 4.10].

- It should be emphasized that MLE (empirical Bayesian estimation) provides asymptotically unbiased estimators. However, in financial applications, this may not be the most useful property because often we have small samples, and we should be more concerned about the balance property, i.e., if our pricing system is not a reallocation of the total costs. Note that the balance property is stronger than unbiasedness, it implies unbiasedness. This unbiasedness is in-sample and needs to be corrected for the pricing of future contracts for changes in the portfolio structure and other non-stationarities. This extrapolation requires additional knowledge which often is not contained in past data.

■

We fit this static random effects Poisson-gamma model to the same synthetic data as used in Section 2.3.4, and the results are directly comparable to Tables 2.5 and 2.7, and Figures 2.4, 2.8 and 2.9. We fit the models using the four different estimators (3.32) and (3.34) and with their homogeneous Bühlmann credibility counterparts. Listing 3.1 gives the R code for fitting the static random effects Poisson-gamma model using the full MLE approach.

The results are presented in Figure 3.1, the left-hand side shows the in-sample figures for the last calendar year observations given by $\mathcal{L}_T^\circ$, and the right-hand side the out-of-sample figures for forecasting the next calendar year $T+1$. We compare them to the true means using the average negative binomial KL divergence, and the null model and the GLM are taken from Figure 2.4. Interestingly, we have a clear ordering of the two fitting methods (a) and (b) proposed in (3.32) and (3.34). The full MLE approach on $\mathcal{L}_T$ does not seem to be competitive, which indicates that our data is non-stationary. Remark that the in-sample figures on the left-hand side have been fitted on $\mathcal{L}_T$ in methods (a) and (a0), and they are evaluated only on $\mathcal{L}_T^\circ$. The reason for this is that want comparability to the null model and the GLM which are only based and evaluated on $\mathcal{L}_T^\circ$, that is, we generally consider the one-year ahead predictions in all model evaluations, also in-sample even if the model has been fitted on the entire past history $\mathcal{L}_T$. The worse behavior then shows non-stationarity in the data (and the annual claim frequencies), and it also indicates that the balance property is less relevant in non-stationary situations (where we also need to predict/extrapolate the change in overall expected frequency).

We have started our synthetic data simulation procedure in year 2011 without any claims



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Listing 3.1: Full MLE approach for fitting the static random effects EDF model.

```
### negative log-likelihood function
neg.SRE.LL <- function(param){
   mu0   <- exp(param[1])
   alpha <- exp(param[2])
   -sum(lgamma(alpha+datT$NN)-lgamma(alpha)+alpha*log(alpha/(alpha+mu0*datN$VV))+
                                datT$NN *log(1-alpha/(alpha+mu0*datN$VV)))
           }
### select data
T <- 2015
datT <- data.frame(dat[which(dat$Year<=T),] %>%  group_by(PolicyID) %>%
                                         summarize(NN = sum(ClaimNb),
                                                   VV = sum(Exposure)))
### MLE
param.init <- c(log(mean(dat$ClaimNb)), log(1))
paramT     <- exp(optim(par=param.init, fn=neg.SRE.LL)$par)
### prediction on data having exposure in year T+1
datT1 <- dat[which(dat$Year<=(T+1)),]
datT2 <- datT1[which(datT1$Year==(T+1)), c("PolicyID","ClaimNb","Exposure","True")]
datT1 <- datT1[which(datT1$PolicyID %in% datT2$PolicyID),]
datT1 <- data.frame(datT1[which(datT1$Year<=T),] %>%  group_by(PolicyID) %>%
                                            summarize(NN = sum(ClaimNb),
                                                      VV = sum(Exposure)))
datT2 <- datT2 %>% left_join(datT1, by = "PolicyID")
datT2$pred <- datT2$Exposure * (1/(datT2$VV+paramT[2]/paramT[1])*datT2$NN
                           + (1-datT2$VV/(datT2$VV+paramT[2]/paramT[1]))*paramT[1])
```

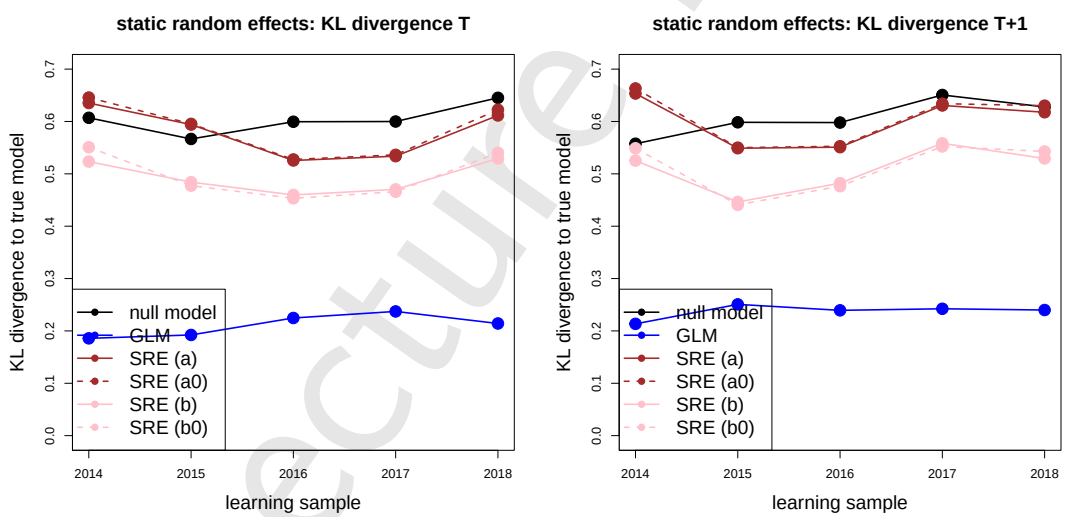


Figure 3.1: Fitted static random effects (SRE) models: (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one-year ahead forecast, the labels (a)-(b) refer to the two different fitting methods (3.32) and (3.34), (a0)-(b0) are their homogeneous Bühlmann credibility counterparts, and the null model and the GLM are taken from Figure 2.4; units are $10^{-2}$.

history, and the negative binomial MLEs $\left(\widehat{\mu}^{\text{MLE}}_{0,1:T}, \widehat{\alpha}^{\text{MLE}}_{1:T}\right)$ on the full data $\mathcal{L}_T$, methods (a) and (a0), tell us that it takes 3 to 5 calendar years to receive a steady state in this simulation and experience rating process. In particular, we still have a rather bad out-

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of-sample model in year 2014, for this year we have estimated $(\mu_0, \alpha)$ with only three years of past claims history. Therefore, we believe that this process needs a burn-in of 3 to 5 years to receive reliable estimates, also benefiting from individual claims history information.

The second fitting approach (3.34), methods (b) and (b0), gives clearly better results. From Figure 3.1 we verify that past claims history $Y_{i,1:T}$ has predictive power because the fitted posterior predictors (b) and (b0) have a clearly better performance than the null model. Only the GLM has a better performance from which we conclude that we should not only consider past claims history $Y_{i,1:T}$, but also covariates $\boldsymbol{x}_{i,T+1}$ for insurance pricing. In fact, Figure 3.1 suggests that these fixed effects covariates $\boldsymbol{x}_{i,T+1}$ are more explanatory than the past claims history $Y_{i,1:T}$ in this example. Of course, this may be a specific of this simulated data, and real motor insurance data may also have an opposite behavior. Comparing the empirical Bayesian estimates (b) and its homogeneous Bühlmann credibility counterparts (b0) we observe some differences, but there is not a clear preference for one of the predictors, but method (b0) can be preferred because of its mathematical properties of the global balance property.

The next natural step is to study *static mixed effects models* which is going to be done in Section 3.3, below. This closes this example.

### 3.2.7 Fitting the static random effects gamma-gamma model

For completeness, we provide the empirical Bayesian fitting approach for the static random effects gamma-gamma model of Example 3.5. This model has three parameters $\varphi > 0$, $\mu_0 > 0$ and $\alpha > 1$ that need to be estimated from the data. Assume we have observations $Y_{1:T}$ for a single insurance policy, we drop index $i$ in this section. We use the additive form $Z_t = v_t Y_t/\varphi$ of the claim sizes, and we assume that this is the observation scale under which the true claims are observed. Set $Z_\bullet = \sum_{t=1}^T Z_t$ and $v_\bullet = \sum_{t=1}^T v_t$. The log-likelihood function is given by

$$
\begin{aligned}
&\ell_{Y_{1:T}}(\varphi, \mu_0, \alpha) \\
&\quad = \log\left(\int_0^\infty \prod_{t=1}^T \left(\frac{(v_t\lambda/\varphi)^{v_t/\varphi}}{\Gamma(v_t/\varphi)} Y_t^{v_t/\varphi-1} e^{-Y_t v_t \lambda/\varphi}\right) \frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\mu_0(\alpha-1)\lambda}\, d\lambda\right) \\
&\quad \propto \log\left(\frac{\left(\prod_{t=1}^T Z_t^{v_t}\right)^{1/\varphi}}{\prod_{t=1}^T \Gamma(v_t/\varphi)} \frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha + v_\bullet/\varphi)}{(\mu_0(\alpha-1) + Z_\bullet)^{\alpha + v_\bullet/\varphi}}\right) \\
&\quad \propto \frac{1}{\varphi}\log\left(\prod_{t=1}^T (Z_t/Z_\bullet)^{v_t}\right) - \sum_{t=1}^T \log\Gamma(v_t/\varphi) + \log((v_\bullet/\varphi)!) \qquad (3.35) \\
&\qquad + \log\left(\frac{\Gamma(\alpha + v_\bullet/\varphi)}{\Gamma(\alpha)(v_\bullet/\varphi)!}\left(\frac{\mu_0(\alpha-1)}{\mu_0(\alpha-1) + Z_\bullet}\right)^\alpha \left(\frac{Z_\bullet}{\mu_0(\alpha-1) + Z_\bullet}\right)^{v_\bullet/\varphi}\right).
\end{aligned}
$$

The last line has for given $\varphi > 0$ the structure of a negative binomial distribution. Similarly we can derive the one-year ahead MLE approach, we refrain from giving the explicit calculations. To comply with the balance property we can then use the homogeneous Bühlmann credibility estimator with MLE estimated credibility weights.



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We come back to analyze (3.35). Assume that $v_1/\varphi = 1$, which implies that

$$Y_1 = Z_1|_\Lambda \sim \Gamma(1, \Lambda) \stackrel{(d)}{=} \text{Expo}(\Lambda),$$

has a conditional exponential distribution. We compute the marginal density of $Y_1$. Its log-likelihood function is given by

$$\begin{aligned}\ell_{Y_1}(\mu_0, \alpha) &= \log\left(\int_0^\infty \lambda e^{-Y_1\lambda} \frac{(\mu_0(\alpha-1))^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\mu_0(\alpha-1)\lambda}\, d\lambda\right) \\ &= \log\left(\frac{\alpha}{\mu_0(\alpha-1)}\left(1 + \frac{Y_1}{\mu_0(\alpha-1)}\right)^{-(\alpha+1)}\right).\end{aligned}$$

This is a Pareto Type II density, also known as the Lomax distribution [140]. Thus, we receive a regularly varying distribution with tail index $\alpha > 1$ by marginalizing (mixing) the exponential distribution over the gamma prior density. This distribution has a finite mean since $\alpha > 1$.

## 3.3 Static mixed effects models

### 3.3.1 Definition of static mixed effects models

We extend the static random effects model of Definition 3.1 to a *static mixed effects model* by complementing the static random effects $\Theta$ with fixed covariates $\boldsymbol{x}_t$. This has been considered in Dionne–Vanasse [68, 69] for a Poisson-gamma claim counts model and in Pinquet [158] for the gamma and the log-normal claim size examples, a more recent reference for the Poisson-gamma case include Duval et al. [71]. A mixed effects extension requires some care because one needs to ensure that the models and the parameter spaces have admissible supports.

**Definition 3.24** (static mixed effects EDF model)**.** *The triplet* $(Y_{1:T+1}, \boldsymbol{x}_{1:T+1}, \Lambda)$ *follows a static mixed effects EDF model if, given* $\Lambda \sim \pi$*, the random vectors* $(Y_t, \boldsymbol{x}_t, v_t)$ *are conditionally independent for* $1 \le t \le T+1$ *and, given* $(\Lambda, \boldsymbol{x}_t, v_t)$*, we have conditional EDF distributions* $Y_t|_{\{\Lambda, \boldsymbol{x}_t, v_t\}} \sim \text{EDF}(\Theta_t; v_t/\varphi, \kappa, \nu_t)$ *for* $1 \le t \le T+1$ *with conditional means*

$$\mathbb{E}\left[Y_t|\, \Lambda, \boldsymbol{x}_t, v_t\right] = \mu(\boldsymbol{x}_t, \Lambda),$$

*for a given regression function* $\mu : \mathcal{X} \times \text{supp}(\pi) \to \mathcal{M}$ *and for (random) canonical parameters* $\Theta_t = h(\mu(\boldsymbol{x}_t, \Lambda)) \in \boldsymbol{\Theta}$*,* $\mathbb{P}$*-a.s., with canonical link* $h = (\kappa')^{-1}$*.*

**Remark 3.25.** Definition 3.1 of the static random effects EDF model has been formulated a bit in a sloppy way to not overload the wording. In fact, we assume that $(Y_t, v_t)$ are conditionally independent for $1 \le t \le T+1$, given $\Theta$, and that $Y_t|_{\{\Theta, v_t\}}$ has the corresponding conditional EDF distribution. ■

If we choose a single point measure for $\pi$ in Definition 3.24 we obtain a classical fixed effects regression model. For the trivial covariate space $\mathcal{X} = \{1\}$ we have a static random effects EDF model with static random effects $\Theta = h(\mu(1, \Lambda))$.



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### 3.3.2 Static mixed effects generalized linear models with log-link

We consider Definition 3.24 under a GLM assumption for the regression function $\mu$. More specifically, we choose GLMs with the log-link $g(\cdot) = \log(\cdot)$ in this section, because this is the most convenient case for modeling claim counts and claim sizes in an actuarial context. Using other links $g$ is not difficult, but they are less commonly used in actuarial modeling, except of the logit link for binary response variables and the identity link for Gaussian response variables.

Throughout, we work under the model in Definition 3.24. We choose the log-link for $g$, and we set

$$\mu(\boldsymbol{x}_t, \Lambda) = \Lambda^k \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle = \exp\left\{\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle + k \log \Lambda\right\}, \tag{3.36}$$

for a strictly positive static latent factor $\Lambda \sim \pi$, fixed parameter $k \in \mathbb{Z} \setminus \{0\}$, covariates $\boldsymbol{x}_t \in \mathcal{X} \subset \{1\} \times \mathbb{R}^q$ and GLM regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$. Parameter $k$ would not be necessary, as we could directly choose $\Lambda^k \sim \pi$ for a fixed $k$. However, it will be convenient in the sequel because $\Lambda$ will commonly be a gamma distributed random variable with a transformation $k \in \{-1, +1\}$ for the gamma and the Poisson case, respectively.

Based on the conditional EDF assumption in Definition 3.24, we can select the canonical link $h = (\kappa')^{-1}$ which provides us with canonical parameters

$$\Theta_t = h\left(\exp\left\{\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle + k \log \Lambda\right\}\right).$$

If the mean parameter space $\mathcal{M}$ contains the positive real line $\mathbb{R}_+$, we receive a well-defined canonical parameter $\Theta_t \in \mathring{\boldsymbol{\Theta}}$, $\mathbb{P}$-a.s.

**Remark 3.26** (Notation)**.** The reader may be confused by the fact that $\Theta_t$ has a lower time index $t$ under Definition 3.24, and we still called it a *static* mixed effects model. In this case, the static refers to the random effects $\Lambda$, which is the only latent random term involved. The time dependent factors in $\Theta_t$ are fully known from the covariates $\boldsymbol{x}_t$. If these covariates do not involve any components that are aging in $t$, e.g., age of policyholder is aging, but only static information, e.g., car brand, then $\Theta_t$ will not have any time related effects.

In Chapter 4, we are going to consider dynamic random effects $\Theta_{1:T}$ being stochastic processes, where each time period $t$ adds some extra randomness to the process. The notation is the same, but these two cases, static vs. dynamic, should not be mixed up. We apologize for this notation, but most convenient one for Definition 3.24. ■

**Example 3.27** (Poisson-gamma GLM)**.** We closely follow the Poisson-gamma model of Example 3.3. Assume that the responses $(N_t, \boldsymbol{x}_t, v_t)$, $1 \le t \le T+1$, are conditionally independent, given the static latent factor $\Lambda$, with Poisson distributions

$$N_t|_{\{\Lambda, \boldsymbol{x}_t, v_t\}} \;\sim\; \mathrm{Poi}\left(v_t \,\Lambda \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle\right),$$

and with static latent factor $\Lambda \sim \Gamma(\alpha, \alpha)$ for a given parameter $\alpha > 0$; this uses $k = 1$ in (3.36). For the cumulant function $\kappa(\theta) = \exp\{\theta\}$ of the Poisson EDF we obtain mean in the reproductive form $Y_t = N_t / v_t$

$$\kappa'(\Theta_t) = \exp\{\Theta_t\} = \Lambda \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle \;\sim\; \Gamma(\alpha, \alpha/\mu_t), \tag{3.37}$$



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where we set

$$\mu_t = \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle . \tag{3.38}$$

Thus, similarly to (3.3), we have a static common latent factor, but now we do not consider a null model (scaling with $\mu_0$), but we have a calendar year dependent scaling $\mu_t$. Importantly, (3.37) gives the marginal view of the static mixed effects model, and these effects are comonotonic across the different calendar years $1 \leq t \leq T+1$.
We have the following first two moments

$$\mathbb{E}\left[N_t \middle| \boldsymbol{x}_t, v_t\right] = \mathbb{E}\left[\mathbb{E}\left[N_t \middle| \Lambda, \boldsymbol{x}_t, v_t\right] \middle| \boldsymbol{x}_t, v_t\right] = v_t \mathbb{E}\left[\Lambda \mu_t \middle| \boldsymbol{x}_t, v_t\right] = v_t \mu_t,$$

and

$$\begin{aligned}
&\mathrm{Cov}\left(N_t, N_s \middle| \boldsymbol{x}_t, v_t, \boldsymbol{x}_s, v_s\right) \\
&\quad = v_t \mathbb{E}\left[\Lambda \mu_t \middle| \boldsymbol{x}_t, v_t\right] \mathbb{1}_{\{t=s\}} + v_t v_s \mathrm{Cov}\left(\Lambda \mu_t, \Lambda \mu_s \middle| \boldsymbol{x}_t, v_t, \boldsymbol{x}_s, v_s\right) \\
&\quad = v_t \mu_t \left(\mathbb{1}_{\{t=s\}} + \frac{v_s \mu_s}{\alpha}\right).
\end{aligned}$$

This covariance structure shows that we have an exchangeable auto-correlation structure, i.e., the time difference $t-s$ does not have any impact on the strength of dependence. This is a critical point that is going to be discussed and changed below.

Next, we calculate the posterior distribution of $\Lambda$, given observations $\mathcal{L}_T$, see (2.49). Using Bayes' rule [18], we have posterior density

$$\begin{aligned}
\pi(\Lambda|\mathcal{L}_T) &\propto f(N_{1:T}|\Lambda, \boldsymbol{x}_{1:T}, v_{1:T})\, \pi(\Lambda) \\
&= \left[\prod_{t=1}^{T} e^{-v_t \Lambda \mu_t} \frac{(v_t \Lambda \mu_t)^{N_t}}{N_t!}\right] \frac{\alpha^\alpha}{\Gamma(\alpha)} \Lambda^{\alpha-1} e^{-\alpha\Lambda} \\
&\propto \Lambda^{\alpha + \sum_{t=1}^T N_t - 1}\, e^{-\left(\alpha + \sum_{t=1}^T v_t \mu_t\right)\Lambda}.
\end{aligned}$$

This implies a posterior gamma distribution

$$\Lambda|_{\mathcal{L}_T} \sim \Gamma\left(\alpha + \sum_{t=1}^T N_t, \alpha + \sum_{t=1}^T v_t \mu_t\right).$$

We have the following Bühlmann credibility (exact predictive mean)

$$\begin{aligned}
\mu^{\text{post}}_{N_{T+1}|N_{1:T}} &= \mathbb{E}\left[N_{T+1} \middle| \mathcal{L}_T, \boldsymbol{x}_{T+1}, v_{T+1}\right] = v_{T+1}\mu_{T+1}\, \mathbb{E}\left[\Lambda \middle| \mathcal{L}_T\right] \\
&= v_{T+1}\mu_{T+1} \left[\frac{\sum_{t=1}^T v_t \mu_t}{\alpha + \sum_{t=1}^T v_t \mu_t} \frac{\sum_{t=1}^T N_t}{\sum_{t=1}^T v_t \mu_t} + \frac{\alpha}{\alpha + \sum_{t=1}^T v_t \mu_t}\right] \\
&= v_{T+1}\mu_{T+1} \left[\omega_T\, \bar{\Lambda}_T + (1-\omega_T)\cdot 1\right],
\end{aligned} \tag{3.39}$$

with credibility weight

$$\omega_T = \frac{\sum_{t=1}^T v_t \mu_t}{\sum_{t=1}^T v_t \mu_t + \alpha} \in (0,1), \tag{3.40}$$

and with weighted scaled empirical mean

$$\bar{\Lambda}_T = \frac{\sum_{t=1}^T N_t}{\sum_{t=1}^T v_t \mu_t} = \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s \mu_s} Y_t = \sum_{t=1}^T \frac{v_t \mu_t}{\sum_{s=1}^T v_s \mu_s} \frac{Y_t}{\mu_t}. \tag{3.41}$$



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Formula (3.39) gives the static mixed effects counterpart to the static random effects Bühlmann credibility (3.4). The main difference is that the homogeneous mean $\mu_0$ has been replaced by the fixed effects (regression terms) $\mu_t = \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle$; this model corresponds to of Dionne–Vanasse [69, formula (13)]. We emphasize that $\bar{Y}_T$ in (3.4) and $\bar{\Lambda}_T$ in (3.39) are scaled differently, indicated by a different notation, namely,

$$\bar{Y}_T = \frac{\sum_{t=1}^T N_t}{\sum_{t=1}^T v_t} = \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s} Y_t \qquad \text{and} \qquad \bar{\Lambda}_T = \frac{\sum_{t=1}^T N_t}{\sum_{t=1}^T v_t \mu_t} = \sum_{t=1}^T \frac{v_t \mu_t}{\sum_{s=1}^T v_s \mu_s} \frac{Y_t}{\mu_t}.$$

The former lives on the expected frequency scale $\mu_t$ and the latter lives on the unit scale, i.e., is scale-free. We highlight this to avoid mistakes and confusion in implementations. ■

**Example 3.28** (gamma-gamma GLM)**.** For the claim sizes we consider a static mixed effects gamma-gamma model. Assume a conditional gamma distribution for the claim sizes $Y_t|_{\{\Lambda, \boldsymbol{x}_t, v_t\}} \sim \Gamma(v_t/\varphi, v_t\Lambda/(\mu_t\varphi))$, with static random effects $\Lambda > 0$, shape parameter $1/\varphi > 0$, exposure $v_t \in \mathbb{N}$ and with GLM regression function

$$\mu_t = \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle,$$

selecting the log-link $g(\cdot) = \log(\cdot)$. The conditional mean and variance satisfy, respectively,

$$\begin{aligned} \mathbb{E}\left[Y_t \middle| \Lambda, \boldsymbol{x}_t, v_t\right] &= \mu_t/\Lambda = \exp\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle / \Lambda = \exp\left\{\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle - \log \Lambda\right\}, \\ \mathrm{Var}\left(Y_t \middle| \Lambda, \boldsymbol{x}_t, v_t\right) &= \frac{\varphi}{v_t} \mu_t^2 \Lambda^{-2}; \end{aligned}$$

this uses $k = -1$ in (3.36). There remains the choice of the static random effects. We choose a gamma density $\Lambda \sim \Gamma(\alpha, \alpha - 1)$ with parameter $\alpha > 1$. This gamma distributed random variable is parametrized as follows

$$\mathbb{E}\left[\Lambda^{-1}\right] = 1 \qquad \text{and} \qquad \mathbb{E}\left[\Lambda^{-2}\right] = \frac{\alpha - 1}{\alpha - 2},$$

for the latter statement we need $\alpha > 2$. We now turn our attention to the gamma EDF parametrization with cumulant function $\kappa(\theta) = -\log(-\theta)$ for canonical parameter $\theta < 0$. For the static random effects $\Theta_t$ we have

$$\kappa'(\Theta_t) = -1/\Theta_t = \mu_t/\Lambda = \exp\left\{\langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle - \log \Lambda\right\}.$$

Thus, $-\Theta_t$ has a gamma distribution $\Gamma(\alpha, \mu_t(\alpha - 1))$.
There remains the posterior distribution of $\Lambda$, given observations $\mathcal{L}_T$. Using Bayes' rule [18], we have posterior density, this uses the reproductive form $Y_t = \varphi Z_t / v_t$ of the original observations $Z_t$,

$$\begin{aligned} \pi(\Lambda|\mathcal{L}_T) &\propto f(Y_{1:T}|\Lambda, \boldsymbol{x}_{1:T}, v_{1:T})\, \pi(\Lambda) \\ &= \left[\prod_{t=1}^T \frac{(v_t\Lambda/(\mu_t\varphi))^{v_t/\varphi}}{\Gamma(v_t/\varphi)} Y_t^{v_t/\varphi - 1} e^{-v_t\Lambda/(\mu_t\varphi)Y_t}\right] \frac{(\alpha-1)^\alpha}{\Gamma(\alpha)} \Lambda^{\alpha-1} e^{-(\alpha-1)\Lambda} \\ &\propto \Lambda^{\alpha + \sum_{t=1}^T v_t/\varphi - 1}\, e^{-\left(\alpha - 1 + \sum_{t=1}^T v_t Y_t/(\mu_t\varphi)\right)\Lambda}. \end{aligned}$$



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This implies a posterior gamma distribution

$$\Lambda|_{\mathcal{L}_T} \sim \Gamma\left(\alpha + \sum_{t=1}^T v_t/\varphi, \alpha - 1 + \sum_{t=1}^T v_t Y_t/(\mu_t\varphi)\right).$$

This provides the Bühlmann credibility (exact predictive mean)

$$\begin{aligned}
\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}} &= \mathbb{E}\left[Y_{T+1}\middle|\,\mathcal{L}_T, \boldsymbol{x}_{T+1}, v_{T+1}\right] \;=\; \mu_{T+1}\,\mathbb{E}\left[\Lambda^{-1}\middle|\,\mathcal{L}_T\right] \\
&= \mu_{T+1}\,\frac{1}{\alpha + \sum_{t=1}^T v_t/\varphi - 1}\left(\alpha - 1 + \sum_{t=1}^T v_t Y_t/(\mu_t\varphi)\right) \\
&= \mu_{T+1}\left[\omega_T \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s}\,\frac{Y_t}{\mu_t} + (1-\omega_T)\cdot 1\right], \qquad (3.42)
\end{aligned}$$

with credibility weight, see also (3.10),

$$\omega_T = \frac{\sum_{t=1}^T v_t}{\sum_{t=1}^T v_t + \varphi(\alpha-1)} \;\in\; (0,1).$$

We observe that the empirical mean in (3.42) is normalized differently in this parametrization compared to the Poisson-gamma static mixed effects model, see (3.41). Here, we first bring the observations to a unit scale $Y_t/\mu_t$ before taking the weighted mean. ■

### 3.3.3 Comparison to the Bühlmann–Straub model

We compare the static mixed effects model to the Bühlmann–Straub Model 3.2. Assuming a multiplicative structure in the random effects $\Lambda$ such that we have the following conditional mean structure

$$\mathbb{E}\left[Y_t\middle|\,\Lambda, \boldsymbol{x}_t, v_t\right] = \mu(\boldsymbol{x}_t, \Lambda) = m(\Lambda)\mu_t, \qquad (3.43)$$

with calendar year dependent mean parameters $\mu_t = \mu(\boldsymbol{x}_t)$ and for some measurable function $m : \text{supp}(\Lambda) \to \mathbb{R}$, see also (3.36) in the case of a GLM. Assuming that we work within the EDF of Definition 3.24, we have conditional variance

$$\text{Var}\left(Y_t\middle|\,\Lambda, \boldsymbol{x}_t, v_t\right) = \frac{\varphi}{v_t}\kappa''\left(h(\mu(\boldsymbol{x}_t, \Lambda))\right) = \frac{\varphi}{v_t}\kappa''\left(h(m(\Lambda)\mu_t)\right).$$

To receive a simple and tractable model, we additionally assume to work within Tweedie's family of the EDF. Tweedie's family has the property for the variance function

$$\mu \;\mapsto\; \kappa''(h(\mu)) = \mu^p,$$

with power variance parameter $p \in \{0\} \cup [1, \infty)$; see Section 2.3.2, Tweedie [192], Bar-Lev–Enis [13], Jørgensen [114] and Wüthrich–Merz [205, Section 2.2.3]. Recall $p = 0$ gives the Gaussian model, $p = 1$ the Poisson model, $p = 2$ the gamma model and $p = 3$ the inverse Gaussian model. Using all these assumptions, we can write the first two conditional moments as follows

$$\begin{aligned}
\mathbb{E}\left[\frac{Y_t}{\mu_t}\middle|\,\Lambda, \boldsymbol{x}_t, v_t\right] &= m(\Lambda), \\
\text{Var}\left(\frac{Y_t}{\mu_t}\middle|\,\Lambda, \boldsymbol{x}_t, v_t\right) &= \frac{\varphi}{v_t\mu_t^{2-p}}\,m(\Lambda)^p.
\end{aligned}$$



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These are Bühlmann–Straub Models 3.2 for $Y_t/\mu_t$ with mean functional $\Lambda \mapsto m(\Lambda)$, variance functional $\Lambda \mapsto \sigma^2(\Lambda) = \varphi m(\Lambda)^p$ and weights $v_t \mu_t^{2-p}$. In fact, this called a Bühlmann–Straub model with prior differences; see Bühlmann–Gisler [41, Section 4.13]. We define the structural parameters

$$m_0 = \mathbb{E}[m(\Lambda)], \qquad \tau_0^2 = \mathrm{Var}(m(\Lambda)) \qquad \text{and} \qquad \sigma^2 = \mathbb{E}\left[\varphi m(\Lambda)^p\right]; \tag{3.44}$$

of course, we assume that all these exist. Theorem 3.9 gives us the inhomogeneous Bühlmann credibility formula

$$\widehat{\widehat{m(\Lambda)}}(Y_{1:T}) = \omega_T \bar{\Lambda}_T^{(p)} + (1-\omega_T)\, m_0,$$

with credibility weights $\omega_T \in [0,1]$ and observation based estimators $\bar{\Lambda}_T^{(p)}$, respectively,

$$\begin{aligned}
\omega_T &= \frac{\sum_{t=1}^T v_t \mu_t^{2-p}}{\sum_{t=1}^T v_t \mu_t^{2-p} + \sigma^2/\tau_0^2}, \\
\bar{\Lambda}_T^{(p)} &= \frac{1}{\sum_{t=1}^T v_t \mu_t^{2-p}} \sum_{t=1}^T v_t \mu_t^{1-p} Y_t \;=\; \sum_{t=1}^T \frac{v_t \mu_t^{2-p}}{\sum_{s=1}^T v_s \mu_s^{2-p}} \frac{Y_t}{\mu_t}.
\end{aligned}$$

In the Poisson and gamma cases with $p=1$ and $p=2$, respectively, we receive the weightings of Examples 3.27 and 3.28, respectively.
Having multiple risks $1 \le i \le n$ following the same model, we can give the homogeneous Bühlmann credibility formula, see Theorem 3.13,

$$\widehat{\widehat{m(\Lambda_i)}}^{\,\mathrm{hom}}(\boldsymbol{Y}) = \omega_{i,T} \bar{\Lambda}_{i,T}^{(p)} + (1-\omega_{i,T})\, \widehat{\widehat{m}}_T^{\,\mathrm{hom}}, \tag{3.45}$$

with collective estimator

$$\widehat{\widehat{m}}_T^{\,\mathrm{hom}} = \frac{1}{\sum_{i=1}^n \omega_{i,T}} \sum_{i=1}^n \omega_{i,T} \bar{\Lambda}_{i,T}^{(p)}.$$

**Corollary 3.29** (balance property in the static mixed effects Poisson case)**.** *Assume we have independent risks following Definition 3.24 with a Poisson distribution ($p=1$), with a multiplicative mean structure* (3.43) *and with structural parameters* (3.44)*. The homogeneous Bühlmann credibility satisfies the global balance property*

$$\sum_{i=1}^n \sum_{t=1}^T v_{i,t} \mu_{i,t} \widehat{\widehat{m(\Lambda_i)}}^{\,\mathrm{hom}}(\boldsymbol{Y}) \;=\; \sum_{i=1}^n \sum_{t=1}^T v_{i,t} Y_{i,t}.$$

**Proof of Corollary 3.29.** We calculate the difference between the left- and the right-hand side of the statement, we set $\bar{\Lambda}_{i,T} = \bar{\Lambda}_{i,T}^{(1)}$,

$$\begin{aligned}
\sum_{i=1}^n \sum_{t=1}^T v_{i,t} \left( \mu_{i,t} \widehat{\widehat{m(\Lambda_i)}}^{\,\mathrm{hom}}(\boldsymbol{Y}) - Y_{i,t} \right) &= \sum_{i=1}^n \sum_{t=1}^T v_{i,t} \mu_{i,t} \left( \omega_{i,T} \bar{\Lambda}_{i,T} + (1-\omega_{i,T}) \widehat{\widehat{m}}_T^{\,\mathrm{hom}} - \frac{Y_{i,t}}{\mu_{i,t}} \right) \\
&= \sum_{i=1}^n \sum_{t=1}^T v_{i,t} \mu_{i,t} \left( \omega_{i,T} \bar{\Lambda}_{i,T} + (1-\omega_{i,T}) \widehat{\widehat{m}}_T^{\,\mathrm{hom}} - \bar{\Lambda}_{i,T} \right) \\
&= \sum_{i=1}^n \sum_{t=1}^T v_{i,t} \mu_{i,t} (1-\omega_{i,T}) \left( -\bar{\Lambda}_{i,T} + \widehat{\widehat{m}}_T^{\,\mathrm{hom}} \right).
\end{aligned}$$



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Note that $(1-\omega_{i,T})\sum_{t=1}^T v_{i,t}\mu_{i,t} = \omega_{i,T}\sigma^2/\tau_0^2$. This provides

$$\sum_{i=1}^n \sum_{t=1}^T v_{i,t}\left(\mu_{i,t}\widehat{\widehat{m(\Lambda_i)}}^{\text{hom}}(\boldsymbol{Y}) - Y_{i,t}\right) = \frac{\sigma^2}{\tau_0^2}\sum_{i=1}^n \omega_{i,T}\left(\widehat{\widehat{m}}_T^{\text{hom}} - \bar{\Lambda}_{i,T}\right) = 0.$$

This proves the claim. □

**Remarks 3.30.**

- The balance property of Corollary 3.29 holds regardless of how we set the parameters $\mu_{i,t} = \mu(\boldsymbol{x}_{i,t})$. Of course, this is an excellent and very useful property because it implies that such models can be fitted in a two step approach. In a first step we estimate $\mu_{i,t}$ from any reasonable regression model in an empirical Bayesian way. In the second step we insert these estimates into the homogeneous Bühlmann credibility formula (3.45) to receive the balance property in the Poisson case.

- The balance property of Corollary 3.29 only holds in the Poisson case, i.e., for power variance parameter $p = 1$, but not, e.g., in the gamma case with $p = 2$. For such other $p$'s we have a balance property

$$\sum_{i=1}^n \sum_{t=1}^T v_{i,t}\mu_{i,t}^{2-p}\widehat{\widehat{m(\Lambda_i)}}^{\text{hom}}(\boldsymbol{Y}) \;=\; \sum_{i=1}^n \sum_{t=1}^T v_{i,t}\mu_{i,t}^{1-p}Y_{i,t},$$

  but this is only useful for pricing in the Poisson case $p = 1$ due to the linearity in $\mu_{i,t}$ in (3.43). Intuitively, this is again related to the fact that we have a multiplicative mean structure (3.43) which aligns with the log-link of the Poisson model.

■

### 3.3.4 Lab: Static mixed effects Poisson-gamma GLM

We revisit the static random effects example of Section 3.2.6, but this time we integrate the fixed effects $\boldsymbol{x}_{i,t}$ using the static mixed effects GLM (3.36) with $k = 1$. Additionally to parameter $\alpha > 0$, we need to fit the GLM parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ in this static mixed effects Poisson-gamma GLM. We first discuss parameter estimation, this closely follows Section 3.2.6 by exploiting an empirical Bayesian approach.

*(a) Full MLE: empirical Bayesian approach.* We perform MLE at time $T$ based on all available data $\mathcal{L}_T$. The log-likelihood function of a single insurance policy for given observations $(N_{1:T}, \boldsymbol{x}_{1:T}, v_{1:T})$ is, set $\mu_t = \exp\langle\boldsymbol{\beta}, \boldsymbol{x}_t\rangle$,

$$\begin{aligned}
\ell_{N_{1:T}}(\boldsymbol{\beta},\alpha) &= \log\left(\int_0^\infty \prod_{t=1}^T \left(e^{-v_t\lambda\mu_t}\frac{(v_t\lambda\mu_t)^{N_t}}{N_t!}\right)\frac{\alpha^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\alpha\lambda}\,d\lambda\right)\\
&\propto \log\left(\prod_{t=1}^T \mu_t^{N_t}\int_0^\infty \frac{\alpha^\alpha}{\Gamma(\alpha)}\lambda^{\alpha+\sum_{t=1}^T N_t-1}e^{-\left(\alpha+\sum_{t=1}^T v_t\mu_t\right)\lambda}\,d\lambda\right)\\
&= \log\left(\frac{\Gamma\left(\alpha+\sum_{t=1}^T N_t\right)}{\Gamma(\alpha)}\frac{\alpha^\alpha\prod_{s=1}^T\mu_s^{N_s}}{\left(\alpha+\sum_{t=1}^T v_t\mu_t\right)^{\alpha+\sum_{s=1}^T N_s}}\right) \qquad (3.46)\\
&\propto \log\left(\frac{\Gamma\left(\alpha+\sum_{t=1}^T N_t\right)}{\Gamma(\alpha)\left(\sum_{t=1}^T N_t\right)!}\left(\frac{\alpha}{\alpha+\sum_{t=1}^T v_t\mu_t}\right)^\alpha\prod_{s=1}^T\left(\frac{v_s\mu_s}{\alpha+\sum_{t=1}^T v_t\mu_t}\right)^{N_s}\right).
\end{aligned}$$



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The static random effects case (3.31) provides a proportionality to a negative binomial distribution, in the static mixed effects case we have a more complex structure because we weight the observations $N_{1:T}$ w.r.t. their (heterogeneous) expected means $\mu_{1:T}$, see last product of the previous formula. The MLE of $(\boldsymbol{\beta}, \alpha)$ is obtained by solving

$$\left(\widehat{\boldsymbol{\beta}}_{1:T}^{\text{MLE}}, \widehat{\alpha}_{1:T}^{\text{MLE}}\right) \;=\; \underset{(\boldsymbol{\beta},\alpha)}{\arg\max} \sum_i \ell_{N_{i,1:T}}(\boldsymbol{\beta}, \alpha). \tag{3.47}$$

This log-likelihood only considers the calendar years for which observations exist, i.e., the observations $N_{i,1:T}$ in the log-likelihoods $\ell_{N_{i,1:T}}(\boldsymbol{\beta}, \alpha)$ may have different lengths, which is not indicated in our notation but sums and products in (3.46) should only consider calendar years with positive exposures.

*(a0) All accident years: homogeneous Bühlmann credibility approach.* Using the MLE from (3.47) we define the empirical Bayesian estimated means by

$$\widehat{\mu}_{i,t} = \exp\left\langle \widehat{\boldsymbol{\beta}}_{1:T}^{\text{MLE}}, \boldsymbol{x}_{i,t} \right\rangle.$$

This gives us estimated credibility weights

$$\widehat{\omega}_{i,T} = \frac{\sum_{t=1}^T v_{i,t}\widehat{\mu}_{i,t}}{\sum_{t=1}^T v_{i,t}\widehat{\mu}_{i,t} + \widehat{\alpha}_{1:T}^{\text{MLE}}} \;\in\; (0,1),$$

see (3.40), and estimated weighted empirical means

$$\bar{\Lambda}_{i,T} = \frac{\sum_{t=1}^T N_{i,t}}{\sum_{t=1}^T v_{i,t}\widehat{\mu}_{i,t}} = \sum_{t=1}^T \frac{v_{i,t}\widehat{\mu}_{i,t}}{\sum_{s=1}^T v_{i,s}\widehat{\mu}_{i,s}} \frac{Y_t}{\widehat{\mu}_{i,t}},$$

see (3.41). From this we can compute the homogeneous Bühlmann credibility formula (3.45) which satisfies the balance property on $\mathcal{L}_T$, see Corollary 3.29.

*(b) One-year ahead MLE approach: empirical Bayesian approach.* The one-year ahead MLE only considers the observations $N_T$ in $\mathcal{L}_T^\circ$, and their histories are used to determine the conditional distribution of $N_T$, given observations $N_{1:T-1}$. We have log-likelihood in this case

$$\begin{aligned}
\ell_{N_T|N_{1:T-1}}(\boldsymbol{\beta}, \alpha) \;&=\; \log\Bigg(\int_0^\infty e^{-v_T\lambda\mu_T} \frac{(v_T\lambda\mu_T)^{N_T}}{N_T!} \frac{(\alpha + \sum_{t=1}^{T-1} v_t\mu_t)^{\alpha+\sum_{t=1}^{T-1} N_t}}{\Gamma(\alpha + \sum_{t=1}^{T-1} N_t)} \\
&\qquad\qquad \times\, \lambda^{\alpha+\sum_{t=1}^{T-1} N_t - 1} e^{-(\alpha+\sum_{t=1}^{T-1} v_t\mu_t)\lambda}\, d\lambda\Bigg) \\
&=\; \log\left(\frac{\Gamma\left(\alpha + \sum_{t=1}^T N_t\right)}{\Gamma(\alpha + \sum_{t=1}^{T-1} N_t) N_T!} \left(\frac{\alpha + \sum_{t=1}^{T-1} v_t\mu_t}{\alpha + \sum_{t=1}^T v_t\mu_t}\right)^{\alpha+\sum_{t=1}^{T-1} N_t} \left(\frac{v_T\mu_T}{\alpha + \sum_{t=1}^T v_t\mu_t}\right)^{N_T}\right).
\end{aligned}$$

Thus, in this case we have a proper negative binomial distribution, and the idea behind its derivation is the same recursive credibility structure as in (3.33). Aggregating over all accident years $i \in \mathcal{I}_T$, we solve the MLE problem

$$\left(\widehat{\boldsymbol{\beta}}_T^{\text{MLE}}, \widehat{\alpha}_T^{\text{MLE}}\right) \;=\; \underset{(\boldsymbol{\beta},\alpha)}{\arg\max} \sum_{i\in\mathcal{I}_T} \ell_{N_{i,T}|N_{i,1:T-1}}(\boldsymbol{\beta}, \alpha). \tag{3.48}$$



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*(b0) One-year ahead: homogeneous Bühlmann credibility approach.* We proceed as in (a0) but using the MLE (3.48) of the empirical Bayesian estimation to compute the homogeneous collective mean.

**Remarks 3.31.** We have derived the two different estimation methods of the full MLE approach (3.47) and the one-year ahead MLE approach (3.48) in the static mixed effects GLM, and we have given the corresponding homogeneous Bühlmann credibility formulas. The static random effects is the special case received by dropping the covariates $\boldsymbol{x}_t$ and setting $\mu_t \equiv \mu_0$, i.e., only including an intercept term to the GLM. On the other hand, the above formulas are not specific to a GLM structure in $\mu_t$, and we can easily replace this GLM by a FNN regression model, see Section 3.3.5, below. ■

Implementation of (3.47) and (3.48) is a bit cumbersome. We are going to use the gradient descent algorithm within the R package `keras` [51]. In the first step, we need to pre-process the data so that it is in the appropriate structure. For this, we arrange the feature information in tensors, and we follow the notation in Wüthrich–Merz [205, Chapters 8 and 9]. Tensors are multi-dimensional arrays. For time-series we can use a tensor of order 2 which is simply a matrix. It is common to have the time labels on the rows of this matrix, thus, we consider the matrix for an individual insurance policyholder

$$\boldsymbol{x}_{1:T}^{\top} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T]^{\top} \ \in \ \mathbb{R}^{T \times (q+1)},$$

see Wüthrich–Merz [205, Section 9.1.3]. The mean vector for a given regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ and under the log-link choice is then easily obtained by

$$\log(\mu_{1:T}) = (\log(\mu_1), \ldots, \log(\mu_T))^{\top} = \boldsymbol{x}_{1:T}^{\top}\boldsymbol{\beta} \ \in \ \mathbb{R}^T.$$

The previous indicates that to all rows of the tensor $\boldsymbol{x}_{1:T}^{\top}$ we apply the same operation (scalar/dot product). In `keras` [51], this can be implemented by using a so-called time-distributed layer that uses multiple times the same FNN layer with the same parameters; this is going to be introduced more formally in (5.4), below. Listing 3.2 gives the corresponding code snippet; the FNN layer uses for any row of $\boldsymbol{x}_{1:T}^{\top}$ the same regression parameter $\boldsymbol{\beta} \in \mathbb{R}^{q+1}$ and the output activation corresponds to the log-link. Listings 3.3, 3.4 and 3.5, below, give a more complete code for fitting a static mixed effects FNN regression model; this is rather similar to the static mixed effects GLM, therefore, we do not give the full code here.

Listing 3.2: Time-distributed GLM layer with log-link in `keras` [51].

```
DesignX = layer_input(shape = c(T,q))
GLMs    = DesignX %>%
            time_distributed(layer_dense(units=1, activation='exponential'))
```

The remaining difficulties then are the implementation of the objective functions and masking the calendar years $t$ for which a specific insurance contract $i$ does not have any observations, i.e., $i \notin \mathcal{I}_t$. The masking is achieved by setting $v_{i,t} = 0$ for all $i \notin \mathcal{I}_t$ in the summations, and by dropping the corresponding terms from the products $\prod_{t=1}^{T}$ in the



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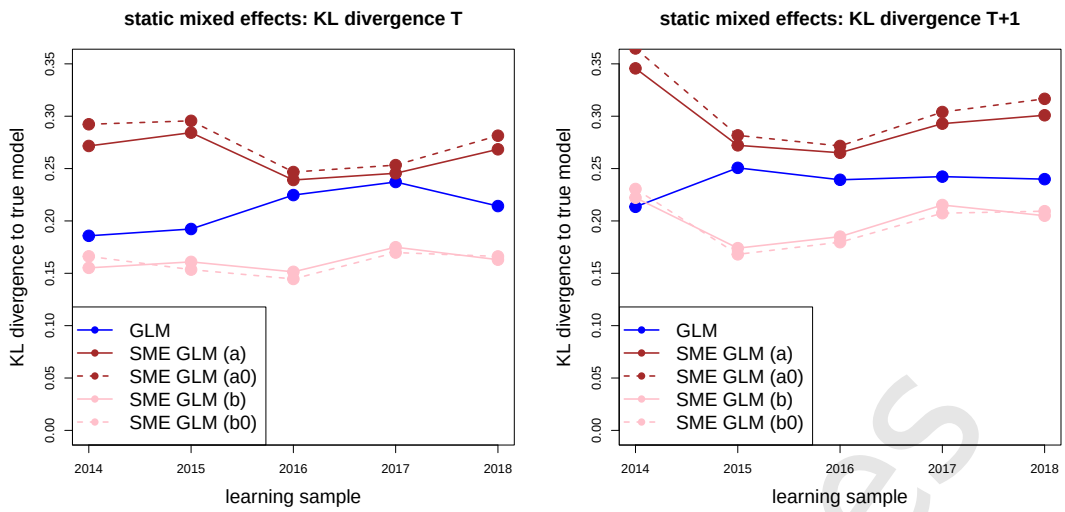


Figure 3.2: Fitted static mixed effects (SME) models: (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one-year ahead forecast, the labels (a)-(b) refer to the two different fitting methods (3.47) and (3.48), (a0)-(b0) are their homogeneous Bühlmann credibility counterparts, and the GLM is taken from Figure 2.4; units are $10^{-2}$.

full MLE approach, this is easily achieved because in these years we have $N_{i,t} = 0$, see, e.g., (3.46).

The results are presented in Figure 3.2, and this figure is comparable to Figure 3.1, only the $y$-axis uses a different scale. Things are getting more interesting now! The one-year ahead approaches (b) and (b0) of the static mixed effects GLM outperform the classical fixed effects GLM. This indicates that both, the past claims history $Y_{i:T}$ and the fixed effects covariates $\boldsymbol{x}_{i,T+1}$, have predictive power. Benefiting from both of them by combining their information outperforms the models that only use either one of them, compare Figures 3.2 and 3.1. We observe that the full MLE approach over all past observations $\mathcal{L}_T$ is not competitive in this parametrization, we again interpret this by the non-stationarity of the data and, thus, the model assumptions do not seem to be fully appropriate if we do not allow for a calendar year effects modeling, we also refer to Figure 2.5. We could introduce a categorical covariate for calendar years, however, this does not allow one for an automated extrapolation. Therefore, we refrain from studying this in detail here; from Figure 2.5 we also observe that a continuous linear or exponential calendar year component will likely not match the data.

By comparing the empirical Bayesian estimate (b) and its homogeneous Bühlmann credibility counterpart (b0), we give preference to the homogeneous Bühlmann credibility predictor. Thus, though the data is non-stationary, the balance property regularization has a positive effect on model accuracy in calendar years 2015-2017, and in the most recent year 2018 it is only slightly worse. This closes the static mixed effects GLM Poisson-gamma example.

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### 3.3.5 Lab: Static mixed effects Poisson-gamma neural network

The step from the static mixed effects GLM to a static mixed effects FNN is a comparably small one in view of the preparatory work we have already done above. The only change is that one replaces the GLM regression function in (3.36) by a FNN regression function

$$\mu(\boldsymbol{x}_t, \Lambda) = \Lambda^k \exp\left\langle \boldsymbol{\beta}, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_t)\right\rangle = \exp\left\{\left\langle \boldsymbol{\beta}, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_t)\right\rangle + k\log\Lambda\right\}, \tag{3.49}$$

for a strictly positive static latent factor $\Lambda \sim \pi$, fixed parameter $k \in \mathbb{Z}\setminus\{0\}$, covariates $\boldsymbol{x}_t \in \mathcal{X} \subset \{1\}\times\mathbb{R}^q$, readout parameter $\boldsymbol{\beta} \in \mathbb{R}^{q_d+1}$, and where $\boldsymbol{z}^{(d:1)} : \mathbb{R}^{q+1} \to \mathbb{R}^{q_d+1}$ is the FNN architecture given in (2.61) of depth $d \in \mathbb{N}$. In the Poisson-gamma case we select $k = 1$ and static latent factor $\Lambda \sim \Gamma(\alpha, \alpha)$.

Listing 3.3: Static mixed effects FNN architecture in `keras` [51].

```
1  NN.SME <- function(seed, lookback, q, b){
2      k_clear_session()
3      set.seed(seed)
4      set_random_seed(seed)
5      #
6      Design   <- layer_input(shape = c(lookback, 7), dtype = 'float32')
7      Exposure <- layer_input(shape = c(lookback),    dtype = 'float32')
8      Const1   <- layer_input(shape = c(1),           dtype = 'float32')
9      Cat      <- layer_input(shape = c(lookback, 1), dtype = 'int32')
10     #
11     CatEmb = Cat %>%
12       time_distributed(layer_embedding(input_dim = c(15), output_dim = b[1],
13                        input_length = 1)) %>% time_distributed(layer_flatten())
14     #
15     Network = list(Design, CatEmb) %>% layer_concatenate() %>%
16           time_distributed(layer_dense(units=q[1], activation='tanh')) %>%
17           time_distributed(layer_dense(units=q[2], activation='tanh')) %>%
18           time_distributed(layer_dense(units=q[3], activation='tanh')) %>%
19           time_distributed(layer_dense(units=1, activation='exponential')) %>%
20           layer_flatten()
21     # estimation of shape parameter
22     Alpha = Const1 %>% layer_dense(units=1, activation='exponential',
23                            use_bias=FALSE, weights=list(array(0, dim=c(1,1))))
24     # estimation of mean parameter
25     Output = list(Network, Exposure) %>% layer_multiply()
26     #
27     Response = list(Output, Alpha) %>% layer_concatenate()
28     #
29     keras_model(inputs = c(Design, Exposure, Cat, Const1), outputs = c(Response))
30     }
```

From a modeling perspective, we replace the time-distributed GLM layer of Listing 3.2 by a time-distributed FNN architecture. Listing 3.3 gives the R code for a static mixed effects FNN, having one categorical covariate that uses an embedding layer. Moreover, lines 22-23 of Listing 3.3 take care of the prior parameter $\alpha > 0$ estimation in the empirical Bayesian estimation procedure.

Listings 3.4 and 3.5 give the negative log-likelihoods for model fitting, because we use SGD that minimizes loss functions. We implement these fitting procedures in `keras` [51]. We use the same early stopping strategy as in Section 2.4.2 using a 20% validation sample from the learning data. Moreover, we ensemble over 10 FNN fits to reduce the randomness coming from SGD fitting.

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Listing 3.4: Loss function empirical Bayesian estimation for the full data (3.47).

```
Q_loss <- function(y_true, y_pred){
      mu       <- y_pred[,1:T]
      alpha    <- y_pred[,T+1]
      YY       <- y_true                    # observed number of claims
      YY.0     <- k_sum(YY[,1:T],axis=2)    # total number of observed claims
      mu.0     <- k_sum(mu[,1:T],axis=2)    # sum_t v_t*\mu_t
      xx       <- k_sum(YY*k_log(mu+.00000001), axis=2)
      # negative log-likelihood all past observations
      -k_sum(+tf$math$lgamma(alpha+YY.0)-tf$math$lgamma(alpha) +
             alpha*k_log(alpha)-(alpha+YY.0)*log(alpha+mu.0) + xx)
          }
```

Listing 3.5: Loss function empirical Bayesian estimation one-year ahead data (3.48).

```
Q_loss <- function(y_true, y_pred){
      mu       <- y_pred[,1:T]
      alpha    <- y_pred[,T+1]
      YY.0     <- y_true[,1]    # sum of claims for t<=T-1
      YY.1     <- y_true[,2]    # claims in the last observed year T
      mu.0     <- k_sum(mu[,1:(T-1)],, axis=2) # v_t*\mu_t for t<=T-1
      mu.1     <- mu[,T]                       # v_T*\mu_T
      # negative log-likelihood one year ahead
      -k_sum(+tf$math$lgamma(alpha+YY.0+YY.1)-tf$math$lgamma(alpha+YY.0) +
             (alpha+YY.0)*k_log((alpha+mu.0)/(alpha+mu.0+mu.1)) +
              YY.1*k_log(mu.1/(alpha+mu.0+mu.1)))
          }
```

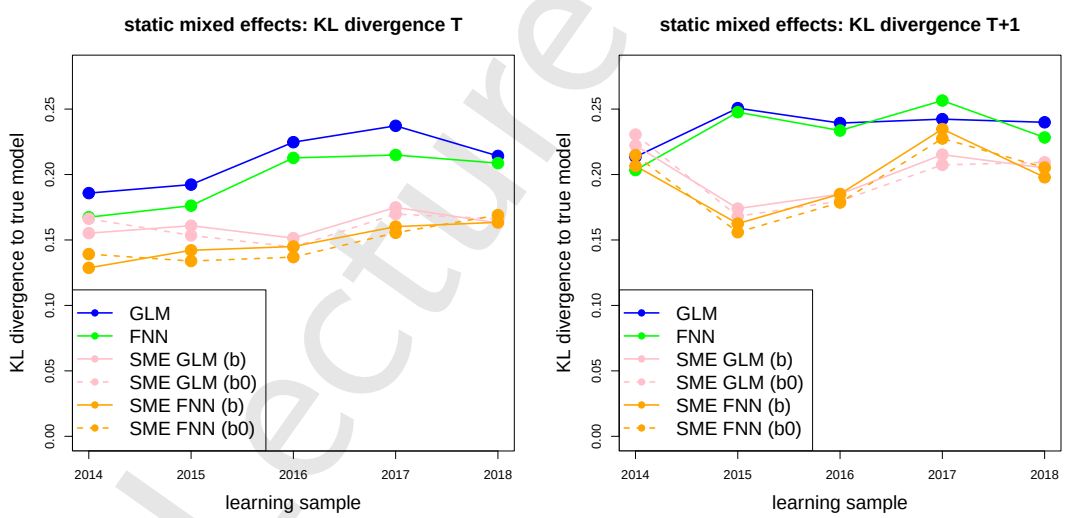


Figure 3.3: Fitted static mixed effects (SME) models: (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one-year ahead forecast, the labels (b)-(b0) refer to fitting method (3.48) and its homogeneous Bühlmann credibility counterpart, and the GLM is taken from Figure 2.4, the FNN is taken from Figure 2.8; units are $10^{-2}$.

Figure 3.3 shows the results from the one-year ahead fitting approach (3.48), both the empirical Bayesian version (b) and its homogeneous Bühlmann credibility counterpart

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(b0). We observe an in-sample improvement of the static mixed effects FNN approach over the GLM version, and also the homogeneous Bühlmann credibility regularization (to ensure the balance property) seems beneficial. Out-of-sample on $\mathcal{T}_{T+1}$ we still give preference to the static mixed effects FNN, but the preference is less clear because the models struggle with the non-stationarity of the data. Nevertheless, we can conclude that the static mixed effects FNN with the homogeneous Bühlmann credibility regularization (b0) has the best performance among the models studied.

When it comes to the full data fitting approaches (a) and (a0), we remark that these are again not competitive here, because of the non-stationarity of the data; these graphs are omitted in Figure 3.3 to not overload the plots.

A next natural idea is to equip the covariate information $\boldsymbol{x}_{i,t}$ with a time component, i.e., to integrate the calendar year information $t$ as a component into $\boldsymbol{x}_{i,t}$. In-sample this will improve the models because this additional covariate component can take care of the non-stationarity in the learning sample $\mathcal{L}_T$. But this does not necessarily give a better out-of-sample model for forecasting on $\mathcal{T}_{T+1}$ as the extrapolation of the regression function beyond the latest observations at time $T$ may not have the correct slope and curvature; we refer to Figure 2.5 for the analysis of a categorical calendar effect variable. In many cases, extrapolation with expert judgment gives better results, therefore, we do not further consider this here and close the example.

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# Chapter 4

# Dynamic Mixed Effects Models

## 4.1 Literature overview

The main deficiency of the static random and mixed effects models of Chapter 3 is that the components of the past claims history $Y_{1:T}$ are exchangeable, meaning that one can arbitrarily permute the components of $Y_{1:T}$ and one receives the same experience rating premium (predictive mean) $\mu^{\text{post}}_{Y_{T+1}|Y_{1:T}}$, see, e.g., Bühlmann credibility formula (3.4). Typically, we believe that more recent observations should be higher weighted than older ones, i.e., the credibility weighting scheme should apply a *seniority weighting* to the observations $Y_{1:T}$. Exactly this point has initiated numerous research, we mention, e.g., Pinquet et al. [163] and Bollancé et al. [26]. This is also our motivation to consider *dynamic random effects models* and *dynamic mixed effects models*, these dynamic latent factor models will assign higher credibility weights to more recent observations.

There are the following different approaches to tackle this seniority weighting problem.

(1) The first possible approach is to use the framework of auto-regressive (AR) time-series models. Popular examples are AR models of order $p$, AR($p$), auto-regressive moving average (ARMA) models, auto-regressive conditional heteroskedastic (ARCH) models and generalized auto-regressive conditional heteroskedastic (GARCH) models. These time-series models are very popular choices in econometrics, we mention the Noble Prize awarded work of Engle [76]. These models are very suitable, e.g., to model log-returns of financial time-series. For claim counts, there are integer-valued (IN) counterparts, such as the INAR($p$) and the INGARCH models; see McKenzie [147], Gourieroux–Jasiak [96] and Gonçalves et al. [94]. These models are so-called *observation-driven* as these processes are directly impacted by past observations, this should be contrasted with *parameter-driven* processes which are going to be discussed below. These observation-driven processes take an observation $Y_t$ at time $t$, this observation is thinned $\varrho\circ Y_t$, where $\varrho\circ$ denotes a *thinning operator*, and then a new innovation $\varepsilon_{t+1}$ is added. The resulting new observation takes the form

$$Y_{t+1} = \varrho \circ Y_t + \varepsilon_{t+1}. \tag{4.1}$$

E.g., a Poisson INAR(1) model starts from a Poisson observation $Y_t \sim \text{Poi}(\lambda_t)$. Then, one takes an independent Poisson innovation $\varepsilon_{t+1}$ with mean $\delta_{t+1}$, and the thinning operator



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$\varrho\circ$ is obtained by considering a compound sum

$$\varrho\circ Y_t=\sum_{j=1}^{Y_t}\varrho_j,$$

where $\varrho_j$ are i.i.d. Bernoulli($q$) random variables, being independent of $Y_t$. The thinned random variable $\varrho\circ Y_t$ has moment generating function for $r\in\mathbb{R}$

$$\begin{aligned}\mathbb{E}\left[\exp\left\{r(\varrho\circ Y_t)\right\}\right] &= \mathbb{E}\left[\prod_{j=1}^{Y_t}\mathbb{E}\left[\exp\left\{r\varrho_j\right\}\right]\right]\\ &= \mathbb{E}\left[\left(qe^r+(1-q)\right)^{Y_t}\right]\\ &= \exp\left\{\lambda_t q\left(e^r-1\right)\right\}.\end{aligned}$$

Thus, $\varrho\circ Y_t\sim\text{Poi}(\lambda_t q)$ and, henceforth, one has a new next (unconditional) Poisson response

$$Y_{t+1}\sim\text{Poi}\left(\lambda_{t+1}=\lambda_t q+\delta_{t+1}\right).$$

This recursive construction leads to a time-series $(Y_t)_{t\geq1}$ with an auto-correlation structure that is decreasing in an increasing time lag. Some references which use such AR time-series processes in an actuarial context are Cosette et al. [53], Karlis [119], Bermúdez et al. [22], Bermúdez–Karlis [23], Pedeli–Karlis [155] or Chen et al. [49, 50]. We will not further study such models in these notes because from an interpretation point of view they do not fully suit an insurance pricing context. We give an example. Assume that at time $t$ there are $Y_t=5$ insurance claims that have been generated by an insurance policyholder with covariate $\boldsymbol{x}_t$ saying that she/he drives a vintage car. Applying the thinning operator $\varrho\circ Y_t=2$, we may carry forward two claims to the next period, and we may add another one, $\varepsilon_{t+1}=1$, which is generated by a sports car $\boldsymbol{x}_{t+1}$, assuming that this policyholder has changed her/his car from time $t$ to $t+1$, this gives three claims $Y_{t+1}=3$ in the next period $t+1$. Such a storyline does not seem to make full sense, which doubts application of such an INAR(1) model in our insurance pricing context, at least it cannot be communicated to stakeholders in a convincing manner.

(2) A second approach are state-space models of Kalman filter type [118]. State-space models are *parameter-driven*. In contrast to their observation-driven counterparts in item (1), they are more tractable allowing one to integrate covariates and fixed effects more easily. These types of models assume that there is a latent stochastic process $(\Theta_t)_{t\geq1}$, and, conditionally given this latent process, we have independent responses $(Y_t)_{t\geq1}$, with a distribution of $Y_t$ being determined by $\Theta_t$, this is illustrated in the directed acyclic graph (DAG) of Figure 4.1.
More formally, we call a latent process $(\Theta_t)_{t\geq1}$ *parameter-driven* if

$$\Theta_t|_{\{\Theta_{1:t-1},Y_{1:t-1}\}}\overset{(d)}{=}\Theta_t|_{\Theta_{1:t-1}}\qquad\text{for all } t\geq2. \tag{4.2}$$

This parameter-driven latent process is Markovian if we additionally have

$$\Theta_t|_{\{\Theta_{1:t-1},Y_{1:t-1}\}}\overset{(d)}{=}\Theta_t|_{\Theta_{1:t-1}}\overset{(d)}{=}\Theta_t|_{\Theta_{t-1}}\qquad\text{for all } t\geq2. \tag{4.3}$$



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$$\begin{array}{lccccccccc}
\text{state-space process } (\Theta_t)_{t\geq 1}: & \cdots & \Theta_{t-1} & \longrightarrow & \Theta_t & \longrightarrow & \Theta_{t+1} & \longrightarrow & \Theta_{t+2} & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
\text{observations } (Y_t)_{t\geq 1}: & & Y_{t-1} & & Y_t & & Y_{t+1} & & Y_{t+2} &
\end{array}$$

Figure 4.1: DAG of a Markovian parameter-driven state-space model of Kalman filter type [118].

The DAG in Figure 4.1 shows a Markovian parameter-driven latent process, and it becomes apparent from this DAG why we call such models *dynamic random effects models*. In contrast to their static counterparts, in this chapter, we consider a (dynamic) stochastic process $(\Theta_t)_{t\geq 1}$, also called state-space process, whereas the models in Chapter 3 use a single static latent factor $\Theta$ which is not evolving in time. If we additionally integrate fixed effects $(\boldsymbol{x}_t)_{t\geq 1}$ we call it a *dynamic mixed effects model*; we indicate again that we do not impose any regression structure on the latent state-space process $(\Theta_t)_{t\geq 1}$ which could also be done. Such dynamic random and mixed effects models are widely studied and applied in actuarial science; actuarial literature in this field compromises, e.g., Gerber–Jones [87, 88], Sundt [187], Bühlmann–Gisler [41, Chapter 9], Pinquet et al. [163], Purcaru–Denuit [165], Brouhns et al. [38], Gourieroux–Jasiak [97], Lu [142], Pinquet [161, 162], Ahn et al. [3] and Verschuren [196]. We highlight Lu [142] who considers an AR gamma dynamic random effects model which seems especially suited for conjugate prior Poisson-gamma problems.

(3) The third approach is an observation-driven state-space approach. This approach connects the previous two approaches (1) and (2), namely, we assume of having a state-space model $(\Theta_t)_{t\geq 1}$ that is not fully parameter-driven, but there is a feedback loop from the observations to the latent state-space process. We call the latent process $(\Theta_t)_{t\geq 1}$ *observation-driven* if there exists $t \geq 2$ such that

$$\Theta_t|_{\{\Theta_{1:t-1},Y_{1:t-1}\}} \overset{(d)}{\neq} \Theta_t|_{\Theta_{1:t-1}}. \tag{4.4}$$

That is, this is the negation of (4.2). This observation-driven latent process is Markovian if we additionally have

$$\Theta_t|_{\{\Theta_{1:t-1},Y_{1:t-1}\}} \overset{(d)}{=} \Theta_t|_{\{\Theta_{t-1},Y_{1:t-1}\}} \qquad \text{for all } t \geq 2. \tag{4.5}$$

Figure 4.2 illustrates the DAG of a Markovian observation-driven state-space model, with the blue arrows being the feedback loops from the observations $Y_t$ to the latent variables $\Theta_{t+s}$, $s \geq 1$.

An observation-driven latent process $(\Theta_t)_{t\geq 1}$ is *purely observation-driven* if additionally

$$\Theta_t|_{\{\Theta_{1:t-1},Y_{1:t-1}\}} \overset{(d)}{=} \Theta_t|_{Y_{1:t-1}} \qquad \text{for all } t \geq 2. \tag{4.6}$$

Purely observation-driven state-space models provide conditionally independent latent state-spaces $(\Theta_t)_{t\geq 1}$, given the observations $(Y_t)_{t\geq 1}$, this can be seen from the DAG of Figure 4.3.



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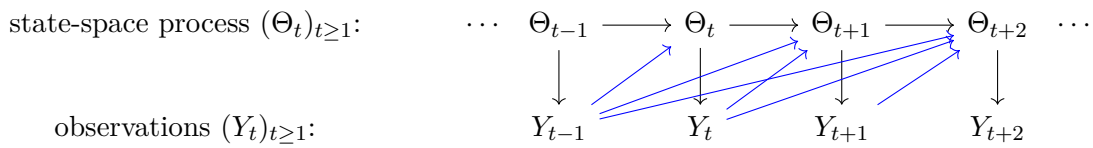


Figure 4.2: DAG of a Markovian observation-driven state-space model; the blue arrows show the feedback loops from the observations $Y_t$ to the latent variables $\Theta_{t+s}$, $s \geq 1$.

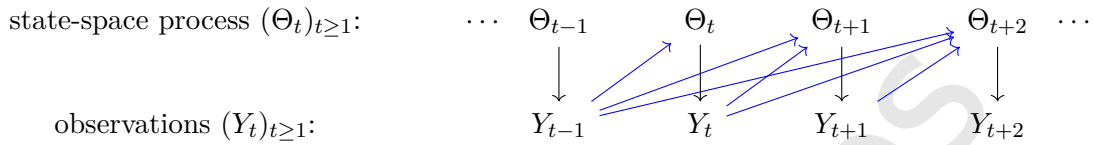


Figure 4.3: DAG of a purely observation-driven state-space model; the blue arrows show the feedback loops from the observations $Y_t$ to the latent variables $\Theta_{t+s}$, $s \geq 1$.

Intuitively, such feedback loops make perfect sense in an insurance modeling context. If one thinks of $(\Theta_t)_{t\geq 1}$ being the unobservable insurance risk factors, e.g., the present (unknown) state of health may influence car driving, then Figure 4.2 emphasizes that the claims experience $(Y_t)_{t\geq 1}$ can influence these latent factors. This may be the case for health risk factors, i.e., an accident may change the driving behavior itself, but it can also have an impact on the state of health. This motivates to extend the parameter-driven state-space models of item (2) by feedback loops to obtain observation-driven dynamic random risk factors. Moreover, if one integrates out the latent process $\Theta_{1:t+1}$, one receives an observation-driven process of type (1) by considering the conditional distribution of $Y_{t+1}|_{Y_{1:t}}$. However, it has no longer the interpretation of claims thinning which has been doubted above for insurance pricing, but it now acts on the latent risk factor process.
Such observation-driven state-space models (with feedback loops) have been introduced by Harrison–Stevens [101], Smith–Miller [182] and Harvey–Fernandes [102], and they have been studied in an actuarial context by Ahn et al. [4, 5] Another actuarial paper with a self-exciting feedback loop is presented in Jeong–Zou [108]. This latter paper is based on Hawkes processes [105], but this is not part of these notes.

## 4.2 Introduction to dynamic random effects models

### 4.2.1 An introductory credibility example

The main deficiency of static random effects models is that all past claims $Y_{1:T}$ are equally weighted, see, e.g., Bühlmann credibility (3.4). This comes from the fact that the static latent factor $\Theta$ acts equally on all past claims. Dynamic random effects modeling replaces this static latent factor by a latent stochastic process $\Theta_{1:T+1} = (\Theta_1, \ldots, \Theta_{T+1})$ that varies over time. The best known example is the Kalman filter [118] which is based on a (latent) state-space model; we refer to Bühlmann–Gisler [41, Chapter 9]. In a nutshell, the latent stochastic process $\Theta_{1:T+1}$ is used to model the (unobservable) random calendar

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year effects that change over time, and an imperfect correlation in this latent stochastic process will result in a seniority weighting of claims, similar to a fading memory property. We start by briefly discussing the Gerber–Jones [88] model which was one of the first models used in actuarial science in this field. This approach is a Markovian parameter-driven state-space model.

**Example 4.1** (Gerber–Jones [88])**.** Assume that the components of $Y_{1:T+1}$ are conditionally independent, given $\Theta_{1:T+1}$, with the first two moments satisfying

$$\begin{aligned} \mathbb{E}\left[Y_t \middle| \Theta_{1:T+1}\right] &= \mu(\Theta_t), \\ \mathrm{Var}\left(Y_t \middle| \Theta_{1:T+1}\right) &= \sigma^2, \end{aligned}$$

for all $1 \leq t \leq T+1$; see Bühlmann–Gisler [41, formulas (9.8)-(9.9)]. The stochastic latent process $(\mu(\Theta_t))_{t=1}^{T+1}$ is assumed to be a random walk with independent and centered increments having finite variance $\delta^2 > 0$. These assumptions give the following recursive Bühlmann credibility formula for experience rating, see Bühlmann–Gisler [41, Example 9.4]. For $1 \leq t \leq T+1$, we have recursive credibility formula

$$\widehat{\widehat{\mu}}^{\text{post}}_{Y_{t+1}|Y_{1:t}} = \widehat{\mathbb{E}}\left[Y_{t+1} \middle| Y_{1:t}\right] = \omega_t Y_t + (1-\omega_t)\widehat{\widehat{\mu}}^{\text{post}}_{Y_t|Y_{1:t-1}}, \tag{4.7}$$

with prior mean initialization $\widehat{\widehat{\mu}}^{\text{post}}_{Y_1|Y_{1:0}} := \mu_0 := \mathbb{E}[\mu(\Theta_1)]$, and with credibility weights

$$\omega_t = \frac{1}{1+\sigma^2/\tau_t^2}, \qquad \text{and} \qquad \tau_t^2 = (1-\omega_{t-1})\tau_{t-1}^2 + \delta^2,$$

being initialized $\tau_0^2 = 0$, and $Y_{1:0}$ generates the trivial $\sigma$-field corresponding to the empty set. In the special case where the initial value $\mu(\Theta_1)$ of the random walk satisfies the (equilibrium) condition

$$\tau^2 := \mathrm{Var}(\mu(\Theta_1)) = \frac{1}{2}\delta^2\left(1+\sqrt{1+4\sigma^2/\delta^2}\right),$$

one receives time-independent credibility weights $\omega \equiv \omega_t$ having the form, see Bühlmann–Gisler [41, Example 9.4],

$$\omega_t \equiv \omega = \frac{1}{1+\sigma^2/\tau^2} \ \in \ (0,1).$$

This equilibrium case provides the exponentially decaying seniority weighting

$$\widehat{\widehat{\mu}}^{\text{post}}_{Y_{T+1}|Y_{1:T}} = \widehat{\mathbb{E}}\left[Y_{T+1} \middle| Y_{1:T}\right] = \omega \sum_{s=0}^{T-1} (1-\omega)^s Y_{T-s} + (1-\omega)^T \mu_0. \tag{4.8}$$

That is, past observations $Y_{T-s}$ receive credibility weights $\omega \exp\{-s\log(1/(1-\omega))\}$, which decay exponentially for $s \to \infty$. This completes this example. ■

We discuss the previous Example 4.1. Pricing formula (4.8) applies an exponentially decaying weighting to the seniority of claims. On purpose we put hats on top of $\widehat{\widehat{\mu}}^{\text{post}}_{Y_{T+1}|Y_{1:T}}$ and $\widehat{\mathbb{E}}$ in (4.8), as this pricing formula is an inhomogeneous Bühlmann credibility approximation to an exact Bayesian formula. Note that we only set assumptions on the first



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two moments of the claims $Y_t$, given the latent process $\Theta_{1:T+1}$. The Bühlmann credibility formula (4.7) should then be understood as an approximation to the full posterior mean, it minimizes the MSE between the observation $Y_{T+1}$ and the closed subspace $\mathcal{L}(Y_{1:T}, 1) \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$ generated by the affine functions of $Y_{1:T}$; we refer to Section 3.2.3. The full Bayesian estimate would consider the subspace $\mathcal{G}(Y_{1:T})$ generated by all measurable functions of $Y_{1:T}$, but to determine this full Bayesian estimate we would need stronger assumptions. Thus, in general, (4.8) does not provide the predictor that has the minimal MSE. As in Example 3.10, the formula is exact in a multivariate Gaussian model, and there it can also easily be derived.

In practice, the model of Example 4.1 seems problematic as it does not comply with stylized facts. The random walk assumption on $(\mu(\Theta_t))_{t=1}^{T+1}$ leads to the property that the variance of this latent process is linearly increasing, and it is asymptotically exploding for $T \to \infty$. Considering an entire portfolio $1 \leq i \leq n$ of such independent risks described by independent stochastic processes $(\mu(\Theta_{i,t}))_{t=1}^{T+1}$ leads to increasing heterogeneity within the portfolio over time $T \geq 1$. Such a behavior cannot be justified by real world observations. In Bühlmann–Gisler [41, Example 9.5] the random walk for $(\mu(\Theta_t))_{t=1}^{T+1}$ is therefore replaced by an AR(1)-process, i.e., a different Markovian parameter-driven state-space dynamics. As described in the remarks after Example 9.5 of Bühlmann–Gisler [41], such a model seems more sensible in (long-term) considerations.

Pinquet et al. [163] and Pinquet [161, 162] discuss choices of latent parameter-driven state-space processes, in particular, the behavior of their auto-correlation coefficients is studied

$$\varrho_\Theta(h) = \mathrm{Corr}\left(\mu(\Theta_t), \mu(\Theta_{t-h})\right),$$

for $1 \leq t-h \leq t \leq T+1$. Extrapolating the fitted models to their long-term behavior, Pinquet et al. [163, Table 4] conclude that the long-term experience rating discount of drivers without any accidents should be around 30% to 40% of the initial premium (on their portfolio and the underlying motor insurance product studied). Moreover, the choice of the latent state-space process needs a careful specification because not every choice will guarantee positive credibility weights, see Pinquet [161, 162]. These are important issues when studying parameter-driven state-space models. Our main focus will be on developing observation-driven state-space models.

### 4.2.2 Interlude: An auto-regressive gamma (state-space) process

Most dynamic random effects models are either based on a Gaussian latent state-space process or they use a Kalman filter approximation [118]. An exception is the AR gamma process studied in Lu [142] which is based on the work Gourieroux–Jasiak [97]. This AR gamma process uses the non-centered gamma distribution. It is a Poisson mixture of gamma distributions, and it can be interpreted as a discrete time version of the continuous time Cox–Ingersoll–Ross (CIR) process [54].

We use a different approach, here. We construct an AR gamma process being based on random thinning using a result of Lukacs [143]. The construction of this process will be useful later on in designing tractable observation-driven dynamic random effects models.

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**Lemma 4.2** (Lukacs [143]). *Assume the random variables $\Lambda$ and $B$ are independent with $\Lambda \sim \Gamma(\alpha, \beta)$ and $B \sim \text{Beta}(p\alpha, (1-p)\alpha)$ for parameters $\alpha > 0$, $\beta > 0$ and $p \in (0,1)$. Then, $\Lambda B \sim \Gamma(p\alpha, \beta)$.*

For later purposes, we can interpret the degenerate cases $p = 0, 1$ as $B$ being identical equal to 0 or 1, respectively.

**Proof of Lemma 4.2.** We have moment generating function of $\log(\Lambda)$

$$\mathbb{E}\left[\exp\{r\log(\Lambda)\}\right] = \mathbb{E}\left[\Lambda^r\right] = \frac{\Gamma(\alpha+r)}{\Gamma(\alpha)}\beta^{-r},$$

for $r > -\alpha$, see also (3.6). For $\log(B)$ with $B \sim \text{Beta}(\gamma, \delta)$ we have for $r > -\gamma$

$$\mathbb{E}\left[\exp\{r\log(B)\}\right] = \mathbb{E}\left[B^r\right] = \int_0^1 x^r \frac{\Gamma(\gamma+\delta)}{\Gamma(\gamma)\Gamma(\delta)} x^{\gamma-1}(1-x)^{\delta-1}\,dx = \frac{\Gamma(\gamma+\delta)\Gamma(\gamma+r)}{\Gamma(\gamma+\delta+r)\Gamma(\gamma)}.$$

Setting $\gamma = p\alpha$ and $\delta = (1-p)\alpha$, the independent random variables $\Lambda$ and $B$ provide moment generating function

$$\begin{aligned}\mathbb{E}\left[\exp\{r\log(\Lambda B)\}\right] &= \mathbb{E}\left[\Lambda^r B^r\right] = \mathbb{E}\left[\Lambda^r\right]\mathbb{E}\left[B^r\right] \\ &= \frac{\Gamma(\alpha+r)}{\Gamma(\alpha)}\beta^{-r}\frac{\Gamma(\alpha)\Gamma(p\alpha+r)}{\Gamma(\alpha+r)\Gamma(p\alpha)} = \frac{\Gamma(p\alpha+r)}{\Gamma(p\alpha)}\beta^{-r}.\end{aligned}$$

This proves the claim. □

This result allows us to define an AR process $(\Lambda_t)_{t\geq 1}$ as follows.

(0) Initialization. Set $\Lambda_1 \sim \Gamma(\alpha_1, \beta_1)$ with initial parameters $\alpha_1 = \beta_1 > 0$; this implies normalization $\mathbb{E}[\Lambda_1] = 1$.

(1) Recursion for $t \geq 1$. Choose two parameter sequences

$$(p_t)_{t\geq 1} \subset (0,1) \text{ and } (q_t)_{t\geq 1} \subset (0,\infty), \text{ with } 0 < p_t q_t < 1 \text{ for all } t \geq 1, \tag{4.9}$$

and define recursively for $t \geq 1$[1][2]

$$\beta_{t+1} = q_t\beta_t \qquad \text{and} \qquad \alpha_{t+1} = p_t q_t \alpha_t + (1-p_t)q_t\beta_t = p_t q_t \alpha_t + (1-p_t)\beta_{t+1}.$$

Set recursively for $t \geq 1$

$$\Lambda_{t+1} = \frac{\Lambda_t B_{t+1}}{q_t} + \eta_{t+1}, \tag{4.10}$$

with independent random variables $\Lambda_1$, $(B_t)_{t\geq 2}$ and $(\eta_t)_{t\geq 2}$ satisfying for $t \geq 1$

$$\begin{aligned} B_{t+1} &\sim \text{Beta}\left(p_t q_t \alpha_t, (1-p_t q_t)\alpha_t\right), \\ \eta_{t+1} &\sim \Gamma((1-p_t)q_t\beta_t, q_t\beta_t). \end{aligned}$$

This AR process $(\Lambda_t)_{t\geq 1}$ has a similar structure as (4.1) used to define an INAR(1) model, applying a thinning and a new innovation, here the thinning of $\Lambda_t$ is received by the multiplicative term $B_{t+1}/q_t$, see (4.10).

[1] Remark that in this chapter $(\beta_t)_{t\geq 1}$ denotes the scale parameter of the gamma process, and this is different form the bold face GLM parameter $\boldsymbol{\beta}$ used, e.g., in the static mixed effects GLM (3.47).

[2] The sequence $(q_t)_{t\geq 1} \subset (0,\infty)$ should not be confused with the number of neurons $(q_m)_{m=1}^d \in \mathbb{N}^d$ in the FNN Section 2.4.1.



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**Lemma 4.3.** *All components of the AR process $(\Lambda_t)_{t\geq 1}$ defined in items (0)-(1) are gamma distributed with shape parameters $(\alpha_t)_{t\geq 1}$ and scale parameters $(\beta_t)_{t\geq 1}$.*

**Proof of Lemma 4.3.** By assumption $\Lambda_1 \sim \Gamma(\alpha_1, \beta_1)$. Lemma 4.2 implies for $t = 2$

$$\frac{\Lambda_1 B_2}{q_1} \sim \Gamma\left(p_1 q_1 \alpha_1, q_1 \beta_1\right).$$

Adding the new noise term gives

$$\Lambda_2 = \frac{\Lambda_1 B_2}{q_1} + \eta_2 \sim \Gamma\left(p_1 q_1 \alpha_1 + (1-p_1) q_1 \beta_1, q_1 \beta_1\right) \overset{(d)}{=} \Gamma\left(\alpha_2, \beta_2\right).$$

Recursive iteration (induction) for $t \geq 3$ proves the claim. □

**Definition 4.4** (AR gamma process)**.** *We call the stochastic process $(\Lambda_t)_{t\geq 1}$ defined in items (0)-(1) an AR gamma process with parameters* (4.9) *and initialization $\alpha_1 = \beta_1 > 0$.*

**Corollary 4.5.** *Under the above assumptions, the AR gamma process $(\Lambda_t)_{t\geq 1}$ is mean-stationary $\mathbb{E}[\Lambda_t] = 1$ for all $t \geq 1$.*

**Proof of Corollary 4.5.** From Lemma 4.3 it follows for $t \geq 1$

$$\mathbb{E}\left[\Lambda_{t+1}\right] = \frac{\alpha_{t+1}}{\beta_{t+1}} = \frac{p_t q_t \alpha_t + (1-p_t) q_t \beta_t}{q_t \beta_t} = p_t \frac{\alpha_t}{\beta_t} + (1-p_t) = p_t \mathbb{E}\left[\Lambda_t\right] + (1-p_t).$$

The proof follows by induction. □

The above construction provides a mean-stationary AR gamma process $(\Lambda_t)_{t\geq 1}$ with variances for $t \geq 1$

$$\begin{aligned}
\operatorname{Var}\left(\Lambda_{t+1}\right) &= \frac{\alpha_{t+1}}{\beta_{t+1}^2} = \frac{1}{\beta_{t+1}} \mathbb{E}\left[\Lambda_{t+1}\right] \\
&= \frac{1}{\beta_{t+1}} = \frac{1}{q_t \beta_t} \\
&= \frac{1}{q_t} \operatorname{Var}\left(\Lambda_t\right) = \operatorname{Var}\left(\Lambda_1\right) \prod_{s=1}^{t} \frac{1}{q_s}.
\end{aligned}$$

The sequence $(q_t)_{t\geq 1} \subset (0, \infty)$ allows us to control (and to model) the variance behavior of this AR gamma process. E.g., for $q_t \equiv 1$ we have a stationary variance behavior.

For the covariance we have, note $\mathbb{E}[B_{t+1}] = p_t q_t < 1$,

$$\operatorname{Cov}\left(\Lambda_t, \Lambda_{t+1}\right) = \operatorname{Cov}\left(\Lambda_t, \mathbb{E}[\Lambda_{t+1} | \Lambda_t]\right) = \frac{\mathbb{E}[B_{t+1}]}{q_t} \operatorname{Var}\left(\Lambda_t\right) = p_t \operatorname{Var}\left(\Lambda_t\right).$$

This gives auto-correlation

$$\operatorname{Corr}\left(\Lambda_t, \Lambda_{t+1}\right) = p_t \sqrt{q_t} = \sqrt{p_t} \sqrt{p_t q_t}, \tag{4.11}$$

with $p_t < 1$ and $p_t q_t < 1$. Thus, the sequence $(p_t)_{t\geq 1} \subset (0, 1)$ is used to control the auto-correlation, and for a variance-stationary process, $q_t \equiv 1$, we have auto-correlation $p_t \in (0, 1)$ in (4.11). We can then also interpret the boundary cases $p_t \equiv 0$ as zero auto-correlation only considering independent innovations $\eta_{t+1}$, and $p_t \equiv 1$ as full correlation, not adding any new innovations $\eta_{t+1} \equiv 0$.



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This AR gamma process $(\Lambda_t)_{t\geq 1}$ could be used to define a parameter-driven dynamic random effects model, see Figure 4.1, and results similar to Pinquet et al. [163] apply. We proceed differently by replacing the parameter-driven state-space process by an observation-driven one. Using the Poisson-gamma conjugacy we derive an analytically tractable model for experience rating; this follows Ahn et al. [4, 5] and it is based on Harvey–Fernandes [102].

## 4.3 Observation-driven dynamic mixed effects model

We consider the proposal of Ahn et al. [5] in this section and define an observation-driven dynamic mixed effects model. The general construction is based on the ideas of Smith–Miller [182] and Harvey–Fernandes [102], and it modifies the AR gamma process of Corollary 4.5 to an observation-driven version.

### 4.3.1 Observation-driven dynamic mixed effects Poisson-gamma model

In all what follows, we consider four deterministic exogenous sequences $(v_t)_{t\geq 1} \subset (0,\infty)$, $(\mu_t)_{t\geq 1} \subset (0,\infty)$, $(p_t)_{t\geq 1} \subset (0,1)$ and $(q_t)_{t\geq 1} \subset (0,\infty)$ with constraint

$$0 < p_t q_t < 1 \qquad \text{for all } t \geq 1.$$

These deterministic sequences are given exogenously, $(\mu_t)_{t\geq 1}$ will be used to model time-dependent fixed effects, e.g., making a GLM assumption (3.38), as usual $(v_t)_{t\geq 1}$ are the exposures, and the sequences $(p_t)_{t\geq 1}$ and $(q_t)_{t\geq 1}$ will keep the meaning of the previous section, see (4.9).
In the following model we define the laws of the claim counts random variables $(N_t)_{t\geq 1}$. Because this model is defined in its additive form, we use the notation $N_t$, and the reproductive form is received by setting $Y_t = N_t/v_t$ for $\varphi = 1$. $N_t$ is a counts random variable, $Y_t$ is a frequency, and $\Lambda_t$ is scale-free; we also refer to the end of Example 3.27.

**Model 4.6.** *For $t \geq 1$, we set as usual $\Lambda_{1:t} = (\Lambda_1, \ldots, \Lambda_t)$ and $N_{1:t} = (N_1, \ldots, N_t)$, and $N_{1:0}$ denotes the empty sequence generating the trivial $\sigma$-field.*

(0) Initialization. *We initialize the state-space process for given starting shape and scale parameters $\alpha_1 = \beta_1 > 0$ by*

$$\Lambda_1 \sim \Gamma(\alpha_1, \beta_1). \tag{4.12}$$

(1) Observation equation for $t \geq 1$. *Conditionally given the state-spaces $\Lambda_{1:t}$ and the past observations $N_{1:t-1}$, we assume a Poisson distribution*

$$N_t|_{\{\Lambda_{1:t}, N_{1:t-1}\}} \;\sim\; \mathrm{Poi}\left(v_t \mu_t \Lambda_t\right). \tag{4.13}$$

(2) Bayesian filtering $t \geq 1$. *Conditionally given the observations $N_{1:t}$, we have posterior distribution*

$$\Lambda_t|_{N_{1:t}} \;\sim\; \Gamma\left(\alpha_t + N_t, \beta_t + v_t \mu_t\right). \tag{4.14}$$



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(3) State-space update $t \geq 1$. *Conditionally given the observations $N_{1:t}$, we have predictive distribution of the new state-space*

$$\Lambda_{t+1}|_{N_{1:t}} \sim \Gamma\left(\alpha_{t+1}, \beta_{t+1}\right), \tag{4.15}$$

*with deterministic scale and $\sigma(N_{1:t})$-measurable shape parameters, respectively,*

$$\beta_{t+1} = q_t\left(\beta_t + v_t\mu_t\right) > 0, \tag{4.16}$$
$$\alpha_{t+1} = p_t q_t\left(\alpha_t + N_t\right) + \left(1 - p_t\right)\beta_{t+1} > 0. \tag{4.17}$$

In the remainder of this section we verify the consistency of the assumptions of Model 4.6 by constructing an explicit process that satisfies these assumptions. This is again based on Lukacs [143], see Lemma 4.2, but this time we need to carefully treat the integration of the observations $N_{1:t}$. Remark that if we set $v_t \equiv 0$ and $N_t \equiv 0$ (degenerate Poisson model) we precisely have the AR gamma process $(\Lambda_t)_{t\geq 1}$ of Lemma 4.3.

We proceed by induction.

For $t = 1$, we have from (4.12) the latent variable $\Lambda_1 = \Lambda_1|_{N_{1:0}} \sim \Gamma(\alpha_1, \beta_1)$. Based on (4.13), we generate claim counts

$$N_1|_{\{\Lambda_{1:1}, N_{1:0}\}} = N_1|_{\Lambda_1} \sim \text{Poi}\left(v_1\mu_1\Lambda_1\right).$$

Bayesian inference gives (4.14) for $t = 1$, see Example 3.3, that is,

$$\Lambda_1|_{N_{1:1}} = \Lambda_1|_{N_1} \sim \Gamma\left(\alpha_1 + N_1, \beta_1 + v_1\mu_1\right).$$

Thus, item (2) is not a model assumption, but it is a natural consequence of Bayes' rule [18].
Finally, we update the state-space as in (4.10) by thinning and adding a new gamma noise term. Choose, conditionally given $N_1 = N_{1:1}$,

$$\begin{aligned} B_2|_{N_1} &\sim \text{Beta}\left(p_1q_1(\alpha_1 + N_1), (1 - p_1q_1)(\alpha_1 + N_1)\right), \\ \eta_2|_{N_1} &\sim \Gamma((1 - p_1)q_1(\beta_1 + v_1\mu_1), q_1(\beta_1 + v_1\mu_1)), \end{aligned}$$

with $\Lambda_1$, $B_2$ and $\eta_2$ assumed to be conditionally independent, given $N_1$. This provides us with, see also proof of Lemma 4.3,

$$\begin{aligned} \Lambda_2|_{N_{1:1}} &= \left.\frac{\Lambda_1 B_2}{q_1} + \eta_2\right|_{N_1} \\ &\sim \Gamma\left(p_1q_1\left(\alpha_1 + N_1\right) + \left(1 - p_1\right)q_1\left(\beta_1 + v_1\mu_1\right), q_1\left(\beta_1 + v_1\mu_1\right)\right), \\ &\overset{(d)}{=} \Gamma\left(\alpha_2, \beta_2\right), \end{aligned} \tag{4.18}$$

given $N_{1:1} = N_1$. This is precisely (4.15) for $t = 1$. This procedure is recursively iterated by simulating the next claim $N_2$, conditionally given $\Lambda_2$, i.e., from (4.13) we have for $t = 2$ the next response

$$N_2|_{\{\Lambda_{1:2}, N_{1:1}\}} = N_2|_{\Lambda_2} \sim \text{Poi}\left(v_2\mu_2\Lambda_2\right).$$



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The Bayesian inference step of $\Lambda_2$, given $N_{1:2}$, needs a bit of care. We have posterior density, given $N_{1:2}$,

$$\begin{aligned}\pi\left(\Lambda_2 \middle| N_{1:2}\right) &\propto f\left(N_2, \Lambda_2 \middle| N_{1:1}\right)\\ &= f\left(N_2 \middle| \Lambda_2, N_{1:1}\right)\pi\left(\Lambda_2 \middle| N_{1:1}\right)\\ &= f\left(N_2 \middle| \Lambda_2\right)\pi\left(\Lambda_2 \middle| N_{1:1}\right).\end{aligned}$$

From this we conclude

$$\Lambda_2|_{N_{1:2}} \sim \Gamma\left(\alpha_2 + N_2, \beta_2 + v_2\mu_2\right),$$

this justifies the Bayesian inference step (4.14) for $t = 2$. For the state-space update we choose thinning and adding a new gamma noise term as follows, conditionally given $N_{1:2}$,

$$\begin{aligned}B_3|_{N_{1:2}} &\sim \text{Beta}\left(p_2q_2(\alpha_2 + N_2), (1 - p_2q_2)(\alpha_2 + N_2)\right),\\ \eta_3|_{N_{1:2}} &\sim \Gamma((1 - p_2)q_2(\beta_2 + v_2\mu_2), q_2(\beta_2 + v_2\mu_2)),\end{aligned}$$

with $\Lambda_2$, $B_3$ and $\eta_3$ assumed to be conditionally independent, given $N_{1:2}$. This provides (4.15) for $t = 2$ using the state-space update

$$\Lambda_3|_{N_{1:2}} = \frac{\Lambda_2 B_3}{q_2} + \eta_3\Big|_{N_{1:2}} \sim \Gamma\left(\alpha_3, \beta_3\right).$$

This is iterated for $t \geq 3$, an illustration is given in Figure 4.2.

This explicit construction verifies that processes of the type of Model 4.6 exist and, additionally, it also shows us how to simulate from this model.

**Remark 4.7.** One may argue that the above verification of the existence of a process of the type of Model 4.6 is too complicated. In fact, there are infinitely many models that comply with these assumptions, and the previous construction gives a Markovian observation-driven state-space model complying with the DAG of Figure 4.2. Let us give a version that complies with the DAG of Figure 4.3. The Bayesian filtering step (4.14) gives us the posterior distribution of $\Lambda_t$, given $N_{1:t}$, and this needs to be coupled to the next state $\Lambda_{t+1}$, given $Y_{1:t}$, see (4.15). Formula (4.18) gives an explicit construction that is based on Lukacs' [143] result, and which has been used in Harvey–Fernandes [102] and Ahn et al. [4, 5]. However, we can use *any* copula to connect the marginals $\Lambda_t|_{N_{1:t}} \sim \Gamma(\alpha_t + N_t, \beta_t + v_t\mu_t)$ and $\Lambda_{t+1}|_{N_{1:t}} \sim \Gamma(\alpha_{t+1}, \beta_{t+1})$. One option is to assume that they are conditionally independent, given $Y_{1:t}$, and then we are in the situation of the DAG of Figure 4.3. All these models provide the same predictive distribution of $N_{T+1}$, given $N_{1:T}$, but they have different state-space dynamics. Another consequence of this identifiability issue of the copula of the state-space updates is that Model 4.6 does not define a full stochastic model that we can simulate from. Nevertheless, this model structure is sufficient for evaluating predictive distributions, we come back to this in Section 4.3.3, and this is further discussed in Ahn et al. [7] ■



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### 4.3.2 Long term model behavior

Table 1 of Ahn et al. [5] gives different long-term behaviors of the observation-driven state-space Model 4.6. We revisit and discuss these. The crucial property that distinguishes the different cases is the behavior of the variance $\mathrm{Var}(\Lambda_t)$ of the state-space process as a function of time $t \geq 1$.

**Lemma 4.8** (Ahn et al. [5])**.** *In Model 4.6 we have mean-stationarity of the state-space process* $(\Lambda_t)_{t\geq 1}$*, that is,* $\mathbb{E}[\Lambda_t] = 1$ *for all* $t \geq 1$*. The variance satisfies for* $t \geq 1$

$$\mathrm{Var}\left(\Lambda_t\right) = \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{q_t}{\beta_{t+1}}, \tag{4.19}$$

$$\mathrm{Var}\left(\Lambda_{t+1}\right) = p_t^2\,\mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{1}{\beta_{t+1}}. \tag{4.20}$$

**Proof of Lemma 4.8.** We first compute the mean of the state-space process. It is for $t \geq 1$ given by

$$\begin{aligned}\mathbb{E}\left[\Lambda_{t+1}\right] &= \mathbb{E}\left[\mathbb{E}\left[\Lambda_{t+1} \middle| N_{1:t}\right]\right] = \mathbb{E}\left[\frac{\alpha_{t+1}}{\beta_{t+1}}\right] = \frac{p_t q_t}{q_t}\mathbb{E}\left[\frac{\alpha_t + N_t}{\beta_t + v_t\mu_t}\right] + (1-p_t)\\ &= p_t\mathbb{E}\left[\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right] + (1-p_t) = p_t\mathbb{E}\left[\Lambda_t\right] + (1-p_t).\end{aligned}$$

From this we see that we can recursively decrease the time index $t \geq 1$, and we have initialization $\mathbb{E}[\Lambda_1] = 1$. This provides mean-stationarity $\mathbb{E}[\Lambda_t] = 1$ for all $t \geq 1$ by induction.
We calculate the variances. We have using that $(\beta_t)_{t\geq 1}$ is a deterministic sequence

$$\begin{aligned}\mathrm{Var}\left(\Lambda_t\right) &= \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \mathbb{E}\left[\mathrm{Var}\left(\Lambda_t \middle| N_{1:t}\right)\right]\\ &= \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \mathbb{E}\left[\frac{1}{\beta_t + v_t\mu_t}\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right]\\ &= \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{1}{\beta_t + v_t\mu_t}\mathbb{E}\left[\Lambda_t\right]\\ &= \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{1}{\beta_t + v_t\mu_t}.\end{aligned}$$

This proves (4.19). On the other hand, we have

$$\begin{aligned}\mathrm{Var}\left(\Lambda_{t+1}\right) &= \mathrm{Var}\left(\mathbb{E}\left[\Lambda_{t+1} \middle| N_{1:t}\right]\right) + \mathbb{E}\left[\mathrm{Var}\left(\Lambda_{t+1} \middle| N_{1:t}\right)\right]\\ &= \mathrm{Var}\left(\frac{\alpha_{t+1}}{\beta_{t+1}}\right) + \mathbb{E}\left[\frac{1}{\beta_{t+1}}\mathbb{E}\left[\Lambda_{t+1} \middle| N_{1:t}\right]\right]\\ &= \mathrm{Var}\left(\frac{p_t(\alpha_t + N_t)}{\beta_t + v_t\mu_t}\right) + \frac{1}{\beta_{t+1}}\mathbb{E}\left[\Lambda_{t+1}\right]\\ &= p_t^2\mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{1}{\beta_{t+1}}.\end{aligned}$$

This proves the last claim (4.20) of the lemma. □

Recall $p_t \in (0,1)$ and $q_t \in (0,\infty)$ with $0 < p_t q_t < 1$ for all $t \geq 1$. We compare (4.19) and (4.20). If we have a strict inequality $p_t < 1$ (i.e., if we do not consider the degenerate case $p_t = 1$), the first term on the right-hand side of (4.20) decreases the variance in $t$ compared to (4.19). On the other hand, if we have a strict inequality $q_t < 1$, the last term on the right-hand side of (4.20) increases in $t$ compared to (4.19). Therefore, we can exhibit different variance behaviors depending on which of the two terms has the more significant change. In the remainder of this section we discuss some possible long-term variance behaviors. For these considerations we throughout assume boundedness

$$0 < \mu^- := \inf_{t\geq 1} v_t\mu_t \leq \sup_{t\geq 1} v_t\mu_t =: \mu^+ < \infty. \tag{4.21}$$



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**Increasing/exploding variance case.** An increasing variance case is obtained by setting $p_t = 1$ (degenerate case) and $q_t = q \in (0,1)$ for all $t \geq 1$; this is the model of Harvey–Fernandes [102], also studied in Ahn et al. [4]. In this case, there is no independent noise $(\eta_{t+1})_{t\geq 1}$ added to the state-space process, see, e.g., (4.18). Therefore, the first terms on the right-hand sides of (4.19) and (4.20) coincide, but the second term is increasing in $t$. In fact, we have in the case $p_t \equiv 1$ and $q_t \equiv q \in (0,1)$

$$\mathrm{Var}\left(\Lambda_{t+1}\right) - \mathrm{Var}\left(\Lambda_t\right) = (1-q)\,\frac{1}{\beta_{t+1}} \geq (1-q)\,\frac{1}{\sup_{t\geq 1}\beta_t} > 0,$$

and the variance explodes asymptotically for $t \to \infty$. The last inequality follows from (4.16) and (4.21)

$$\beta_{t+1} = q\left(\beta_t + v_t\mu_t\right) \leq q\left(\beta_t + \mu^+\right) < \beta_1 + \mu^+ q/(1-q) < \infty.$$

We have just proved the following statement, see Ahn et al. [5, Lemma 3].

**Proposition 4.9.** *Under the assumptions $p_t \equiv 1$, $q_t \equiv q \in (0,1)$ and (4.21) we have*

$$\lim_{t\to\infty} \mathrm{Var}\left(\Lambda_t\right) = \infty.$$

At the same time we have mean-stationarity $\mathbb{E}[\Lambda_t] = 1$ for all $t \geq 1$, see Lemma 4.8. This says that the distribution of $\Lambda_t$ is becoming non-informative for $t \to \infty$.

**Decreasing/vanishing variance case.** A decreasing variance case is obtained by setting $p_t = p \in [0,1)$ and $q_t = 1$ for all $t \geq 1$, see (4.19) and (4.20). In this case we have

$$\mathrm{Var}\left(\Lambda_{t+1}\right) - \mathrm{Var}\left(\Lambda_t\right) = (p-1)\,\mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) < 0. \tag{4.22}$$

We have the following result, see Ahn et al. [5, Lemma 5].

**Proposition 4.10.** *Under the assumptions $p_t \equiv p \in [0,1)$, $q_t \equiv 1$ and (4.21) we have*

$$\lim_{t\to\infty} \mathrm{Var}\left(\Lambda_t\right) = 0.$$

**Proof.** Since in the assumptions of this proposition the variance $\mathrm{Var}(\Lambda_t)$ is decreasing in $t \to \infty$, see (4.22), and bounded below by zero, we know that the last term in (4.22) has to converge to zero

$$\lim_{t\to\infty} \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) = 0.$$

In view of Lemma 4.8 we have

$$\lim_{t\to\infty} \mathrm{Var}\left(\Lambda_t\right) = \lim_{t\to\infty} \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{1}{\beta_{t+1}} = \lim_{t\to\infty} \frac{1}{\beta_{t+1}} = 0,$$

because for $q_t \equiv 1$ the following converges to $\infty$ for $t \to \infty$

$$\beta_{t+1} = \beta_t + v_t\mu_t \geq \beta_t + \mu^- \geq \beta_1 + (t-1)\mu^-.$$

This proves the claim. □

At the same time we have mean-stationarity $\mathbb{E}[\Lambda_t] = 1$ for all $t \geq 1$, see Lemma 4.8. Proposition 4.10 then says that the distribution of $\Lambda_t$ is becoming more and more concentrated in 1 for $t \to \infty$, i.e., the uncertainty of the dynamic random effects process is asymptotically vanishing and we can perfectly forecast this process.



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**Bounded variance case.** A bounded variance case is obtained by the choices $p_t = p \in (0,1)$ and $q_t = q \in (0,1)$ for all $t \geq 1$. This allows one to balance the growth and decrease of the two terms in (4.19) and (4.20). The following result is given in Ahn et al. [5, Lemma 7].

**Proposition 4.11.** *Under the assumptions* $p_t \equiv p \in [0,1)$, $q_t \equiv q \in (0,1)$ *and* (4.21) *we have*

$$\inf_{t\geq 1} \mathrm{Var}\left(\Lambda_t\right) > 0 \qquad \textit{and} \qquad \sup_{t\geq 1} \mathrm{Var}\left(\Lambda_t\right) < \infty.$$

**Proof.** We first note that

$$\sup_{t\geq 1} \beta_{t+1} = \sup_{t\geq 1} q\left(\beta_t + v_t\mu_t\right) \leq \sup_{t\geq 1} q\left(\beta_t + \mu^+\right) < \beta_1 + \mu^+ q/(1-q) < \infty,$$

and

$$\inf_{t\geq 1} \beta_{t+1} = \inf_{t\geq 1} q\left(\beta_t + v_t\mu_t\right) \geq \inf_{t\geq 1} q\left(\beta_t + \mu^-\right) \geq q\mu^- > 0.$$

Using Lemma 4.8, we have

$$\mathrm{Var}\left(\Lambda_t\right) = \mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{q}{\beta_{t+1}} \geq \frac{q}{\sup_{t\geq 1}\beta_{t+1}} > 0.$$

This proves the first claim. For the second claim we have (recursively) from Lemma 4.8

$$\begin{aligned}
\mathrm{Var}\left(\Lambda_{t+1}\right) &= p^2\,\mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{1}{\beta_{t+1}} \\
&= p^2\left(\mathrm{Var}\left(\mathbb{E}\left[\Lambda_t \middle| N_{1:t}\right]\right) + \frac{q}{\beta_{t+1}}\right) - \frac{p^2 q}{\beta_{t+1}} + \frac{1}{\beta_{t+1}} \\
&= p^2\mathrm{Var}\left(\Lambda_t\right) + \frac{1}{\beta_{t+1}}\left(1 - p^2 q\right) \qquad (4.23)\\
&\leq p^2\mathrm{Var}\left(\Lambda_t\right) + \frac{1-p^2 q}{q\mu^-} \;\leq\; \frac{1}{\beta_1} + \frac{1-p^2 q}{q\mu^-}\frac{1}{1-p^2} \;<\; \infty,
\end{aligned}$$

where $\mathrm{Var}(\Lambda_1) = 1/\beta_1$ is the initial variance. This proves the claim. □

Proposition 4.11 gives the bounded variance case. We could also study the constant variance case which needs to satisfy for all $t \geq 1$, see (4.23),

$$\mathrm{Var}\left(\Lambda_{t+1}\right) = p_t^2\mathrm{Var}\left(\Lambda_t\right) + \frac{1}{\beta_{t+1}}\left(1 - p_t^2 q_t\right) = \mathrm{Var}\left(\Lambda_t\right) = \mathrm{Var}\left(\Lambda_1\right) = \frac{1}{\beta_1} = \frac{1}{\alpha_1}.$$

That is, mean- and variance-stationarity of the state-space process $(\Lambda_t)_{t\geq 1}$ is obtained by requiring

$$\frac{1-p_t^2 q_t}{1-p_t^2} = \frac{\beta_{t+1}}{\alpha_1} = \frac{q_t(\beta_t + v_t\mu_t)}{\alpha_1} = \prod_{s=1}^{t} q_s + \frac{1}{\alpha_1}\sum_{s=1}^{t} v_s\mu_s \prod_{u=s}^{t} q_u,$$

for all $t \geq 1$. This model is not of practical relevance if we need to estimate the model parameters from data. For this reason we will not further study this variance-stationary case. However, if one has a homogeneous case $\mu_t \equiv \mu_0$ and $v_t \equiv 1$, variance-stationarity may be a reasonable property. There is an example in Ahn et al. [7] of a mean- and variance-stationary observation-driven state-space model, which easily allows to identify the parameters to make this model also covariance-stationary. We do not further consider this here, but refer to Ahn et al. [7].

Table 4.1 summarizes the different variance behaviors for constant $p$ and $q$ parameters. This table also contains the special case $p_t \equiv 0$ which corresponds to the independent mixed effects case, see below, and we can extend it by the static mixed effects case by setting $p_t \equiv q_t \equiv 1$, see next item.



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| | $q_t \equiv q \in (0,1)$ | $q_t \equiv 1$ |
|---|---|---|
| $p_t \equiv p \in [0,1)$ | bounded variance | decreasing variance |
| $p_t \equiv 1$ | increasing variance | static mixed effects |

Table 4.1: Different behavior of Model 4.6 under (4.21).

**Static mixed effects case.** Setting $p_t = q_t = 1$ for all $t \geq 1$ can be obtained by a deterministic update $\Lambda_t | N_{1:t} = \Lambda_{t+1} | N_{1:t}$, $\mathbb{P}$-a.s. Thus, in this case we are back in the static mixed effects case with $\Lambda_t \equiv \Lambda \sim \Gamma(\alpha_1, \alpha_1)$.

**Independent mixed effects case.** If $p_t \equiv 0$, we completely neglect past information $N_{1:t}$ in the state-space sequence. In that case we can select an independent state-space sequence $(\Lambda_t)_{t\geq 1}$ with $\Lambda_t \sim \Gamma(\alpha_t, \beta_t)$ having equal shape and scale parameters $\alpha_{t+1} = \beta_{t+1} = q_t(\beta_t + v_t\mu_t)$ for $t \geq 1$, and with initialization $\alpha_1 = \beta_1 > 0$. Moreover, if for all $t \geq 1$ we set $q_t = \beta_1/(\beta_1 + v_t\mu_t) = \alpha_1/(\alpha_1 + v_t\mu_t)$, the state-space sequence is i.i.d. with $\Lambda_t \sim \Gamma(\alpha_1, \alpha_1)$.

Propositions 4.9, 4.10 and 4.11 concern the variance behavior of the state-space process $(\Lambda_t)_{t\geq 1}$. Using this for the responses we obtain a similar behavior from the variance decomposition

$$\begin{aligned} \operatorname{Var}(N_t) &= \operatorname{Var}(\mathbb{E}[N_t | \Lambda_t]) + \mathbb{E}[\operatorname{Var}(N_t | \Lambda_t)] \\ &= v_t^2 \mu_t^2 \operatorname{Var}(\Lambda_t) + v_t \mu_t, \end{aligned}$$

where we have used mean-stationarity of $(\Lambda_t)_{t\geq 1}$ in the last term, see Lemma 4.8. We conclude that Model 4.6 provides us with a rich family of observation-driven state-space models for dynamic claim counts modeling that undergo different variance behaviors.

### 4.3.3 Log-likelihood functions

Model 4.6 allows for a closed form solution of the log-likelihood function w.r.t. the observations $N_{1:t}$. That is, if we integrate out the state-spaces, we obtain a (unique) predictive distribution, though there is an identifiability issue in the copula choice under Model 4.6, see Remark 4.7. We have the following negative binomial distribution result.

**Theorem 4.12.** *Let $t \geq 0$. Under Model 4.6, the predictive distribution of $N_{t+1}$, given $N_{1:t}$, is negative binomial with probability weights for $k \in \mathbb{N}_0$*

$$\mathbb{P}[N_{t+1} = k | N_{1:t}] = \frac{\Gamma(\alpha_{t+1} + k)}{\Gamma(\alpha_{t+1})k!} \left( \frac{\beta_{t+1}}{\beta_{t+1} + v_{t+1}\mu_{t+1}} \right)^{\alpha_{t+1}} \left( \frac{v_{t+1}\mu_{t+1}}{\beta_{t+1} + v_{t+1}\mu_{t+1}} \right)^k.$$

**Proof of Theorem 4.12.** The case $t = 0$ has been proved in Example 3.20. Choose $t \geq 1$. We calculate the conditional probability weights of $N_{t+1}$, given $N_{1:t}$. For $k \in \mathbb{N}_0$

$$\begin{aligned} \mathbb{P}[N_{t+1} = k | N_{1:t}] &= \mathbb{E}[\mathbb{E}[N_{t+1} = k | \Lambda_{1:t+1}, N_{1:t}] | N_{1:t}] \\ &= \mathbb{E}\left[ e^{-v_{t+1}\mu_{t+1}\Lambda_{t+1}} \frac{(v_{t+1}\mu_{t+1}\Lambda_{t+1})^k}{k!} \middle| N_{1:t} \right] \\ &= \int_0^\infty e^{-v_{t+1}\mu_{t+1}\lambda} \frac{(v_{t+1}\mu_{t+1}\lambda)^k}{k!} \frac{\beta_{t+1}^{\alpha_{t+1}}}{\Gamma(\alpha_{t+1})} \lambda^{\alpha_{t+1}-1} e^{-\beta_{t+1}\lambda} \, d\lambda \\ &= \frac{\Gamma(\alpha_{t+1} + k)}{\Gamma(\alpha_{t+1})k!} \left( \frac{\beta_{t+1}}{\beta_{t+1} + v_{t+1}\mu_{t+1}} \right)^{\alpha_{t+1}} \left( \frac{v_{t+1}\mu_{t+1}}{\beta_{t+1} + v_{t+1}\mu_{t+1}} \right)^k. \end{aligned}$$



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This proves the theorem. □

**Remarks 4.13.**

- We have introduced this chapter by discussing observation-driven INAR models. Theorem 4.12 proves that the observation-driven dynamic state-space Model 4.6 is a negative binomial INAR model with the first two conditional moments for $t \geq 1$ given by

$$\begin{aligned}\mathbb{E}\left[N_{t+1} \middle| N_{1:t}\right] &= v_{t+1}\mu_{t+1}\frac{\alpha_{t+1}}{\beta_{t+1}},\\ \operatorname{Var}\left(N_{t+1} \middle| N_{1:t}\right) &= v_{t+1}\mu_{t+1}\frac{\alpha_{t+1}}{\beta_{t+1}}\left(1+\frac{v_{t+1}\mu_{t+1}}{\beta_{t+1}}\right).\end{aligned}$$

  and we observe the classical over-dispersed behavior of the mixed Poisson distribution. There is one important difference tough between the negative binomial INAR model of Theorem 4.12 and the Poisson INAR model constructed in (4.1). For the negative binomal INAR model we apply thinning to the state-space process $(\Lambda_t)_{t\geq 1}$, whereas in the Poisson INAR model we apply thinning directly to the process of observations $(N_t)_{t\geq 1}$ and $(Y_t)_{t\geq 1}$, respectively. While we have argued in the introduction that the latter does not lead to an explainable modeling framework for insurance pricing, see Section 4.1, the observation-driven dynamic state-space Model 4.6 has a convincing explanation for an insurance pricing model.

- From Theorem 4.12, we observe that for the predictive distribution of $N_{t+1}$, given $N_{1:t}$, the specific copula choice of $(\Theta_t, \Theta_{t+1})$, given $N_{1:t}$, does not influence the results, once the marginal of $\Theta_{t+1}$, given $N_{1:t}$, is fixed. Thus, for a fixed marginal, Theorem 4.12 holds for both a Markovian version of the observation-driven state-space model, see DAG of Figure 4.2, but also for a purely observation-driven state-space model according to the DAG of Figure 4.3. This seems a bit counter-intuitive, but it reflects the fact that the same predictive distribution can originate from different state-space models, in this sense, the state-space dynamics is only a work horse that makes the predictive distribution tractable and explainable.

■

Theorem 4.12 allows us to recursively derive the joint log-likelihood function for given observations $N_{1:t}$.

**Corollary 4.14.** *Under Model 4.6, the joint log-likelihood function for given observations $N_{1:t}$ is given by*

$$\begin{aligned}\ell_{N_{1:t}}(\alpha_1, p_{1:t-1}, q_{1:t-1}, \mu_{1:t}) &= \sum_{s=1}^{t}\log f\left(N_s \middle| N_{1:s-1}\right) \qquad (4.24)\\ &= \sum_{s=1}^{t}\log\left(\frac{\Gamma\left(\alpha_s+N_s\right)}{\Gamma\left(\alpha_s\right)N_s!}\left(1-\frac{v_s\mu_s}{\beta_s+v_s\mu_s}\right)^{\alpha_s}\left(\frac{v_s\mu_s}{\beta_s+v_s\mu_s}\right)^{N_s}\right).\end{aligned}$$

The joint log-likelihood (4.24) is a function of the parameters $\alpha_1 = \beta_1 > 0$, $p_{1:t-1} \in (0,1)^{t-1}$, $q_{1:t-1} \in \mathbb{R}_+^{t-1}$ and $\mu_{1:t} \in \mathbb{R}_+^t$. The shape parameters $\alpha_s = \alpha_s(N_{1:s-1})$ encode the past claims history $N_{1:s-1}$, and the scale parameters $\beta_s$ are deterministic and do not depend on the past claims history, see definitions (4.16)-(4.17). We are now at the stage



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where we can fit this observation-driven dynamic mixed effects Poisson-gamma model to data. This requires to choose a regression function $\boldsymbol{x}_t \mapsto \mu_t = \mu(\boldsymbol{x}_t)$, and we can select, e.g., (1) the null model $\mu_t = \mu(\boldsymbol{x}_t) \equiv \mu_0$ giving us a dynamic random effects model, (2) a GLM with log-link $\log(\mu_t) = \log(\mu(\boldsymbol{x}_t)) = \langle \boldsymbol{\beta}, \boldsymbol{x}_t \rangle$, or (3) a FNN regression model with log-link $\log(\mu_t) = \log(\mu(\boldsymbol{x}_t)) = \langle \boldsymbol{\beta}, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_t) \rangle$. These regression models can be fitted with empirical Bayesian methods either with (a) full MLE approach using the log-likelihood (4.24) or with (b) the one-year ahead MLE approach using Theorem 4.12, we refer to Section 3.2.6. The main difficulty is to efficiently encode these shape parameters $\alpha_s = \alpha_s(N_{1:s-1})$ and scale parameters $\beta_s$ so that we can perform gradient descent because the regression terms $\mu_t = \mu(\boldsymbol{x}_t)$ enter all these (recursive) shape and scale parameters, see (4.16)-(4.17).

### 4.3.4 Experience rating and credibility formula

Above, we have presented all the necessary tools to provide the Bühlmann credibilities of the observation-driven dynamic mixed effects Model 4.6. The following corollary is an immediate consequence of Theorem 4.12 and Remark 4.13.

**Corollary 4.15** (recursive Bühlmann credibility formula)**.** *For experience rating, we have the following scale-free conditional expectations in Model 4.6*

$$\lambda_{t+1|1:t} := \mathbb{E}\left[ \frac{N_{t+1}}{v_{t+1}\mu_{t+1}} \middle| N_{1:t} \right] = \frac{\alpha_{t+1}}{\beta_{t+1}} \qquad \text{for } t \geq 0.$$

*For $t \geq 1$ we can expand this recursively providing a Bühlmann credibility formula*

$$\lambda_{t+1|1:t} = p_t \left( \gamma_t \frac{N_t}{v_t \mu_t} + (1-\gamma_t)\, \lambda_{t|1:t-1} \right) + (1-p_t)\,, \tag{4.25}$$

*with deterministic (one-year ahead prediction) credibility weights*

$$\gamma_t = \frac{v_t \mu_t}{v_t \mu_t + \beta_t} = \frac{q_t}{\beta_{t+1}}\, v_t \mu_t \ \in \ (0,1). \tag{4.26}$$

The term in (4.26) can be interpreted in terms of the last term in (4.19).

We recall the remark on the notation. $N_t$ gives the claim counts in its additive form, and $Y_t = N_t/v_t$ in its reproductive form in the Poisson model with $\varphi = 1$. This leads to the weighted empirical frequency at time $T$, see (3.4),

$$\bar{Y}_T = \frac{1}{\sum_{t=1}^T v_t} \sum_{t=1}^T v_t Y_t = \frac{1}{\sum_{t=1}^T v_t} \sum_{t=1}^T N_t = \sum_{t=1}^T \frac{v_t}{\sum_{s=1}^T v_s} \frac{N_t}{v_t}.$$

This random variable lives on the scale of the means $\mu_{1:T}$ (expected frequencies). We have also considered the random variables $Y_t/\mu_t = N_t/(v_t\mu_t)$. This lead to the normalized empirical frequencies, see (3.41),

$$\bar{\Lambda}_T = \frac{1}{\sum_{t=1}^T v_t \mu_t} \sum_{t=1}^T v_t Y_t = \frac{1}{\sum_{t=1}^T v_t \mu_t} \sum_{t=1}^T N_t = \sum_{t=1}^T \frac{v_t \mu_t}{\sum_{s=1}^T v_s \mu_s} \frac{N_t}{v_t \mu_t}.$$

This random variable lives on the unit scale, which is the same as the scale-free conditionally expected mean quantities $(\lambda_{t|1:t-1})_{t=1}^T$.



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**Proof of Corollary 4.15.** The first claim follows from Theorem 4.12 and Remark 4.13. Consider the following ratio is for $t \geq 1$ given by, see (4.16)-(4.17),

$$\begin{aligned}\frac{\alpha_{t+1}}{\beta_{t+1}} &= \frac{p_t q_t(\alpha_t + N_t) + (1-p_t)\beta_{t+1}}{\beta_{t+1}} \\ &= p_t \frac{\alpha_t + N_t}{\beta_t + v_t\mu_t} + 1 - p_t \\ &= p_t \left[\frac{v_t\mu_t}{\beta_t + v_t\mu_t}\frac{N_t}{v_t\mu_t} + \frac{\beta_t}{\beta_t + v_t\mu_t}\frac{\alpha_t}{\beta_t}\right] + 1 - p_t.\end{aligned}$$

This proves the claim. □

In the previous derivations we have carefully distinguished between the posterior means in the additive and in the reproductive form, respectively,

$$\mu^{\text{post}}_{N_{t+1}|N_{1:t}} = \mathbb{E}\left[N_{t+1}|\, N_{1:t}\right] \qquad \text{and} \qquad \mu^{\text{post}}_{Y_{t+1}|Y_{1:t}} = \mathbb{E}\left[Y_{t+1}|\, Y_{1:t}\right].$$

Corollary 4.15 considers the scale-free posterior mean $\lambda_{t+1|1:t}$ which scales as follows

$$\mu^{\text{post}}_{N_{t+1}|N_{1:t}} = \frac{v_{t+1}}{\varphi}\mu^{\text{post}}_{Y_{t+1}|Y_{1:t}} = v_{t+1}\mu_{t+1}\,\lambda_{t+1|1:t},$$

with $\varphi = 1$ in the Poisson case.

**Interpretation of Corollary 4.15.** The past claims $N_{1:t}$ may receive a seniority weighting in the experience rating premium $\lambda_{t+1|1:t}$. Namely, any claim $N_s$, $1 \leq s \leq t$, provides a contribution

$$\prod_{u=s}^{t} p_u\,\gamma_s \prod_{u=s+1}^{t}(1-\gamma_u)\,\frac{N_s}{v_s\mu_s} = \omega_{t,s}\,\frac{N_s}{v_s\mu_s},$$

to the experience rating premium $\lambda_{t+1|1:t}$, this is similar to (4.7); and where we have defined the credibility weights

$$\omega_{t,s} = p_s\gamma_s \prod_{u=s+1}^{t} p_u\,(1-\gamma_u) \;\in\; (0,1). \tag{4.27}$$

Note that $\omega_{t,s} \in (0,1)$ are the unfolded credibility weights, whereas $\gamma_t \in (0,1)$ are the one-year ahead forecast credibility weights in the recursive formula (4.25). The prior mean $\lambda_{1|1:0} = \alpha_1/\beta_1 = 1$ gets an unfolded credibility weight of, set $p_0 = 0$,

$$1 - \sum_{s=1}^{t}\omega_{t,s} = \sum_{s=0}^{t}(1-p_s)\prod_{u=s+1}^{t} p_u\,(1-\gamma_u) \;\in\; (0,1).$$

Corollary 4.15 then gives the inhomogeneous Bühlmann credibility formula, given observations $N_{1:T}$,

$$\lambda_{t+1|1:t} = \sum_{s=1}^{t}\omega_{t,s}\frac{N_s}{v_s\mu_s} + \left(1 - \sum_{s=1}^{t}\omega_{t,s}\right)\cdot 1. \tag{4.28}$$

Depending on the (deterministic) choices of the sequences $(v_t\mu_t)_{t\geq 1}$, $(p_t)_{t\geq 1}$ and $(q_t)_{t\geq 1}$, the contributions in (4.28) can have different behaviors. For example, if $p_t \equiv p \in (0,1)$, we have at least an exponential decrease in seniority weighting because

$$\omega_{t,s} = p^{t-s+1}\,\gamma_s \prod_{u=s+1}^{t}(1-\gamma_u) \leq e^{-(t-s+1)\log(1/p)}.$$



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## 4.4 Dynamic random effects Poisson-gamma model

We start this section by discussing the different cases of Table 4.1 in the dynamic random effects case, i.e., not considering any covariates $\boldsymbol{x}_t$. For simplicity, in this first step, we assume unit exposure $v_t \equiv 1$. Under a null model assumption, we then have

$$v_t \mu_t = \mu_0 > 0 \qquad \text{for all } t \geq 1. \tag{4.29}$$

We analyze the cases of Table 4.1, and for this first step we assume time-homogeneity by setting for all $t \geq 1$

$$p_t \equiv p \in [0,1] \qquad \text{and} \qquad q_t \equiv q \in (0,1]. \tag{4.30}$$

We call this time-homogeneous null case the *dynamic random effects null model*.

### 4.4.1 Credibility formula in the dynamic random effects null model

We analyze the Bühlmann credibility formulas (4.25) and (4.28), respectively, under the different cases of Table 4.1 in the dynamic random effects null model.

**Proposition 4.16.** *Assume the dynamic random effects null model* (4.29)-(4.30)*. We have Bühlmann credibility formula for* $t \geq 1$

$$\lambda_{t+1|1:t} = \sum_{s=1}^{t} \omega_{t,s} \frac{N_s}{\mu_0} + \left(1 - \sum_{s=1}^{t} \omega_{t,s}\right) \cdot 1,$$

*with credibility weights for* $1 \leq s \leq t$

$$\omega_{t,s} = p^{t+1-s} \frac{q^{1-s}}{\sum_{u=1}^{t} q^{1-u} + \alpha_1/\mu_0} \ \in \ (0,1).$$

**Proof of Proposition 4.16.** We first expand the scale parameters $(\beta_{t+1})_{t\geq 1}$ under the homogeneous null model assumptions (4.29)-(4.30)

$$\beta_{t+1} = q(\beta_t + \mu_0) \ = \ q^2(\beta_{t-1} + \mu_0) + q\mu_0 = \cdots = q^t \beta_1 + \mu_0 \sum_{s=1}^{t} q^s \ \overset{(*)}{=} \ q^t \beta_1 + \mu_0 q \frac{1-q^t}{1-q}, \tag{4.31}$$

the last identity $\overset{(*)}{=}$ requires $q < 1$, however, this last identity is not used any further in this proof, but we give it for later reference. Analogously, we have for the shape parameters $(\alpha_{t+1})_{t\geq 1}$

$$\begin{aligned}
\alpha_{t+1} &= q\left[p(\alpha_t + N_t) + (1-p)(\beta_t + \mu_0)\right] \ = \ qp\alpha_t + qpN_t + (1-p)\beta_{t+1} \\
&= qp(qp\alpha_{t-1} + qpN_{t-1} + (1-p)\beta_t) + qpN_t + (1-p)\beta_{t+1} \ = \ \cdots \\
&= (qp)^t \beta_1 + \sum_{s=1}^{t} (pq)^{t+1-s} N_s + (1-p) \sum_{s=1}^{t} (pq)^{t-s} \beta_{s+1},
\end{aligned}$$

using $\alpha_1 = \beta_1$. We introduce the following weights dropping the $p$-deflation

$$\begin{aligned}
\widetilde{\omega}_{t,s} &= p^{-(t+1-s)} \omega_{t,s} \ = \ \frac{\mu_0 q^{t+1-s}}{q^t \beta_1 + \mu_0 \sum_{u=1}^{t} q^{t+1-u}} \ \in \ (0,1), \qquad \textit{and} \\
\widetilde{\omega}_{t,0} &= \frac{q^t \beta_1}{q^t \beta_1 + \mu_0 \sum_{u=1}^{t} q^u} \ = \ 1 - \sum_{s=1}^{t} \widetilde{\omega}_{t,s} \ \in \ (0,1).
\end{aligned}$$



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We now divide the shape by the scale parameter to obtain

$$\begin{aligned}\frac{\alpha_{t+1}}{\beta_{t+1}} &= \frac{(qp)^t\beta_1 + \sum_{s=1}^t (pq)^{t+1-s} N_s + (1-p)\sum_{s=1}^t (pq)^{t-s}\beta_{s+1}}{q^t\beta_1 + \mu_0 \sum_{s=1}^t q^s} \\ &= p^t\,\widetilde{\omega}_{t,0} + \sum_{s=1}^t \frac{(pq)^{t+1-s} N_s}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u} + (1-p)\sum_{s=1}^t \frac{(pq)^{t-s}\beta_{s+1}}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u}.\end{aligned}$$

We analyze the last two terms. We have for the middle term

$$\sum_{s=1}^t \frac{(pq)^{t+1-s} N_s}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u} = \sum_{s=1}^t p^{t+1-s}\frac{\mu_0 q^{t+1-s}}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^{t+1-u}}\frac{N_s}{\mu_0} = \sum_{s=1}^t \omega_{t,s}\,\frac{N_s}{\mu_0}.$$

For the last term plug in (4.31) and compute the following

$$\begin{aligned}(1-p)\sum_{s=1}^t \frac{(pq)^{t-s}\beta_{s+1}}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u} &= (1-p)\sum_{s=1}^t \frac{(pq)^{t-s}\left(q^s\beta_1 + \mu_0\sum_{k=1}^s q^k\right)}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u} \\ &= (1-p)\sum_{s=1}^t p^{t-s}\frac{q^t\beta_1}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u} + p^{t-s}\frac{\mu_0\sum_{k=1}^s q^{t+k-s}}{q^t\beta_1 + \mu_0\sum_{u=1}^t q^u} \\ &= (1-p)\,\widetilde{\omega}_{t,0}\sum_{s=0}^{t-1} p^s + (1-p)\sum_{s=1}^t p^{t-s}\sum_{k=1}^s \widetilde{\omega}_{t,k} \\ &= (1-p^t)\,\widetilde{\omega}_{t,0} + (1-p)\sum_{s=1}^t p^{t-s}\sum_{k=1}^s \omega_{t,k}\,p^{-(t+1-k)} \\ &= (1-p^t)\,\widetilde{\omega}_{t,0} + (1-p)\sum_{k=1}^t \omega_{t,k}\,p^{-(t+1-k)}\sum_{u=0}^{t-k} p^u \\ &= (1-p^t)\,\widetilde{\omega}_{t,0} + \sum_{k=1}^t \omega_{t,k}\left(p^{-(t+1-k)} - 1\right).\end{aligned}$$

Collecting all terms gives us

$$\begin{aligned}\frac{\alpha_{t+1}}{\beta_{t+1}} &= \widetilde{\omega}_{t,0} + \sum_{s=1}^t \omega_{t,s}\,\frac{N_s}{\mu_0} + \sum_{s=1}^t \omega_{t,s}\left(p^{-(t+1-s)} - 1\right) \\ &= \widetilde{\omega}_{t,0} + \sum_{s=1}^t \widetilde{\omega}_{t,s} + \sum_{s=1}^t \omega_{t,s}\left(\frac{N_s}{\mu_0} - 1\right) \\ &= 1 + \sum_{s=1}^t \omega_{t,s}\left(\frac{N_s}{\mu_0} - 1\right) \;=\; \sum_{s=1}^t \omega_{t,s}\,\frac{N_s}{\mu_0} + \left(1 - \sum_{s=1}^t \omega_{t,s}\right).\end{aligned}$$

This completes the proof. □

The credibility weights $\omega_{t,s}$ can further be computed, in the static random effects case ($p = q = 1$) we obtain the weights for observations $N_s/\mu_0$

$$\omega_{t,s} = \frac{t}{t + \alpha_1/\mu_0}\,\frac{1}{t} \;\in\; (0,1).$$

This is precisely the result of the inhomogeneous Bühlmann–Straub credibility formula [42] for exposures $v_s \equiv 1$; see (3.4).

**Corollary 4.17.** *The credibility weights of Proposition 4.16 fulfill for $s \geq 1$ and $\Delta \geq 0$*

$$\omega_{s+\Delta,s} = \begin{cases} (1-q)\,p\,\frac{p^\Delta}{(q^{-\Delta}-1)+(1-q^s)+q^{s-1}(1-q)\alpha_1/\mu_0} & \text{for } q < 1, \\ p\,\frac{p^\Delta}{\Delta+s+\alpha_1/\mu_0} & \text{for } q = 1. \end{cases}$$



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*In the case $q < 1$ we have asymptotic behavior (exponential decay), for fixed $s \in \mathbb{N}$,*

$$\lim_{t\to\infty} (pq)^{-t}\, \omega_{t,s} = (1-q)p\,(pq)^{-s},$$

*and in the case $q = 1$ we have asymptotic behavior, for fixed $s \in \mathbb{N}$,*

$$\lim_{t\to\infty} tp^{-t}\, \omega_{t,s} = p^{1-s}.$$

**Proof of Corollary 4.17.** From Proposition 4.16 we have for $q < 1$

$$\begin{aligned}
\omega_{t,s} &= p^{t+1-s}\, \frac{q^{1-s}}{\sum_{u=1}^{t} q^{1-u} + \alpha_1/\mu_0} = p^{t+1-s}\, \frac{q^{t+1-s}}{q\sum_{u=0}^{t-1} q^{u} + q^t\alpha_1/\mu_0} \\
&= p^{t+1-s}\, \frac{q^{1-s}}{q\frac{q^{-t}-1}{1-q} + \alpha_1/\mu_0} = p^{t+1-s}\, \frac{q^{-s}(1-q)}{(q^{-s}-1) + q^{-s}(q^{-(t-s)}-1) + q^{-1}(1-q)\alpha_1/\mu_0} \\
&= p^{\Delta+1}\, \frac{1-q}{(1-q^s) + (q^{-\Delta}-1) + q^{s-1}(1-q)\alpha_1/\mu_0}.
\end{aligned}$$

This proves the first claim. For $q = 1$ we have

$$\omega_{t,s} = p^{t+1-s}\, \frac{q^{1-s}}{\sum_{u=1}^{t} q^{1-u} + \alpha_1/\mu_0} = p^{t+1-s}\, \frac{1}{t + \alpha_1/\mu_0}.$$

This proves the second claim. The asymptotic behavior is straightforward. □

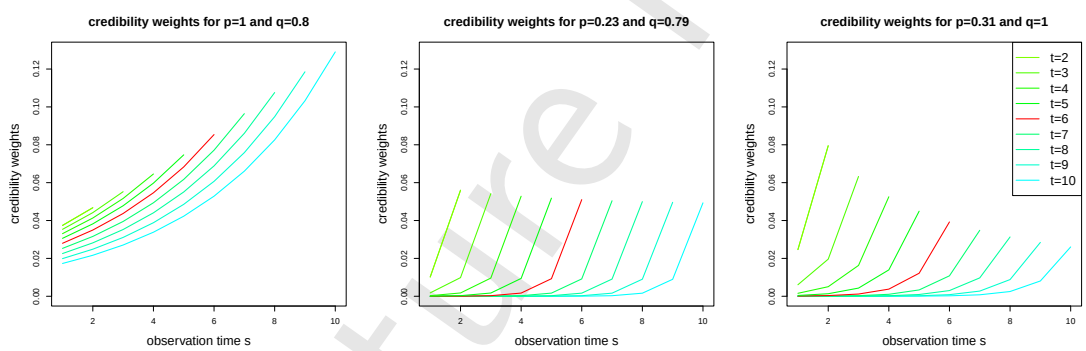


Figure 4.4: Dynamic random effects null model with credibility weights $(\omega_{t,s})_{s=1}^{t}$ for $2 \le t \le 10$ (in different colors): (lhs) increasing variance case, (middle) bounded variance case, and (rhs) decreasing variance case for the different choices of $\alpha_1/\mu_0$, $p$ and $q$; see also Example 4.19, below; the $y$-axis has the same scale in all plots.

Corollary 4.17 shows that we have an exponential decay of the credibility weights if $pq < 1$ (increasing, bounded and decreasing variance cases of Propositions 4.9, 4.10 and 4.11). Figure 4.4 illustrates the three cases for selected parameters for $p$ and $q$ and $\alpha_1/\mu_0$; for the specific selection we refer to Example 4.19, below. The red color shows the case $(\omega_{6,s})_{s=1}^{6}$, i.e., for fixed time $t = 6$. In the bounded variance case (middle), we observe a comparably fast decay of the credibility weights for the selected parameters, basically only the last 2 observations contribute with 5% and 1%, respectively, whereas in the increasing variance case we have a longer memory behavior of past claims experience; note that the selected parameters in Figure 4.4 have been received from fitting the three cases to the same data.

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### 4.4.2 Implementation of the full exposure null model

We provide an implementation of the dynamic random effects null model assuming full (equal) exposures, i.e., assuming $v_t\mu_t \equiv \mu_0 > 0$, see (4.29). This dynamic random effects model is analytically tractable and parameter estimation has a nice structure. This should be seen as a warm-up computation before studying more complex cases.

There are the following parameters to be estimated in this model: $\mu_0 > 0$, $\alpha_1 > 0$, $p_t \equiv p \in (0,1]$ and $q_t \equiv q \in (0,1]$, see (4.29)-(4.30), and we set for these four parameters

$$\vartheta = (\mu_0, \alpha_1, p, q).$$

The log-likelihood function for observation $N_{1:t}$ is given by, see (4.24),

$$\vartheta \;\mapsto\; \ell_{N_{1:t}}(\vartheta) = \sum_{s=1}^{t} \log\left(\frac{\Gamma(\alpha_s + N_s)}{\Gamma(\alpha_s)\, N_s!}\left(1 - \frac{\mu_0}{\beta_s + \mu_0}\right)^{\alpha_s}\left(\frac{\mu_0}{\beta_s + \mu_0}\right)^{N_s}\right).$$

**Lemma 4.18.** *The scale and shape parameters in the dynamic random effects null model with full exposures are given by, recall $\alpha_1 = \beta_1 > 0$,*

$$\beta_{t+1} = \begin{cases} q^t\alpha_1 + \mu_0 q\,\frac{1-q^t}{1-q} & \text{for } q < 1, \\ \alpha_1 + t\mu_0 & \text{for } q = 1, \end{cases}$$

*and*

$$\alpha_{t+1} = \begin{cases} \sum_{s=1}^{t}(pq)^{t+1-s}N_s + \alpha_1 q^t + \frac{(1-p)\mu_0 q}{1-q}\left[\frac{1-(pq)^t}{1-pq} - q^t\frac{1-p^t}{1-p}\right] & \text{for } q < 1 \text{ and } p < 1, \\ \sum_{s=1}^{t} q^{t+1-s}N_s + \alpha_1 q^t & \text{for } q < 1 \text{ and } p = 1, \\ \sum_{s=1}^{t} p^{t+1-s}N_s + \alpha_1 + t\mu_0 - \mu_0 p\,\frac{1-p^t}{1-p} & \text{for } q = 1 \text{ and } p < 1. \end{cases}$$

**Proof of Lemma 4.18.** The formula of the scale parameters is proved in (4.31). We turn our attention to the shape parameters. For $q < 1$ we have, see displayed formula after (4.31),

$$\begin{aligned} \alpha_{t+1} &= (qp)^t\beta_1 + \sum_{s=1}^{t}(pq)^{t+1-s}N_s + (1-p)\sum_{s=1}^{t}(pq)^{t-s}\beta_{s+1} \\ &= (qp)^t\beta_1 + \sum_{s=1}^{t}(pq)^{t+1-s}N_s + (1-p)\sum_{s=1}^{t}(pq)^{t-s}\left(q^s\beta_1 + \mu_0 q\frac{1-q^s}{1-q}\right) \\ &= (qp)^t\beta_1 + \sum_{s=1}^{t}(pq)^{t+1-s}N_s + (1-p)\beta_1 q^t\sum_{s=0}^{t-1}p^s + \frac{(1-p)\mu_0 q}{1-q}\sum_{s=0}^{t-1}(pq)^s - \frac{(1-p)\mu_0 q^{t+1}}{1-q}\sum_{s=0}^{t-1}p^s. \end{aligned}$$

For $q < 1$ and $p < 1$ this is computed as follows

$$\begin{aligned} \alpha_{t+1} &= (qp)^t\beta_1 + \sum_{s=1}^{t}(pq)^{t+1-s}N_s + (1-p^t)\beta_1 q^t + \frac{(1-p)\mu_0 q}{1-q}\frac{1-(pq)^t}{1-pq} - \frac{(1-p^t)\mu_0 q^{t+1}}{1-q} \\ &= \sum_{s=1}^{t}(pq)^{t+1-s}N_s + \beta_1 q^t + \frac{(1-p)\mu_0 q}{1-q}\left[\frac{1-(pq)^t}{1-pq} - q^t\frac{1-p^t}{1-p}\right]. \end{aligned}$$



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For $q < 1$ and $p = 1$ the claim is immediate. For $q = 1$ and $p < 1$ we have

$$\begin{aligned}
\alpha_{t+1} &= p^t\beta_1 + \sum_{s=1}^{t} p^{t+1-s}N_s + (1-p)\sum_{s=1}^{t} p^{t-s}(\beta_1 + s\mu_0) \\
&= p^t\beta_1 + \sum_{s=1}^{t} p^{t+1-s}N_s + (1-p^t)\beta_1 + (1-p)\mu_0\sum_{s=0}^{t-1} p^s(t-s) \\
&= \sum_{s=1}^{t} p^{t+1-s}N_s + \beta_1 + (1-p^t)t\mu_0 - (1-p)\mu_0\sum_{s=0}^{t-1} p^s s \\
&= \sum_{s=1}^{t} p^{t+1-s}N_s + \beta_1 + (1-p^t)t\mu_0 - (1-p)\mu_0\sum_{s=0}^{t-1} p^s(s+1-1) \\
&= \sum_{s=1}^{t} p^{t+1-s}N_s + \beta_1 + (1-p^t)t\mu_0 - (1-p)\mu_0\frac{d}{dp}\sum_{s=0}^{t-1} p^{s+1} + (1-p)\mu_0\sum_{s=0}^{t-1} p^s \\
&= \sum_{s=1}^{t} p^{t+1-s}N_s + \beta_1 + (1-p^t)t\mu_0 - (1-p)\mu_0\frac{d}{dp}\frac{p-p^{t+1}}{1-p} + \mu_0(1-p^t) \\
&= \sum_{s=1}^{t} p^{t+1-s}N_s + \beta_1 + (1-p^t)t\mu_0 - \mu_0 + \mu_0(t+1)p^t - \mu_0\frac{p-p^{t+1}}{1-p} + \mu_0(1-p^t) \\
&= \sum_{s=1}^{t} p^{t+1-s}N_s + \beta_1 + t\mu_0 - \mu_0\frac{p-p^{t+1}}{1-p}.
\end{aligned}$$

This completes the proof of the lemma. □

For the implementation of the shape parameters of Lemma 4.18, it is useful to notice that the first sums can be written as a matrix multiplication

$$\begin{pmatrix} \sum_{s=1}^{1}(pq)^{2-s}N_s \\ \sum_{s=1}^{2}(pq)^{3-s}N_s \\ \vdots \\ \sum_{s=1}^{t}(pq)^{t+1-s}N_s \end{pmatrix} = \begin{pmatrix} pq & 0 & 0 & \cdots & 0 & 0 \\ (pq)^2 & pq & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ (pq)^t & (pq)^{t-1} & (pq)^{t-2} & \cdots & (pq)^2 & pq \end{pmatrix} \begin{pmatrix} N_1 \\ N_2 \\ \vdots \\ N_t \end{pmatrix}.$$

The big advantage of this is that it allows one to write the log-likelihood function (4.24) in a non-recursive form, and maximization in parameter $\vartheta$ can be performed using, e.g., the general purpose R optimizer `optim` [166].

This implementation starts to get cumbersome if one makes the parameters $v_{1:t}$, $\mu_{1:t}$, $p_{1:t}$ and $q_{1:t}$ time-dependent. This latter case is going to be solved using the gradient descent method within the R package `keras` [51]. This requires that we code a tailor-made recurrent neural network (RNN) layer which precisely performs the recursions (4.16)-(4.17). This tailor-made RNN layer is also helpful to implement the general regression case that considers the regression functions $\boldsymbol{x}_t \mapsto \mu_t = \mu(\boldsymbol{x}_t)$, being functions of the covariates $\boldsymbol{x}_t$ at time $t$; this is discussed in detail in Section 4.4.3, below.

**Example 4.19.** We consider an example. Since we have assumed full exposures $v_t = 1$ for all considered calendar years $2011 \leq t \leq 2019$ in this section, we fit the dynamic random effects null model only to the insurance policies $i$ that are fully exposed during the entire time period considered. In our example, these are $n = 56,806$ insurance policies. We perform this analysis to get some intuition whether we should expect a decreasing, bounded or increasing variance behavior, see Table 4.1. This is empirically



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tested by analyzing the resulting estimates of $p$ and $q$. However, there is one caveat (besides a potential model error), namely, considering only insurance policies that are exposed during the entire time period may lead to a (survival) bias, because policyholders without claims likely stay longer within a portfolio, as a claim event is a potential trigger for changing the insurer.

| | dynamic random effects null models | | | |
|---|---|---|---|---|
| | increasing variance | bounded variance | decreasing variance | static effect |
| $\widehat{\mu}_0$ | 0.0825 | 0.0823 | 0.0808 | 0.0825 |
| $\widehat{\alpha}_1$ | 2.0199 | 0.2414 | 0.1535 | 1.1351 |
| $\widehat{p}$ | 1 | 0.2301 | 0.3100 | 1 |
| $\widehat{q}$ | 0.8021 | 0.7885 | 1 | 1 |
| $\ell_{N_{1:T}}(\widehat{\vartheta})/n$ | -2.6199 | -2.5853 | -2.5939 | -2.6271 |
| AIC | 297,662 | 293,724 | 294,709 | 298,474 |

Table 4.2: Results of the dynamic random effects null models under model assumptions (4.29)-(4.30).

We implement the log-likelihood (4.24) using Lemma 4.18 for its analytical calculation. Optimization is performed by the R optimizer `optim` [166]. Table 4.2 presents the results. In view of Akaike's information criterion (AIC) [8], we give a clear preference to the bounded variance case. In this case the parameter estimates $\widehat{p} = 0.2301$ and $\widehat{q} = 0.7885$ are well bounded away from zero and one. In Figure 4.4 we have plotted precisely these numerical examples, and we observe that we prefer a short memory and the seniority weights decay very quickly in this estimated model. The static random effects case $p = q = 1$ is obtained from (3.30) by subtracting the term $\sum_{i,t} \log(N_{i,t}!)$ which has been dropped through the proportionality sign $\propto$. Some further analysis (not shown here) indicates that this model does not fit the data well and we need more sophisticated models, also considering the covariates $\boldsymbol{x}_t$. ■

**Remark 4.20.** There is one issue though that needs careful consideration. Namely, one should also study the credibility weight assigned to the prior mean $\lambda_{1|1:0} = 1$. For $q < 1$ it is given by

$$\omega_{t,0} := 1 - \sum_{s=1}^{t} \omega_{t,s} = 1 - \frac{q^{-t} \sum_{s=1}^{t} (pq)^{t+1-s}}{q^{-t} \sum_{u=1}^{t} q^{t+1-u} + \alpha_1/\mu_0} = 1 - \frac{q^{1-t} p \, \frac{1-(pq)^t}{1-pq}}{q^{1-t} \, \frac{1-q^t}{1-q} + \alpha_1/\mu_0}, \tag{4.32}$$

see Proposition 4.16. A customer would expect that the sequence $(\omega_{t,0})_{t \geq 1}$ is monotonically decreasing in $t$. However, this is not necessarily the case. For instance, for the parameters $\alpha_1/\mu_0$, $p$ and $q$ selected in Figure 4.4 (middle) these weights assigned to the prior mean are not monotonically decreasing in time $t$. Thus, this parameter selection should be questioned in practical applications, though it is empirical Bayesian MLE optimal. ■



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### 4.4.3 Implementation of the dynamic random effects model

We relax the full exposure assumption (4.29) and (4.30) by letting $v_t > 0$, $p_t \in (0,1]$ and $q_t \in (0,1]$ be time-dependent, we only keep the prior mean $\mu_0 > 0$ as time-independent (and of course the initial value $\alpha_1 = \beta_1 > 0$). This is the general observation-driven dynamic random effects Model 4.6. In this case, we can no longer use Lemma 4.18 for model fitting. We design a RNN layer to fit this model recursively using the gradient descent method. The R code of this tailor-made RNN layer is shown in Listing 4.1.

Listing 4.1: Recurrent neural network (RNN) layer for $(\beta_t)_{t\geq 1}$ and $(\alpha_t)_{t\geq 1}$.

```
1  library(keras)
2  library(tensorflow)
3  ########
4  # This code is based on "Working with RNNs", date rendered 2022-12-16
5  # Authors Scott Zhu, Francois Chollet, Tomasz Kalinowski
6  # https://tensorflow.rstudio.com/guides/keras/working_with_rnns
7  ########
8  BetaAlphaCell(keras$layers$Layer) %py_class% {
9    initialize <- function(unit_1, unit_2, ...) {
10     self$unit_1      <- unit_1
11     self$unit_2      <- unit_2
12     self$state_size  <- list(shape(unit_1), shape(unit_2))
13     self$output_size <- list(shape(unit_1), shape(unit_2))
14     super$initialize(...)
15   }
16   call <- function(inputs, states) {
17     c(input_1, input_2, input_3, input_4) %<-% tf$nest$flatten(inputs)
18     c(s1, s2) %<-% states
19     # beta sequence
20     output_1   <- input_1 * (s1 + input_2)
21     state_1    <- output_1
22     # alpha sequence
23     output_2   <- input_3 * input_1 * (s2 + input_4) + (1-input_3) * output_1
24     state_2    <- output_2
25     #
26     output     <- tuple(output_1, output_2)
27     new_states <- tuple(state_1, state_2)
28     tuple(output, new_states)
29   }
30   get_config <- function(){list("unit_1" = self$unit_1, "unit_2" = self$unit_2)}
31 }
32 ########
33 cell      <- BetaAlphaCell(1, 1)
34 RNN       <- layer_rnn(cell = cell, return_sequence=TRUE)
35 BetaAlpha <- RNN(tuple(q, mu, p, N))
```

The R code in Listing 4.1 is based on Zhu et al. [209]. It designs a RNN cell that recursively computes $(\beta_t, \alpha_t)_{t\geq 1}$. Line 33 defines a `cell` that has one unit for $(\beta_t)_{t\geq 1}$ and one unit for $(\alpha_t)_{t\geq 1}$. These are one-dimensional processes. Lines 12-13 and 26-28 of `BetaAlphaCell` define the memory `state` that is forwarded to the next loop and the output variable of loop $t$. The first output defined on line 20 corresponds to $\beta_{t+1} = q_t(\beta_t + v_t\mu_t)$. The new first state component on line 21 corresponds to the same $\beta_{t+1}$. The second output defined on line 23 corresponds to $\alpha_{t+1} = p_t q_t(\alpha_t + N_t) + (1-p_t)\beta_{t+1}$, and the new second state component on line 24 carries $\alpha_{t+1}$ forward to the next time period. Line 34 defines this RNN layer in a variable `RNN` returning the entire output sequence

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$(\beta_t, \alpha_t)_{t=1}^T$, and line 35 states the input variables to this RNN layer. This code allows us to fit flexible time-dependent models to the data using gradient descent methods.

Listing 4.2: RNN for dynamic random effects null model fitting.

```
network.null.model <- function(seed, q, init){
    k_clear_session()
    set.seed(seed)
    set_random_seed(seed)
    ##
    NN      = layer_input(shape = c(q[1]),   dtype = 'float32')
    one     = layer_input(shape = c(1),      dtype = 'float32')
    mu0     = layer_input(shape = c(q[1],1), dtype = 'float32')
    timep   = layer_input(shape = c(q[1],1), dtype = 'float32')
    timeq   = layer_input(shape = c(q[1],1), dtype = 'float32')
    ##
    alpha = one %>% layer_dense(units=1, use_bias=FALSE, activation='exponential',
                                weights=list(array(init[2], dim=c(1,1))))
    NN1 = list(alpha, NN) %>% layer_concatenate() %>%
                               layer_reshape(target_shape=c(q[1]+1,1))
    ##
    p0 = timep %>% layer_embedding(input_dim=q[1], output_dim=1, input_length=1,
                   weights=list(array(init[3], dim=c(q[1],1)))) %>%
                   layer_flatten() %>% layer_lambda(function(x) k_sigmoid(x)) %>%
                                       layer_reshape(target_shape=c(q[1]))
    p1 = list(one, p0) %>% layer_concatenate() %>%
                           layer_reshape(target_shape=c(q[1]+1,1))
    ##
    q0 = timeq %>% layer_embedding(input_dim=q[1], output_dim=1, input_length=1,
                   weights=list(array(init[4], dim=c(q[1],1)))) %>%
                   layer_flatten() %>% layer_lambda(function(x) k_sigmoid(x)) %>%
                                       layer_reshape(target_shape=c(q[1]))
    q1 = list(one, q0) %>% layer_concatenate() %>%
                           layer_reshape(target_shape=c(q[1]+1,1))
    ##
    mu = mu0 %>% layer_embedding(input_dim=q[1], output_dim=1, input_length=1,
                 weights=list(array(init[1], dim=c(q[1],1)))) %>%
                 layer_flatten() %>% layer_lambda(function(x) k_exp(x)) %>%
                                     layer_reshape(target_shape=c(q[1]))
    mu1 = list(alpha, mu) %>% layer_concatenate() %>%
                              layer_reshape(target_shape=c(q[1]+1,1))
    ##
    cell    = BetaAlphaCell(1, 1)
    rnn     = layer_rnn(cell = cell, return_sequence=TRUE)
    Network = rnn(tuple(q1, mu1, p1, NN1)) %>% layer_concatenate()
    outputs = list(mu1, p1, q1, Network)  %>% layer_concatenate()
    ##
    keras_model(inputs = c(NN, one, mu0, timep, timeq), outputs = list(outputs))
    }
```

Listing 4.2 shows the application of our RNN layer of Listing 4.1 to fit the same dynamic random effects null model as in Section 4.4.2, i.e., for this null model we set $v_t\mu_t \equiv \mu_0 > 0$, $p_t \equiv p \in (0,1)$ and $q_t \equiv q \in (0,1)$, see (4.29)-(4.30). The points to be highlighted are that we choose the activation functions so that all parameters are restricted to their domains, and we initialize the sequence $(N_t, \mu_0, p_t, q_t)_{t\geq 1}$ with a 0th state $(\beta_1, \alpha_1, 1, 1)$ with $\alpha_1 = \beta_1 > 0$ because the `BetaAlphaCell` of Listing 4.1 has initial `states` zero. If we choose as inputs for `mu0`, `timep` and `timeq` tensors that only contain 0, we are precisely in the dynamic random effects null model of Section 4.4.2; note that in Listing



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4.1, lines 17, 24 and 31, we use one-dimensional embedding layers for these null model parameters, and setting their values to zero assigns the first embedding parameters to them. We fit this model to the data using the gradient descent algorithm and we would like to verify that everything runs properly by replicating the results of Table 4.2. The first two columns of Table 4.3 show the two different fitting methods, `optim` and RNN layer, and it is verified that they provide the same results (up to small numerical errors).

| | models (4.29)-(4.30) | | flexible non-parametric | | | flexible | |
|---|---|---|---|---|---|---|---|
| | `optim` | RNN | $p_t \in (0,1)$ | $q_t \in (0,1)$ | $\mu_t > 0$ | $(p_t, q_t)$ | $(p_t, q_t, \mu_t)$ |
| $\widehat{\mu}_0$; $\widehat{\mu}_t$ | 0.0823 | 0.0824 | 0.0824 | 0.0826 | – | 0.0826 | – |
| $\widehat{\alpha}_1$ | 0.2414 | 0.2414 | 0.2391 | 0.2062 | 0.2579 | 0.2009 | 0.2177 |
| $\widehat{p}$; $\widehat{p}_t$ | 0.2301 | 0.2304 | – | 0.2325 | 0.2365 | – | – |
| $\widehat{q}$; $\widehat{q}_t$ | 0.7885 | 0.7884 | 0.7913 | – | 0.7824 | – | – |
| $\ell_{\boldsymbol{Y}}(\widehat{\vartheta})/n$ | -2.5853 | -2.5853 | -2.5852 | -2.5848 | -2.5829 | -2.5846 | -2.5823 |
| ♯parameters | 4 | 4 | 12 | 12 | 12 | 20 | 28 |
| AIC | 293,724 | 293,724 | 293,731 | 293,691 | 293,469 | 293,687 | 293,438 |

Table 4.3: Bounded variance cases with non-parametric estimation of $(p_t)_t$, $(q_t)_t$ and/or $(\mu_t)_t$.

The RNN fitting approach has the advantage that we can now easily relax the constant parameter assumptions $\mu_t \equiv \mu_0 > 0$, $p_t \equiv p \in (0,1)$ and/or $q_t \equiv q \in (0,1)$ by fitting these parameters individually (non-parametrically) for all considered calendar years $t \geq 1$. This is achieved by inputting tensors with entries $0, \ldots, T-1$ for the variables `mu0`, `timep` and `timeq` on line 43 of Listing 4.2. On lines 17, 24 and 31 they are then embedded so that each time point $t$ receives its own parameter(s). We first make only one variable time-dependent at a time. Columns 3-5 of Table 4.3 show the results. The biggest decrease in AIC is observed by letting the expected frequency $\mu_t$ be time-dependent. Letting $q_t$ be time-dependent leads to a smaller decrease in AIC, and a flexible modeling of $p_t$ is not justified by AIC. This indicates that non-stationarity of the expected frequency may be the biggest issue in this time-series data.

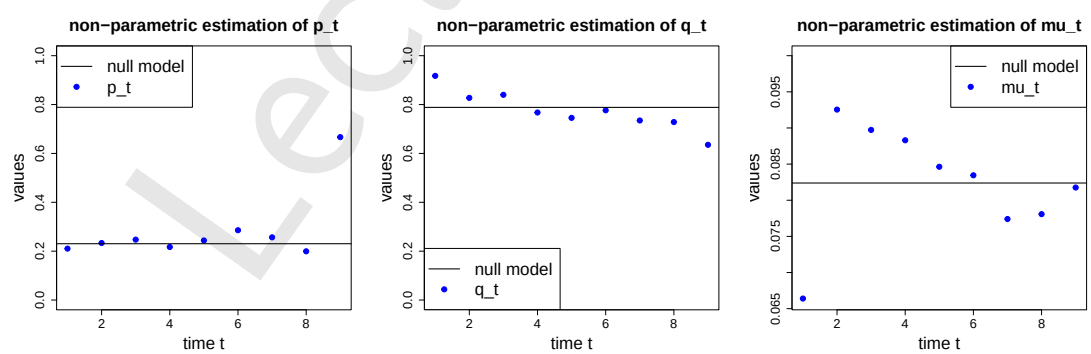


Figure 4.5: Bounded variance cases with non-parametric estimation of $(p_t)_t$, $(q_t)_t$ or $(\mu_t)_t$; this corresponds to columns (3)-(5) of Table 4.3.

Figure 4.5 shows the estimated time-series $(\widehat{p}_t)_t$, $(\widehat{q}_t)_t$ and $(\widehat{\mu}_t)_t$ if we estimate these



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parameters non-parametrically time-dependent. The sequence $(\widehat{p}_t)_t$ is quite stable, only the last estimation stands out, this proposes to model $p_t \equiv p$ not time-dependent. The sequence $(\widehat{q}_t)_t$ has a downward trend, and the sequence $(\widehat{\mu}_t)_t$ shows the non-stationarity of the empirical frequency, compare to Figure A.1 (rhs); the first value $\widehat{\mu}_1$ does not seem to be a reliable estimate, may be this comes from the fact that this expected frequency has been estimated without any history, i.e., this corresponds to the initial calendar year of our observations.

If we keep the frequency estimate time-independent, and only estimate the values $(p_t)_t$ and $(q_t)_t$ time-dependent, we receive the second last column of Table 4.3. This has a lower AIC than the other versions and hence this joint time-dependent modeling of $(p_t)_t$ and $(q_t)_t$ seems preferable. Finally, the last column Table 4.3 shows the case where we let all three sequences be time-dependent. This results in the best of these models w.r.t. AIC. Below, we will not study such non-parametric models because they are more difficult to extrapolate as this needs to be done manually. This completes this example and, in particular, we are now equipped with a RNN layer that allows us to flexibly fit more advanced dynamic mixed effects claim count models.

## 4.5 Lab: Dynamic mixed effects Poisson-gamma model

We extend the dynamic random effects model studied in the previous section to a dynamic mixed effects model. To have direct comparability to the results of Tables 4.2 and 4.3, we start in Example 4.21 by only considering the insurance contracts that are exposed during the entire observation period, these are $n = 56,806$ insurance policies, see Example 4.19. The fully general case considering all available data is going to be presented in Example 4.22, below.

**Example 4.21.** In the previous section we have studied the dynamic random effects (null) model in the Poisson-gamma case. In particular, we have assumed covariate independent parameters $\mu_0$ (or $\mu_t$), $\alpha_1$, $p_t$ and $q_t$, and a time dependence has been integrated non-parametrically, i.e., not considering any covariates $\boldsymbol{x}_t$. The step from this dynamic random effects model to a dynamic mixed effects model is a comparably small one, we only need to set $\mu_t = \mu(\boldsymbol{x}_t)$ for a regression function $\mu : \mathcal{X} \to \mathbb{R}$ on the covariate space $\mathcal{X}$. In this example, we choose a FNN architecture of depth 3 and with log-link, i.e., this is precisely the same architecture as depicted in Listing 2.5. The corresponding changes are integrated into the code of Listing 4.2. First, this requires that the input `mu0` on line 8 of Listing 4.2 is replaced by the input on lines 9-11 of Listing 2.5 adapted to a tensor (time-series) structure, the code is provided on lines 1-3 of Listing 4.3. Second, we add a time-distributed FNN architecture, see lines 5-17 of Listing 4.3, which plays the role of lines 31-34 in Listing 4.2. The rest is then identical to the latter code.

We fit this model on precisely the same data as in Example 4.19, i.e., only insurance policies are considered that have been active during the entire observation period from 2011 to 2019. These are $n = 56,806$ insurance policies, see Example 4.19. The results are then directly comparable to Tables 4.2 and 4.3.

Table 4.4 shows the results of the fitted model and it is compared to the dynamic random effects null model of Example 4.19, see Table 4.3. We observe a substantial decrease of (in-



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Listing 4.3: Code snippet for a dynamic mixed effects FNN model.

```
Design     = layer_input(shape = c(9,7), dtype = 'float32')
Cat        = layer_input(shape = c(9,1), dtype = 'int32')
Exposure   = layer_input(shape = c(9),   dtype = 'float32')
###
CatEmb = Cat %>%
    time_distributed(layer_embedding(input_dim = 15, output_dim = b[1],
                     input_length = 1)) %>% layer_reshape(target_shape=c(9,b[1]))
#
mu0 = list(Design, CatEmb) %>% layer_concatenate() %>%
           time_distributed(layer_dense(units=q[1], activation='tanh')) %>%
           time_distributed(layer_dense(units=q[2], activation='tanh')) %>%
           time_distributed(layer_dense(units=q[3], activation='tanh')) %>%
           time_distributed(layer_dense(units=1, activation='exponential',
                weights=list(array(0,dim=c(q[3],1)), array(init, dim=c(1))))) %>%
           layer_reshape(target_shape=c(9))
#
mu <- list(mu0, Exposure) %>% layer_multiply()
```

| | models (4.29)-(4.30) | | mixed effects |
|---|---|---|---|
| | `optim` | RNN | FNN for $\mu(\boldsymbol{x}_t)$ |
| $\widehat{\mu}_0$ | 0.0823 | 0.0824 | – |
| $\widehat{\alpha}_1$ | 0.2414 | 0.2414 | 0.3297 |
| $\widehat{p}$ | 0.2301 | 0.2304 | 0.2204 |
| $\widehat{q}$ | 0.7885 | 0.7884 | 0.7836 |
| $\ell_{\boldsymbol{Y}}(\widehat{\vartheta})/n$ | -2.5853 | -2.5853 | -2.5384 |
| ♯parameters | 4 | 4 | 735 |
| AIC | 293,724 | 293,724 | *289,859* |

Table 4.4: Bounded variance cases: dynamic random effects null model and dynamic mixed effects FNN model.

sample) deviance loss (or increase of in-sample log-likelihood). If we adjust these values by the number of estimated parameters 735, we obtain an AIC value which is substantially lower than the one in the null model, supporting the integration of covariates. This AIC value of *289,859* is written in italic font as it does not have any solid mathematical foundation (we early stop gradient descent and therefore we do not consider a MLE in the FNN approach). The magnitudes of the estimated parameters $\widehat{p}$ and $\widehat{q}$ is roughly the same in the random effects null and the mixed effects FNN approaches. Only $\widehat{\alpha}_1$ becomes bigger which decreases the initial volatility in the state-space process $(\Lambda_t)_{t\geq 1}$.

Figure 4.6 shows the resulting credibility weights $(\omega_{t,s})_{s=1}^t$ on the past observations $N_{1:t}$ and the credibility weight $\omega_{t,0}$ on the prior value $\lambda_{1:1:0} = 1$ for the estimated parameters $\widehat{\alpha}_1 = 0.3297$, $\widehat{p} = 0.2204$ and $\widehat{q} = 0.7836$, and we set $\mu_t \equiv \mu_0 = 8.5\%$, see (4.32) for computing $\omega_{t,0}$. Similar to Figure 4.4 (middle), we observe a comparably fast decay of the credibility weights suggesting a short memory of past claims. Figure 4.6 (rhs) tells us that if $N_{1:15} = 0$, i.e., if a customer has zero claims over 15 years of claims experience, her/his insurance premium gets a discount of 6%. This is much less than the 30%-40% suggest by Pinquet et al. [163, Table 4]. Our value is entirely data driven, and it does not contain any commercial considerations. However, since we work with a simulated motor



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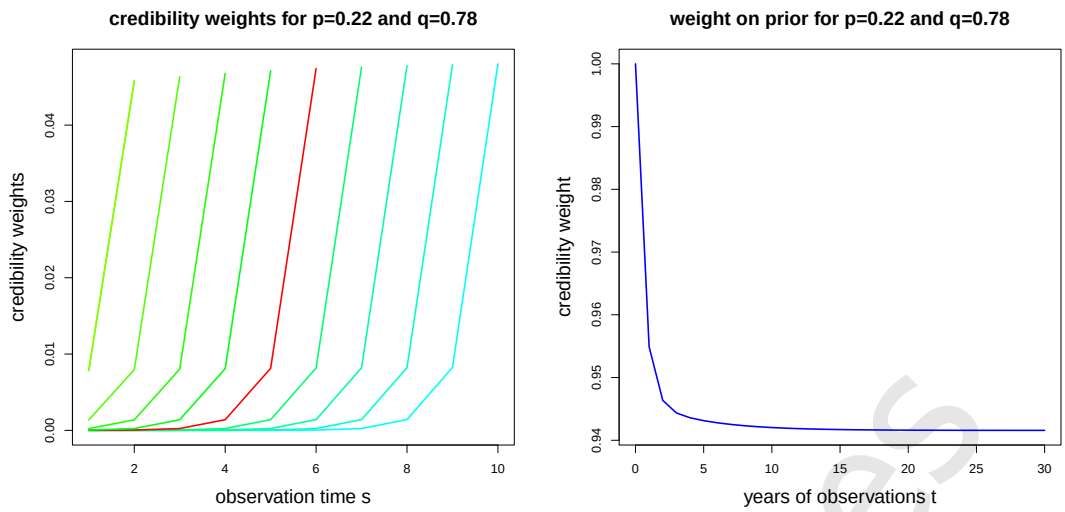


Figure 4.6: (lhs) Dynamic mixed effects FNN model credibility weights $(\omega_{t,s})_{s=1}^{t}$ for $2 \leq t \leq 10$ (in different colors), (rhs) credibility weight $\omega_{t,0}$ on the prior value $\lambda_{1,1:0} = 1$ for $2 \leq t \leq 30$.

insurance data set, our data may have less extreme features than real data. Moreover, Figure 4.6 uses a constant $\mu_t$, whereas in real world applications this expected frequency is changing in time because, e.g., driver's age and age of car is increasing in time. This may also have an impact on the long-term premium discount. This closes this example. ■

In the previous example, we have only considered the insurance policies that have been exposed during the entire observation period from 2011 to 2019. That is, these insurance policies had claims histories all of the same length. The general case of different observation lengths needs a small adaptation to the code in Listing 4.2. The insurance policy dependent sequences $(\alpha_{i,t})_{t\geq 1}$ and $(\beta_{i,t})_{t\geq 1}$ may have different initial years, depending on when the policies $1 \leq i \leq n$ have been issued. From (4.16) and (4.17) we see that we receive $\beta_{i,t+1} = \beta_{i,t}$ and $\alpha_{i,t+1} = \alpha_{i,t}$ if we set

$$v_{i,t} = 0, \qquad N_{i,t} = 0, \qquad p_{i,t} = 1 \qquad \text{and} \qquad q_{i,t} = 1.$$

This is the way how we select these parameters for all calendar years before the insurance policy has been signed (issued) to pad all sequences to equal length. The former two identities can be integrated into the data, the latter two identities require some changes on lines 21 and 28 of Listing 4.2, i.e., we set $p_{i,t} = q_{i,t} = 1$ if the contract has not been issued before year $t + 1$. This choice corresponds to the static (common) latent factor case and, basically, our choice implies that this latent factor does not change before we have any claims history.

There may also be the case of a policyholder temporarily suspending her/his insurance policy, and later on reactivates it. In that case, we may also have $v_{i,t} = N_{i,t} = 0$ for some calendar years $t$, and in this case we can either let the latent state-space process $(\Lambda_{i,s})_{s\geq 1}$ evolve by choosing $p_{i,t} < 1$ and/or $q_{i,t} < 1$, or we can keep it constant by setting $p_{i,t} = q_{i,t} = 1$. Our data does not have insurance policies with this behavior, therefore, we cannot further exploit temporarily suspended insurance policies.



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**Example 4.22.** We study a dynamic mixed effects Poisson-gamma GLM. This continues the example of Section 3.2.6, but it replaces the static mixed effects GLM by an observation-driven dynamic mixed effects GLM. The difference between the two models is fully expressed in the choices of $p_{i,t} \equiv p$ and $q_{i,t} \equiv q$, see Table 4.1, and we choose these two parameters insurance policy independent and time independent. Theoretically, these two assumptions could be relaxed, however, this leads to two issues. First, an extrapolation method needs to be considered. Second, fitting becomes difficult/impossible because the model is too over-parametrized. In fact, to be able to fit this observation-driven dynamic mixed effects model we need reasonable starting values for the gradient descent algorithm, otherwise the fitting cannot be performed successfully. We choose as initial values the optimal solution found in the static mixed effects model, and then apply gradient descent to this initial parameters together with the fitting of the parameters $p$ and $q$.

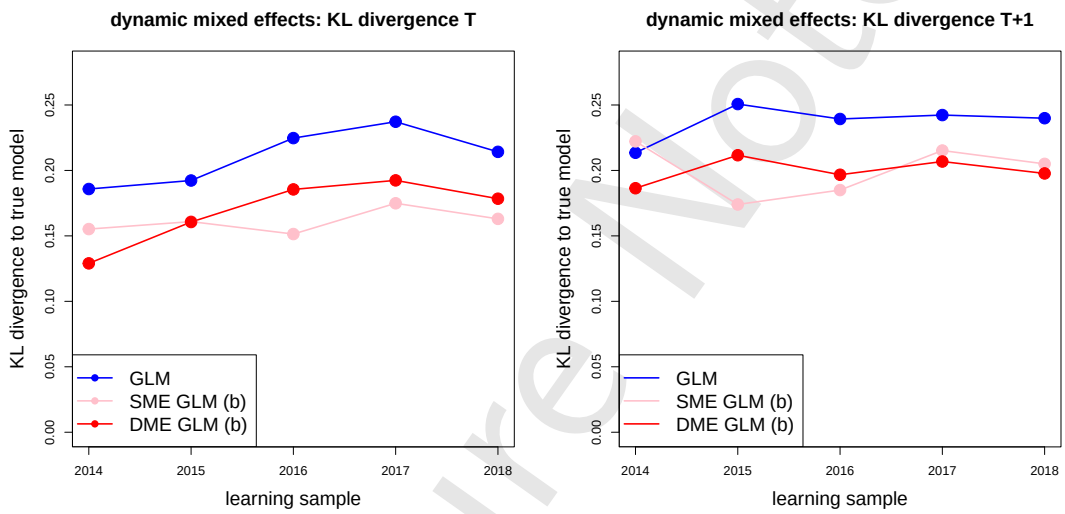


Figure 4.7: Fitted dynamic mixed effects (DME) models: (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one-year ahead forecast, the label (b) refers to the fitting one-year ahead method (3.47), and the SME and the GLM are taken from Figure 3.2; units are $10^{-2}$.

The results are shown in Figure 4.7; this follows up Figure 3.2. We observe that relaxing from a static to a dynamic mixed effects GLM, generally makes the in-sample losses smaller (this is not shown in the graphs), but unfortunately this is not reflected in a more accurate model, see Figure 4.7 (lhs) where we compare the fitted models to the true one (in negative binomial KL divergence). In fact, in view of Figure 4.7 (lhs), we should rather give preference to the static version of the mixed effects GLM. For forecasting the next calendar year this relaxation seems to provide a small model improvement, at least for some of the calendar years, see Figure 4.7 (rhs). However, we conclude that in terms of predictive power, these results do not give full evidence that the dynamic mixed effects GLM performs better than the static one, which, of course, is a small disappointment.
Figure 4.8 shows the estimates $\widehat{p}$, $\widehat{q}$ and $\widehat{\alpha}_1$ fitted on the different learning samples $\mathcal{L}_T$, for $2014 \leq T \leq 2018$. We observe that these estimates are rather stable over time, and the

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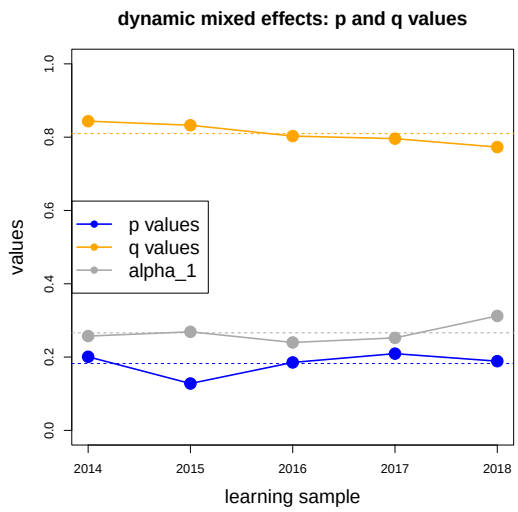


Figure 4.8: Fitted dynamic mixed effects (DME) models: estimates $\widehat{p}$, $\widehat{q}$ and $\widehat{\alpha}_1$ fitted on the different learning samples $\mathcal{L}_T$, for $2014 \leq T \leq 2018$.

estimates of $\widehat{p}$ and $\widehat{q}$ are clearly less than 1. The average estimate of $p$ is 0.18 (without any trend) and the average estimate of $q$ is 0.81 (with a slightly decreasing trend). This suggests to give preference to the dynamic mixed effects GLM. Note that these values are similar to the ones in Table 4.4, though, this table has been computed on slightly different data, only considering insurance policies that have been exposed during the entire observation period.

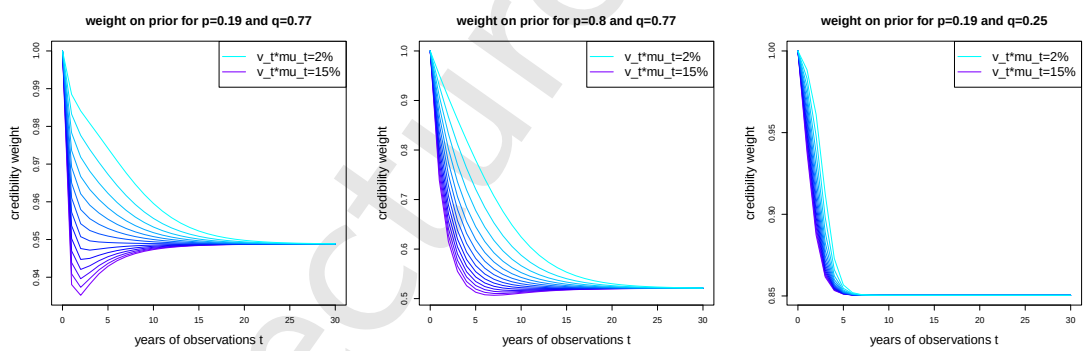


Figure 4.9: Credibility weights $(\omega_{t,0})_{t=1}^{30}$ on the prior value $\lambda_{1,1:0} = 1$ for (lhs) $\widehat{p} = 0.19$ and $\widehat{q} = 0.77$, (middle) $\widehat{p} = 0.80$ and $\widehat{q} = 0.77$, (rhs) $\widehat{p} = 0.19$ and $\widehat{q} = 0.25$, and $\widehat{\alpha}_1 = 0.31$, $v_t\mu_t \equiv 2\%, \ldots, 15\%$ (constant for all $t$).

Figure 4.9 shows the credibility weights $(\omega_{t,0})_{t=1}^{30}$ assigned to the prior value $\lambda_{1,1:0} = 1$ as a function of calendar years $1 \leq t \leq 30$. This corresponds to the experience rating premiums when accounting for $t-1$ years (in a row) of being free of accidents. The figure on the left-hand side shows the behavior for the parameters $\widehat{p} = 0.19$ and $\widehat{q} = 0.77$ obtained from the learning sample $\mathcal{L}_{2018}$. For the plot in the middle we change to $\widehat{p} = 0.80$ and $\widehat{q} = 0.77$, and for the plot on the right-hand side we change to $\widehat{p} = 0.19$ and $\widehat{q} = 0.25$. In all these graphs we set $\widehat{\alpha}_1 = 0.31$ (estimate at time 2018), and we assume a constant

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expected number of claims $v_t\mu_t = v_t\mu(\boldsymbol{x}_t) \equiv \mu_0 \in \{2\%, \ldots, 15\%\}$.
We first discuss Figure 4.9 (lhs). In the long-term behavior, this model only suggests a discount of 5% for being free of accidents, this is similar to Figure 4.6 (rhs). Note that this number is much smaller than typically granted to customers. It does not contain any commercial considerations, but predictive power is the only objective in deriving these credibility weights. This figure also shows that for small expected numbers of claims, e.g., $v_t\mu_t \equiv 2\%$, we have a monotone decrease, whereas for higher expected numbers of claims, e.g., $v_t\mu_t \equiv 15\%$, we do not receive monotonicity. Again, this does not consider any commercial aspects but only predictive power has been considered in deriving these values. There is one more factor that may influence the picture, namely, typically, $v_t\mu_t = v_t\mu(\boldsymbol{x}_t)$ is not constant in time $t$. Often, the covariates $\boldsymbol{x}_t$ include calendar year dependent components, such as driver's age or age of car, and these naturally change with time. This may also impact the monotonicity behavior of the credibility weights $(\omega_{t,0})_{t=1}^{30}$ because it enters the weighting of past claims, see (4.25). However, clearly a non-decreasing behavior cannot be explained to customers.
For a bigger value of $p$ we get a much bigger long-term discount of roughly 50% for being free of accidents, and for a smaller value of $q$ we also get a bigger discount (of roughly 15%) and a faster decay towards this value, see Figure 4.9 (middle, rhs).

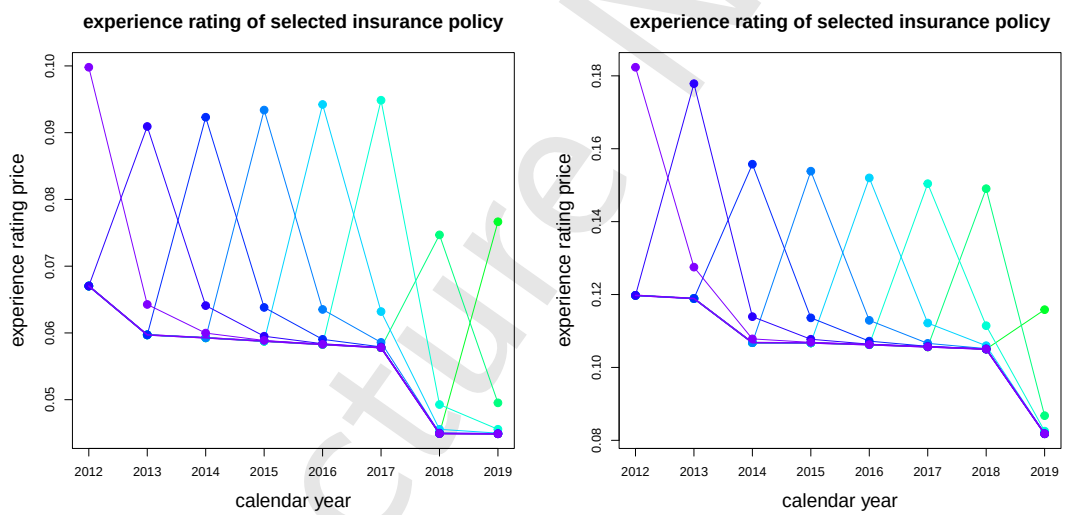


Figure 4.10: Experience rating prices for two selected insurance policies.

Figure 4.10 shows time-series of experience rating prices $\mu^{\text{post}}_{N_{i,t+1}|N_{i,1:t}}$ of two selected insurance policies $i$ for different calendar years $t$, and under different past claims experience $N_{i,1:t}$. The solid lower line shows the experience rating premiums for the past history $N_{i,1:t} = (0, \ldots, 0)$ and all the other lines with the peaks show the premiums if there is exactly one accident $N_{i,1:T} = (0, \ldots, 0, 1, 0, \ldots, 0)$ among all observed calendar years. We observe that an accident increases the experience rating price by 0.03 to 0.06, see Figure 4.10. However, since the credibility weights $\omega_{t,s}$ are comparably small and fast decaying, this increase vanishes rather fast in the experience rating price.
Another interesting observation is that these experience rating prices seem to have discontinuities because there are jumps in the prices of Figure 4.10. This is an artifact of the

selected GLM. For modeling the regression function $\boldsymbol{x}_t \mapsto \mu(\boldsymbol{x}_t)$ we use categorical age classes for vehicle age, see Listings 2.1, 2.2 and 2.4, and this implies jumps in the prices when the car is aging. This seems unpleasant, but within a GLM framework such discontinuities in time are rather common, i.e., if a vehicle is aging it naturally moves across different age classes, each one having a different GLM regression parameter. Only models that treat continuous variables with smooth functions such as generalized additive models (GAMs) or FNNs avoid this problem, but this problem will also be present in regression trees and gradient boosting machines (GBMs). This completes the considerations of this observation-driven mixed effects Poisson-gamma GLM. ■

## 4.6 Outlook

We give a couple of further considerations that may be of interest.

**Observation-driven dynamic mixed effects Poisson-gamma FNN.** A next natural step would be to extend the observation-driven mixed effects GLM of Example 4.22 to its FNN counterpart. Mathematically and technically there is no further difficulty in this extension. However, the main problem lies in fitting such models. Already the dynamic mixed effects GLM has been showing some signs of over-fitting, and applying the same methodology the a FNN architecture, we did not receive any competitive model. A main issue is that gradient descent fitting with early stopping tends to stop the algorithm too early, resulting in a poor model. An option would be to use more sophisticated algorithms, stopping rules, or to complement fitting with regularization techniques such as drop-outs; see Srivastava et al. [186] and Wager et al. [197] for drop-outs. Alternatively, also a two step fitting strategy may be useful where not all parameters are fitted in the same step. We did not exploit such more sophisticated fitting strategies, but it may be worth to do so.

**Bonuses and bonus-malus systems.** Clearly, what we have developed in the previous sections was a statistical and not a commercial solution. Parameter selection has been done by optimizing an objective function related to predictive power, that is, an empirical Bayesian estimation in our case. The resulting bonuses for accident-free claims histories have turned out to be comparably small, in the range of 5% to 15% depending on the specific parameters. Let us give some remarks on this observation.

- First, we work with simulated data from a model that has been fitted to a real motor insurance data set. We cannot exclude that this simulated data does not entirely reflect the features of real data, and that real data leads to bigger bonuses for accident-free drivers. In our case, the credibility weights are comparably small and rapidly decreasing in claims seniority which leads to these small bonuses. Statistically, this is also justified by the fact that the predictive power of past claims does not seem to be very large, e.g., the static random effects model is outperformed by a simple GLM, see Figure 3.1, saying that fixed effects are more important than (linearly included) past claims in this data. This seems a bit surprising, but it is what our data tells us.



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- To commercialize our findings we would have to design a BMS where we can also choose different weightings for past claims, with corresponding transition rules. Stationary and steady-state considerations will provide the overall premium level, similar to the balance property considerations. Using a BMS one compromises predictive power, but one may find a convincing commercial solution. We will not further consider BMS here.

- Considering time-consistency in pricing will also be important, i.e., how do we deal with model refitting and how do we deal with situations like in Figure 4.9 (lhs). Statistically this seems all fine, however, commercially it cannot be explained to customers. A way to deal with such problems is to consider regularized optimization, where regularization acts on the time axis. This may be implement in a similar way as the ICEnet proposal of Richman–Wüthrich [173] to achieve, e.g., monotonicity.

**Beyond the Poisson-gamma model.** In this chapter, we have only considered the Poisson-gamma model which is fully tractable due to the Poisson-gamma conjugacy, and the marginalized process is an INAR negative binomial one. Besides the Poisson-gamma conjugacy we have also used the the construction of the gamma process using the result of Lukacs [143], see Lemma 4.2. There should not be any difficulty to translate this to other cases, e.g., a gamma-gamma model for claim amounts. Moreover, at the end we need to compound the claim count and the claim amount picture for experience rating, which poses another challenging problem.

**Other data sources.** Our simulated data only contains classical covariate information. In the era of big data, commonly telematics car driving information is collected. Such telematics sources will increasingly impact and determine insurance pricing. We can either model them dynamically as the claim counts process, or we may integrate such information (summary statistics and telematics scores) into the covariates. The latter is more in the spirit of the next chapter. For further reading on using advanced data types, we also refer to Corradin et al. [52], Duval et al. [71], Peiris et al. [156] or Yanez et al. [207].

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# Chapter 5

# Deep Experience Rating

## 5.1 Literature overview

The previous chapters have been focusing on developing analytical experience rating models that have fully tractable posterior distributions. In case they were not fully tractable, linear credibility approximations have been considered. Full distributional models have the advantage that one is able to simulate from them, that one can exploit MLE, and that one can study any other quantity that requires the knowledge of the underlying distributional model. On the contrary, machine learning models simply focus on algorithms that provide best predictive performance, and based on the specific choice of the algorithm, one receives more or less tractable models. In this chapter, we consider such machine learning models, and we call them *deep experience rating models* as we do not start from a full stochastic model but rather from a machine learning algorithm that is based on deep learning. To the best of our knowledge, at the time of writing these notes, such a machine learning approach has not been considered in the actuarial literature, and the following models are new to the actuarial community.

Since the following approaches are new for actuarial pricing, we can only give references from the wider machine learning literature. The general philosophy that we are going to pursue in this chapter is to have models that provide optimal predictive performance in the sense of the fixed effects models studied in Chapter 2 and for selected objective functions. For this, we try to find optimal regression functions $\boldsymbol{x}_{t+1} \mapsto \mu(\boldsymbol{x}_{t+1})$ to accurately predict the responses $Y_{t+1}$, but at the same time we would like to integrate past claims experience $(Y_{1:t}, \boldsymbol{x}_{1:t})$ in a linear credibility fashion similar to (4.28). The former prediction module is just going to use the FNN architectures presented in Section 2.4.1. The integration of past claims experience $(Y_{1:t}, \boldsymbol{x}_{1:t})$ involves time-series considerations. Our main approach is to estimate *attention weights* that are assigned to the individual past observations $Y_s$, $1 \leq s \leq t$. These attention weights play the role of credibility weights. More generally, this follows the philosophy of the *attention layer* mechanism and *Transformers* introduced by Vaswani et al. [194]. Attention layers are the building blocks of large language models (LLMs) that celebrate huge success these days. Rather similarly to our derivations below, these attention layers have also inspired the architecture of the LocalGLMnet of Richman–Wüthrich [170, 171] which aims at making FNN regression models more explainable, by giving more or less attention to the components of the

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covariates. That is, instead of estimating GLM regression parameters, the LocalGLMnet is learning FNN regression-based parameters that provide models that locally look like GLMs. We also mention the work of Kuo–Richman [127] and Brauer [34], these authors take similar attention weight and Transformer approaches on tabular actuarial data.

## 5.2 Attention and credibility weights

In the previous chapter we have considered dynamic mixed effects models for claim counts modeling that were fully analytically tractable. These provided linear credibility formulas of the type

$$\mu^{\text{post}}_{N_{t+1}|N_{1:t}} = v_{t+1}\mu_{t+1}\lambda_{t+1|1:t} = v_{t+1}\mu_{t+1}\left(\sum_{s=1}^{t}\omega_{t,s}\frac{N_s}{v_s\mu_s} + \left(1-\sum_{s=1}^{t}\omega_{t,s}\right)\cdot 1\right),$$

with credibility weights $\omega_{t,s} \in [0,1]$ such that $\sum_{s=1}^{t}\omega_{t,s} \leq 1$, see (4.28). These credibility weights $\omega_{t,s}$ were given functions of the inputs $v_{1:t}$ and $\mu_{1:t}$ through recursive formulas. In the era of machine learning, one is tempted to replace these analytical credibility weights by non-parametric weights computed by deep learning models.

Assume that each observation $Y_t = N_t/v_t$ is described by a covariate $\boldsymbol{x}_t$. Typically, its mean $\mu_t = \mu_t(\boldsymbol{x}_t)$ is assumed to be a function of this covariate $\boldsymbol{x}_t$. This assumption motivates us to rewrite the previous linear credibility formula as

$$\mu^{\text{post}}_{N_{t+1}|N_{1:t}} = v_{t+1}\left(\sum_{s=1}^{t}\omega_{t,s}\frac{\mu_{t+1}(\boldsymbol{x}_{t+1})}{\mu_s(\boldsymbol{x}_s)}\,\frac{N_s}{v_s} + \left(1-\sum_{s=1}^{t}\omega_{t,s}\right)\mu_{t+1}(\boldsymbol{x}_{t+1})\right).$$

To avoid too much complication in the following (machine learning) modeling exercise, we moreover assume

$$\frac{\mu_{t+1}(\boldsymbol{x}_{t+1})}{\mu_s(\boldsymbol{x}_s)} \equiv 1 \qquad \text{for all } 1 \leq s \leq t. \tag{5.1}$$

This assumption usually is not (entirely) true, e.g., the insurance policyholder may change her/his car which impacts the covariates and, hence, the expected frequency; or similarly aging will impact this consideration. Thus, (5.1) seems a bit prohibitive. However, this assumption has the big advantage that it does not require us to estimate a regression function $\mu_s(\cdot)$ for every time point $1 \leq s \leq t$, and we still leave the possibility open to enter covariates into weights $\omega_{t,s}$. Thus, using simplification (5.1), we set a linear credibility assumption

$$\mu^{\text{post}}_{N_{t+1}|N_{1:t}} = v_{t+1}\left(\sum_{s=1}^{t}\omega_{t,s}Y_s + \left(1-\sum_{s=1}^{t}\omega_{t,s}\right)\mu_{t+1}(\boldsymbol{x}_{t+1})\right). \tag{5.2}$$

This only involves the regression function $\boldsymbol{x}_{t+1} \mapsto \mu_{t+1}(\boldsymbol{x}_{t+1})$ and the weights $(\omega_{t,s})_{s=1}^{t}$, and the linearity statement applies to the inclusion of the past observed frequencies $Y_{1:t}$. That is, the experience rating premium (5.2) is a (positive) affine function of past observations $Y_{1:t}$, with observed frequencies $Y_s = N_s/v_s$.

The idea now is to model these (credibility) weights $(\omega_{t,s})_{s=0}^{t}$ by a multi-output FNN, thus, we are searching for a suitable function

$$\boldsymbol{x}_{1:t+1} \;\mapsto\; \Omega_t(\boldsymbol{x}_{1:t+1}) := \left(\omega_{t,1}(\boldsymbol{x}_{1:t+1}), \ldots, \omega_{t,t}(\boldsymbol{x}_{1:t+1}), \omega_{t,0}(\boldsymbol{x}_{1:t+1})\right)^{\top} \;\in\; [0,1]^{t+1}, \tag{5.3}$$



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being normalized $\sum_{s=0}^{t} \omega_{t,s}(\boldsymbol{x}_{1:t+1}) = 1$ for all covariates $\boldsymbol{x}_{1:t+1}$.[1] In other words, the credibility weighting function $\Omega_t$ maps the covariates $\boldsymbol{x}_{1:t+1}$ to the unit $t$-simplex in $\mathbb{R}^{t+1}$. We consider the input $\boldsymbol{x}_{1:t+1}$ in tensor structure

$$\boldsymbol{x}_{1:t+1}^{\top} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{t+1}]^{\top} \;\in\; \mathbb{R}^{(t+1)\times(q+1)}.$$

To (the rows) of this input tensor we apply a time-distributed FNN which is going to be described next. Choose a FNN of depth $d$

$$\boldsymbol{z}^{(d:1)} : \{1\} \times \mathbb{R}^q \to \mathbb{R}, \qquad \boldsymbol{x}_t \mapsto \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_t).$$

A time-distributed FNN $\boldsymbol{z}_{\text{td}}^{(d:1)}$ is obtained by applying the identical FNN $\boldsymbol{z}^{(d:1)}$ (with the same network weights/parameters) to all time points $1 \le s \le t+1$ providing the *time-distributed* (td) FNN

$$\boldsymbol{x}_{1:t+1}^{\top} \;\mapsto\; \boldsymbol{z}_{\text{td}}^{(d:1)}(\boldsymbol{x}_{1:t+1}^{\top}) = \left(\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_1), \ldots, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_{t+1})\right)^{\top} \;\in\; \mathbb{R}^{t+1}. \tag{5.4}$$

We map this $(t+1)$-dimensional vector to the unit $t$-simplex by applying the $\mathsf{softmax}$ function. This gives us the *attention weights* in the unit $t$-simplex

$$\boldsymbol{x}_{1:t+1} \;\mapsto\; \Omega_t(\boldsymbol{x}_{1:t+1}) = \mathsf{softmax}\left(\boldsymbol{z}_{\text{td}}^{(d:1)}(\boldsymbol{x}_{1:t+1}^{\top})\right), \tag{5.5}$$

where the $u$-th component of the $\mathsf{softmax}$ function is given by

$$\mathsf{softmax}_u\left(\boldsymbol{z}_{\text{td}}^{(d:1)}(\boldsymbol{x}_{1:t+1}^{\top})\right) \;=\; \frac{\exp\left\{\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_u)\right\}}{\sum_{s=1}^{t+1} \exp\left\{\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_s)\right\}} \;\in\; [0,1], \tag{5.6}$$

for $1 \le u \le t+1$.

We give some remarks.

**Remarks 5.1.**

- As mentioned in the literature overview, these attention (credibility) weights $\Omega_t(\boldsymbol{x}_{1:t+1}) \in [0,1]^{t+1}$ are inspired by the attention layer construction of Vaswani et al. [194]. Attention layers are more complex because they allow covariates $\boldsymbol{x}_s$ and $\boldsymbol{x}_u$ from different time points $s \neq u$ to interact. However, in essence, the underlying idea is quite similar. We are going to discuss attention layers in Section 5.3, below.

- Observe from (5.2) and (5.4) that the components $\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_s)$, $1 \le s \le t$, of the time-distributed FNN architecture encode the attention weights on the past observations $Y_{1:t}$, up to the $\mathsf{softmax}$ transformation (5.5)-(5.6). Since we use a time-distributed FNN $\boldsymbol{z}_{\text{td}}^{(d:1)}$ which is identical for all considered time points $1 \le s \le t+1$, it is essential that the covariates are equipped with time information, i.e., one has to encode claim seniority into the covariates $\boldsymbol{x}_s$; in Vaswani et al. [194, Section 3.5] this is called *positional encoding*. Without a positional encoding there is no notion of time in the available information, and a seniority weighting of past claims is

[1] In (5.3) we use notation $\Omega_t(\boldsymbol{x}_{1:t+1})$ for the $(t+1)$-dimensional credibility vector; this is an element in the unit $t$-simplex in $\mathbb{R}^{t+1}$. This should not be confused with the notation $(\Omega, \mathcal{F}, \mathbb{P})$ of the underlying probability space, where $\Omega$ is a non-empty set.



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not possible. We add a time component $(t+1-s)/(t+1)$ to the covariates $\boldsymbol{x}_s$, $1 \leq s \leq t+1$, encoding the time lag between observations $Y_s$ (predictors) and $Y_{t+1}$ (response). That is, we *revert time* which seems more natural when updating the models over time; in fact, we are going to apply a rolling window approach as going to be explained below.

- One may face the difficulty that different insurance policies have different lengths of past claims history. Assume we have an insurance policy with observations only starting at time $k > 1$, and, thus, $v_{1:k-1} = 0$. In this case we only have a claims history of length $t-k+1$. For this reason, we replace (5.4) in our applications by

$$\Big(\boldsymbol{z}^{(d:1)}(\boldsymbol{x}_1) - \infty \cdot \mathbb{1}_{\{v_1=0\}}, \ldots, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_t) - \infty \cdot \mathbb{1}_{\{v_t=0\}}, \boldsymbol{z}^{(d:1)}(\boldsymbol{x}_{t+1})\Big)^\top. \tag{5.7}$$

  That is, we generally replace all past components without an exposure by $-\infty$. Applying the softmax transformation (5.5) to this modified vector (5.7) gives a weight of 0 to all components (calendar years) without exposures. In the extreme case of no past claims history, $v_{1:t} = 0$, we assign a weight of 1 to the new covariate information and regression function, respectively, $\mu_{t+1}(\boldsymbol{x}_{t+1})$, see (5.2). Moreover, we equip each covariate component $\boldsymbol{x}_s$ with the information of the length of the claims history, as this may also impact the selection of the attention weights $\Omega_t(\boldsymbol{x}_{1:t+1})$.

■

Using the linear credibility assumption (5.2), and modeling these credibility weights by attention weights using a time-distributed deep FNN (5.4) with softmax output activation (5.5) will be called *deep attention FNN experience rating model* in these notes. There remains the modeling of the expected frequencies $\mu_{t+1}(\boldsymbol{x}_{t+1})$. For this we simply choose another deep FNN, which essentially looks the same as the one in Listing 2.5. In Listing 5.1 we give the entire code for modeling (5.2) with a deep attention FNN experience rating model, in particular, this listing depicts the precise architecture used.

**Example 5.2** (deep attention FNN experience rating model)**.** We revisit Example 4.22. This example uses an observation-driven dynamic mixed effects GLM for claim counts modeling, which we now replace by the deep attention FNN experience rating model presented in this section. Note that so far the description in this section has been fully non-distributional, we have only described the experience rating premium as a function of input variables $(N_{1:t}, \boldsymbol{x}_{1:t+1}, v_{1:t+1})$

$$\begin{aligned}
\mu^{\text{post}}_{N_{t+1}|N_{1:t}} &= \mathbb{E}\left[N_{t+1} \middle| N_{1:t}, \boldsymbol{x}_{1:t+1}, v_{1:t+1}\right] \\
&= v_{t+1}\left(\sum_{s=1}^{t} \omega_{t,s}(\boldsymbol{x}_{1:t+1}) \frac{N_s}{v_s} + \omega_{t,0}(\boldsymbol{x}_{1:t+1}) \mu_{t+1}(\boldsymbol{x}_{t+1})\right).
\end{aligned}$$

This is encoded in Listing 5.1. Our next step is to fit this regression function to the learning data $\mathcal{L}_T$, see (2.49), and we set $T = t+1$. Since this regression function estimates (conditional) means of random variables, we should choose a strictly consistent loss function for mean estimation, see Section 2.3.3. We may choose any Bregman divergence, but having the knowledge that we model claim counts, the Poisson unit deviance



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$\mathfrak{d}_{\text{Poi}}$ is a natural candidate, we refer to (2.54). If $\boldsymbol{\vartheta}$ collects all involved parameters, we thus try to find

$$\begin{aligned}\widehat{\boldsymbol{\vartheta}} &\in \arg\min_{\boldsymbol{\vartheta}} \sum_{i\in\mathcal{I}_T} v_{i,T}\, \mathfrak{d}_{\text{Poi}}\left(Y_{i,T},\, \frac{1}{v_{i,T}}\, \mu^{\text{post}}_{N_{i,T}|N_{i,1:T-1}}(\boldsymbol{\vartheta})\right) \\ &= \arg\min_{\boldsymbol{\vartheta}} \sum_{i\in\mathcal{I}_T} \mathfrak{d}_{\text{Poi}}\left(N_{i,T},\, \mu^{\text{post}}_{N_{i,T}|N_{i,1:T-1}}(\boldsymbol{\vartheta})\right).\end{aligned} \tag{5.8}$$

The optimizer(s) $\widehat{\boldsymbol{\vartheta}}$ are the MLE(s) of $\boldsymbol{\vartheta}$ in a conditional Poisson model for the responses $\boldsymbol{N}_T = (N_{i,T})_{i\in\mathcal{I}_T}$. This is quite similar to the fitting procedure used in Section 2.4.2. A suitable parameter is then determined through SGD with early stopping and, again, we ensemble over 10 fitted network models (with different seeds) to limit the influence of the randomness of SGD.

For out-of-sample prediction on the test data $\mathcal{T}_{T+1}$ we can unfortunately not simply extrapolate, as the model has been fitted to a maximal length of past claims history. Similarly to Sections 2.3.4 and 2.4.2, we therefore shift the time index by one unit and we discard the oldest observations $(N_{i,1})_{i\in\mathcal{I}_{T+1}}$, i.e., we apply a *rolling window methodology*. Remark that we essentially benefit here from the fact that we have reverted time, i.e., we measure the time lag w.r.t. the time period to be predicted, see Remarks 5.1. This results in applying the fitted mean model $\mu^{\text{post}}_{N_T|N_{1:T-1}}(\widehat{\boldsymbol{\vartheta}})$ to forecast on $\mathcal{T}_{T+1}$ by considering the time-shifted forecasts $\mu^{\text{post}}_{N_{T+1}|N_{2:T}}(\widehat{\boldsymbol{\vartheta}})$, where the parameter $\widehat{\boldsymbol{\vartheta}}$ has been fitted on the learning data $\mathcal{L}_T$, see (5.8).

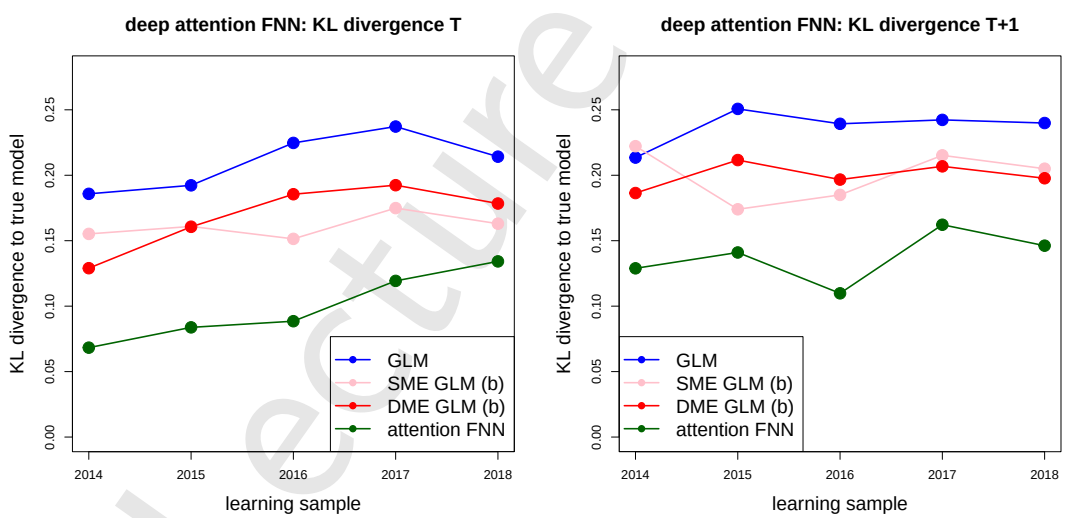


Figure 5.1: Deep attention FNN experience rating models: (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one year ahead forecast; the GLM, SME and DME GLMs are taken from Figure 4.7; units are $10^{-2}$.

Figure 5.1 shows the accuracy of the fitted deep attention FNN experience rating models compared to the true means, again measured in negative binomial KL divergence. Moreover, we compare those accuracies to the ones of the competing models of Figure 4.7, these are the fixed effects GLM of Section 2.3.4 (which has roughly the same performance

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as the fixed effects FNN of Section 2.4.2), the static mixed effects GLM of Section 3.3.4, and the dynamic mixed effects GLM of Example 4.22. We observe that the deep attention FNN approach clearly outperforms all these other models having a much smaller model error. It seems that the flexibility of modeling the weights $\omega_{t,s}(\boldsymbol{x}_{1:t+1})$ by a multi-output time-distributed FNN is very beneficial and necessary to get good predictors. That is, it seems that the previous static and dynamic mixed model assumptions are too rigid for this data. We could also achieve more modeling flexibility in the observation-driven dynamic mixed effects model by letting $p_t$ and $q_t$ be $t$-dependent. We did not do that because we face similar difficulties in extrapolating beyond the latest observation point, i.e., this would have to be done manually. Alternatively and similarly to the positional encoding it may be beneficial to not use time $t$ in $p_t$ and $q_t$, but rather time lag to the latest observation, to maintain consistency in the use of parameters. This then requires to discard the first observed calendar year when extrapolating (shifting) like a rolling time window. However, if we choose this time-reverted notation in the dynamic mixed effects model, it implies that we have to pre-specify the end point of the latent process, which does not seem to be natural because this would require a new model for every newly observed calendar year, likely violating time-consistency across these models.

The resulting attention weights $(\omega_{T,s}(\boldsymbol{x}_{i,1:T+1}))_{s=2011}^{T}$ now depend on the insurance policies $i \in \mathcal{I}_{T+1}$ through the covariates $\boldsymbol{x}_{i,1:T+1}$. In Figure 5.2 (top-lhs) we show the average attention weights as a function of time $s$ (or rather as the time lag to the latest observation), and the average is taken only over the policies that have been fully exposed during the entire observation period. The reason is that we apply the softmax function only to the calendar years with exposures, and these figures should not be distorted by different observation lengths. For the latest observed period $T = 2017$ we obtain an average attention weight of 4.0%, assigned to the claim counts $\boldsymbol{N}_{2017}$ to forecast calendar year $T + 1 = 2018$. The weights then drop to roughly 3.0% remaining (almost) constant back to calendar year 2011, and the average weight assigned to the prior means $\mu_{i,T+1}$ is 78%. Thus, being accident-free over the last 7 past years gives a bonus of 22% in this example.

There is the following observation. The attention weights remain (almost) constant over the calendar years with a time lag of more than one year. This is rather surprising, and it may have different reasons. First, starting with a negative suspicion, the gradient descent algorithm may simply not have been able to find weights that discriminate past claims appropriately w.r.t. their seniority. In this case, this would be a major deficiency of the gradient descent algorithm used, and one should try to find a better fitting strategy. Second, these constant attention weights may indicate that there is long-range dependence and only a slowly decreasing memory of past claims history. Consequently, the observation-driven dynamic mixed effects model may not be fully suitable for this data set. In fact, this result of Figure 5.2 (top-lhs) rather then suggests to consider a latent process $(\Lambda_0 + \Lambda_t)_{t\geq 1}$, where $(\Lambda_t)_{t\geq 1}$ can be an observation-driven (dynamic) state-space process and $\Lambda_0$ is a static latent factor of Bühlmann–Straub type, see Chapter 3. In view of Figure 5.2 (top-lhs), this static latent factor $\Lambda_0$ may account for a credibility/attention weight of roughly 3.0% on past claims, and the remaining differences in weights may be covered by the dynamic state-space process $(\Lambda_t)_{t\geq 1}$. This conclusion also indicates why the observation-driven dynamic mixed effects model does not perform significantly better



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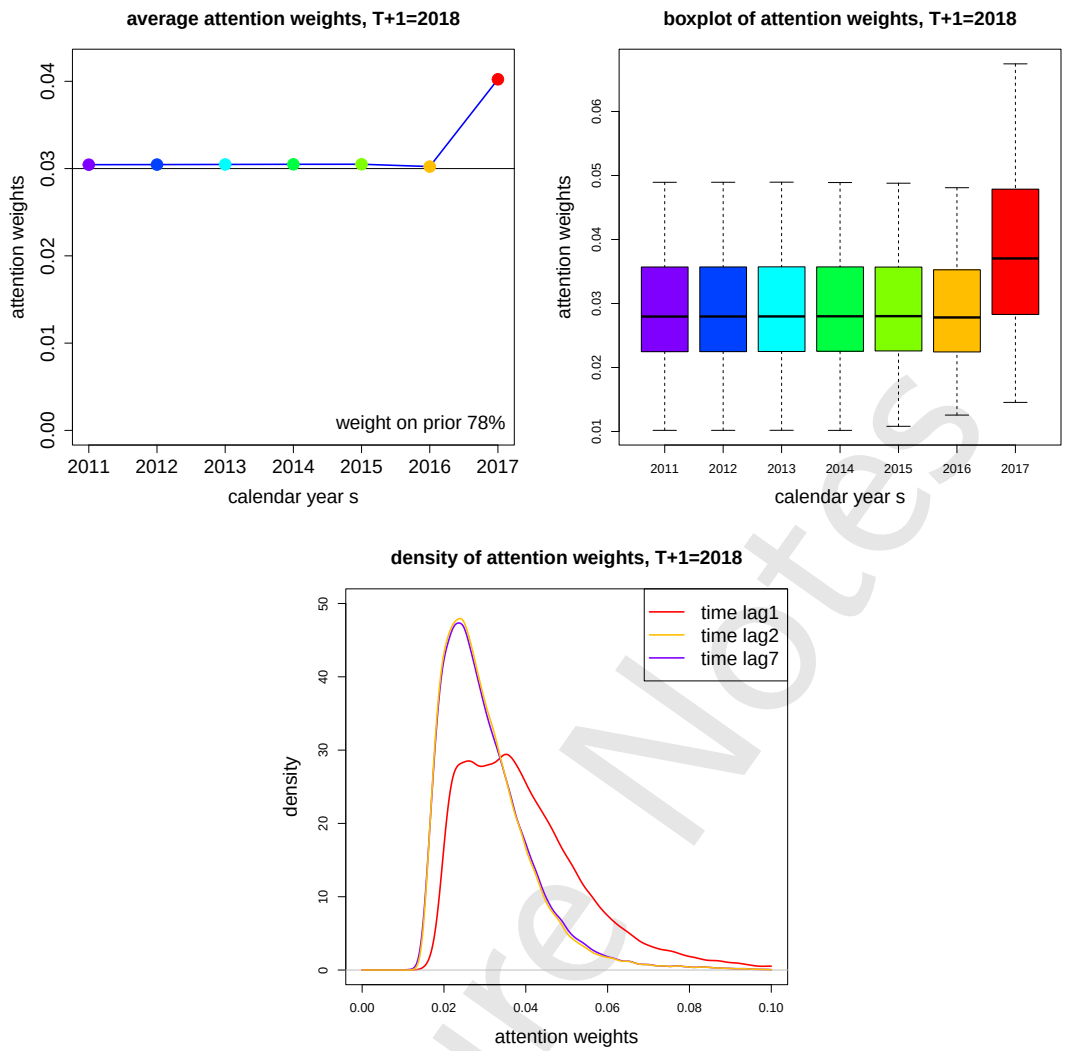


Figure 5.2: Deep attention FNN experience rating models: (top-lhs) average attention weights $(\omega_{T,s}(\boldsymbol{x}_{i,1:T+1}))_{s=2011}^{T}$ at time $T = 2017$; (top-rhs, bottom) boxplot and empirical density of the distribution of the attention weights across the insurance policies $i \in \mathcal{I}_{T+1}$.

than the static mixed effects models in predictive performance.

Figure 5.2 (top-rhs) shows a boxplot of the attention weights $(\omega_{T,s}(\boldsymbol{x}_{i,1:T+1}))_{s=2011}^{T}$ across the insurance policies $i \in \mathcal{I}_{T+1}$ for fixed time lags $s$, and Figure 5.2 (bottom) gives the corresponding empirical densities (we only show. The whiskers in the boxplot correspond to 1 interquartile range, and the outliers are not shown for illustrative reasons. We observe right-skewed densities that are fairly concentrated. From this we may also conclude that the network faces some issues in appropriately discriminating past claims with a time lag bigger than 1, on the other hand, it outperforms any other method studied in these notes. From these graphs, we conclude that an accident-free driver has a premium that decreases 3.0% every year, except in the first year the decrease is around 4.0%. This closes this example. ■

In the previous example we have found a deep learning result that clearly outperforms

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all previously studied models. This is exciting because we are getting closer to the true model, but it is also a bit frustrating because we have invested a lot of time and efforts to develop fully tractable analytical models, and now we find that these are not fully competitive. In any case, the previous deep attention FNN experience rating results inspire us to develop more deep learning approaches. Certainly, Figure 5.2 is not fully convincing, yet, as it seems that the deep learning algorithm cannot discriminate past claims with bigger time lags. Therefore, we will exploit other deep learning algorithms below.

The previous example considers a claim counts case. We emphasize that the techniques studied in this section can be applied to any other pricing situation, e.g., considering claim sizes. The basics in this section are distribution-free, we only consider conditional expectations. For fitting we then select a strictly consistent loss function for mean estimation. A natural choice for claim counts is the Poisson unit deviance (5.8). For claim amounts modeling this can be replaced by any other Bregman divergence, e.g., by a gamma unit deviance which is more suitable for claim amounts modeling, see Table 2.1, and for compound models Tweedie's compound Poisson unit deviances seem to be natural choices.

## 5.3 Attention layer weights

In the previous section, we have used a time-distributed FNN layer to calculate the attention weights $\omega_{t,s}(\boldsymbol{x}_{1:t+1})$ on past observations $Y_s = N_s/v_s$, $1 \leq s \leq t$, see (5.4)-(5.5). A disadvantage of this approach is that the covariates $\boldsymbol{x}_s$ and $\boldsymbol{x}_u$ of different time periods $s \neq u$ do not directly communicate/interact. E.g., if an insurance policyholder purchases a new car in calendar year $s$, then the joint consideration of $\boldsymbol{x}_s$ and $\boldsymbol{x}_{s+1}$ contains this information. If a new car impacts the claims behavior, this interaction should also manifest in the predictive model. Vaswani et al. [194] have introduced *attention layers*, which provide an extension of (5.4) suitable for modeling interactions across time. We present such an attention layer approach, and this introduction of attention layers closely follows Wüthrich–Buser [204, Section 5.2.4]

We introduce an attention layer that processes the input tensor $\boldsymbol{x}_{1:t+1}^\top \in \mathbb{R}^{(t+1)\times(q+1)}$. An attention layer is based on a *key*, a *query* and a *value* defined as follows. For $1 \leq s \leq t+1$ we set

$$\begin{aligned}
\boldsymbol{k}_s &= \phi_K\left(W_K \boldsymbol{x}_s\right) \in \mathbb{R}^{b_1}, \\
\boldsymbol{q}_s &= \phi_Q\left(W_Q \boldsymbol{x}_s\right) \in \mathbb{R}^{b_1}, \\
\boldsymbol{v}_s &= \phi_V\left(W_V \boldsymbol{x}_s\right) \in \mathbb{R}^{b_2},
\end{aligned} \tag{5.9}$$

with given weight matrices $W_K \in \mathbb{R}^{b_1\times(q+1)}$, $W_Q \in \mathbb{R}^{b_1\times(q+1)}$ and $W_V \in \mathbb{R}^{b_2\times(q+1)}$, $b_1, b_2 \in \mathbb{N}$, and activation functions $\phi_K$, $\phi_Q$ and $\phi_V$. Thus, we apply a time-distributed FNN layer to each time component of the input tensor $\boldsymbol{x}_{1:t+1}^\top$; for simplicity we choose zero bias vectors in (5.9). The intuition now is the following: the key $\boldsymbol{k}_s \in \mathbb{R}^{b_1}$ encodes the characteristics of covariate $\boldsymbol{x}_s$, and the query $\boldsymbol{q}_s \in \mathbb{R}^{b_1}$ states for which characteristics the covariate $\boldsymbol{x}_s$ is looking for to have a match. The components $\boldsymbol{x}_s$ and $\boldsymbol{x}_u$ then communicate



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to see whether their keys and queries match. We build the key, query and value tensors

$$\begin{aligned} K &= K(\boldsymbol{x}_{1:t+1}) &&= [\boldsymbol{k}_1, \ldots, \boldsymbol{k}_{t+1}]^\top &&\in \mathbb{R}^{(t+1)\times b_1}, \\ Q &= Q(\boldsymbol{x}_{1:t+1}) &&= [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{t+1}]^\top &&\in \mathbb{R}^{(t+1)\times b_1}, \\ V &= V(\boldsymbol{x}_{1:t+1}) &&= [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{t+1}]^\top &&\in \mathbb{R}^{(t+1)\times b_2}. \end{aligned}$$

The matching problem is solved by applying the softmax function (5.6) to the scalar products between the queries $\boldsymbol{q}_s$ and the keys $\boldsymbol{k}_u$. These scalar products are received by the matrix multiplication $QK^\top \in \mathbb{R}^{(t+1)\times(t+1)}$. Applying the softmax function to the *rows* of this matrix $QK^\top$ provides us with the *attention weight matrix*

$$\boldsymbol{\Omega} = \boldsymbol{\Omega}(\boldsymbol{x}_{1:t+1}) = \mathsf{softmax}\left(\frac{QK^\top}{\sqrt{b_1}}\right) \in [0,1]^{(t+1)\times(t+1)}; \tag{5.10}$$

usually the scaling $\sqrt{b_1}$ is applied to have a dimension-free expression. Thus, each row of the attention weight matrix $\boldsymbol{\Omega}(\boldsymbol{x}_{1:t+1})$ gives a (weighted) vector in the unit $t$-simplex. If the key $\boldsymbol{k}_u$ of covariate $\boldsymbol{x}_u$ matches the query $\boldsymbol{q}_s$ of covariate $\boldsymbol{x}_s$, then their scalar product

$$\langle \boldsymbol{q}_s, \boldsymbol{k}_u \rangle = \boldsymbol{q}_s^\top \boldsymbol{k}_u = (QK^\top)_{s,u} \quad \text{should be large,}$$

and consequently, through the softmax application, their attention weight $\boldsymbol{\Omega}_{s,u} \in [0,1]$ is large. The last step is to multiply that attention weight matrix with the value matrix providing the *attention layer*

$$\boldsymbol{x}_{1:t+1} \mapsto \boldsymbol{z}^{\text{att}}(\boldsymbol{x}_{1:t+1}) = \boldsymbol{\Omega}(\boldsymbol{x}_{1:t+1})V(\boldsymbol{x}_{1:t+1}) \in \mathbb{R}^{(t+1)\times b_2}. \tag{5.11}$$

The attention layer $\boldsymbol{z}^{\text{att}}(\boldsymbol{x}_{1:t+1})$ contains in each element a weighted average (through the rows of $\boldsymbol{\Omega}$) of the columns of the value matrix $V$.

**Remarks 5.3.**

- Similarly to Remarks 5.1, to apply the attention layer successfully one needs to add a positional encoding because the attention layer itself does not have any notion of time (position). This is unlike RNNs that respect time-causality through the recursive structure.
- We are going to mask keys $\boldsymbol{k}_s$ and queries $\boldsymbol{q}_s$ without exposures $v_s = 0$ with a mask value of $-\infty$ so that these receive a zero attention weight; this is similar to (5.7).
- The key, query and value are FNN layer transformations of the inputs in (5.9) using a time-distributed FNN layer with given weight matrices $W_K$, $W_Q$ and $W_V$, and activation function $\phi_K$, $\phi_Q$ and $\phi_V$. We are going to use time-distributed (deep) FNNs $\boldsymbol{z}^{(d:1)}_{\text{td}(K)}$ and $\boldsymbol{z}^{(d:1)}_{\text{td}(Q)}$ for the key and the query, and for attention layers there is some preference for the Gaussian error linear unit (GELU) activation function given by $x \mapsto \phi(x) = x\Phi(x)$, where $\Phi$ denotes the standard Gaussian distribution function. For the value we use the identity for the activation function $\phi_V$, which allows us to include past observations linearly.

There remains the explicit specifications of the key, query and value. We start with the key and the query. Using two (different) time-distributed multi-output FNNs with



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output dimension $b_1$, we choose the key and the query as follows

$$\begin{aligned} K(\boldsymbol{x}_{1:t+1}) &= \boldsymbol{z}^{(d:1)}_{\mathrm{td}(K)}(\boldsymbol{x}^\top_{1:t+1}) = \left(\boldsymbol{z}^{(d:1)}_K(\boldsymbol{x}_1), \ldots, \boldsymbol{z}^{(d:1)}_K(\boldsymbol{x}_{t+1})\right)^\top \in \mathbb{R}^{(t+1)\times b_1}, \\ Q(\boldsymbol{x}_{1:t+1}) &= \boldsymbol{z}^{(d:1)}_{\mathrm{td}(Q)}(\boldsymbol{x}^\top_{1:t+1}) = \left(\boldsymbol{z}^{(d:1)}_Q(\boldsymbol{x}_1), \ldots, \boldsymbol{z}^{(d:1)}_Q(\boldsymbol{x}_{t+1})\right)^\top \in \mathbb{R}^{(t+1)\times b_1}, \end{aligned}$$

with two deep FNNs $\boldsymbol{z}^{(d:1)}_K : \mathbb{R}^{q+1} \to \mathbb{R}^{b_1}$ and $\boldsymbol{z}^{(d:1)}_Q : \mathbb{R}^{q+1} \to \mathbb{R}^{b_1}$ having GELU activation functions, we also refer to (5.4). This encodes the covariate information $\boldsymbol{x}_{1:t+1}$ non-linearly. Then, we let this encoded information interact across time through the attention weight matrix $\boldsymbol{\Omega}(\boldsymbol{x}_{1:t+1})$, and this attention weight matrix has row sums equal to one

$$\sum_{s=1}^{t+1} \boldsymbol{\Omega}_{k,s}(\boldsymbol{x}_{1:t+1}) = 1 \qquad \text{for all } 1 \le k \le t+1, \tag{5.12}$$

see (5.10). Thus, every row of of the attention weight matrix $\boldsymbol{\Omega}$ can be seen as a sequence of attention weights $(\omega_{t,s})_{s=0}^t$ in the unit $t$-simplex.
Targeting for an affine mean function in the past claims history $Y_{1:t}$ motivates to choose a value different from (5.9), namely, for our problem we choose the value

$$V(Y_{1:t}) = (Y_1, \ldots, Y_t, 0)^\top \in \mathbb{R}^{t+1}.$$

We set the last value entry equal to zero, so that it does not influence the prediction; we do not carry this last (normalizing) value forward. In fact, we use it to balance the weights between the observations $Y_{1:t}$. Using this attention layer we receive the first part of the prediction

$$\boldsymbol{z}^{\mathrm{att}}(\boldsymbol{x}_{1:t+1}, Y_{1:t}) = \boldsymbol{\Omega}(\boldsymbol{x}_{1:t+1})V(Y_{1:t}) = \left(\sum_{s=1}^{t} \boldsymbol{\Omega}_{k,s}(\boldsymbol{x}_{1:t+1})Y_s\right)_{k=1}^{t+1}.$$

There remains the fixed effects part $\mu^\omega_{t+1}(\boldsymbol{x}_{t+1})$ to predict $Y_{t+1} = N_{t+1}/v_{t+1}$. For this we simply choose another deep FNN. This then motivates the *attention layer weighting experience premium*

$$\begin{aligned} \mu^{\mathrm{post}}_{N_{t+1}|N_{1:t}} &= v_{t+1}\left(\frac{(1,\ldots,1)}{t+1}\boldsymbol{z}^{\mathrm{att}}(\boldsymbol{x}_{1:t+1}, Y_{1:t}) + \mu^\omega_{t+1}(\boldsymbol{x}_{t+1})\right) \\ &= v_{t+1}\left(\sum_{s=1}^{t}\left(\frac{1}{t+1}\sum_{k=1}^{t+1}\boldsymbol{\Omega}_{k,s}(\boldsymbol{x}_{1:t+1})\right)Y_s + \mu^\omega_{t+1}(\boldsymbol{x}_{t+1})\right). \end{aligned} \tag{5.13}$$

This is an affine function of the past observations $Y_{1:t}$. Listing 5.2 shows the corresponding code. We give some remarks.

**Remarks 5.4.**

- The inner big round bracket in (5.13) gives the weights for the affine function in the observations $Y_{1:t}$. Since the chosen attention layer has $t+1$ rows of weighted entries, see (5.12), we rescale them back to unit weight in (5.13) by dividing by $t+1$. As a result, all elements of this rescaled attention weight matrix add up to one, and the row sums are equal to $1/(t+1)$. This provides the credibility weighting interpretation with normalized weights. For later reference we set

$$\omega_{t,s} = \omega_{t,s}(\boldsymbol{x}_{1:t+1}) = \frac{1}{t+1}\sum_{k=1}^{t+1}\boldsymbol{\Omega}_{k,s}(\boldsymbol{x}_{1:t+1}).$$



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- We exclude the FNN fixed effects $\mu^{\omega}_{t+1}(\boldsymbol{x}_{t+1})$ from the weighting scheme, and therefore this should be understood as a weighted quantity, (carefully) indicated by the upper index $^{\omega}$ in $\mu^{\omega}_{t+1}(\boldsymbol{x}_{t+1})$. The reason for excluding this fixed effects term from the attention layer weighting is that we did not receive very good fitting results by integrating also the fixed effects into the attention layer mechanism; this was our experience, however, we did not exploit this in much depth. In the next Example 5.5, we observe that this proposal similar results than the previous example.

■

**Example 5.5** (attention layer weighting experience rating model)**.** We continue Example 5.2, and we implement the attention layer weighting approach (5.13); the detailed R code is given in Listing 5.2. Similarly to Example 5.2, in this non-parametric modeling approach we have only specified the conditional means of the model, i.e., at this stage it is fully distribution-free. We choose the Poisson deviance loss (5.8) for model fitting.

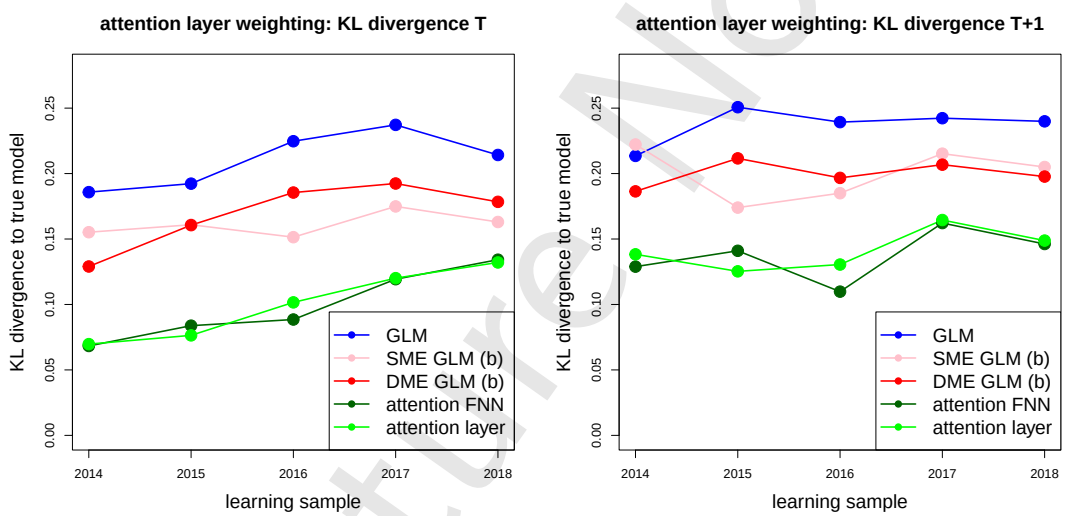


Figure 5.3: Attention layer weighting experience rating premiums: (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one year ahead forecast; the other results are taken from Figure 5.1; units are $10^{-2}$.

Figure 5.3 shows the results, and we note that these results are very similar. From this we conclude that covariate interactions do not seem to be an essential feature to improve predictive performance.

In Figure 5.4 we show the resulting attention weights on past claims. Interestingly, we come to a rather similar conclusion to our previous example, see Figure 5.2. Namely, it seems that attention weights for claims with bigger time lags should be quite constant. Again, this is in support of a mixture of a static and a dynamic random effects model supporting the inclusion of a Bühlmann–Straub type static latent component. This closes this example. ■

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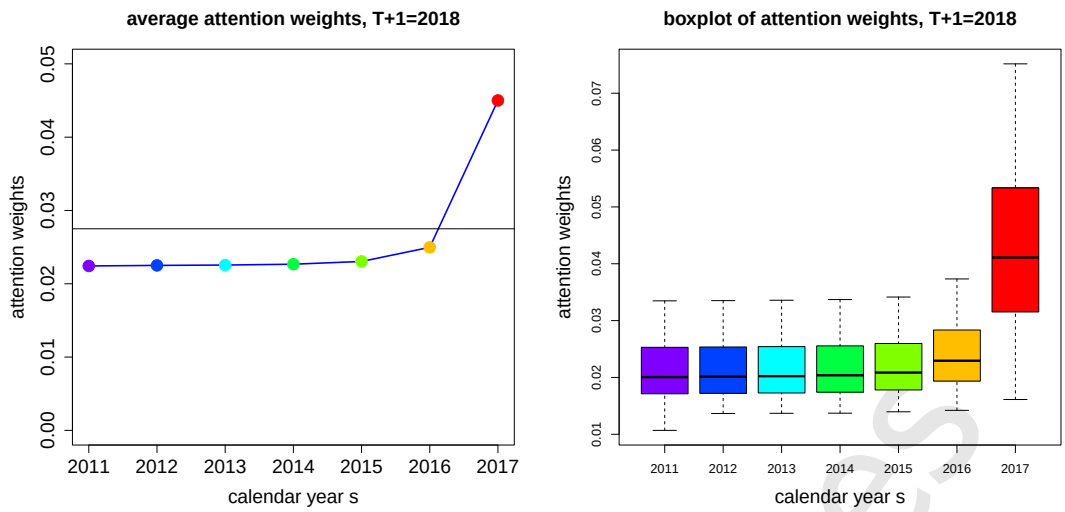


Figure 5.4: Attention layer weighting experience rating: (lhs) average attention weights $(\omega_{T,s}(\boldsymbol{x}_{i,1:T+1}))_{s=2011}^{T}$ at time $T = 2017$; (rhs) boxplot of the distribution of the attention weights across the insurance policies $i \in \mathcal{I}_{T+1}$.

## 5.4 Deep experience rating models

The previous two sections have been focusing on affine models in the past claims history $N_{1:t}$ and $Y_{1:t}$, respectively. This has been inspired by the Bühlmann–Straub framework of Chapter 3 which tried to optimally approximate the posterior mean $\mu^{\text{post}}_{N_{t+1}|N_{1:t}}$ by a predictor that is linear in this past claims history. In fact, under conjugate priors this approximation is even optimal (exact), see Proposition 3.6. In the present section we sacrifice this linearity to see whether this leads to an improvement of prediction accuracy. We call such models *deep experience rating models*.

There are many different ways in considering the claims history $(N_{1:t}, \boldsymbol{x}_{1:t})$. The most common approaches for considering such time-series inputs are RNNs, convolutional neural networks (CNNs) or attention layers and Transformers. In the previous section we have already set the ground for considering an attention layer approach, thus, we are going generalize this previous approach by also letting past observations entering the regression function in a non-linear way.

We consider the attention layer mechanism introduced in (5.11), but we modify the original input tensor $\boldsymbol{x}_{1:t+1}^{\top}$, by equipping this tensor also with claims information. We define the claims history by

$$N_{1:t+1}^{+} = \Big(N_1 \mathbb{1}_{\{v_1>0\}} - 2\mathbb{1}_{\{v_1=0\}}, \ldots, N_t \mathbb{1}_{\{v_t>0\}} - 2\mathbb{1}_{\{v_t=0\}}, -2\Big)^{\top} \;\in\; (\mathbb{N}_0 \cup \{-2\})^{t+1}. \tag{5.14}$$

This sets past claims history to a mask value of $-2$ if there has not been any exposure in the corresponding calendar year, and we also set the $(t+1)$-st component equal to $-2$, because this component has not been observed at time $t$, i.e., this is the response $N_{t+1}$ that needs to be forecasted. We define the extended input tensor by

$$(\boldsymbol{x}_{1:t+1}^{+})^{\top} = \Big[[\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{t+1}]^{\top}, N_{1:t+1}^{+}, v_{1:t+1}\Big] \;\in\; \mathbb{R}^{(t+1)\times(q+3)}. \tag{5.15}$$



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Thus, we equip the input covariate $\boldsymbol{x}_t$ with the corresponding claim $N_t$ and exposure $v_t \geq 0$ resulting in an extended input $\boldsymbol{x}_t^+ \in \mathbb{R}^{q+3}$, and we input both to the attention layer. This provides us with key, query and value for $1 \leq s \leq t+1$

$$\begin{aligned} \boldsymbol{k}_s^+ &= \boldsymbol{z}_K^{(d:1)}\left(\boldsymbol{x}_s^+\right) \in \mathbb{R}^{b_1}, \\ \boldsymbol{q}_s^+ &= \boldsymbol{z}_Q^{(d:1)}\left(\boldsymbol{x}_s^+\right) \in \mathbb{R}^{b_1}, \\ \boldsymbol{v}_s^+ &= \boldsymbol{z}_V^{(d:1)}\left(\boldsymbol{x}_s^+\right) \in \mathbb{R}^{b_2}, \end{aligned} \tag{5.16}$$

with three time-distributed deep FNNs. Correspondingly, we replace the attention layer (5.11) by its extended version also including past claims history

$$\boldsymbol{x}_{1:t+1}^+ \mapsto \boldsymbol{z}^{\text{att}}(\boldsymbol{x}_{1:t+1}^+) = \boldsymbol{\Omega}(\boldsymbol{x}_{1:t+1}^+)V(\boldsymbol{x}_{1:t+1}^+) \in \mathbb{R}^{(t+1)\times b_2}, \tag{5.17}$$

where the attention weight matrix $\boldsymbol{\Omega}$ is based on the extended keys $\boldsymbol{k}_{1:t+1}^+$ and queries $\boldsymbol{q}_{1:t+1}^+$ given in (5.16), and also the value matrix $V$ is based on the extended values $\boldsymbol{v}_{1:t+1}^+$. This attention layer $\boldsymbol{z}^{\text{att}}(\boldsymbol{x}_{1:t+1}^+)$ now encodes the entire past history $(N_{1:t}, \boldsymbol{x}_{1:t})$, and we need to decode it to optimally to predict the next response $N_{t+1}$. Note that the encoding of $N_{1:t}$ is non-linear if we choose non-trivial deep FNNs in (5.16). Therefore, we are no longer in the (linear) Bühlmann–Straub framework of Chapter 3. Moreover, having large deep FNNs we can approximate a fairly large function class, and we are fairly flexible in approximating the conditional expectations; see the universality theorems, e.g., by Cybenko [56] and Hornik et al. [106].

In a next step, we apply a flatten operator to this attention layer output (5.17) which maps the matrix $\boldsymbol{z}^{\text{att}}(\boldsymbol{x}_{1:t+1}^+) \in \mathbb{R}^{(t+1)\times b_2}$ to a vector $\vec{\boldsymbol{z}}^{\text{att}}(\boldsymbol{x}_{1:t+1}^+) \in \mathbb{R}^{(t+1)b_2}$. We concatenate this vector with the covariates $\boldsymbol{x}_{t+1}$ of response $N_{t+1}$, and we use another deep FNN $\boldsymbol{z}^{(d:1)}$ to decode this information

$$\left(\boldsymbol{x}_{t+1}, \vec{\boldsymbol{z}}^{\text{att}}(\boldsymbol{x}_{1:t+1}^+)\right) \mapsto \mu_{t+1} = \boldsymbol{z}^{(d:1)}\left(\boldsymbol{x}_{t+1}, \vec{\boldsymbol{z}}^{\text{att}}(\boldsymbol{x}_{1:t+1}^+)\right). \tag{5.18}$$

This gives us a deep attention experience rating model; its R code and further specifications are given in Listing 5.3.

**Remarks 5.6.**

- We believe that this architecture (5.18) is quite exciting because it flexibly processes past claims experience $N_{1:t}$. But this construction is not limited to past claims experience, e.g., if we have telematics data of extreme braking events and speeding, this can easily be integrated into the tensor (5.15) which then leads to very flexible experience rating models, that is, this could be way of modifying, e.g., Corradin et al. [52] to a deep learning experience rating model including telematics information.

- Our considerations have naturally assumed an annual time grid. Of course, this is not imperative and we can choose any other time scale. E.g., moving more to real time pricing, a weekly time scale may be suitable which is more in the spirit of the pay-how-you-drive pricing schemes including telematics information as studied, e.g., in Yanez et al. [207].

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- Our deep experience rating model (5.18) only includes one attention layer, thus, only one building block of the Transformer architecture of Vaswani et al. [194]. Transformers are more sophisticated, see Vaswani et al. [194, Figure 1]. First, a Transformer architecture considers multiple parallel attention layers (called mulit-head attention). Second, their output is added to its tensor input (using a skip/residual connection), and this merged information is then further processed by FNNs and residual connections. We leave it to the reader to exploit such more complicated network architectures.

■

**Example 5.7** (deep experience rating models)**.** We continue Examples 5.2 and 5.5. We proceed similarly as in these two previous examples, and we only change the network architecture. Our choice is depicted in Listing 5.3. We mask all claim counts that are non-existent according to (5.14), and time lag is integrated as a positional encoding to the attention layer input. We then fit this deep (attention) experience rating model to the learning data $\mathcal{L}_T$ using the Poisson deviance loss (5.8). The results are presented in Figure 5.5.

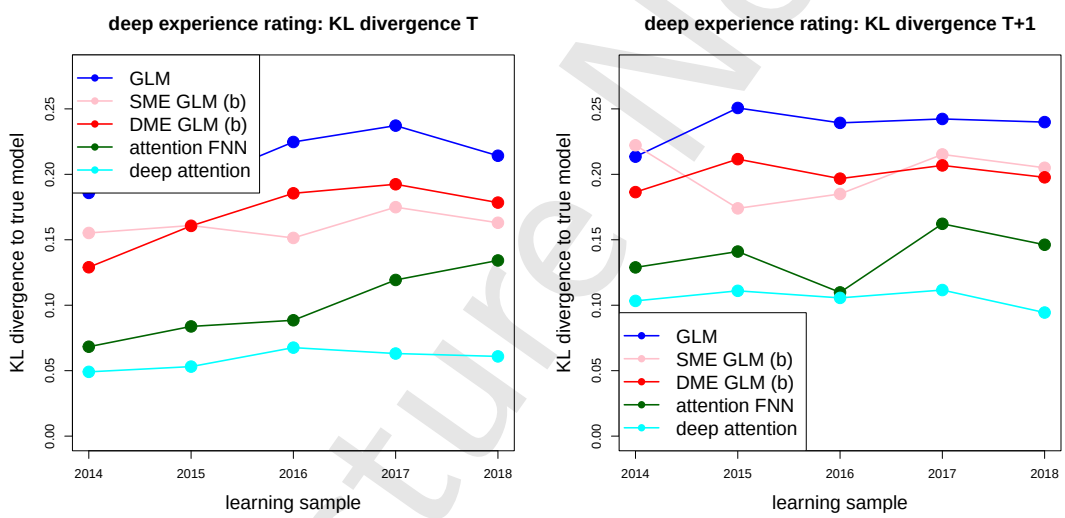


Figure 5.5: Deep experience rating models (labeled 'deep attention'): (lhs) year $T$ average KL divergences to the true models, (rhs) year $T+1$ average KL divergences to the true models, this is the one year ahead forecast; the other results are taken from Figure 5.1; units are $10^{-2}$.

We observe that this non-linear inclusion of past claims history $(N_{1:t}, \boldsymbol{x}_{1:t})$ gives the best predictive model among all options considered in these notes, i.e., the following regression functions is at the end the preferred one across all models studied

$$(N^+_{1:t+1}, \boldsymbol{x}_{1:t+1}) \;\mapsto\; \mu^{\text{post}}_{N_{t+1}|N_{1:t}} = v_{t+1}\mu_{t+1} = v_{t+1}\, \boldsymbol{z}^{(d:1)}\left(\boldsymbol{x}_{t+1}, \bar{\boldsymbol{z}}^{\text{att}}(\boldsymbol{x}^+_{1:t+1})\right).$$

The average in-sample KL divergence to the true model is roughly $6 \cdot 10^{-4}$ and the average out-of-sample KL divergence to the true model roughly doubles to $9 \cdot 10^{-4}$. Thus, this is by far the smallest value that we have found across all studied models.



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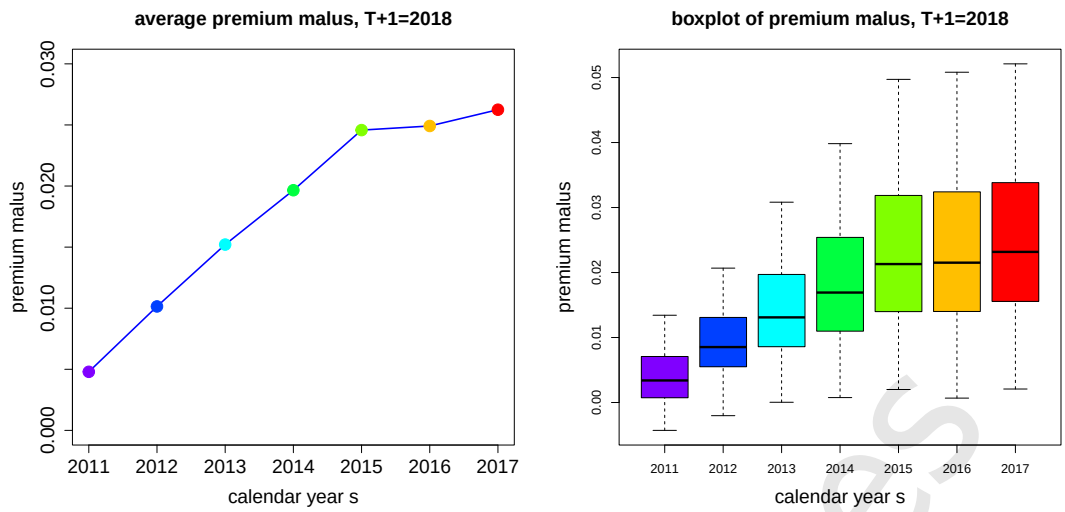


Figure 5.6: Deep experience rating: (lhs) average premium malus if a single claim in one of the past years at time $T = 2017$; (rhs) boxplot of the distribution of the premium malus across the insurance policies $i \in \mathcal{I}_{T+1}$.

Since we now have a non-linear model in past claims experience we can no longer plot the attention weights $(\omega_{t,s})_{s=1}^{t}$ on past claims $N_{1:t}$ and $Y_{1:t}$, respectively, similar to Figures 5.2 and 5.4. However, we can input different past claims histories into the deep experience rating premium computation. The base case is given by not having any claims in the past $N_{i,1:t} = (0, \ldots, 0)$, and we can compare this to the claims histories of having exactly one accident $N_{i,1:T} = (0, \ldots, 0, 1, 0, \ldots, 0)$ in one of the past years. This is precisely how Figure 5.6 has been computed on all policies $i$ which have been exposed during the entire observation period, i.e., the same policies used in Figures 5.2 and 5.4. We now see a more interesting picture compared to the previous two plots, this picture suggest a clear seniority weighting of past claims, however, not differentiating much the most recent three years. This seems interesting, and the underlying reasons may be manifold which would certainly require more time to investigate.

Our final analysis tests the resulting deep experience rating premiums $\mu^{\text{post}}_{N_{T+1}|N_{1:T}}/v_{T+1}$ for auto-calibration. We consider this test in-sample on the last fitted year $T+1 = 2018$; for auto-calibration we refer to Section 2.5.2. In order to test for auto-calibration we perform the isotonic recalibration step (2.64), with one essential difference though, namely, we replace the observations $Y_{i,T+1}$ by the true expected frequencies $\mu^*(\boldsymbol{x}_{i,T+1})$ because they are available for our synthetic data examples. That is, we aim at benchmarking against the true model. Figure 5.7 shows the results, and their interpretation is similarly to Figure 2.10. First, the blue empirical density shows the distribution of the fitted deep experience rating premiums (this corresponds to the $y$-axis on the right-hand side). The isotonic recalibration step is then illustrated by the black dots, Figure 5.7 (lhs) shows the results on the original scale and Figure 5.7 (rhs) on the log-scale. If we have perfect auto-calibration, the black dots should all lie on the diagonal orange line. In view of Figure 5.7 we conclude that for very small expected frequencies, we have some under-estimation. In the upper tail we have a slight S-shape, first slightly under-estimating and then (may

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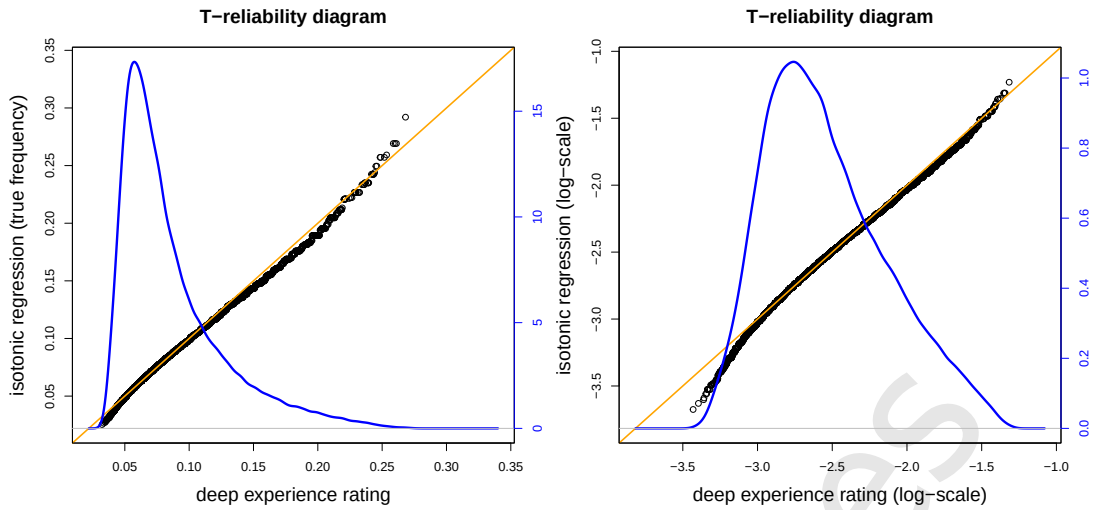


Figure 5.7: $T$-reliability diagram based on an isotonic recalibration step: (lhs) original scale, (rhs) log-scale.

be) slightly over-estimating. However, in the main body of the expected frequencies we seem to have an excellent fit, which is clearly in support of this model, and we have to admit that fitting tails is a generally difficult problem. We can conclude that this model has led to the best predictive performance. This closes our series of examples. ■

## 5.5 Summary and outlook

We have studied fixed effects regression models and experience rating models, where complexity has been increasing throughout these notes. Finally, we came up with an attention-based deep learning model that provided the best predictive performance. Predictive performance has always been benchmarked against the true model. This attention-based deep learning 'model' is not really a full model, because we do not have a distributional model, but we only designed regression functions in the last modeling approaches. These regression functions have been fitted using the Poisson deviance loss. Going forward there are many item that require further investigation. We list some of them.

- It should be confirmed that real data has similar features because our results are based on simulated (synthetic) data. The simulation model has been designed and fitted based on real data.
- Our deep learning architecture can be benchmark with other models, e.g., a proper multi-head attention Transformer model.
- We only study claim counts, and literature emphasizes that in these time-series models, also claim amounts may play a significant role to get good forecast models.
- Our study is based on a small feature set, and increased feature sets and telematics data should be integrated, too. This will be challenging as telematics data often

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has a poor quality, but it will certainly be exciting to see how new data sources can further improve predictive performance.

- Real time pricing is another exciting possible next step.
- Explainability of models and results is another challenge that we have not touched upon in these notes; see Lorentzen–Mayer [141] and Mayer et al. [145]. The same, of course, applies to commercial premium schemes, because stakeholders should understand the product and prices they purchase.

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Listing 5.1: Deep attention FNN experience rating model.

```
network.attention.weighting <- function(seed, q0, lambda0, cat0, b0){
    k_clear_session()
    set.seed(seed)
    set_random_seed(seed)
    Design   <- layer_input(shape = c(q0[1], q0[2]), dtype = 'float32')
    Cat      <- layer_input(shape = c(q0[1]), dtype = 'int32')
    Exposure <- layer_input(shape = c(q0[1]), dtype = 'float32')
    Obs      <- layer_input(shape = c(q0[1]), dtype = 'float32')
    PosT     <- layer_input(shape = c(q0[1]), dtype = 'float32')
    YY       <- layer_input(shape = c(q0[1]-1), dtype = 'float32')
    ### credibility weights for past claims
    CatEmb0 = Cat %>% layer_reshape(c(q0[1],1)) %>%
        time_distributed(layer_embedding(input_dim=cat0[1], output_dim=b0[1],
                         input_length=1)) %>%  time_distributed(layer_flatten())
    #
    PosT0  = PosT %>% layer_reshape(c(q0[1],1))
    Obs0   = Obs  %>% layer_reshape(c(q0[1],1))
    Obs1   = Obs  %>% k_sum(axis=2) %>% layer_reshape(c(1)) %>% layer_repeat_vector(c(q0[1]))
    #
    Network = list(Design, CatEmb0, PosT0, Obs1/q0[1]) %>%
          layer_concatenate() %>% time_distributed(layer_layer_normalization()) %>%
          time_distributed(layer_dense(units=q0[3], activation='gelu')) %>%
          time_distributed(layer_dense(units=q0[5], activation='gelu')) %>%
          time_distributed(layer_layer_normalization()) %>%
          time_distributed(layer_dense(units=1, activation='gelu'))
    Weights = Obs0 * Network - 100 * (1-Obs0)  # setting off periods without observations
    Weights = Weights %>% layer_flatten() %>% layer_activation_softmax()
    ### regression function for expected frequency
    CatEmb1 = Cat[,q0[1]] %>%
      layer_embedding(input_dim = cat0[1], output_dim = b0[1], input_length = 1) %>%
                      layer_flatten()
    #
    Mu = list(Design[,q0[1],], CatEmb1) %>% layer_concatenate() %>%
          layer_dense(units=q0[2], activation='tanh') %>%
          layer_dense(units=q0[3], activation='tanh') %>%
          layer_dense(units=q0[4], activation='tanh') %>%
          layer_dense(units=1, activation='exponential',
                    weights=list(array(0, dim=c(q0[4],1)), array(log(lambda0), dim=c(1))))
    #
    YY = YY/(Exposure[,1:(q0[1]-1)]+0.001) # normalize with exposure
    Vector = list(YY, Mu) %>% layer_concatenate()
    Response = list(Weights, Vector) %>% layer_multiply() %>% k_sum(axis=2) %>%
                  layer_reshape(c(1))
    Response = list(Response, Exposure[,q0[1]]) %>% layer_multiply()
    #
    keras_model(inputs = c(Design, Cat, Exposure, Obs, PosT, YY), outputs = c(Response))
    }
```

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Listing 5.2: Deep attention layer weighting experience rating model.

```
attention.layer.weighting <- function(seed, q0, lambda0, cat0, b0, mask){
    k_clear_session()
    set.seed(seed)
    set_random_seed(seed)
    Design   <- layer_input(shape = c(q0[1], q0[2]), dtype = 'float32')
    Cat      <- layer_input(shape = c(q0[1]), dtype = 'int32')
    Exposure <- layer_input(shape = c(q0[1]), dtype = 'float32')
    Obs      <- layer_input(shape = c(q0[1]), dtype = 'float32')
    PosT     <- layer_input(shape = c(q0[1]), dtype = 'float32')
    # the last response needs to identical to zero (and not carry any information)
    YY0      <- layer_input(shape = c(q0[1]), dtype = 'float32')
    ### expected frequency new response
    CatEmb1 = Cat[,q0[1]] %>%
        layer_embedding(input_dim=cat0[1], output_dim=b0[1], input_length=1) %>%
        layer_flatten()
    #
    Mu = list(Design[,q0[1],], CatEmb1) %>% layer_concatenate() %>%
          layer_dense(units=q0[3], activation='tanh') %>%
          layer_dense(units=q0[4], activation='tanh') %>%
          layer_dense(units=q0[5], activation='tanh') %>%
          layer_dense(units=1, activation='exponential',
                weights=list(array(0, dim=c(q0[5],1)), array(log(lambda0), dim=c(1))))
    #
    ### attention weights
    CatEmb0 = Cat %>% layer_reshape(c(q0[1],1)) %>%
        time_distributed(layer_embedding(input_dim=cat0[1], output_dim=b0[1], input_length=1)) %>%
        time_distributed(layer_flatten())
    #
    PosT0  = PosT %>% layer_reshape(c(q0[1],1))
    Obs0   = Obs %>% layer_reshape(c(q0[1],1))
    Obs1   = Obs %>% k_sum(axis=2) %>% layer_reshape(c(1)) %>% layer_repeat_vector(c(q0[1]))
    Input  = list(Design, CatEmb0, PosT0, Obs1/q0[1]) %>% layer_concatenate() %>%
                   time_distributed(layer_layer_normalization())
    ### query and key
    Q  =  Input %>%
            time_distributed(layer_dense(units = q0[5], activation = "gelu")) %>%
            time_distributed(layer_layer_normalization()) %>%
            time_distributed(layer_dense(units = 1, activation = "gelu"))
    Q = Obs0 * Q - mask * (1-Obs0)
    K = Input %>%
             time_distributed(layer_dense(units = q0[5], activation = "gelu")) %>%
             time_distributed(layer_layer_normalization()) %>%
             time_distributed(layer_dense(units = 1, activation = "gelu"))
    K = Obs0 * K - mask * (1-Obs0)
    V = YY0/(Exposure+0.001)
    V = V %>% layer_reshape(c(q0[1],1))
    #
    Attention = layer_attention(list(Q,V,K), use_scale = FALSE, trainable = TRUE) %>%
                layer_reshape(q0[1])
    Experience = Attention %>% k_sum(axis=2) %>% layer_reshape(c(1))
    Response   = list(Experience/q0[1], Mu) %>% layer_add()
    Response   = list(Response, Exposure[,q0[1]]) %>% layer_multiply()
    #
    keras_model(inputs = c(Design, Cat, Exposure, Obs, PosT, YY0), outputs = c(Response))
    }
```

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Listing 5.3: Deep attention experience rating model.

```
deep.attention <- function(seed, q0, lambda0, cat0, b0){
    k_clear_session()
    set.seed(seed)
    set_random_seed(seed)
    Design   <- layer_input(shape = c(q0[1], q0[2]), dtype = 'float32')
    Cat      <- layer_input(shape = c(q0[1]), dtype = 'int32')
    Exposure <- layer_input(shape = c(q0[1]), dtype = 'float32')
    Obs      <- layer_input(shape = c(q0[1]), dtype = 'float32')
    PosT     <- layer_input(shape = c(q0[1]), dtype = 'float32')
    # unobserved responses should be masked
    YY0      <- layer_input(shape = c(q0[1]), dtype = 'float32')
    ### attention weights
    CatEmb = Cat %>% layer_reshape(c(q0[1],1)) %>%
        time_distributed(layer_embedding(input_dim=cat0[1], output_dim=b0[1], input_length=1)) %>%
        time_distributed(layer_flatten())
    #
    Expo   = Exposure %>% layer_reshape(c(q0[1],1))
    PosT0  = PosT %>% layer_reshape(c(q0[1],1))
    Obs0   = Obs %>% layer_reshape(c(q0[1],1))
    YY00   = YY0 %>% layer_reshape(c(q0[1],1))
    #
    Input  = list(Design, CatEmb, Expo, PosT0, Obs0, YY00) %>% layer_concatenate()
    ### query and key
    Q =  Input %>%
            time_distributed(layer_dense(units = q0[5], activation = "gelu")) %>%
            time_distributed(layer_dense(units = q0[6], activation = "gelu"))
    K =  Input %>%
            time_distributed(layer_dense(units = q0[5], activation = "gelu")) %>%
            time_distributed(layer_dense(units = q0[6], activation = "gelu"))
    V =  Input %>%
            time_distributed(layer_dense(units = q0[5], activation = "gelu")) %>%
            time_distributed(layer_dense(units = q0[6], activation = "gelu"))
    Attention = layer_attention(list(Q,V,K), use_scale = FALSE, trainable = TRUE) %>%
            layer_flatten()
    ### expected frequency new response
    CatEmb1 = Cat[,q0[1]] %>%
        layer_embedding(input_dim=cat0[1], output_dim=b0[1], input_length=1) %>%
        layer_flatten()
    ##
    Network = list(Design[,q0[1],], CatEmb1, Attention) %>% layer_concatenate() %>%
          layer_dense(units=q0[3], activation='tanh') %>%
          layer_dense(units=q0[4], activation='tanh') %>%
          layer_dense(units=q0[5], activation='tanh') %>%
          layer_dense(units=1, activation='exponential',
                weights=list(array(0, dim=c(q0[5],1)), array(log(lambda0), dim=c(1))))

    Response = list(Network, Exposure[,q0[1]]) %>% layer_multiply()
    # %>% layer_reshape(c(1))
    #
    keras_model(inputs = c(Design, Cat, Exposure, Obs, PosT, YY0), outputs = c(Response))
    }
```

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# Appendix A

# Description of the Data

We consider a synthetic motor insurance data set. This data set has been generated from a stochastic model that has been fitted to real motor insurance claims data. In a first step we have simulated the covariates of an insurance portfolio with $150,000$ insurance policies, and based on this individual car driver information we have simulated claims histories over the calendar years 2011-2019. An excerpt of the simulated panel data is given in Listing A.1.

Listing A.1: Excerpt of the simulated motor insurance data.

```
'data.frame':   1350000 obs. of  17 variables:
 $ PolicyID  : int  100001 100001 100001 100001 100002 100002 100002 ...
 $ CarID     : num  1 1 1 1 1 1 1 1 1 2 ...
 $ StartYear : num  2010 2010 2010 2010 2010 2010 2010 2010 2010 2010 ...
 $ Year      : num  2011 2012 2013 2014 2011 ...
 $ Exposure  : num  1 1 1 1 1 1 1 1 1 1 ...
 $ ClaimNb   : num  0 0 0 0 0 0 0 0 1 0 ...
 $ DrivAge   : num  31 32 33 34 49 50 51 52 53 54 ...
 $ Gender    : Factor w/ 2 levels "female","male": 1 1 1 1 2 2 2 2 2 2 ...
 $ Region    : Factor w/ 15 levels "R11","R12","R13",..: 13 13 13 13 4 4 ...
 $ Leasing   : num  0 0 0 0 0 0 0 0 0 1 ...
 $ LowMileage: num  1 1 1 1 0 0 0 0 0 0 ...
 $ CatPrice  : num  16200 16200 16200 16200 50000 50000 50000 50000 ...
 $ VehAge    : num  5 6 7 8 1 2 3 4 5 0 ...
 $ Deductible: num  0 0 0 0 0 0 0 0 0 0 ...
 $ TermCar   : Factor w/ 2 levels "NO","YES": 1 1 1 1 1 1 1 1 2 1 ...
 $ EndYear   : num  2014 2014 2014 2014 NA ...
 $ True      : num  0.0797 0.0982 0.0798 0.0629 0.0754 ...
```

We start by describing the data of Listing A.1 and discuss how it has been constructed.

- `PolicyID`. Unique insurance policy identifier running from $100,001$ to $366,975$.
- `CarID`. Every insurance policy, `PolicyID`, has unique identifiers `CarID` for the vehicles used. If the car on a certain insurance policy is the same over the entire insurance period, then `CarID` will be 1. However, a car driver may change the vehicle during the insurance period, which will result in a new `CarID`. The variable `TermCar` indicates whether the vehicle has been changed during the considered calendar year, `Year`, and in calendar year `Year` $+1$ the new vehicle details will be available.

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- `StartYear`. Year of insurance contract inception; `StartYear` being equal to 2010 means that the contract has been incepted before our observation period that starts in calendar year 2011.
- `Year`. Actual calendar year considered.
- `Exposure`. Insurance exposure $v \in (0, 1]$ in yearly units in calendar year `Year`. Typically, the exposure is equal to 1, except in `StartYear` because insurance contracts can be signed continuously over the entire calendar year. Contract termination is always set to the end of a calendar year.
- `ClaimNb`. Number of insurance claims in calendar year `Year`.
- `DrivAge`. Driver's age in `Year`. `DrivAge` is between 18 and 90 years; it is censored at 90.
- `Gender`. Binary indicator for the gender of the policyholder.
- `Region`. Living place of the policyholder, this is a categorical variable with 15 levels having the labels `R11` to `R25`.
- `Leasing`. Binary indicator for car leasing.
- `LowMileage`. Binary indicator for low mileage.
- `CatPrice`. Catalogue price of the vehicle being between $9,000$ and $100,000$.
- `VehAge`. Vehicle age in `Year`. `VehAge` is between 0 and 20 years; it is censored at 20.
- `Deductible`. Binary indicator for deductible.
- `TermCar`. Indicator whether the vehicle has been changed during `Year`, see `CarID`; this variable is not tracked if the contract is terminated in the same calendar year, see `EndYear`.
- `EndYear`. Calendar year of contract termination, if `NA` the contract is still active at the end of calendar year 2019.
- `True`. True expected number of claims that has been used to generate `ClaimNb`; `True` already includes `Exposure`. Note that `True` is *not available* in a real insurance modeling situation.

The simulated numbers of claims, `ClaimNb`, are based on the true expected numbers of claims, `True` $= v\mu^*$, where $\mu^* > 0$ is the true expected claim frequency and $v \in (0, 1]$ is the `Exposure` of the insurance policy in the corresponding calendar year. The true expected claim frequencies have been constructed by a regression function

$$\boldsymbol{x} \;\mapsto\; \mu^*(\boldsymbol{x}),$$



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where the covariates $\boldsymbol{x}$ reflect policyholder information, calendar year information, as well as past claims history. Based on this we have simulated independent claim numbers using a negative binomial distribution

$$N\,|_{\{\boldsymbol{x},v\}} \;\sim\; \text{NegBin}\,(v\mu^*(\boldsymbol{x}), \varphi^*)\,, \tag{A.1}$$

having conditional mean and variance, respectively,

$$\mathbb{E}\,[\,N|\,\boldsymbol{x}, v] = v\mu^*(\boldsymbol{x}) \qquad \text{and} \qquad \text{Var}\,(\,N|\,\boldsymbol{x}, v) = \varphi^*\, v\mu^*(\boldsymbol{x}), \tag{A.2}$$

with (over-)dispersion parameter $\varphi^* = 1.3 > 1$. In the usual gamma parametrization of the negative binomial distribution we have, see Wüthrich [200, Proposition 2.21],

$$\text{Var}\,(\,N|\,\boldsymbol{x}, v) = v\mu^*(\boldsymbol{x}) + (v\mu^*(\boldsymbol{x}))^2/\alpha(\boldsymbol{x}),$$

with shape parameter $\alpha(\boldsymbol{x}) = v\mu^*(\boldsymbol{x})/(\varphi^* - 1) > 0$.

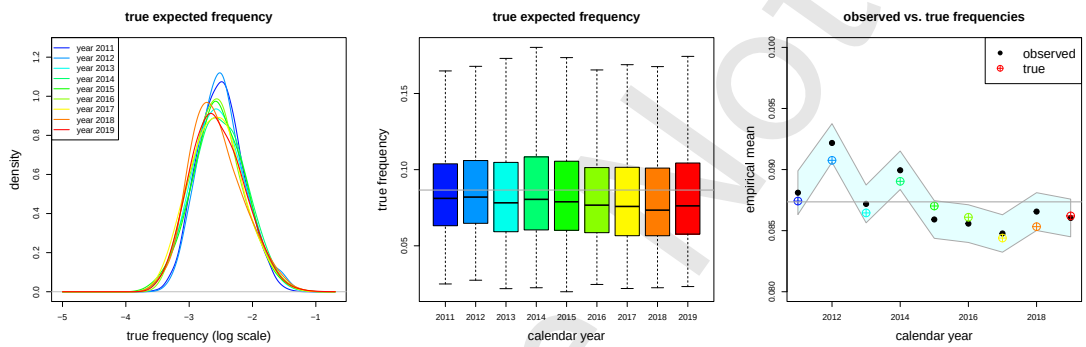


Figure A.1: True expected frequencies $\mu^*(\boldsymbol{x})$: (lhs) densities on log scale for each calendar year, (middle) boxplots, and (rhs) empirical averages vs. true averages, the shaded area shows estimated two standard deviations confidence bounds based on an independent Poisson assumption.

Figure A.1 (lhs) shows the empirical densities of the true expected frequencies $\mu^*(\boldsymbol{x}_{i,t})$ over all policies $(\boldsymbol{x}_{i,t})_i$ in each calendar year $t = 2011, \ldots, 2019$ (different colors). Figure A.1 (middle) gives the corresponding boxplots (excluding outliers outside of the whiskers). Figure A.1 (rhs) shows the empirical frequencies from the observations $N_{i,t}$ (black dots) vs. the true weighted frequencies $\mu^*(\boldsymbol{x}_{i,t})$ (colored dots) with weights/`Exposure` $v_{i,t} \in (0, 1]$ for the calendar years $t = 2011, \ldots, 2019$. The cyan band gives two standard deviations confidence bounds being based on an independent Poisson assumption. These confidence bounds are too narrow because the claim counts $N_{i,t}$ have been generated by (over-dispersed) negative binomial distributions (A.1). However, in practice, this is not known, and one therefore just operates with a standard Poisson assumption for the construction of these confidence bounds. We observe business cycles/non-stationarity which may make it difficult to forecast and model this data.

Next, we are going to analyze the covariates $\boldsymbol{x}_{i,t}$. We start with $150,000$ insurance policies in calendar year $t = 2011$. Some of these insurance policies are renewed in calendar year $t + 1 = 2012$, some of these insurance policies are renewed but for a new vehicle (`CarID` changes), and some of these insurance policies are terminated (indicated

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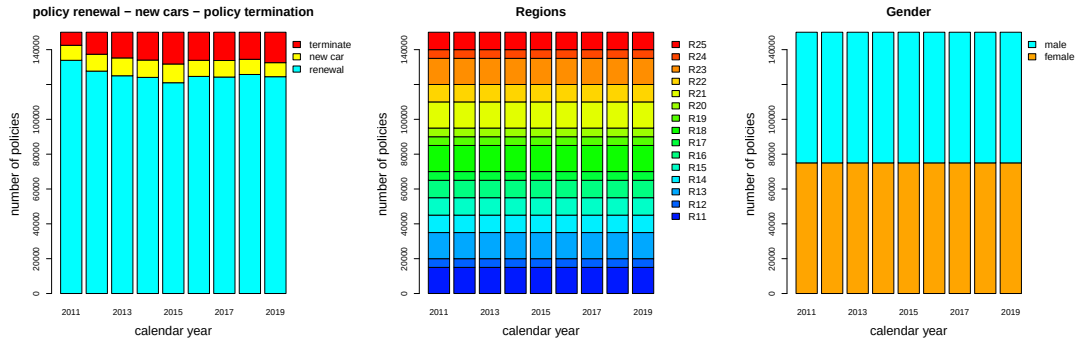


Figure A.2: (lhs) policy renewal, new cars and policy termination, (middle) `Region` and (rhs) `Gender` over the different calendar years from 2011 to 2019.

by `EndYear` $= t = 2011$). Figure A.2 (lhs) shows the time-series of renewals, new cars and termination, respectively. For the policies with new cars (yellow), we just randomly simulate a new vehicle with resulting covariates $\boldsymbol{x}_{i,t+1}$ for the next calendar year $t+1$. Terminated policies (red) stop being part of the insurance portfolio, however, at the same time we sign new insurance policies. The simulation process has been designed such that every terminated policy is replaced by a new contract. This new contract is assumed to be in the same `Region` and the new insurance policyholder is assumed to have the same `Gender`. The remaining covariates in $\boldsymbol{x}_{i,t+1}$, like `DrivAge` or the `VehAge`, are simulated at random. This (special) simulation process implies that in all calendar years $t = 2011, \ldots, 2019$ we have $150,000$ active insurance contracts. It is also verified from Figure A.2 (middle, rhs) that the numbers of insurance policies per `Region` and per `Gender` remain constant over time, in fact, the regions have either 5,000, 10,000 or 15,000 insurance policies, and the gender split is 50% in all regions.

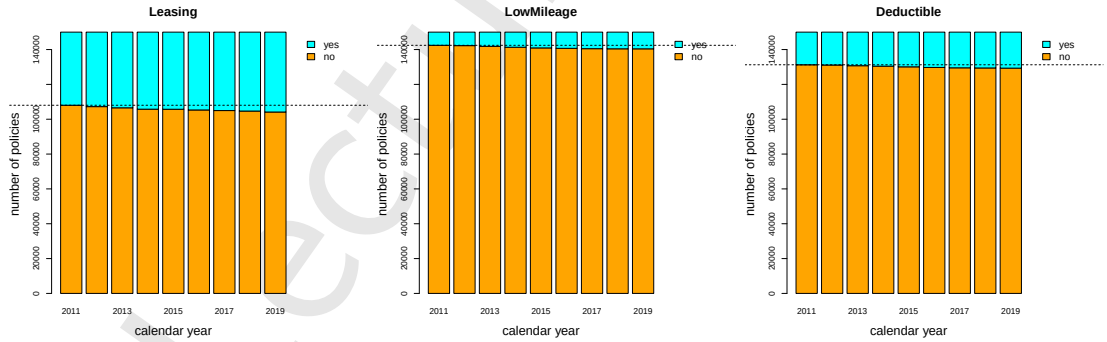


Figure A.3: Development of the binary covariates (lhs) `Leasing`, (middle) `LowMileage` and (rhs) `Deductible` over the different calendar years from 2011 to 2019.

Figure A.3 shows the development of the binary covariates: (lhs) `Leasing`, (middle) `LowMileage` and (rhs) `Deductible` over the different calendar years from 2011 to 2019. We observe in all plots that the ratio of indicator 1 ("yes") is slightly increasing. This increase stems from newly signed contracts and policy renewals with a change of vehicle. The average leasing ratio is 30%, the average low mileage ratio is 6%, and the average deductible ratio is 15%.

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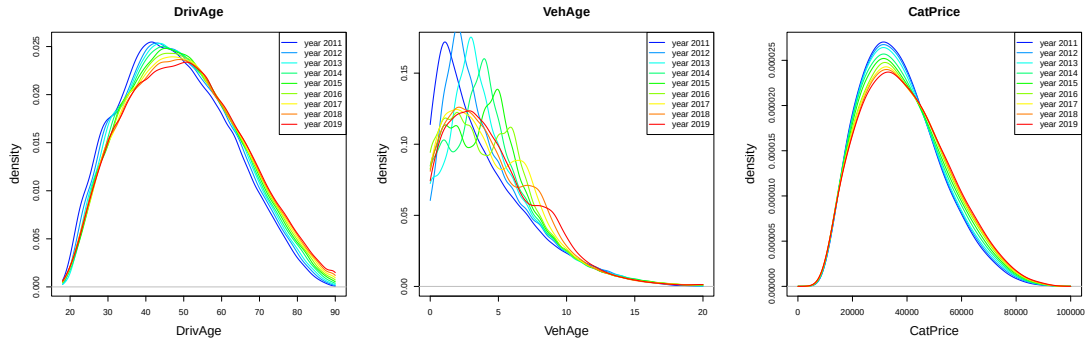


Figure A.4: Density of the continuous covariates for each calendar year: (lhs) `DrivAge`, (middle) `VehAge`, and (rhs) `CatPrice`.

Figure A.4 shows the development of the continuous covariates `DrivAge`, `VehAge` and `CatPrice` over the considered calendar years. We observe that we model an aging portfolio, i.e., both the driver's age distribution and the vehicle age distribution are aging over time, because we do not have sufficiently many new contracts with young people and young new cars that would keep these two covariates in an equilibrium over time. Also the `CatPrice` distribution is increasing, this is rather natural if we think of having price inflation making more recent vehicles more expensive.

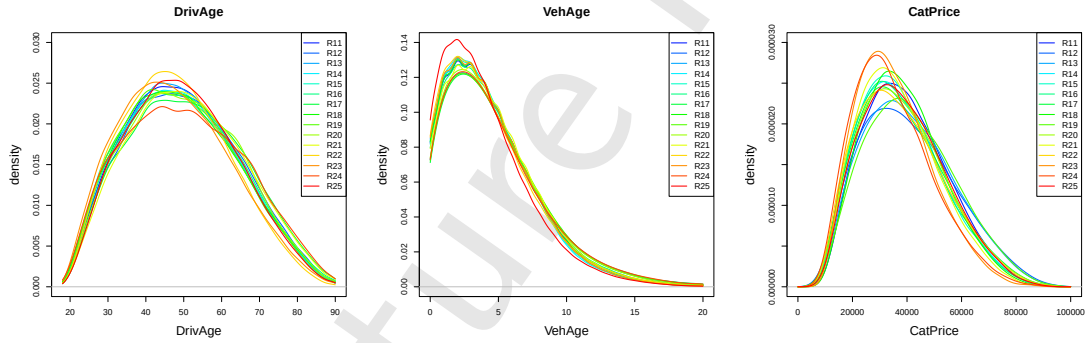


Figure A.5: Density of the continuous covariates for each `Region`: (lhs) `DrivAge`, (middle) `VehAge`, and (rhs) `CatPrice`.

Figure A.5 shows the densities of the continuous covariates per `Region`. We see some differences, especially, in `CatPrice` which says that in some regions cheaper or more expensive cars are preferred. There are more dependencies between the covariates of a similar nature.

Next, we focus on the observed frequencies for the different levels of the categorical covariates. Figure A.6 shows these observed frequencies. We observe bigger differences between the regions, region `R25` having a significantly higher observed frequency than all other regions (the region labels have been ordered w.r.t. the observed frequencies). Then there are three regions `R22`, `R23` and `R24` that have a similar observed frequency, and all remaining regions `R11`–`R21` have much lower observations.

There are no significant differences between `Gender`, see Figure A.6 (top-rhs). `Leasing` has a much higher frequency than policies with non-leased vehicles, see Figure A.6

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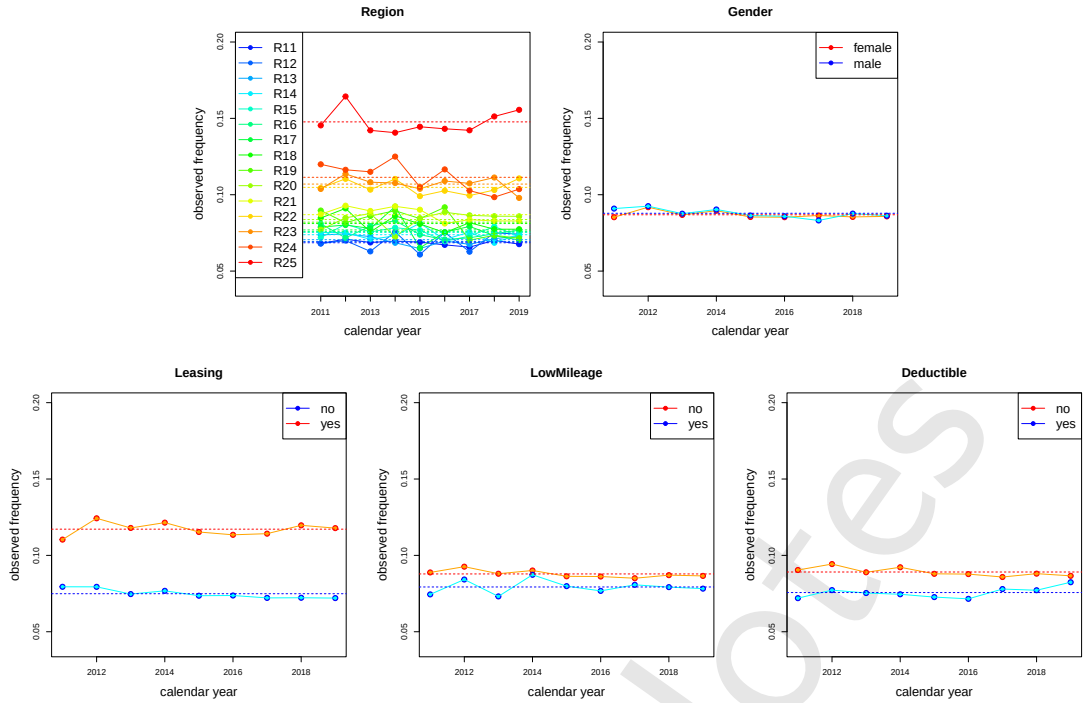


Figure A.6: Observed claim frequencies for different levels of the categorical covariates; the $y$-scale is identical in all graphs.

(bottom-lhs), of course, for this covariate we should also study interactions with `VehAge`. Low mileage and a deductible positively affect the observed frequencies, see Figure A.6 (bottom-middle/rhs).

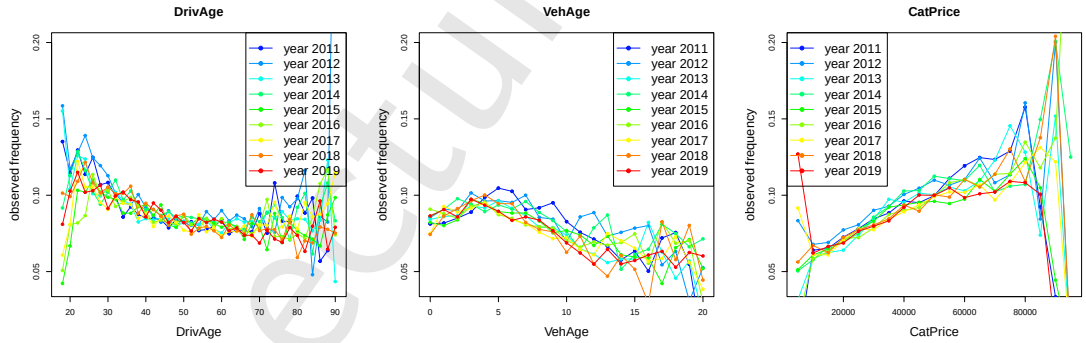


Figure A.7: Observed claim frequencies of the continuous covariates; the $y$-scale is identical to Figure A.6.

Figure A.7 shows the observed frequencies as a function of the continuous covariates. We observe a decreasing observed frequency in `DrivAge` and `VehAge`, and an increasing behavior in `CatPrice`. The volatility in these graphs is induced by scarcity of observations, e.g., we do not have many cars with a `CatPrice` above $80,000$.

Figure A.8 presents interesting graphs. On the left-hand side we show the empirical frequencies on insurance policies with a change of vehicle vs. the ones that do not have a change of vehicle. Policies with a change of vehicle have a significantly higher observed

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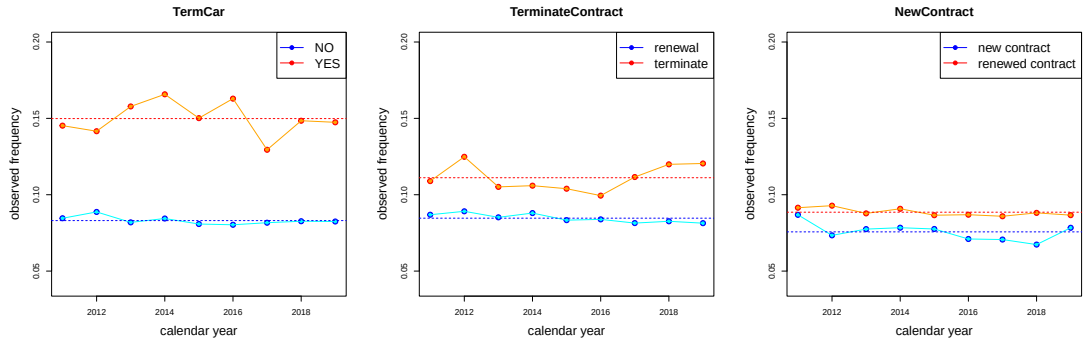


Figure A.8: Observed claim frequencies of the contracts with (lhs) a change of vehicle, (middle) terminated contracts, and (rhs) new contracts; the $y$-scale is identical to Figure A.6.

frequency which indicates that an accident is an (obvious) reason for changing a vehicle. The same applies to contract termination, i.e., an accident is a reason for terminating an insurance contract, see Figure A.8 (middle). Either this policyholder completely stops driving after an accident, or she/he purchases a new car, but insures this new car with another insurance company. On the other hand, we observe that newly signed contracts have a lower observed frequency than renewed contracts, see Figure A.8 (rhs). The explanation of this phenomenon is more difficult. We observe that very new cars have lower frequencies, see Figure A.8 (middle), and new contracts often have new cars. The specific reason for this can be manifold and would require further investigation, e.g., one would need to have more information about the specific type of insurance cover like comprehensive vs. collision cover, etc.

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